

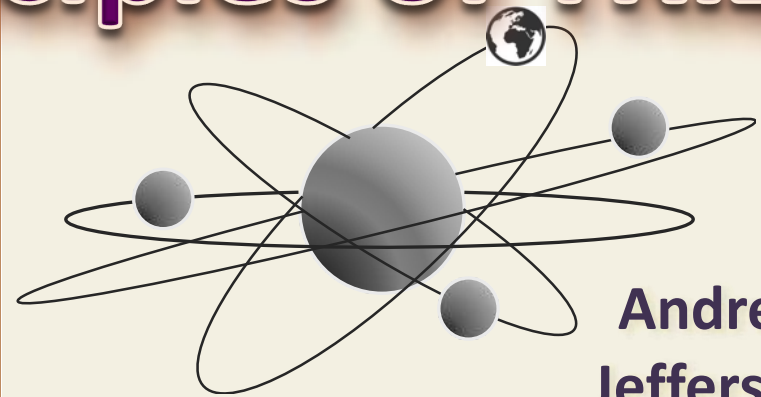


Accelerator Science and Technology via

Inventive Principles of TRIZ



LHC sketches by Sergio Cittolin (CERN)

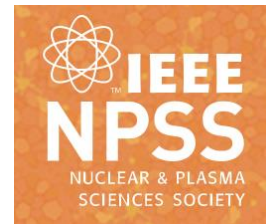


Andrei Seryi
Jefferson Lab

NA-PAC 2022, 10 August 2022

Acknowledgements

- **Jefferson Lab**
 - Continuing support of graduate educational program
- **Old Dominion University**
 - Materials presented here will be used as part of VITA graduate and undergraduate courses
 - VITA – Virginia Innovative Traineeship in Accelerators
 - ODU, NSU, HU – three universities – VITA
- **NA-PAC 2022**
 - For the opportunity to present this tutorial
- **IEEE NPSS Distinguished Lecture Program**
 - <https://ieee-npss.org/distinguished-lecturers/>
- **My co-author for the 1st and oncoming 2nd edition of the book “Unifying Physics of Accelerators, Lasers and Plasma” Elena Seraia**
 - <https://www.unifyingphysics.com/>
- **CERN for “eBook for all!” program that enabled conversion of the 1st edition of “Unifying Physics...” to open access**



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Look for the IEEE Membership booth!

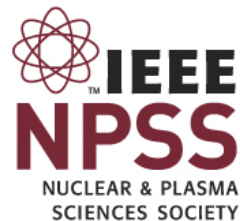
NPSS Journals

IEEE Transactions on Nuclear Science

IEEE Transactions on Plasma Science

IEEE Transactions on Medical Imaging

IEEE Transactions on Radiation and Plasma Medical Sciences



Tutorial materials

- Slides are available at
 - <https://www.unifyingphysics.com/>



- See section Resources
- You can also access the 1st edition of the book which is now Open Access

Scientific revolutions – what drives them?

Two points of view:

Philosopher Thomas Kuhn:

scientific revolutions are concept-driven

“paradigm shifts”

Physicist Freeman Dyson:

scientific revolutions are tool-driven

“The human heritage that gave us toolmaking hands and inquisitive brains did not die. In every human culture, the hand and the brain work together to create the style that makes a civilization....

Science will continue to generate unpredictable new ideas and opportunities. And human beings will continue to respond to new ideas and opportunities with new skills and inventions. We remain toolmaking animals, and science will continue to exercise the creativity programmed into our genes.”

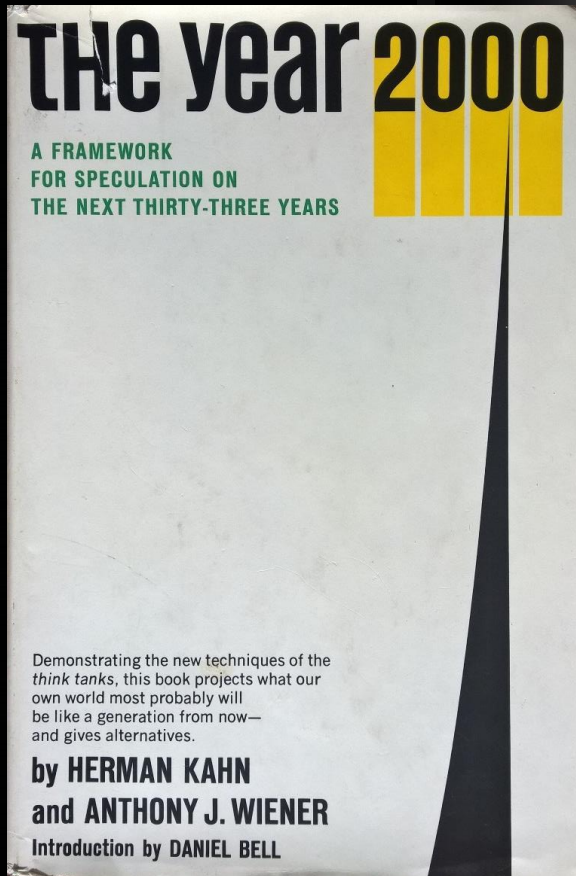


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**We would like
to make
positive and
proactive
impact on the
evolution of
science and
technology**

**Can we learn
from past
efforts to make
our impact
more reliable
and efficient?**

Predictions made in 1968 for the year 2000



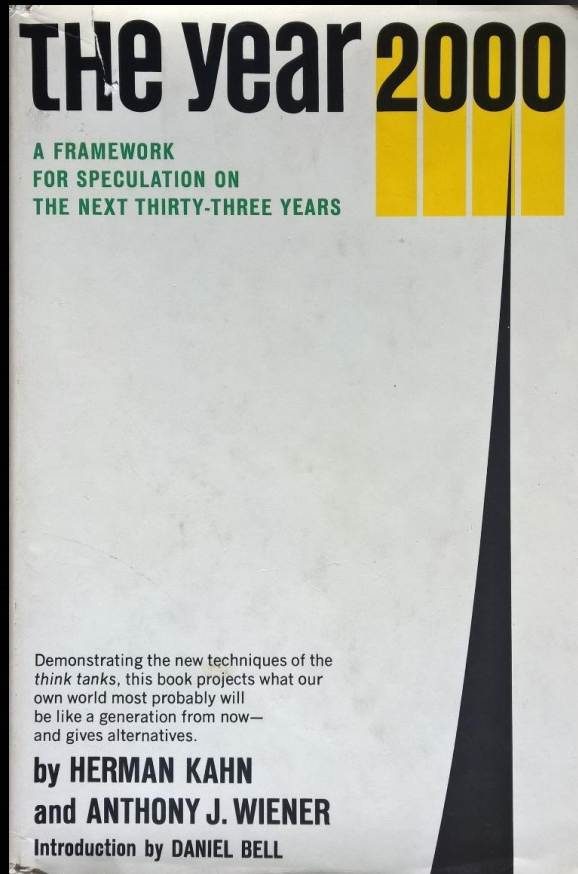
**“The Year 2000”, 1968
K. Herman, A. Wiener
ISBN 978-0025604407**



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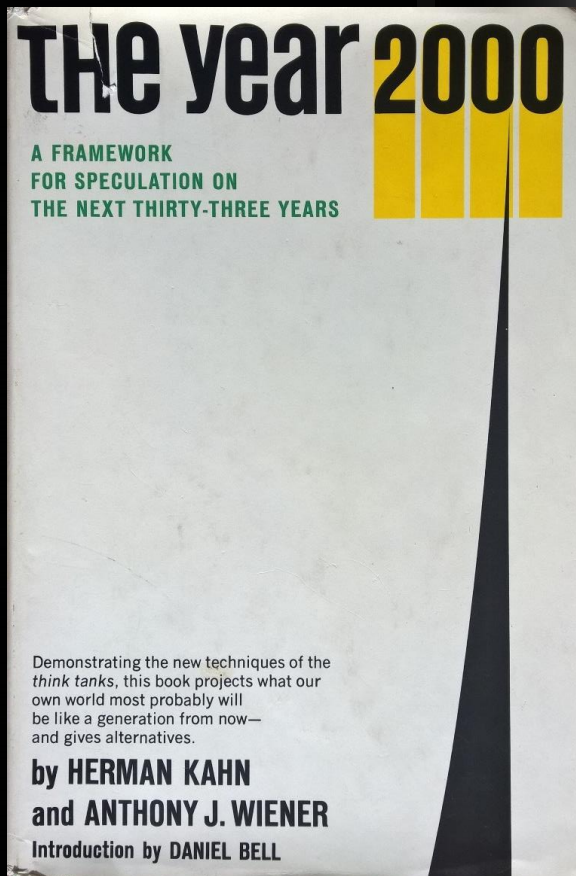


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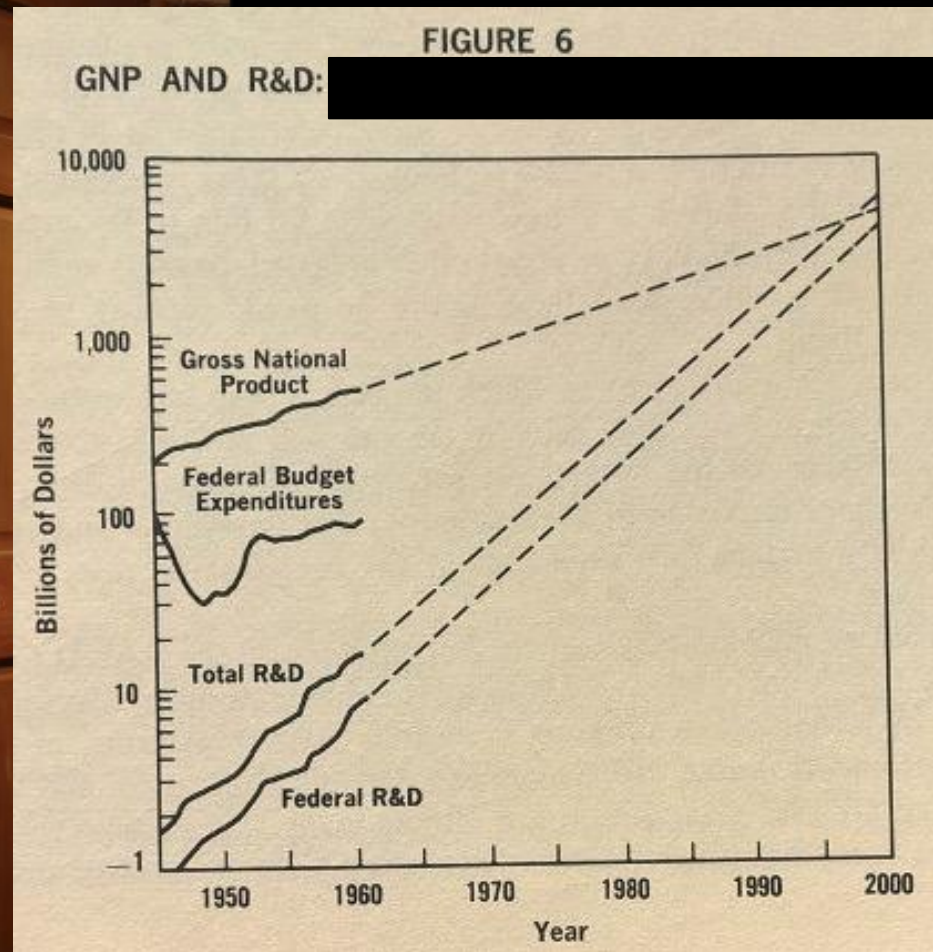
**Can the
methodology
of predictions
be reliable?**

Predictions made in 1968 for the year 2000

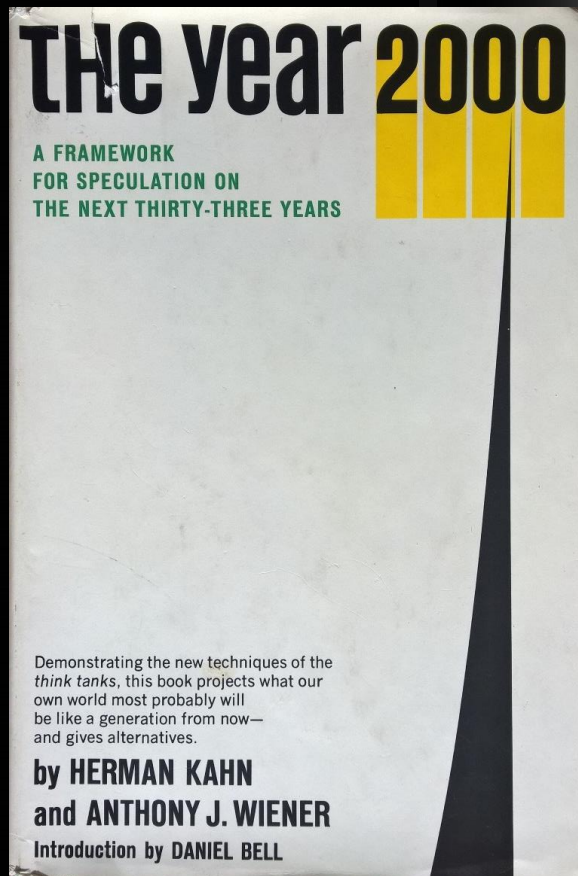


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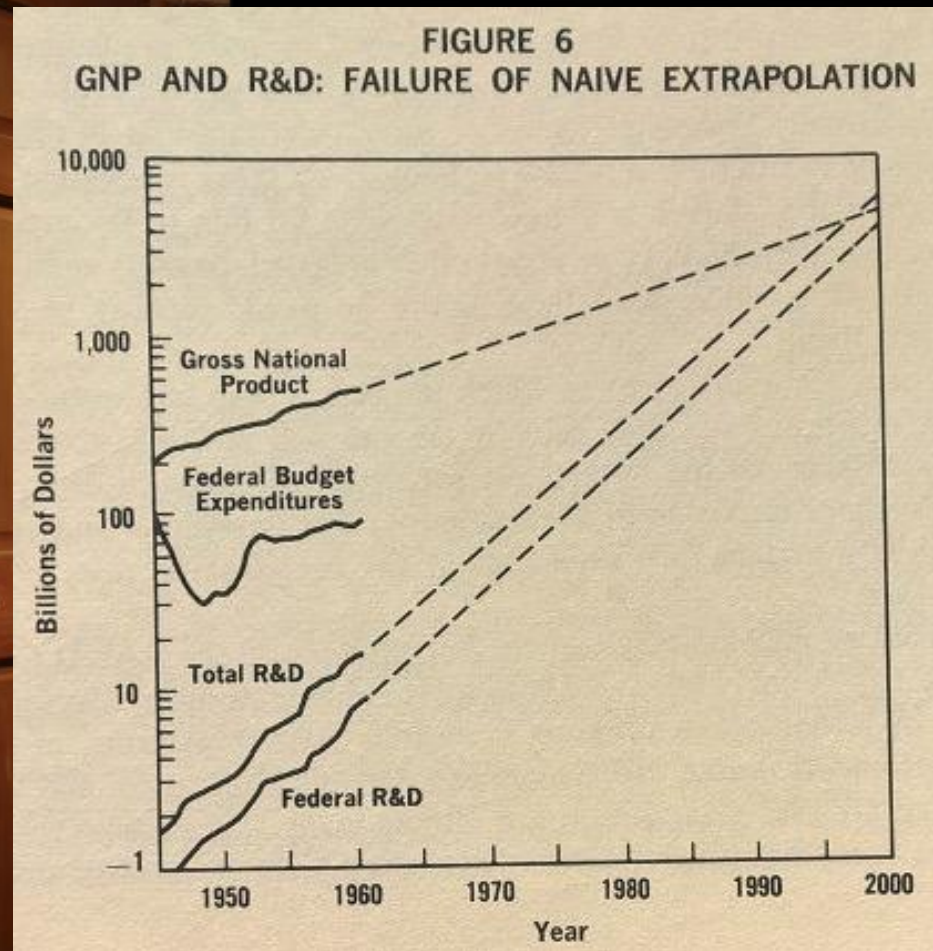


Predictions made in 1968 for the year 2000



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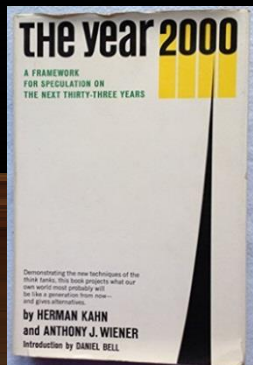
Lesson: avoid naïve
extrapolations

Predictions made in 1968 for the year 2000, examples:

1- Multiple applications of
lasers for sensing,
communication, cutting,
welding... ✓

31- Some control of weather
and/or climate

35 – human hibernation for
extensive periods (months to
years)



“The Year 2000”, 1968
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58- Chemical methods for
improving memory and
learning ~✓

67- Commercial extraction of
oil from shale ✓

81- Personal “pagers” and
perhaps even two-way pocket
phones ✓

99- Artificial moon for
lighting large areas at night

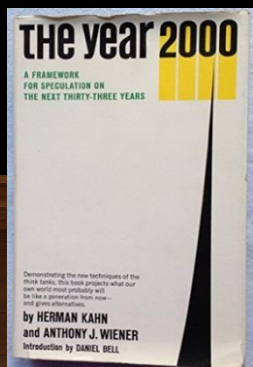
Some predictions
were accurate,
some not

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phones ✓

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lighting large areas at night ~✓

Some predictions
were accurate,
some not

Chengdu to launch "artificial moon" in 2020

(People's Daily Online) 09:06, October 16, 2018



“Southwestern China’s city of Chengdu plans to launch its illumination satellite, also known as the “artificial moon”, in 2020, according to Wu Chunfeng, chairman of Chengdu Aerospace Science and Technology Microelectronics System Research Institute Co., Ltd. ... The illumination satellite is designed to complement the moon at night. Wu introduced that the brightness of the “artificial moon” is eight times that of the real moon, and will be bright enough to replace street lights. The satellite will be able to light an area with a diameter of 10 to 80 kilometers, while the precise illumination range can be controlled within a few dozen meters. ...The testing of the illumination satellite started years ago, and now the technology has finally matured, explained Wu. Some people expressed concern that the lights reflected from space could have adverse effects on the daily routine of certain animals and astronomical observation. Kang Weimin, director of the Institute of Optics, School of Aerospace, Harbin Institute of Technology, explained that the light of the satellite is similar to a dusk-like glow, so it should not affect animals’ routines.”

<http://en.people.cn/n3/2018/1016/c90000-9508748.html>

To make viable predictions and efficient research plans:

Learn from the past of this particular area of science...

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...but also look around, across different disciplines and areas of science...

And, possibly, use “Breakthrough By Design” approach...

...not only for prediction, but for pro-actively shaping the future

Evolution laws and inventive principles

Are there some patterns in evolution of scientific and engineering systems and instruments?

Are there some general inventive principles that connect different instruments/systems in different areas?

Let's look at some examples from a new angle

Two scientific instruments



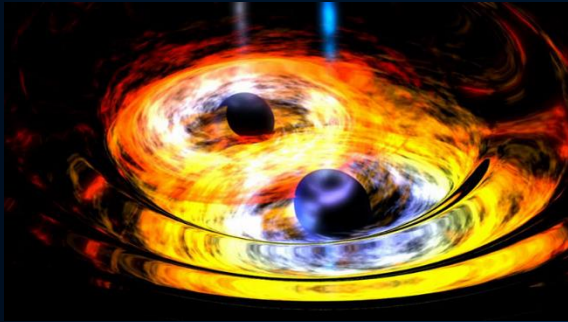
LIGO, Hanford



SLC, Stanford

What is in common?

2017 Nobel Prize in Physics – gravitational waves

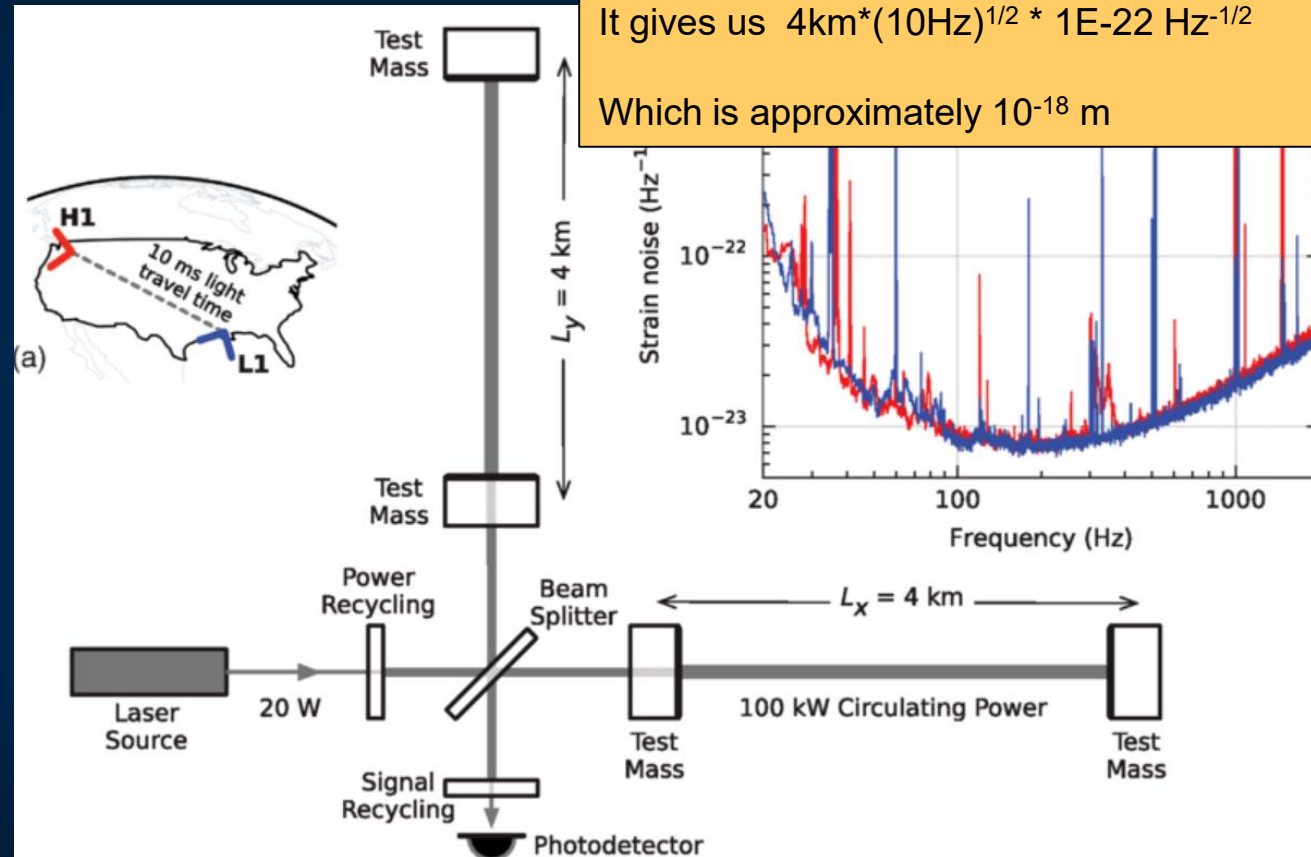
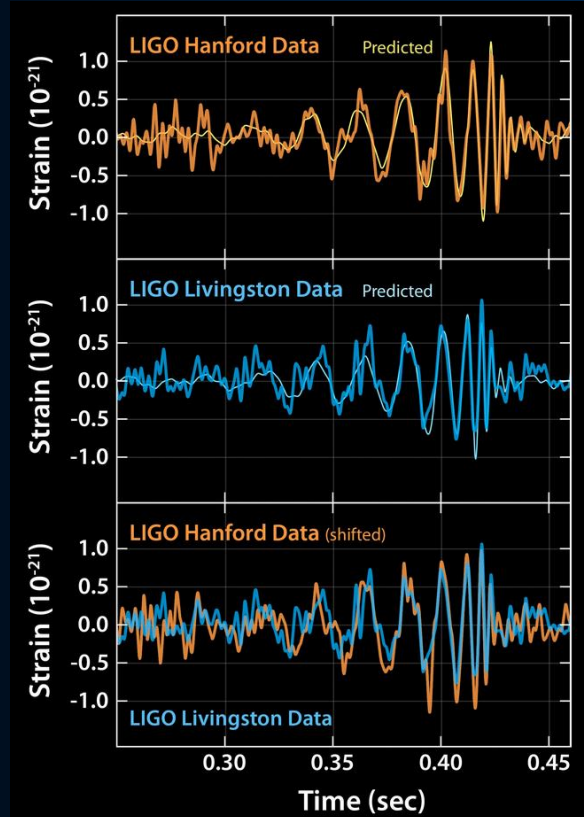


What are these numbers?

Let's say we would like to evaluate noise between 20Hz and 30Hz (i.e. $\Delta f = 10\text{Hz}$), where strain noise is about $1\text{E-}22 \text{ Hz}^{-1/2}$

It gives us $4\text{km} \cdot (10\text{Hz})^{1/2} \cdot 1\text{E-}22 \text{ Hz}^{-1/2}$

Which is approximately 10^{-18} m



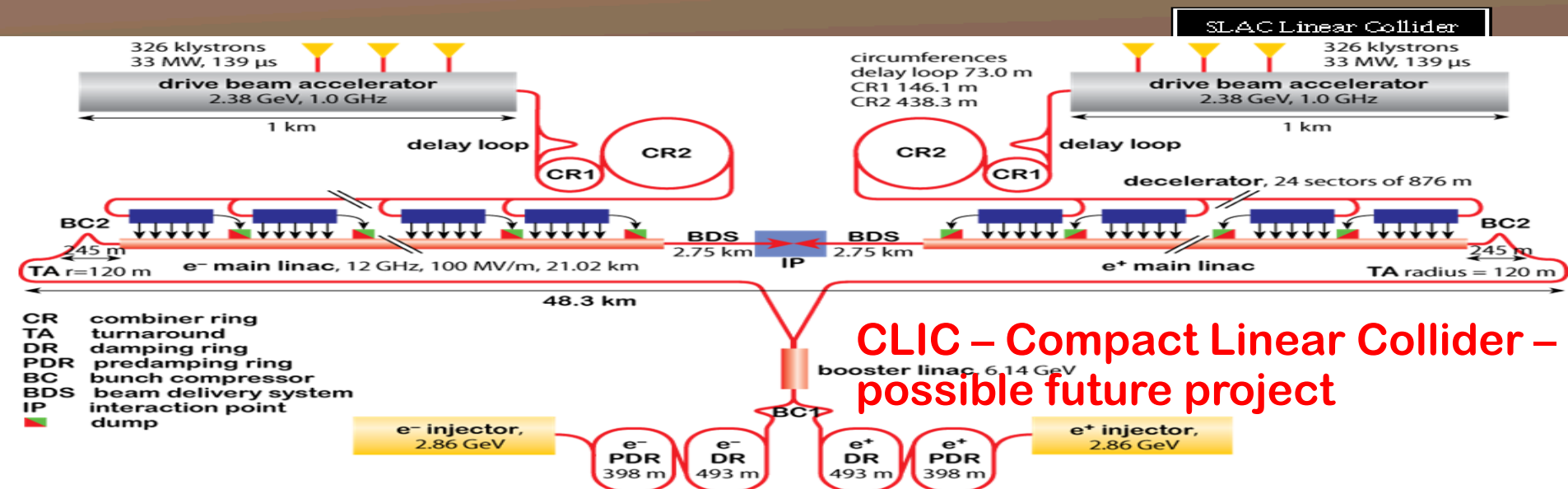
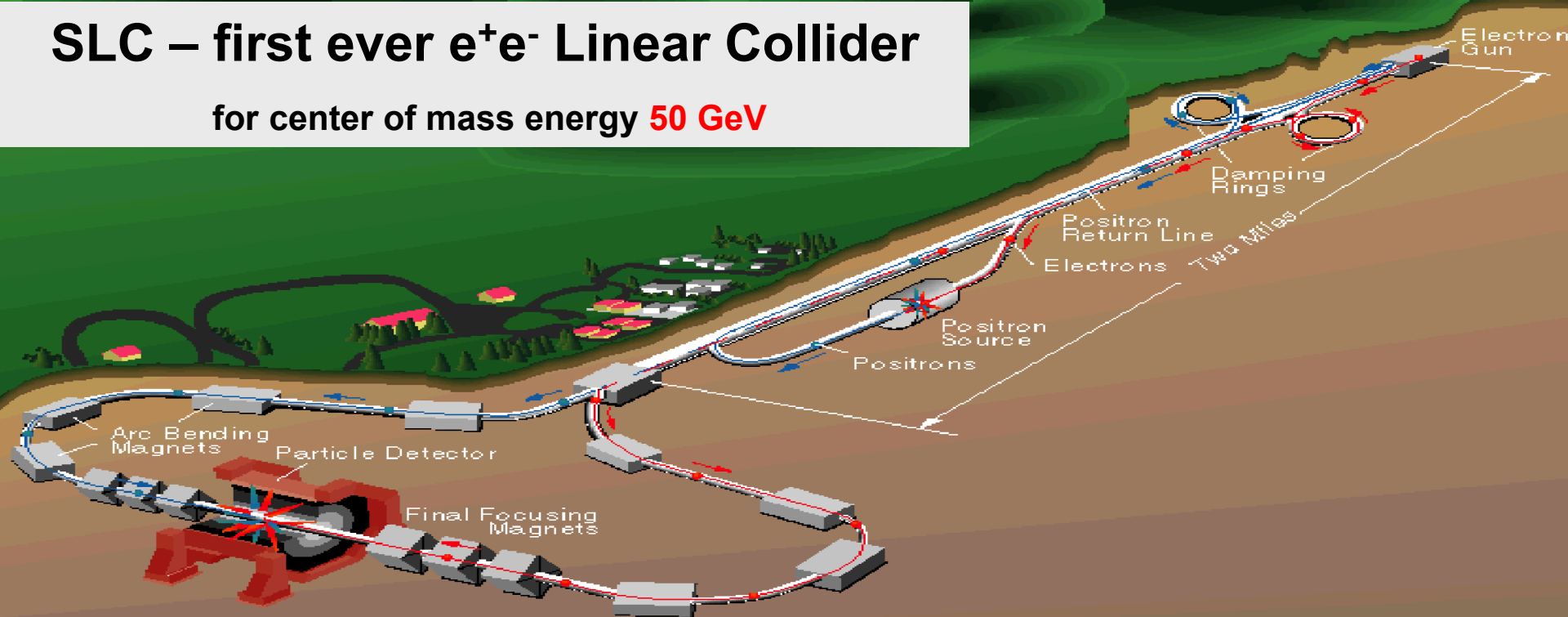
LIGO layout and sensitivity curve

Source: PRL 116, 061102 (2016)

Image: Caltech/MIT/LIGO Lab

SLC – first ever e^+e^- Linear Collider

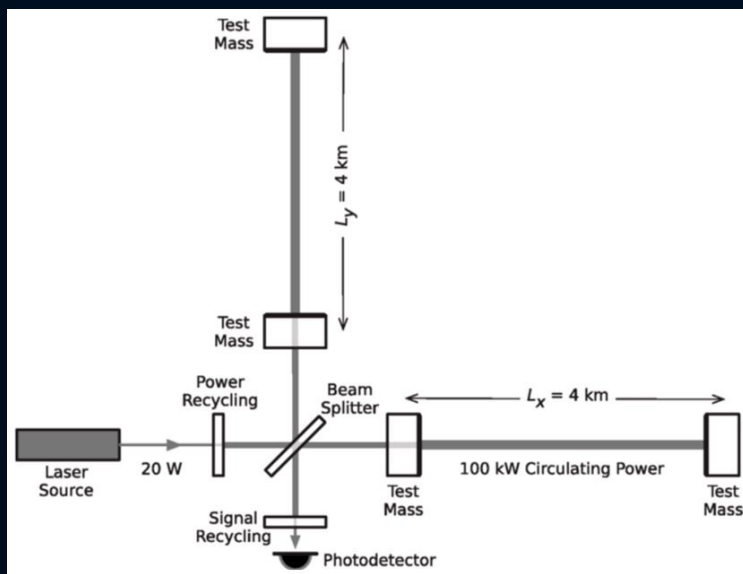
for center of mass energy **50 GeV**



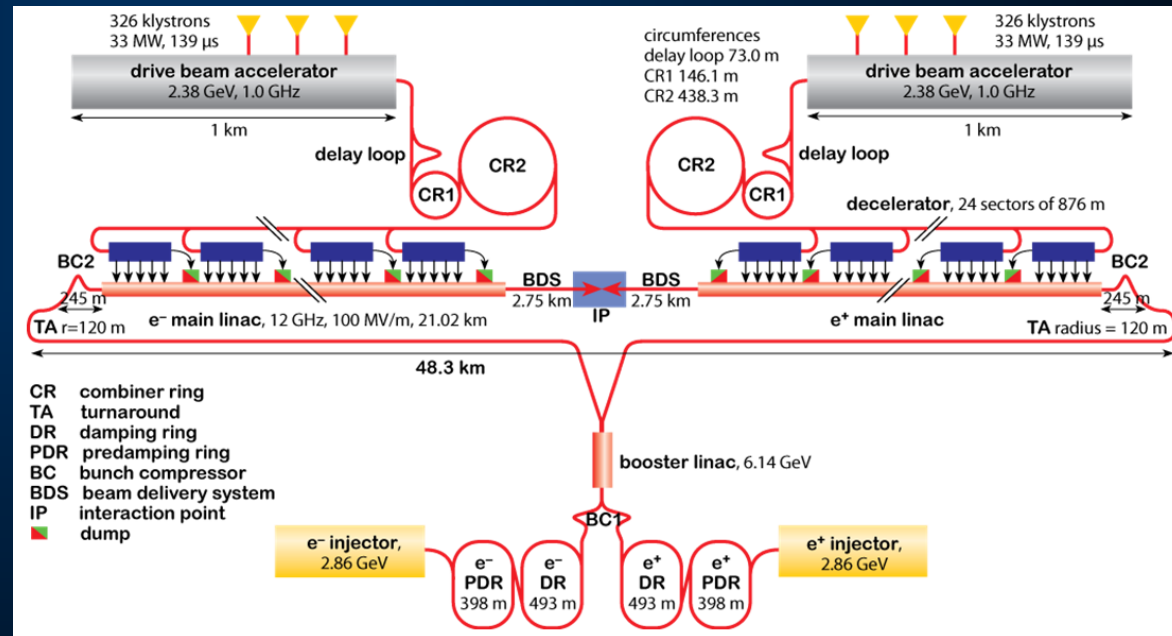
CLIC – Compact Linear Collider – possible future project

These two instruments

LIGO: keep two objects placed 4km apart stable* to about $1\text{e-}9\text{ nm}$

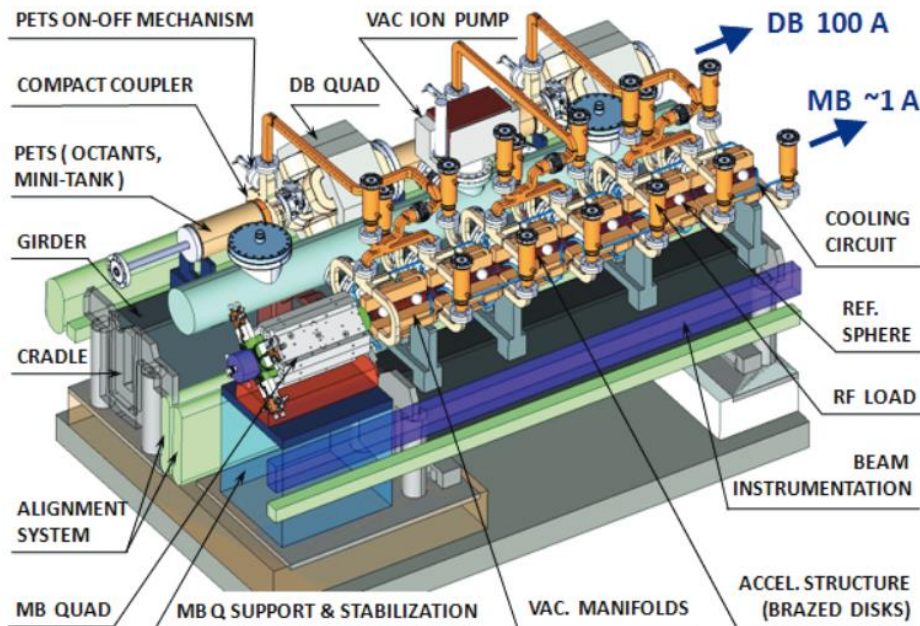


CLIC –
Compact Linear Collider:
keep 100,000 objects
distributed over 50km
stable* to about 10 nm

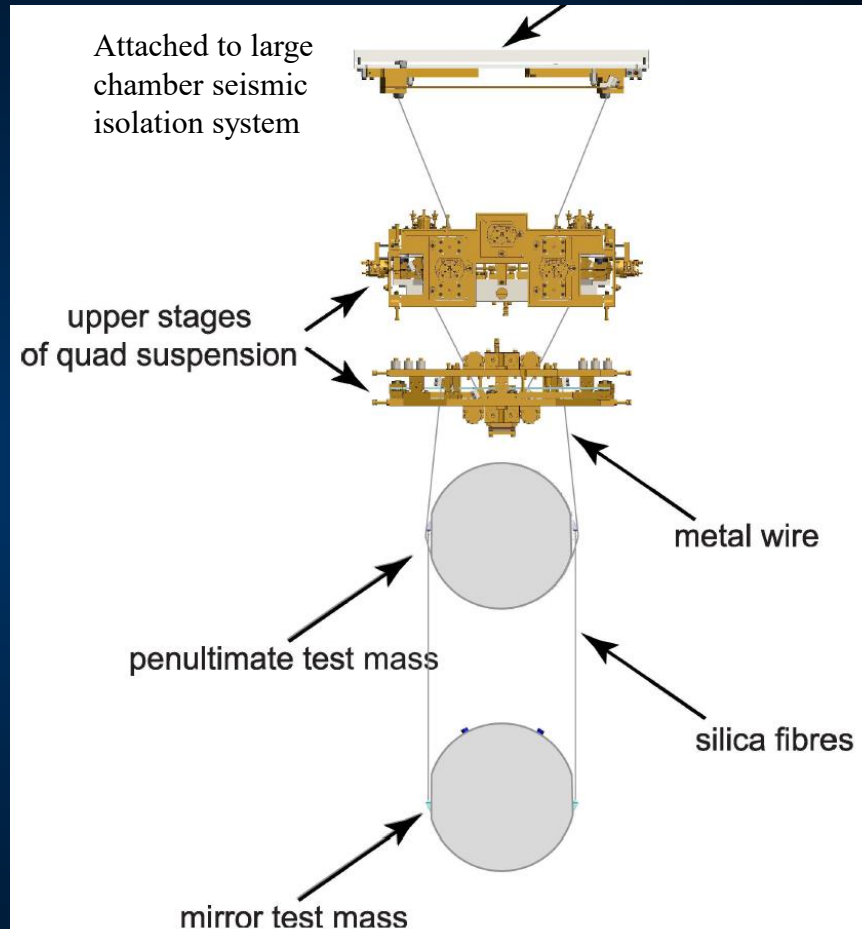


*) approximately, and in certain frequency range

CLIC stability & LIGO test mass isolation



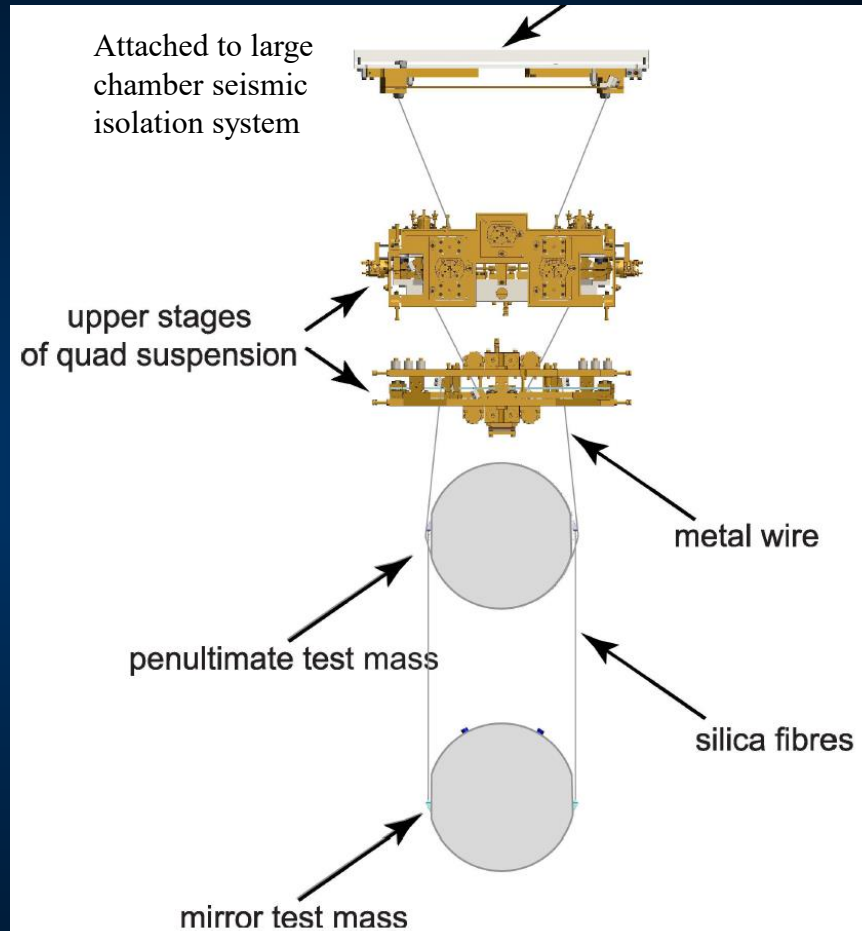
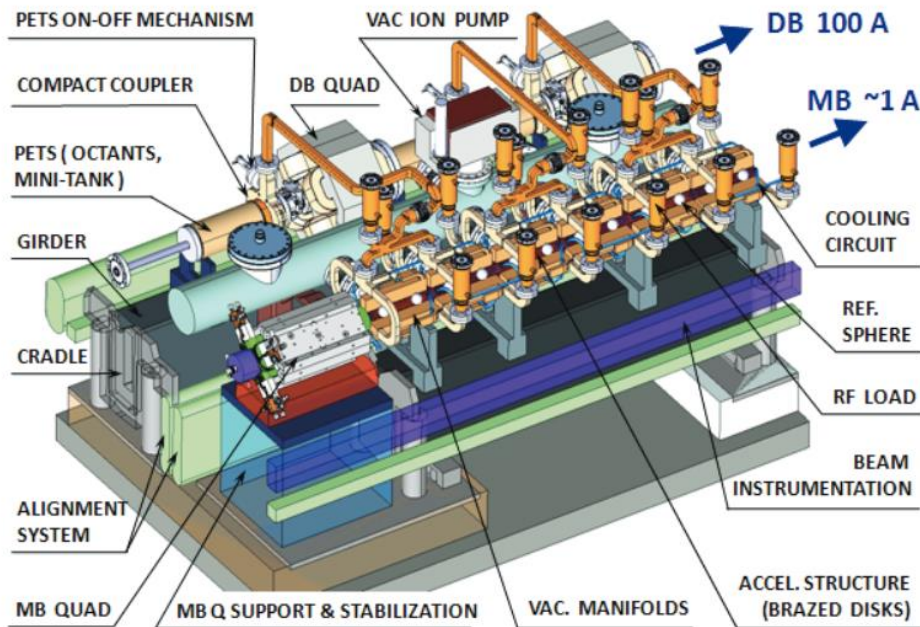
Nested stabilization/alignment systems of CLIC two-beam module



Nested pendulums of LIGO

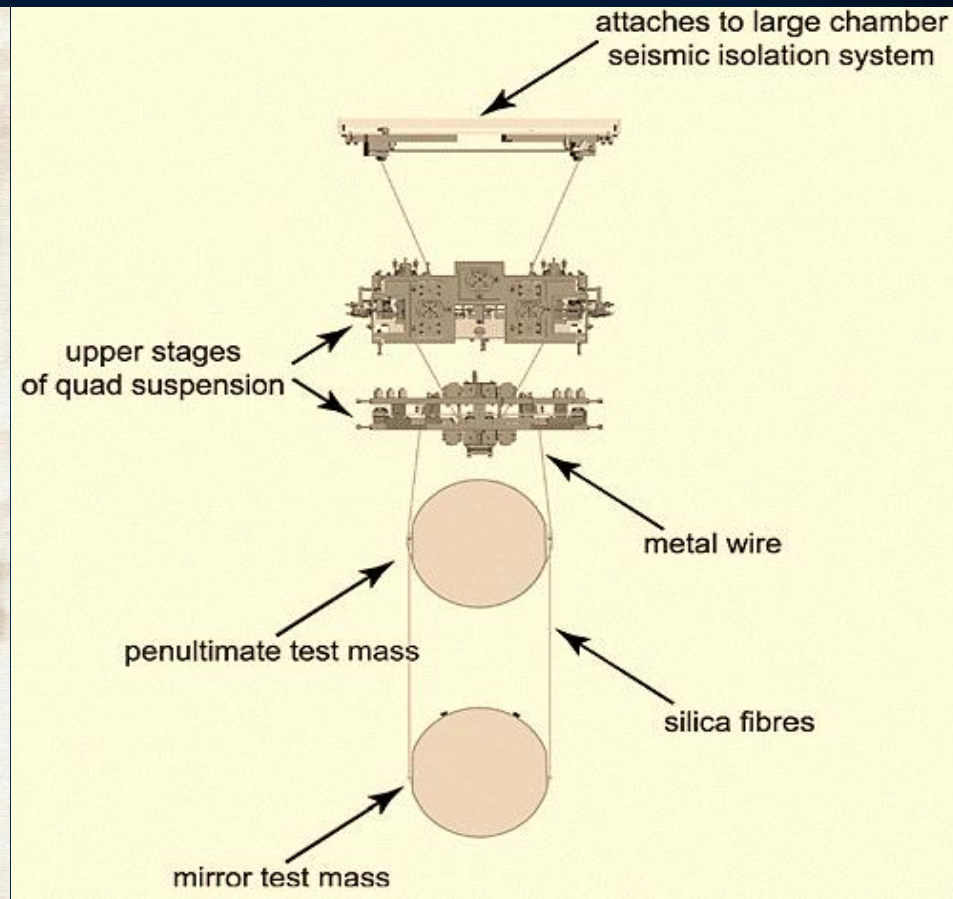
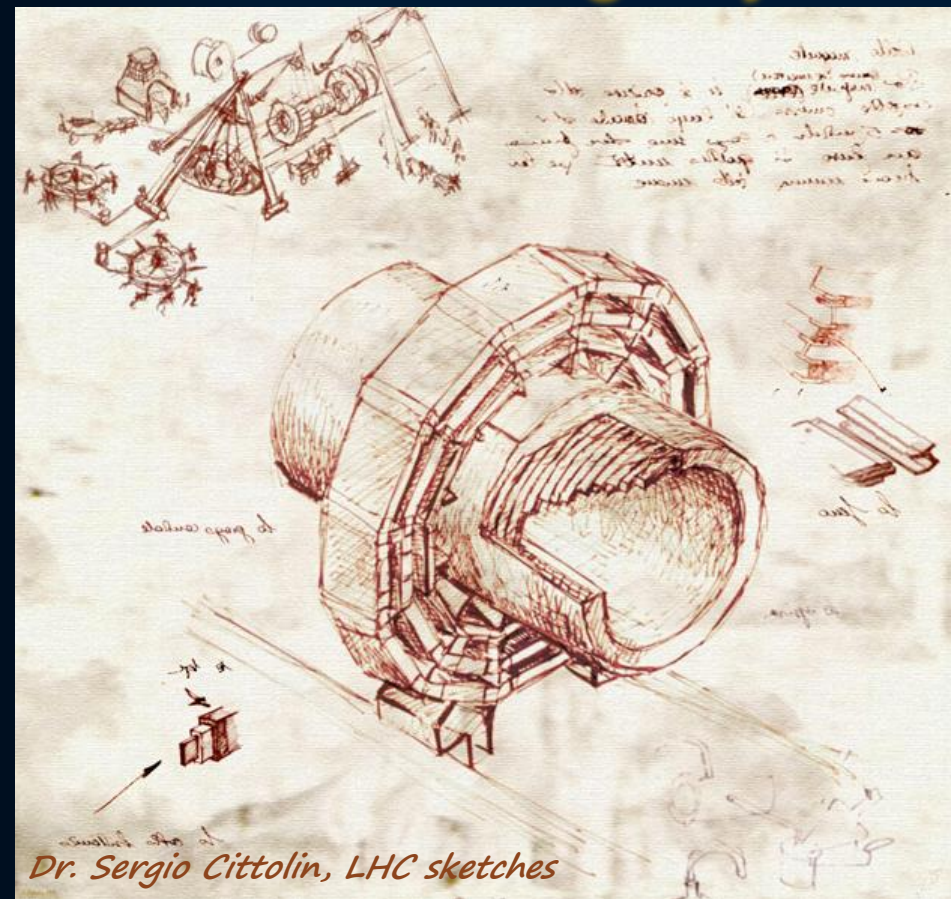
Source: [arXiv:1102.3355](https://arxiv.org/abs/1102.3355)

CLIC stability & LIGO test mass isolation



... connected via an inventive principle – let's call it the principle of "nested dolls"

Particle or gravitational waves detectors are arranged just as nested dolls...



Examples of “nested dolls” *inventive principle* can be found in various areas

The principle of “nested dolls” in poetry

“This is the house that Jack built”

This is the house that Jack built.

This is the malt
That lay in the house that Jack built.

This is the rat,
That ate the malt
That lay in the house that Jack built.

This is the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the cow with the crumpled horn,
That tossed the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.



Illustration by Olga Rubtsova (Atroshenko)



This is the maiden all forlorn,
That milked the cow with the crumpled horn,
That tossed the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the man all tattered and torn,
That kissed the maiden all forlorn,
That milked the cow with the crumpled horn,
That tossed the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the priest all shaven and shorn,
That married the man all tattered and torn,
That kissed the maiden all forlorn,
That milked the cow with the crumpled horn,
That tossed the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the cock that crowed in the morn,
That waked the priest all shaven and shorn,
That married the man all tattered and torn,
That kissed the maiden all forlorn,
That milked the cow with the crumpled horn,
That tossed the dog,
That worried the cat,
That killed the rat,
That ate the malt
That lay in the house that Jack built.

This is the farmer sowing his corn,
That kept the cock that crowed in the morn,
That waked the priest all shaven and shorn,
That married the man all tattered and torn,
That kissed the maiden all forlorn,
That milked the cow with the crumpled horn,
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Mother Goose Rhymes

The principle of “nested dolls” in poetry

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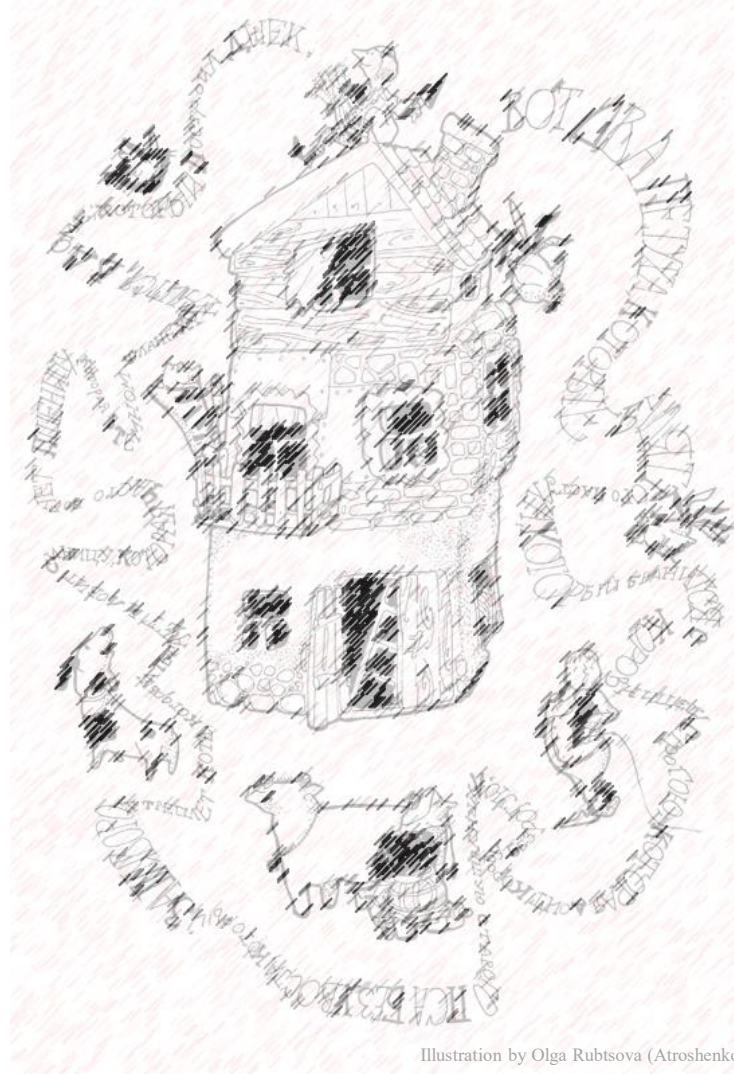


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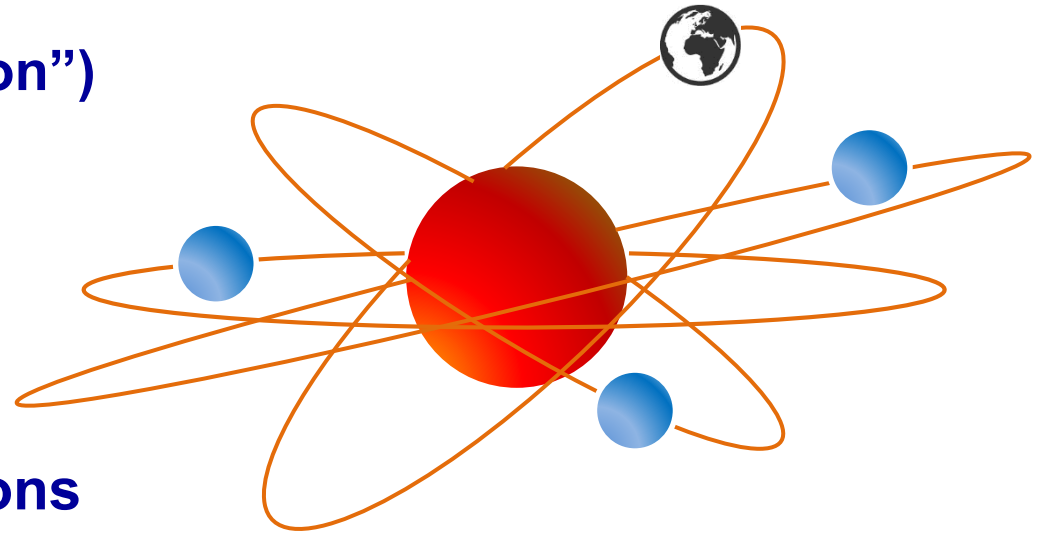
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Mother Goose Rhymes

Is there any example of this principle in science fiction?

The principle of “nested dolls” in sci-fi poetry

Valery Bryusov – 1920 poem
“Atom” (“The World of Electron”)



Can you imagine that electrons
Are planets circling their Suns?
Space exploration, wars, elections
And hundreds of computer tongues

...

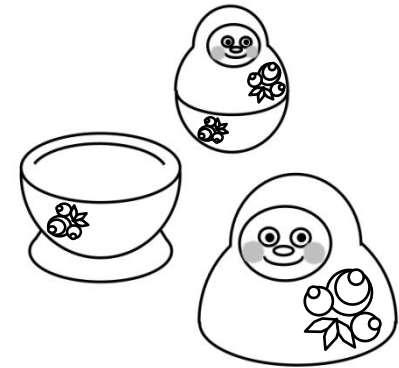
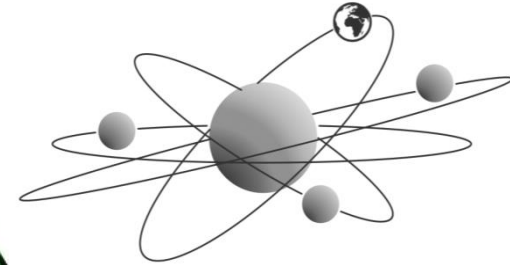
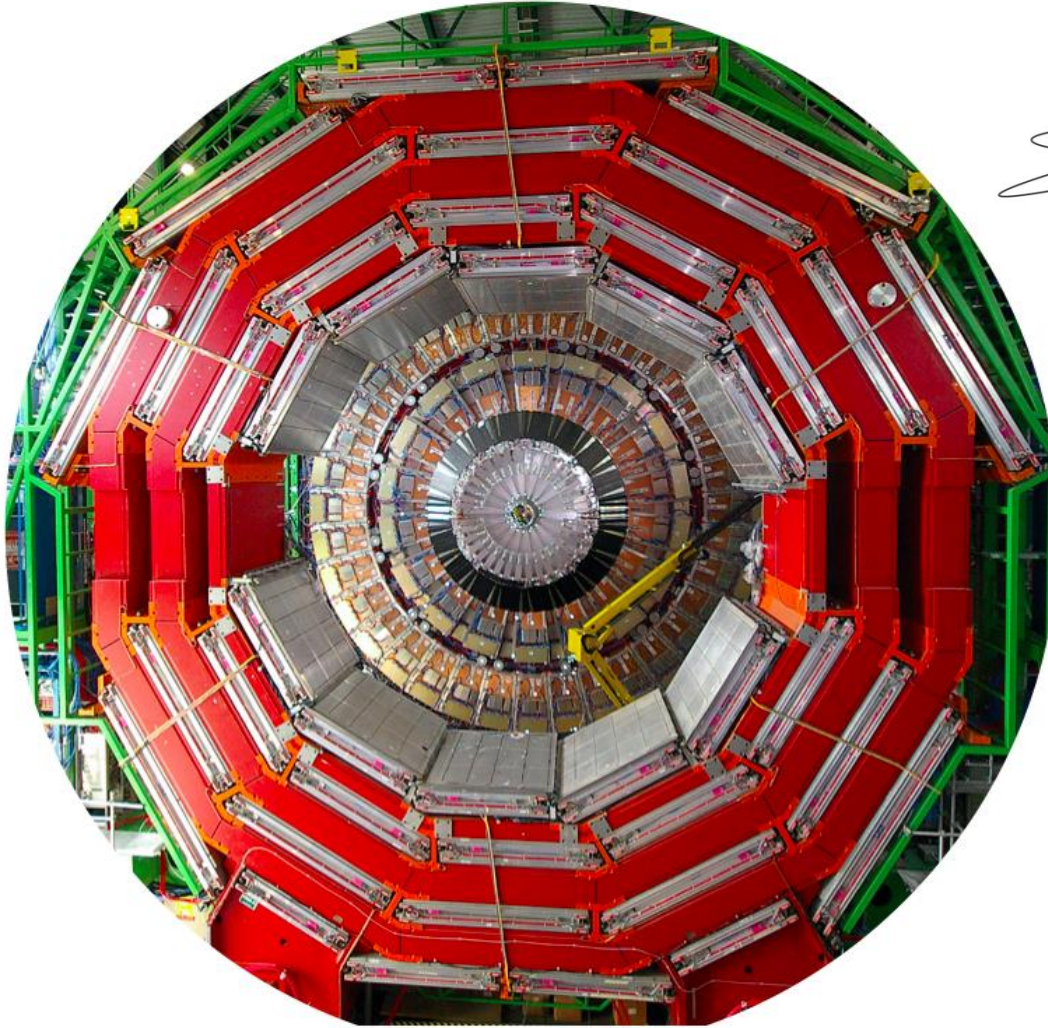
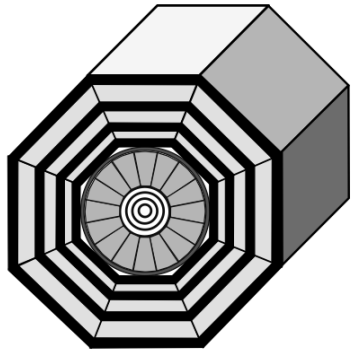
Remake-translation by A.Seryi

Быть может, эти электроны
Миры, где пять материков,
Искусства, знанья, войны, троны
И память сорока веков!

Ещё, быть может, каждый атом —
Вселенная, где сто планет;
Там — всё, что здесь, в объёме сжатом,
Но также то, чего здесь нет.

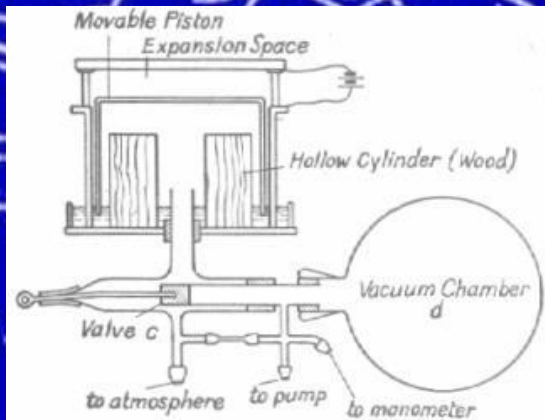
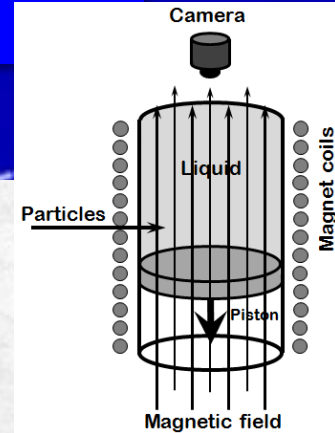
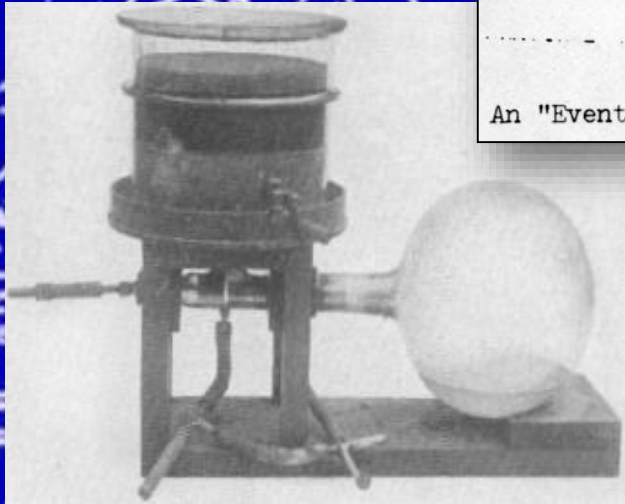
...

Is there world inside of an electron?



Accelerators and detectors can help to understand whether there is a world inside of an electron

Cloud and bubble chambers

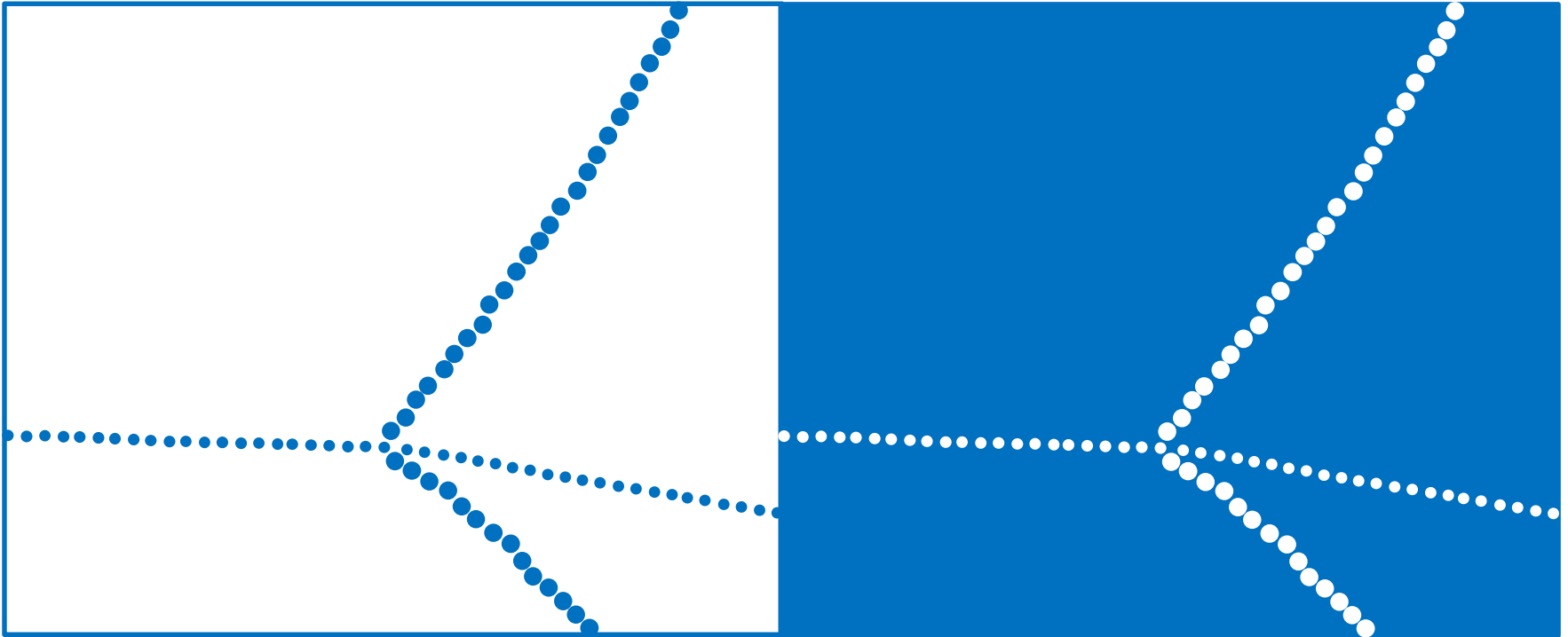


**Wilson's Cloud chamber
invented in 1911**

**Bubble Chamber (invented in 1952
by D. Glaser – Nobel prize 1960)**

**On the photo Bubble chamber
being installed near Fermilab**

Cloud and bubble chambers

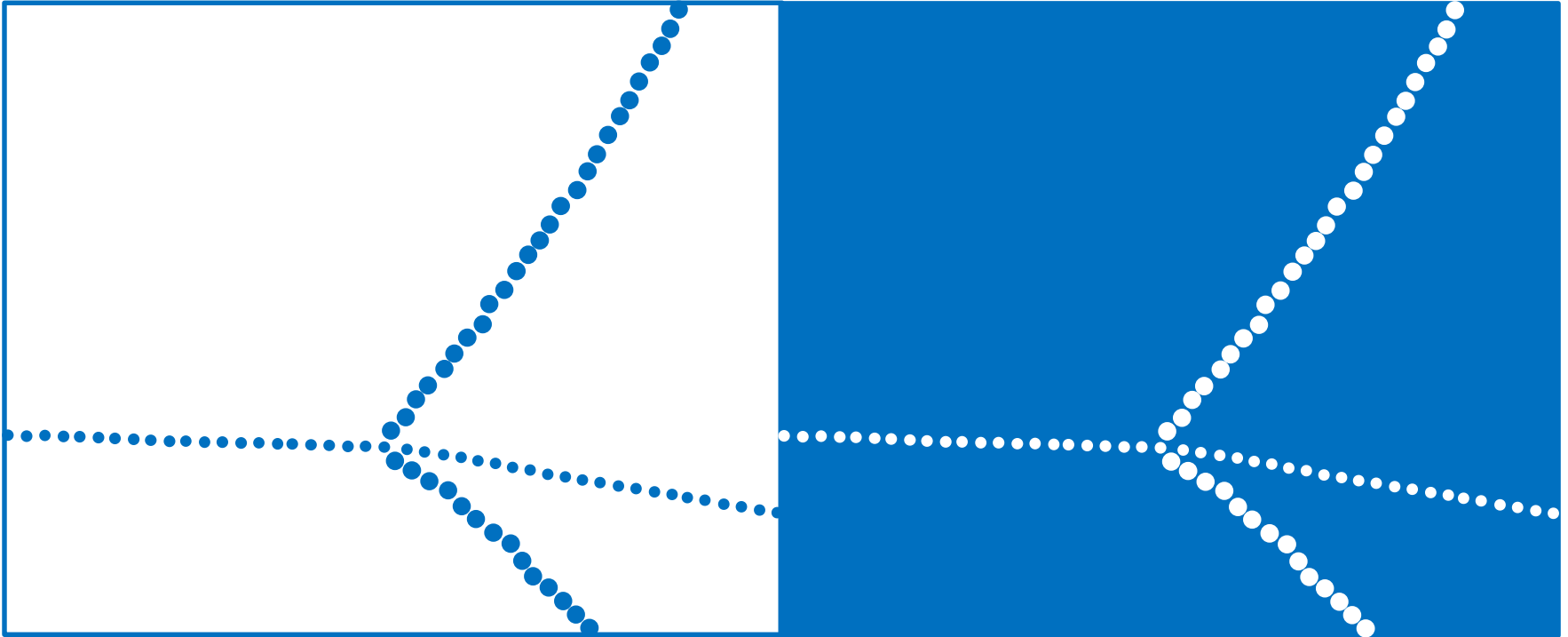


Wilson's Cloud chamber invented in 1911
Bubbles of liquid in a gas

Glaser's Bubble chamber, invented in 1952
Bubbles of gas in a liquid

**These two instruments are connected via another inventive principle –
“the other way around” or “system and anti-system”**

Cloud and bubble chambers



Wilson's Cloud chamber invented in 1911
Bubbles of liquid in a gas

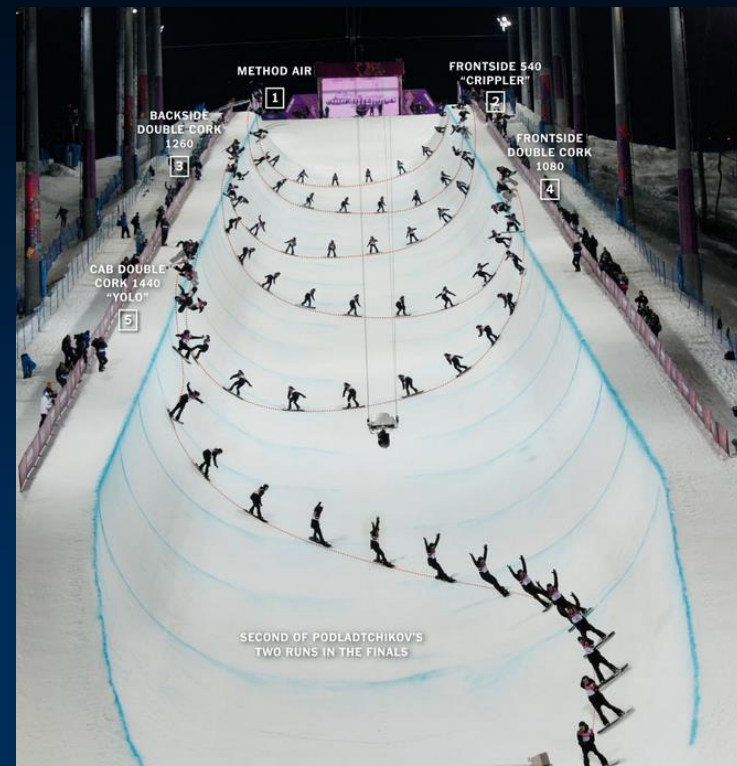
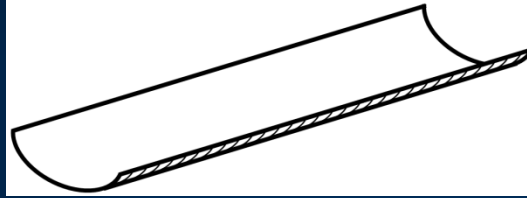
Glaser's Bubble chamber, invented in 1952
Bubbles of gas in a liquid

Bubble chamber could have been invented immediately, and not 40 years later after the cloud chamber, if we would have applied the principle of “system and anti-system”

System-anti-system and focusing in accelerators

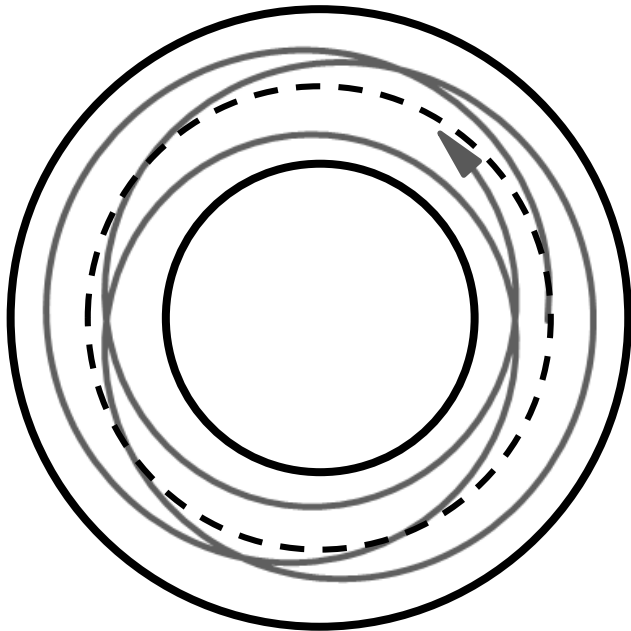
Focusing is needed to keep the particle trajectories near the centre

The analogy with the motion in the gutter



The first accelerators had **weak focusing** with spatial period greater than the perimeter of the accelerator

The trajectories of particles in an accelerator with weak focusing



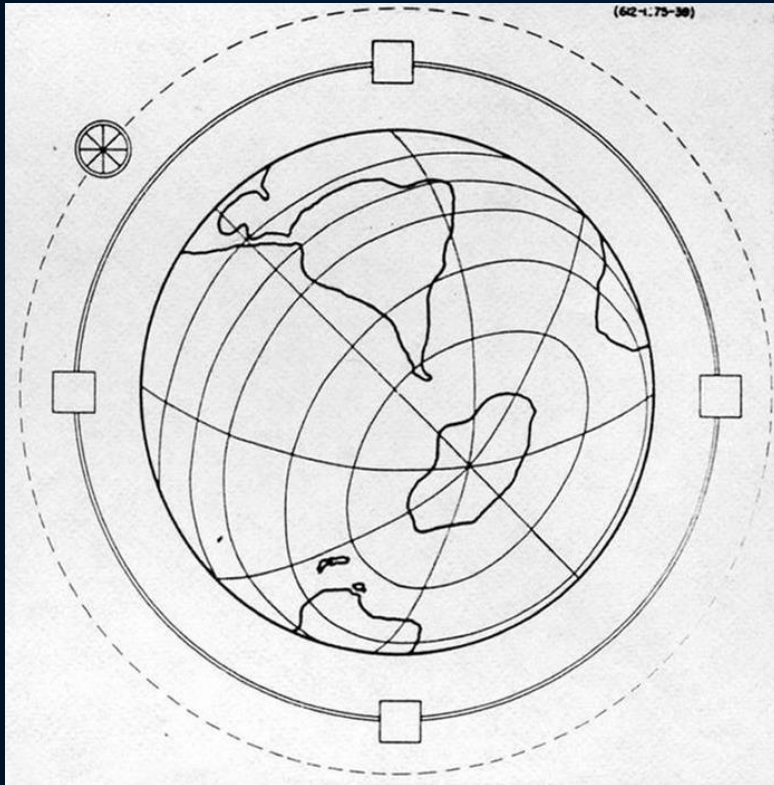
Weak focusing accelerator

10 GeV weak-focusing Synchrophasotron built in Dubna in 1957, the biggest and the most powerful for its time. It is ~60m diameter ring, and its magnets weigh 36,000 tons and it was registered in the Guinness Book of Records as the heaviest in the world.

View inside of the magnets. Vacuum chamber, which occupied all this space, now removed.



Dreaming big



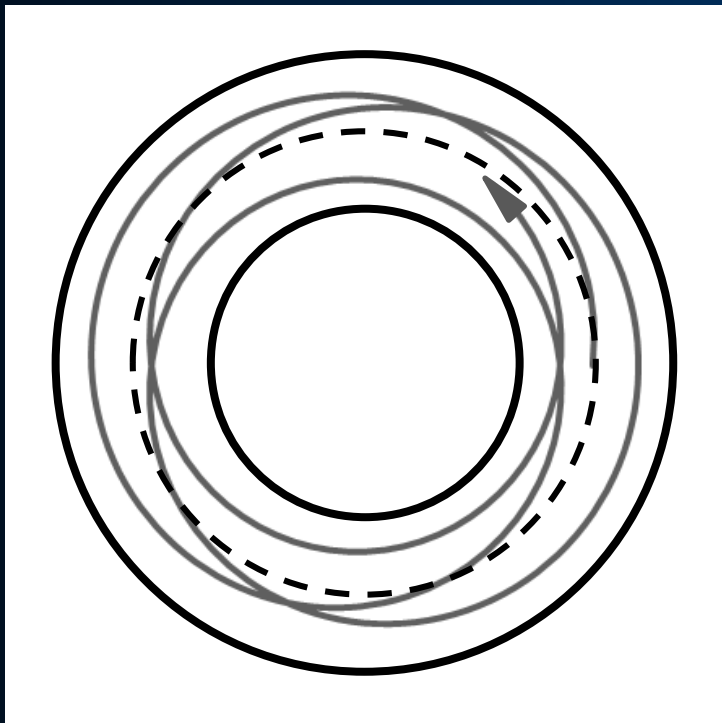
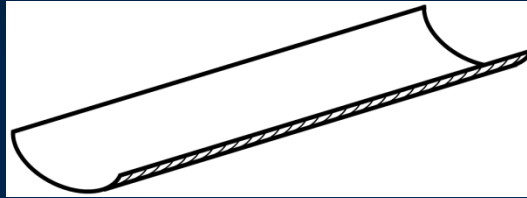
In 1954 Enrico Fermi presented, in his lecture, a vision of an accelerator that would encircle the Earth, and would attain highest possible energies

Imagine how humongous it would be if it would be built as a weak focusing machine!

Focusing

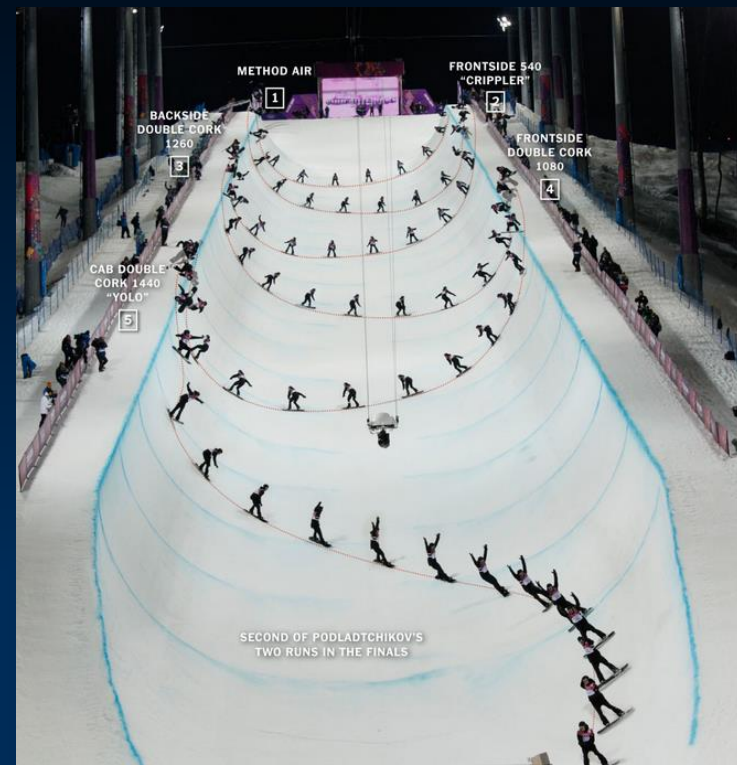
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The analogy with the motion in the gutter



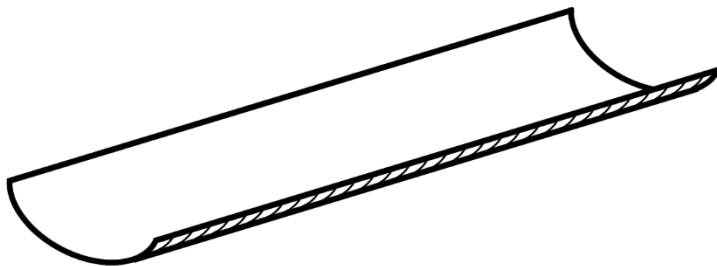
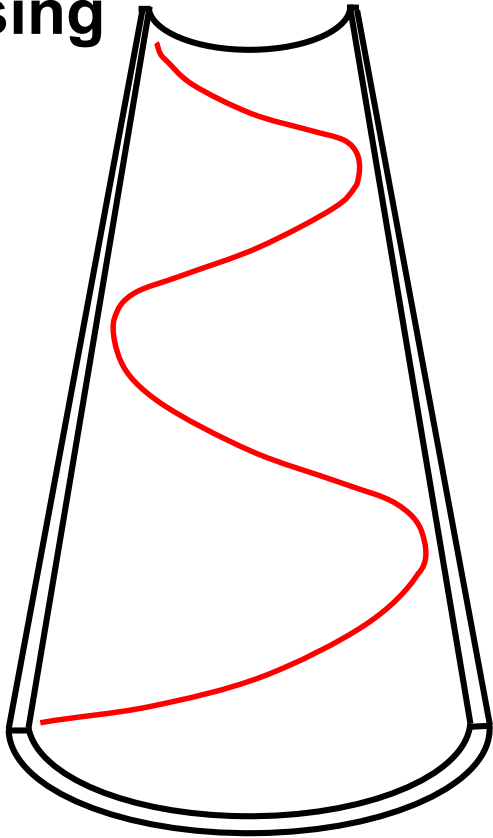
In this analogy, can we bend the gutter stronger, to achieve strong focusing?

The trajectories of particles in an accelerator with weak focusing

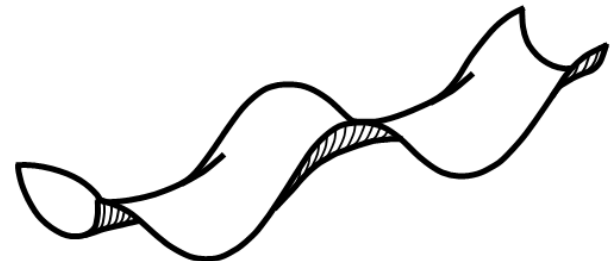
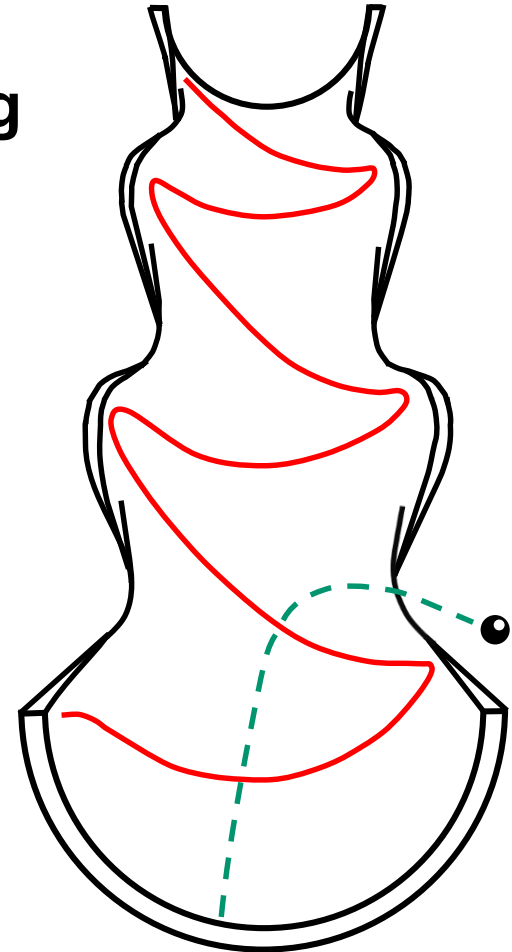


Weak and strong focusing

**Weak
focusing**

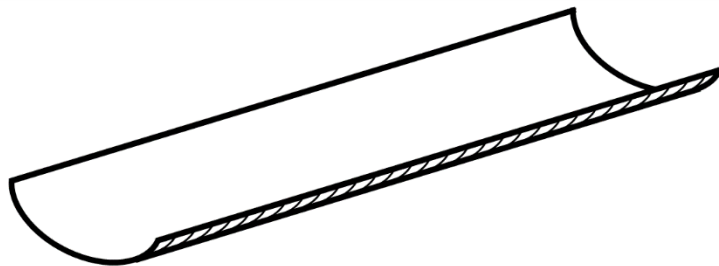
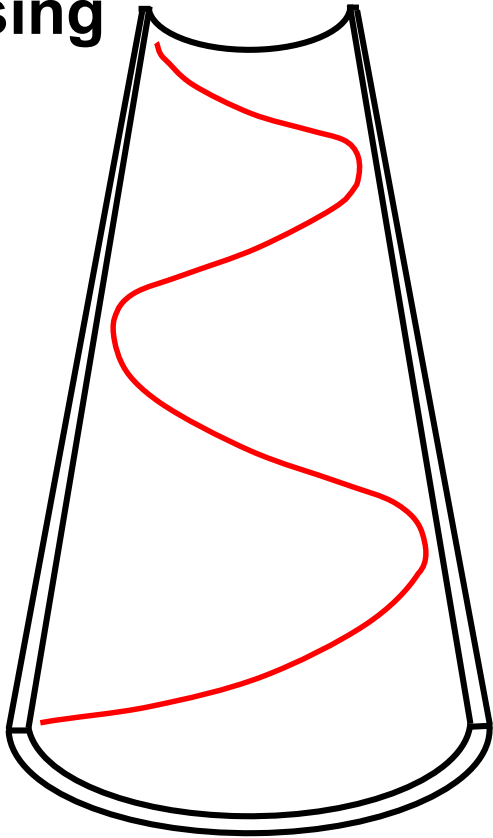


**Strong
focusing**

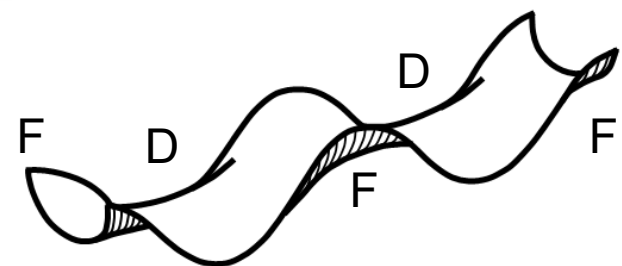
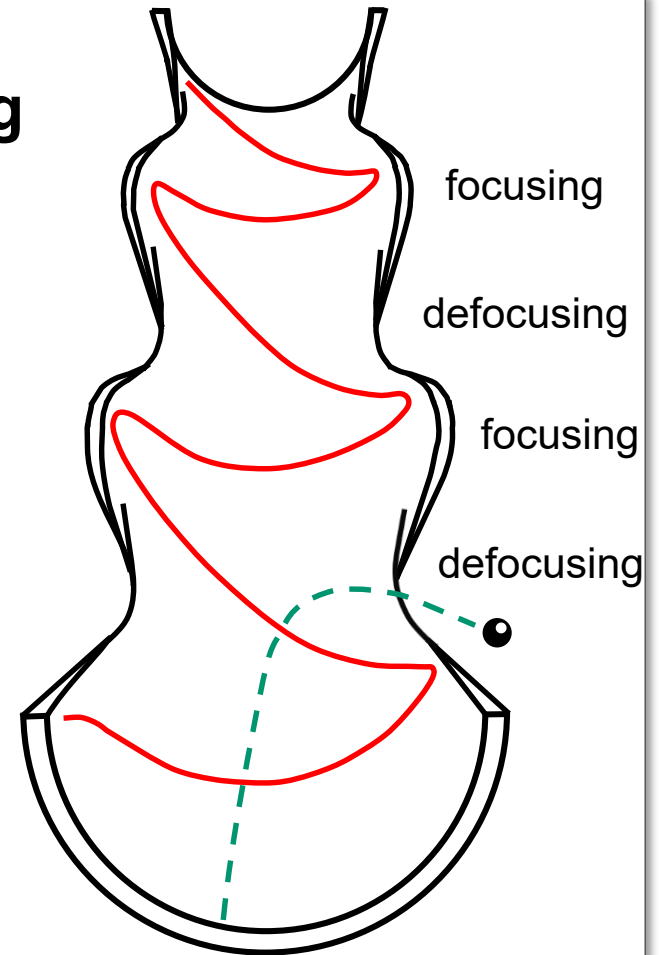


Weak and strong focusing

Weak focusing

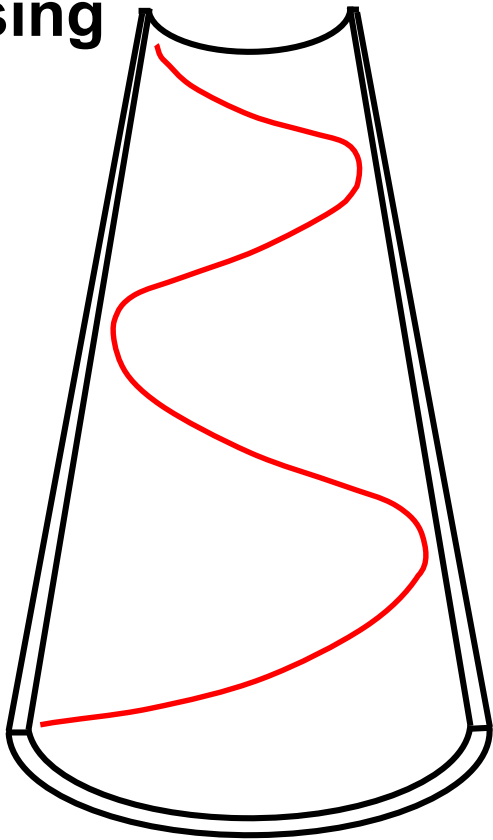


Strong focusing



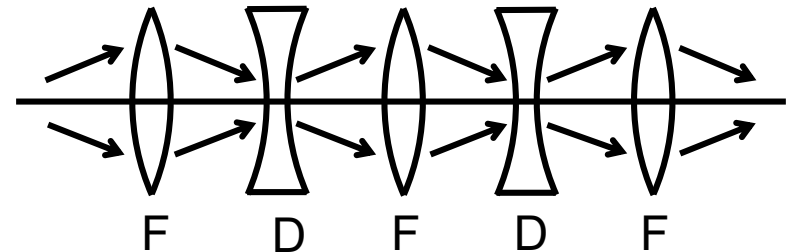
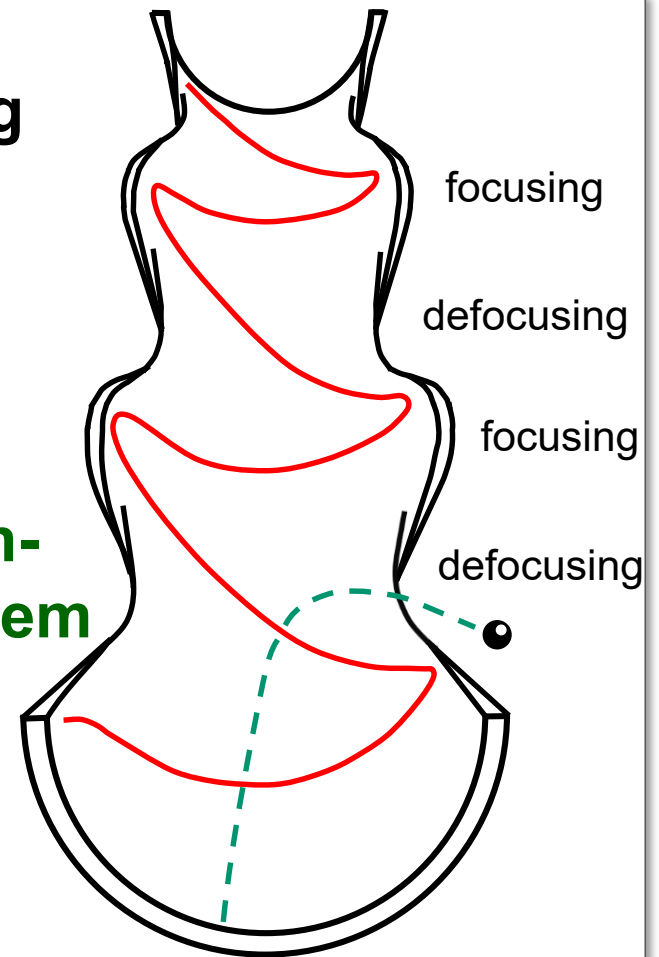
Weak and strong focusing

Weak focusing



Strong focusing

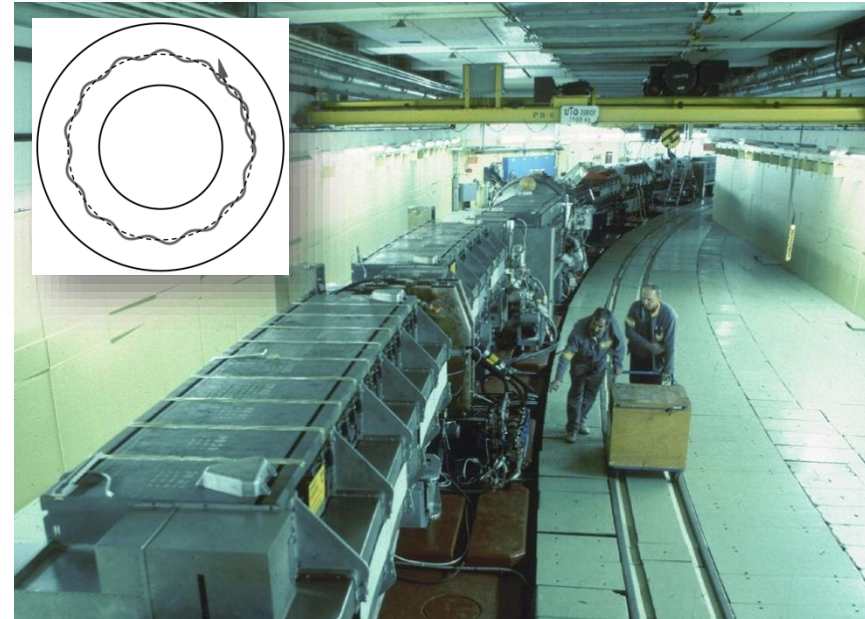
**System-
anti-system**



Weak and strong – compare them

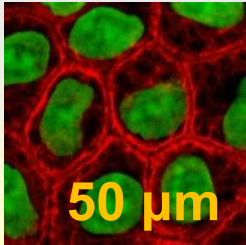


10 GeV weak-focusing Synchrophasotron built in Dubna in 1957, the biggest and the most powerful for his time. It is ~60m diameter ring, and its magnets weigh 36,000 tons and it was registered in the Guinness Book of Records as the heaviest in the world.



CERN's Proton Synchrotron, the first operating strong-focusing accelerator, reached 24 GeV in 1959. It is a ~200-m diameter ring, weight of magnets 3,800 tons.

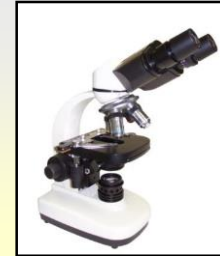
The structure of matter...



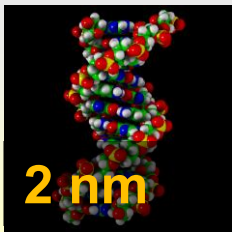
Extra
magnification?

CELLS

Twenty
per mm



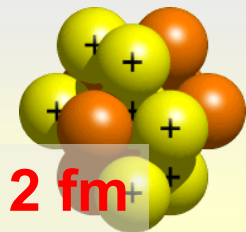
Microscope



x 25 thousand

DNA

Five hundred
thousand
per mm



x 1 million

Nucleus

Five hundred
billion
per mm

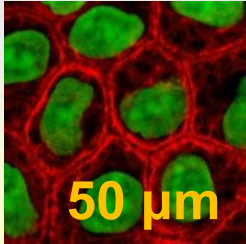


x 2 thousand

Quarks

More than one
million billion
per mm

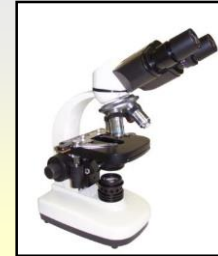
...use particles



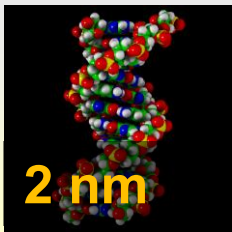
Extra
magnification?

CELLS

Twenty
per mm



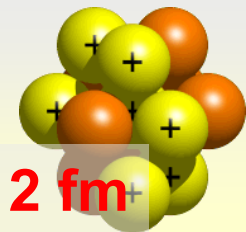
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million billion
per mm

Particles & their wave properties

Wavelength corresponding to the particle:

$$\lambda = h / p$$

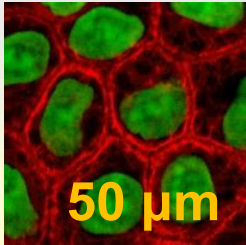
De Broglie
Wavelength

Planck
Constant

Momentum

See small? Use particles
and increase their energy

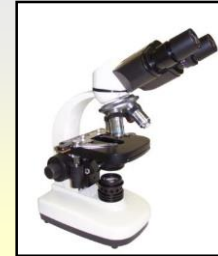
...use particle accelerators



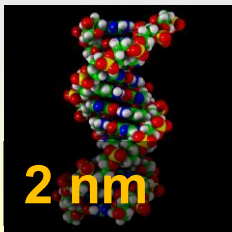
Extra
magnification?

CELLS

Twenty
per mm



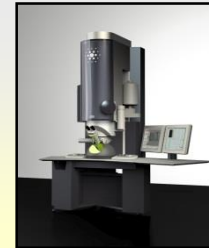
Microscope



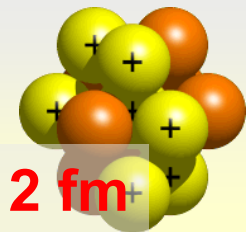
x 25 thousand

DNA

Five hundred
thousand
per mm



Electron
microscope



x 1 million

Nucleus

Five hundred
billion
per mm

Particle Accelerators

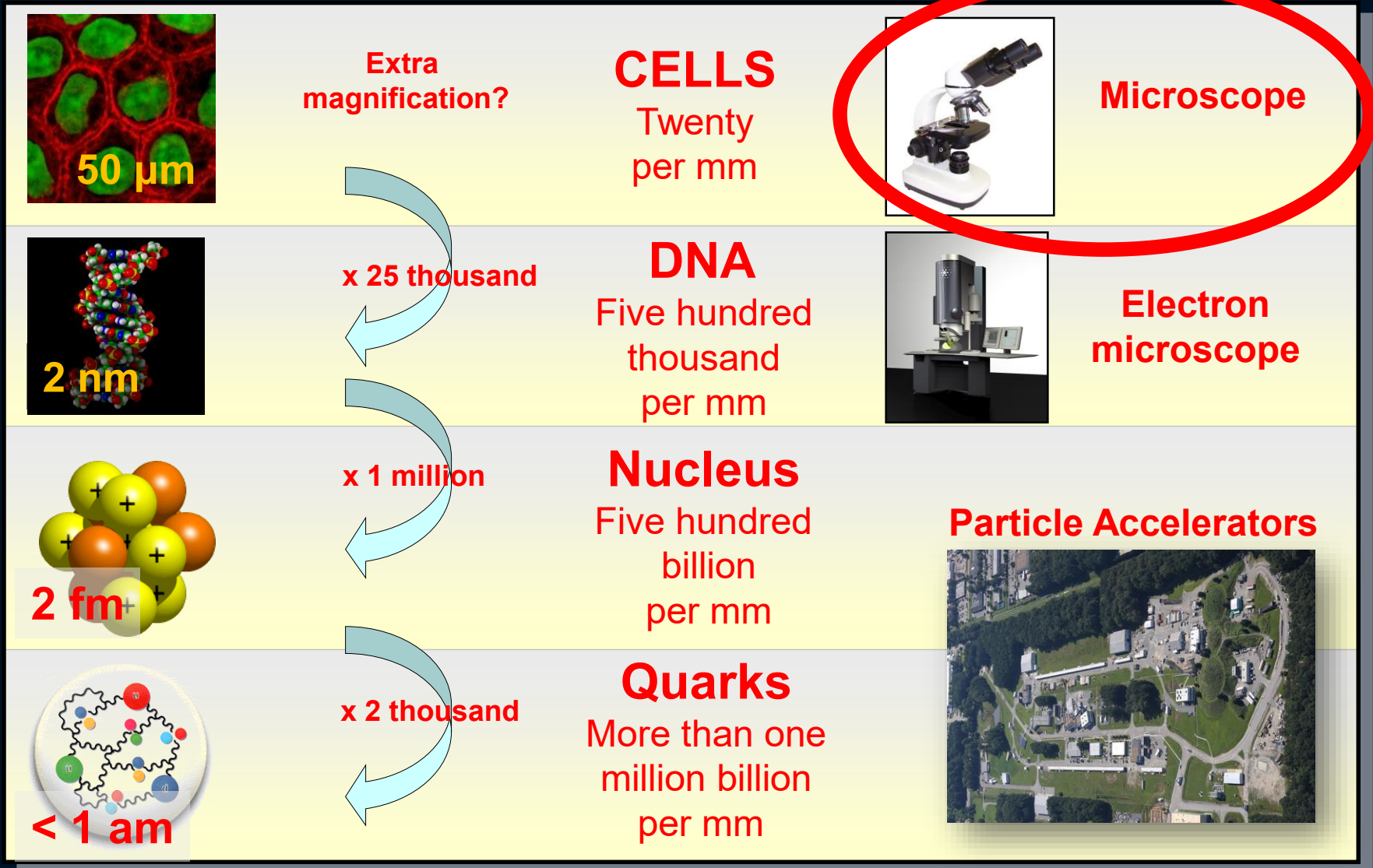


x 2 thousand

Quarks

More than one
million billion
per mm

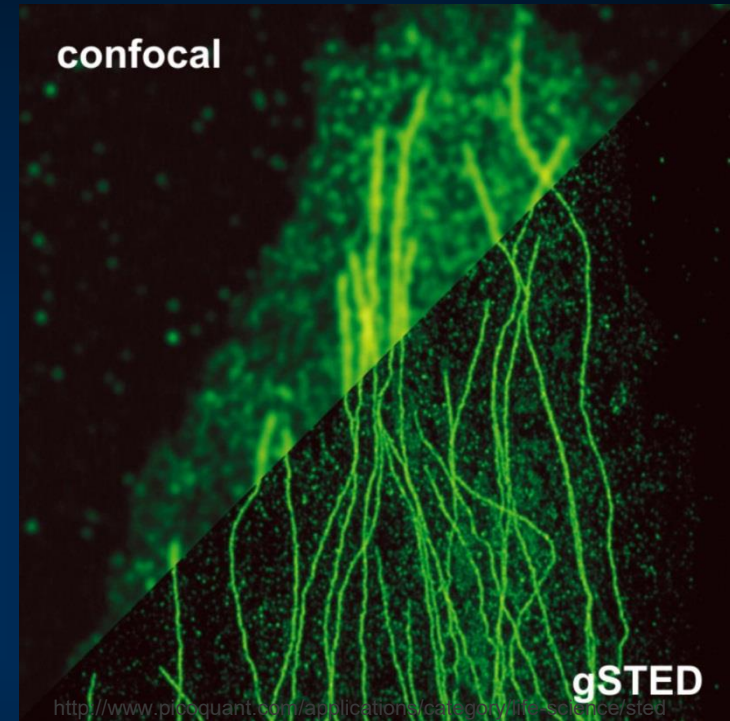
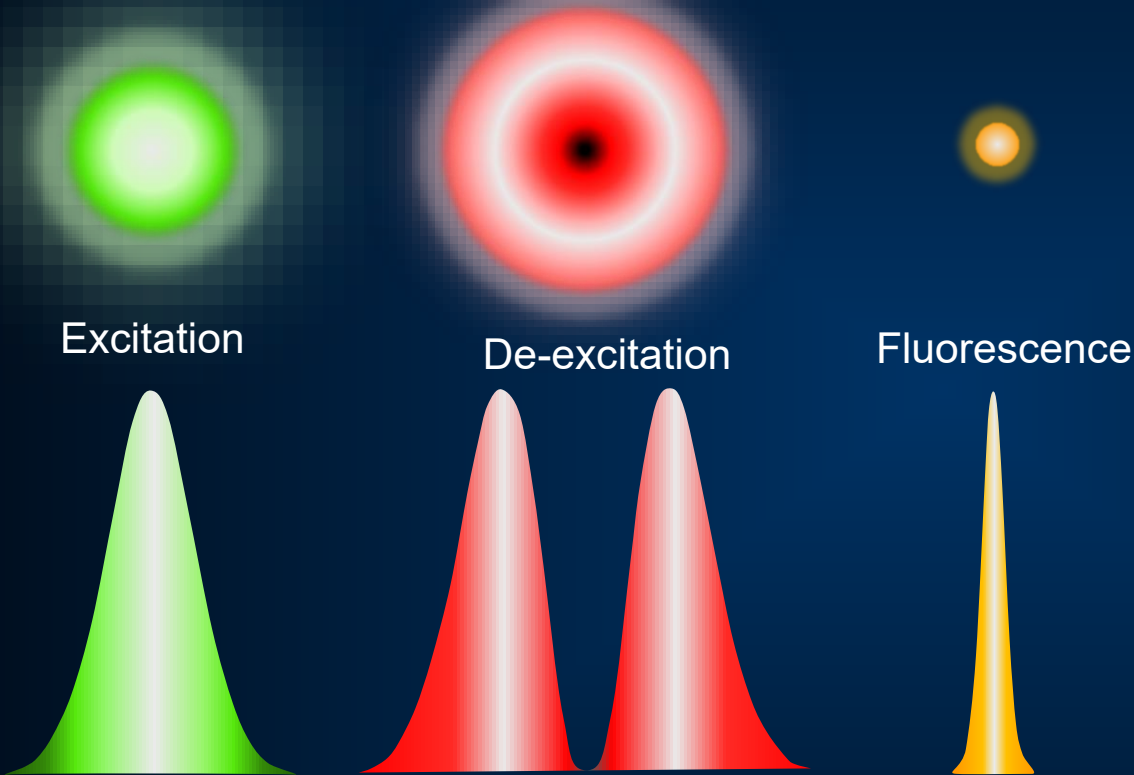
Chemistry Nobel 2014 & inventive principles?



Chemistry Nobel 2014 ...

Stimulated Emission Depletion microscopy (STED)

Stefan W. Hell



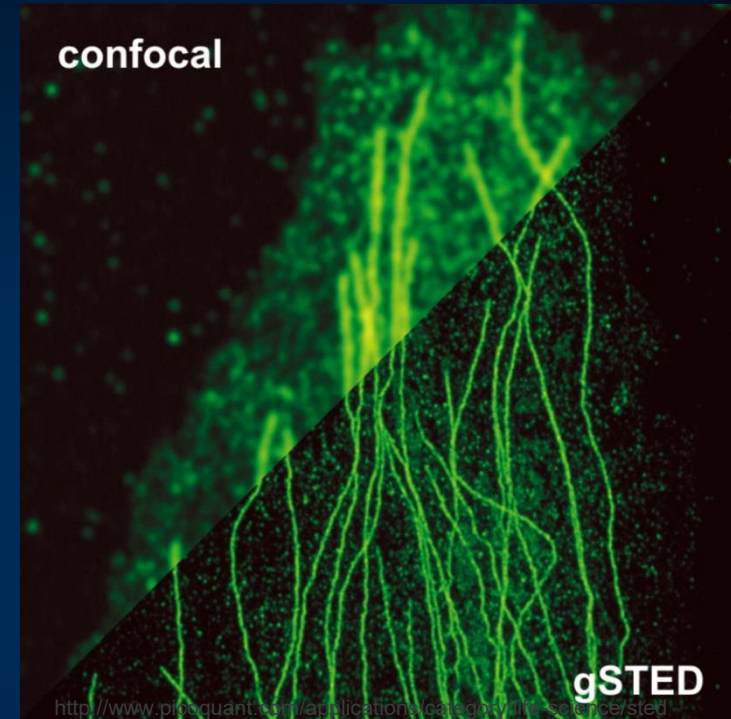
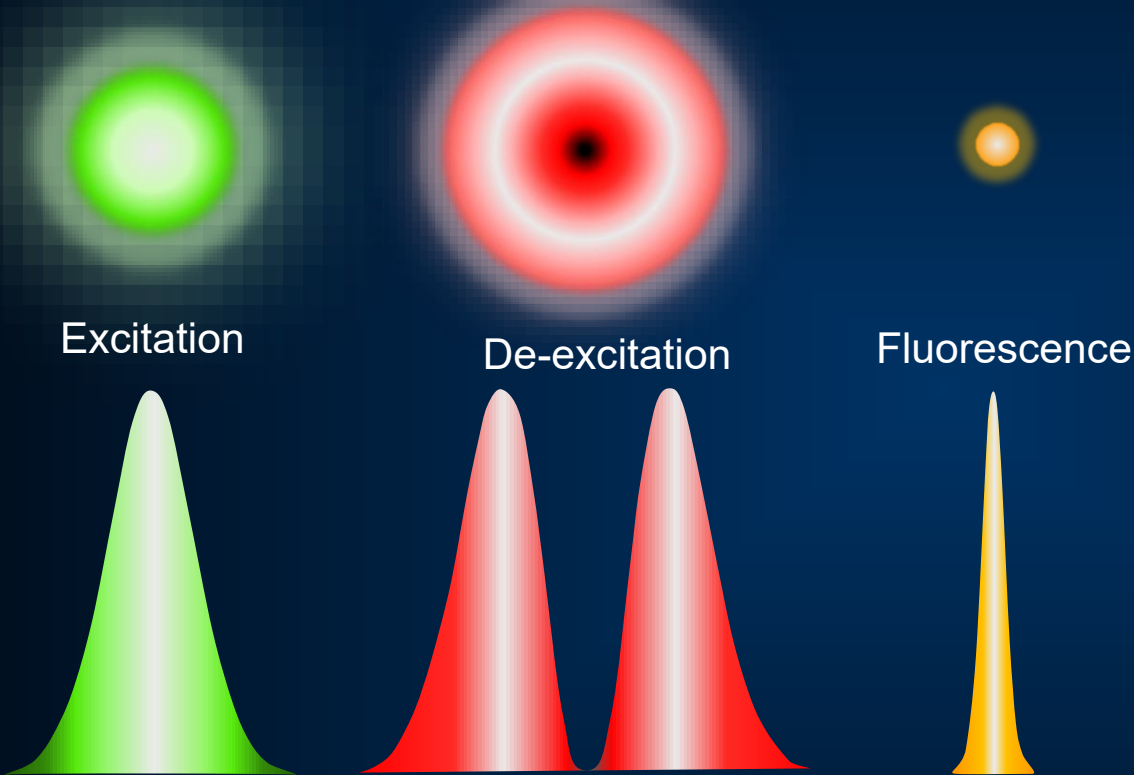
(gated) STED image of Tubulin vs
standard confocal image

**This can improve the resolution to be a factor of
several below the wavelength of light**

Chemistry Nobel 2014 & inventive principles

Stimulated Emission Depletion microscopy (STED)

Stefan W. Hell



(gated) STED image of Tubulin vs standard confocal image

And this can be viewed as a combination of the inventive principles “system and anti-system” and “nested dolls”

Nobel prize 2018 – CPA

Arthur Ashkin, Gérard Mourou and Donna Strickland

CPA - chirped pulse amplification

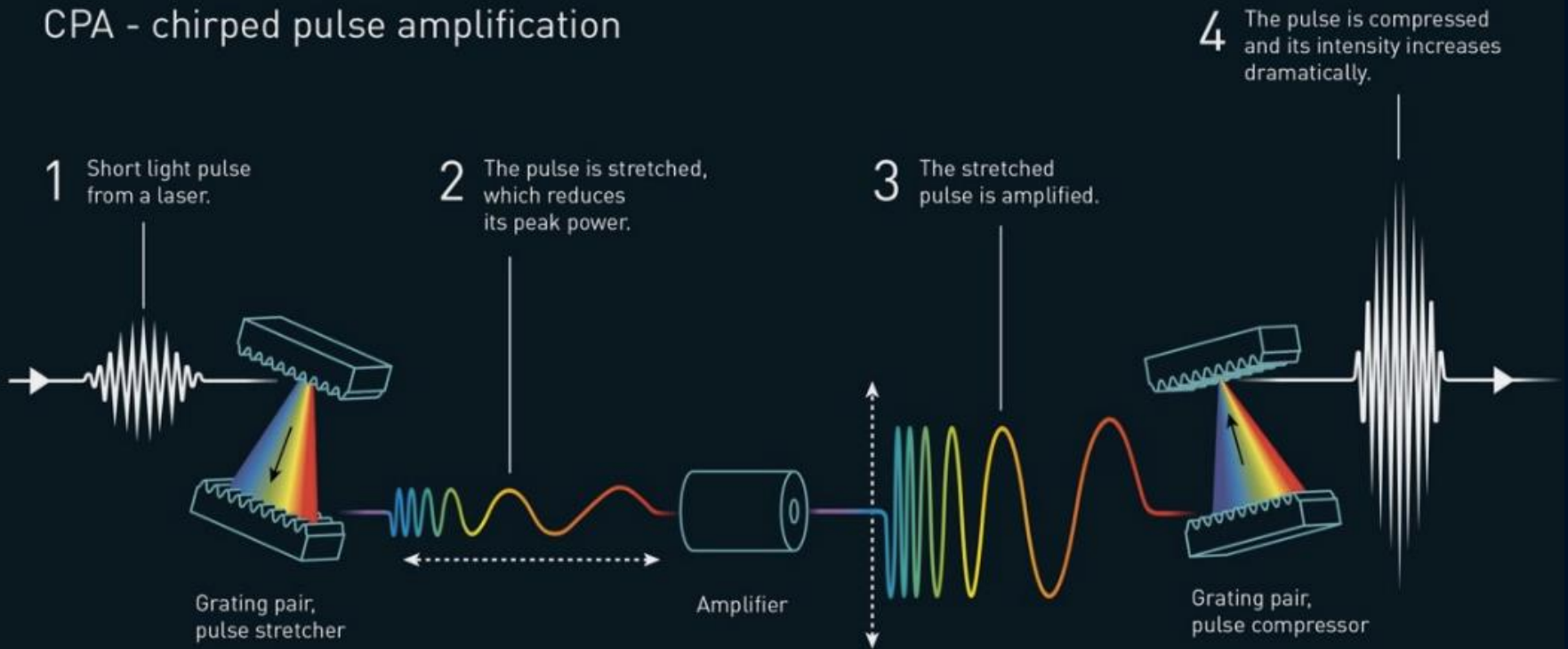


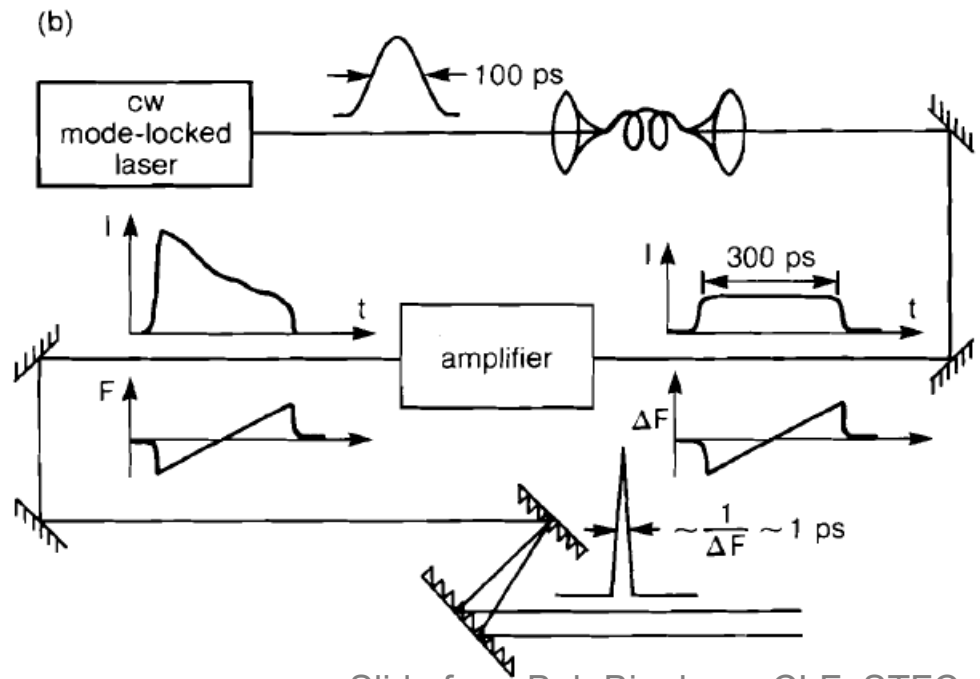
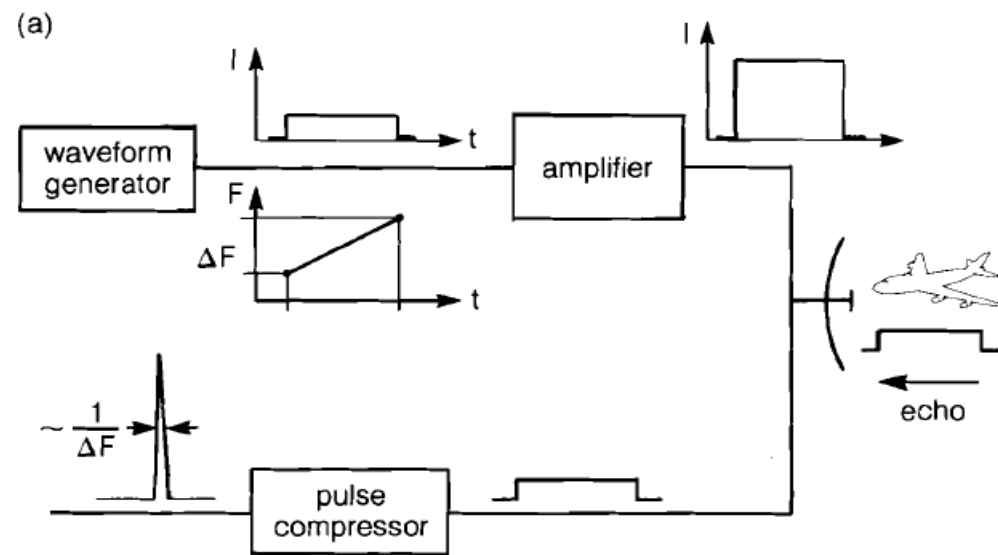
Image: Johan Jarnestad/Royal Swedish Academy of Sciences

Is there an inventive principle here that is used in other areas?

Chirped pulse amplification from Radar to Lasers (CPA)

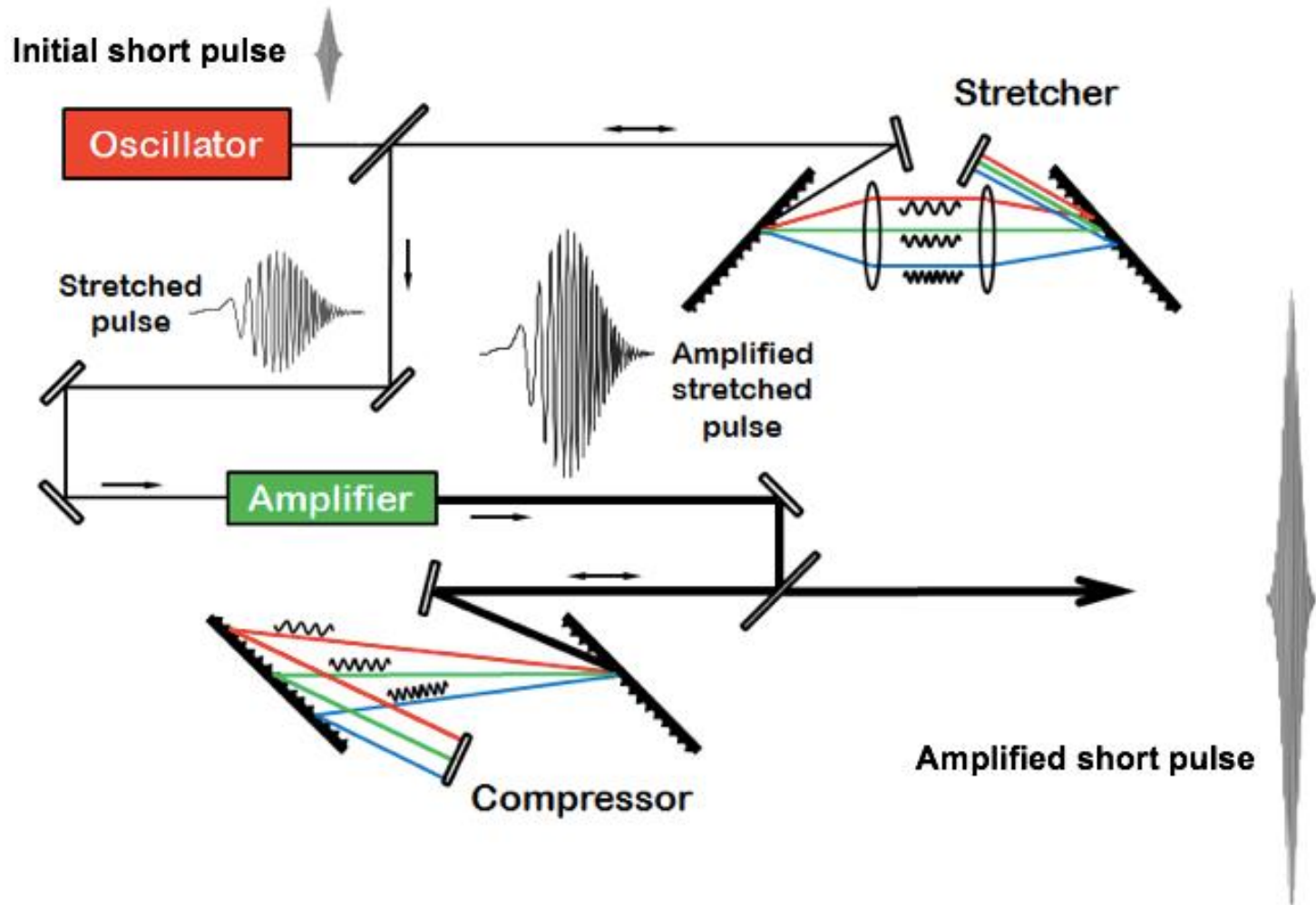
Diagrams taken from early LLE review
On the comparison between RADAR
chirped pulse amplification from the
1940 onwards upper diagram and
laser chirped pulse amplification bottom
diagram carried out at the
LLE Rochester.

LLE Review 25 3B 1985.



Slide from Bob Bingham, CLF, STFC

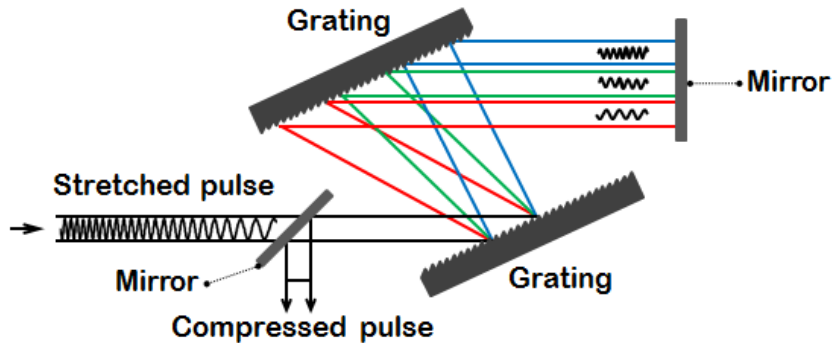
CPA – Chirped Pulse Amplification



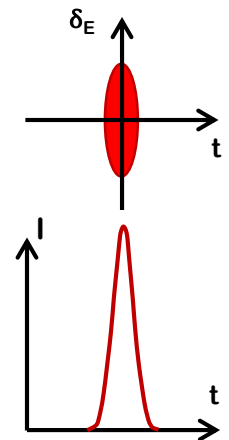
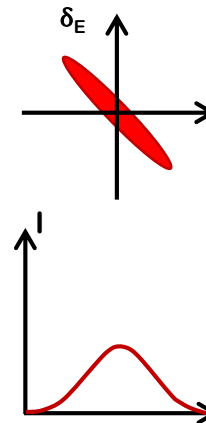
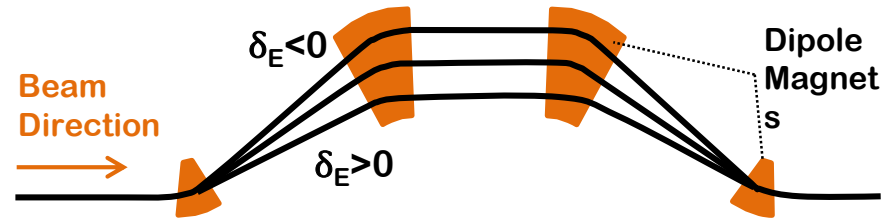
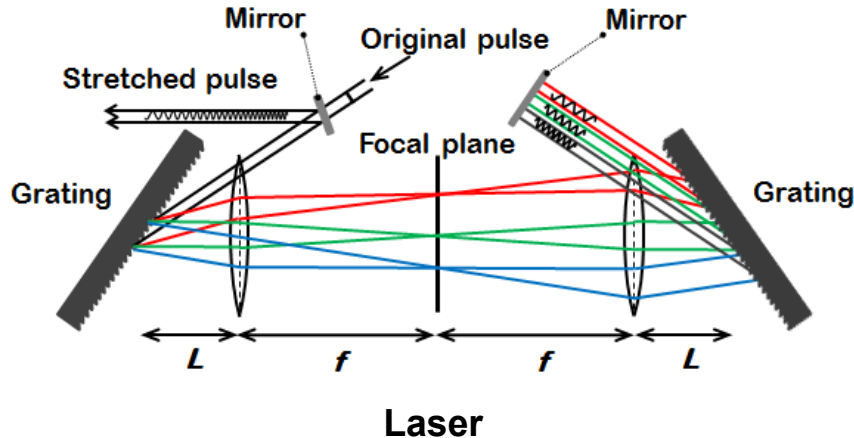
- CPA: pulse stretching and compressing using time-energy correlation

Beam and laser bunch/pulse compression

Compressor



Stretcher



Beam

Both in laser and beam use z-Energy correlation to compress/stretch the pulse – this seem to be one more general inventive principle – connecting laser CPA and particle accelerators

Radar and CPA

Evolution from chirped pulse amplification in radars to lasers, and connection of CPA with bunch compression in accelerators, seem to demonstrate that ...

The same Problems and Inventions appear again and again but in different areas of science and technology

Let's now talk about inventiveness

We have seen several examples of what seem to be some general inventive principles and evolution laws

It happens that many of such inventive principles and evolution laws are known for half a century and widely used...

...but so far not in science

Let's discuss methodologies of inventiveness

How to invent more efficiently?

Forbes



Haydn Shaughnessy, Contributor

I write about enterprise innovation.

TECH | 3/07/2013 @ 6:32AM | 72,570 views

What Makes Samsung Such An Innovative Company?

*What was that magic bullet?
...wait a few slides...*

But it was [REDACTED] that became the bedrock of innovation at Samsung. And it was introduced at Samsung by [REDACTED] whom Samsung had hired into its Seoul Labs in the early 2000s.

In 2003 [REDACTED] led to 50 new patents for Samsung and in 2004 one project alone, a DVD pick-up innovation, saved Samsung over \$100 million. [REDACTED] is now an obligatory skill set if you want to advance within Samsung.

How to invent – evolution of the methods

- **Brute-force or exhaustive search**
 - **consider any possible ideas**
- **Brainstorming**
 - **psychological method which helps to solve problems and to invent**
 - **The main feature of brainstorming – separate the process of idea generation from the process of their critical analysis**
 - **The method of brainstorming did not meet expectations**
 - **the absence of feedback, which is the power of the method, is simultaneously its handicap, as feedback is needed for development and adjusting of an idea**



**Alex Osborn
(1888 – 1966)**

The author of
brainstorming Alex
Osborn introduced
the method around
1950s

How to invent – evolution of the methods

- **Synectics – improved Brainstorming**

- **Features of Synectics:**

- **Permanent groups for problem solving**

- whose members with time become less sensitive to critics and more efficient in problem solving

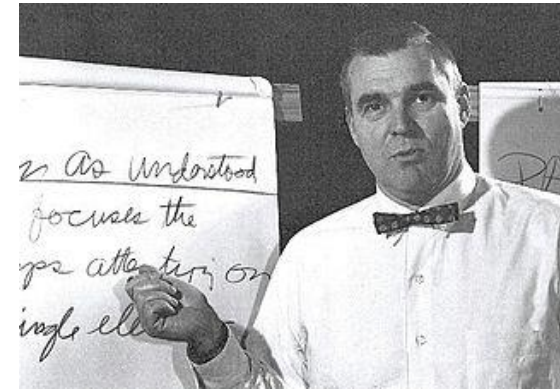
- **Emphasis on the importance to see familiar behind unknown and vice versa**

- which should help to solve a new and unfamiliar problem with known methods

- **Importance of a fresh view at a problem**

- **Use of analogies to generate fresh view**

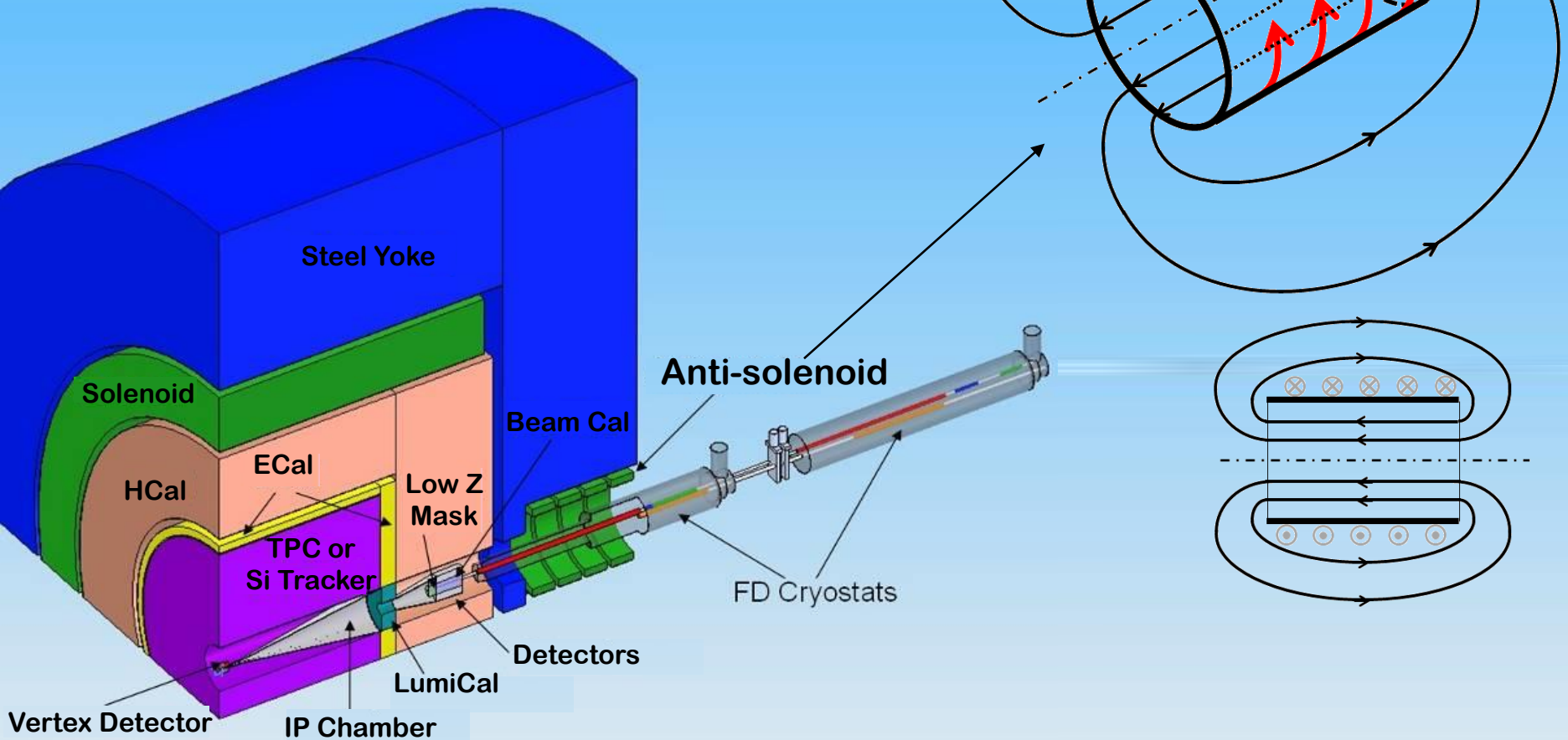
- direct (any analogy, e.g. from nature);
 - empathic (attempting to look at the problem identifying yourself with the object);
 - symbolic (finding a short symbolic description of the problem and the object);
 - metaphorical (describing the problem in terms of fairy-tales and legends);



Attempting to improve brainstorming, George Prince (on the photo) and William Gordon introduced the method of Synectics

ILC Interaction Region...

Anti-solenoid is needed, but it would be pulled into the main solenoid with humongous force

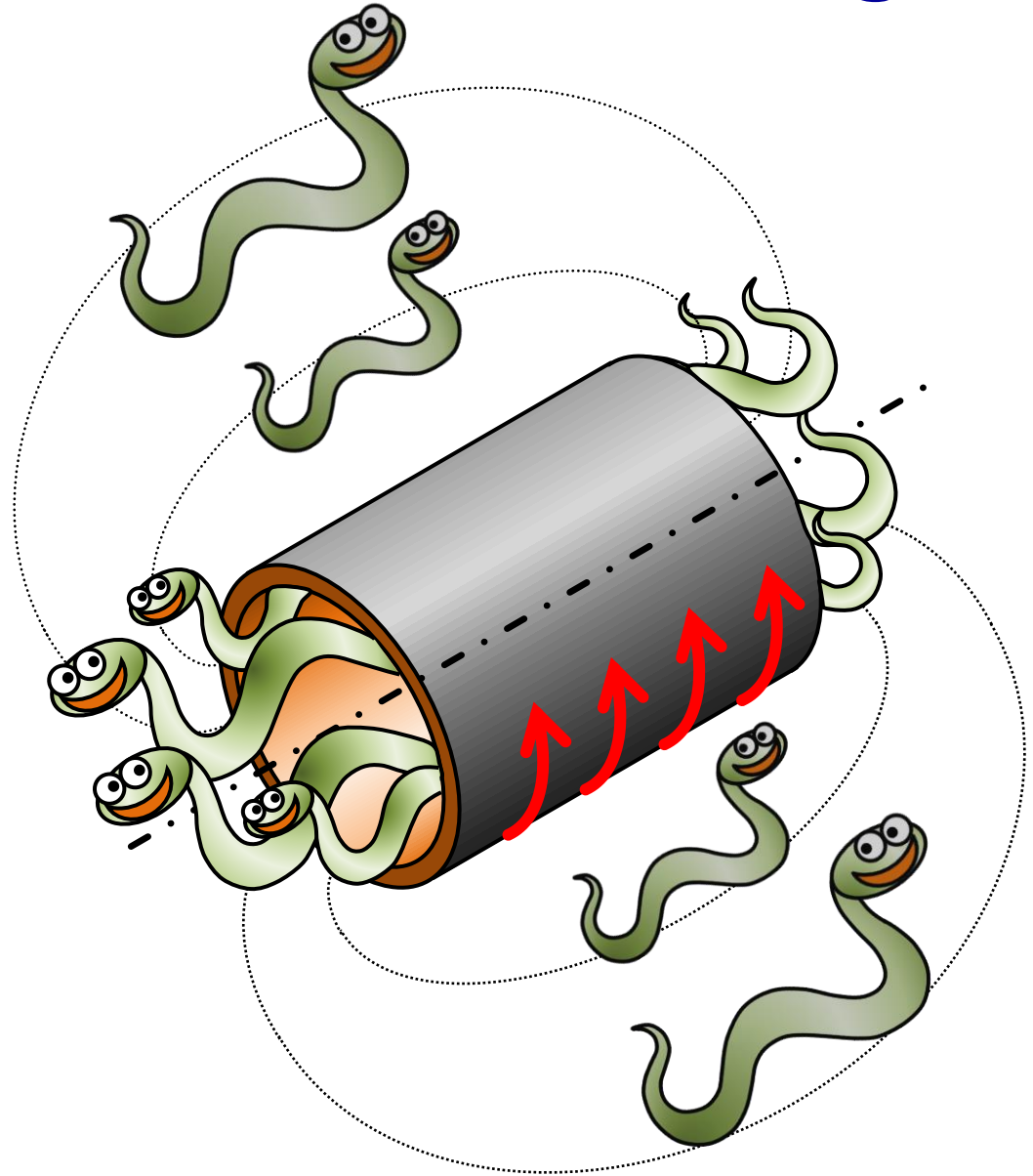


Synectics : use of analogies

– Use of analogies to generate fresh view

- ...
- empathic (attempting to look at the problem identifying yourself with the object);
- ...
- metaphorical (describing the problem in terms of fairy-tales and legends);

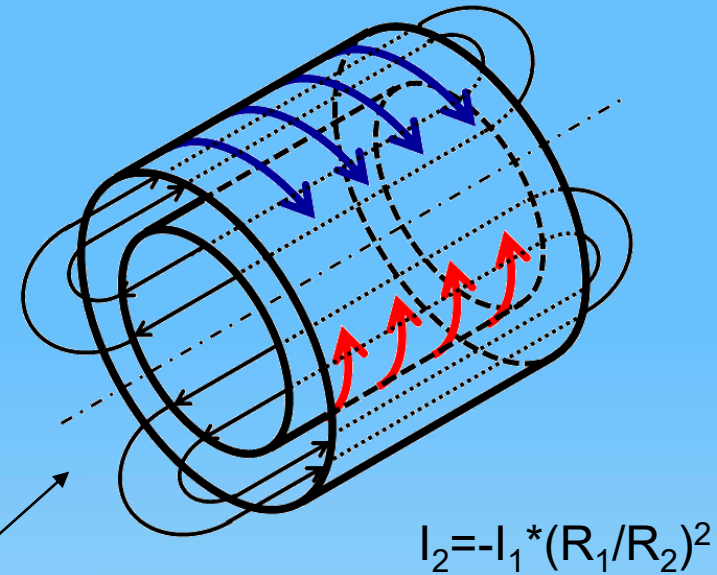
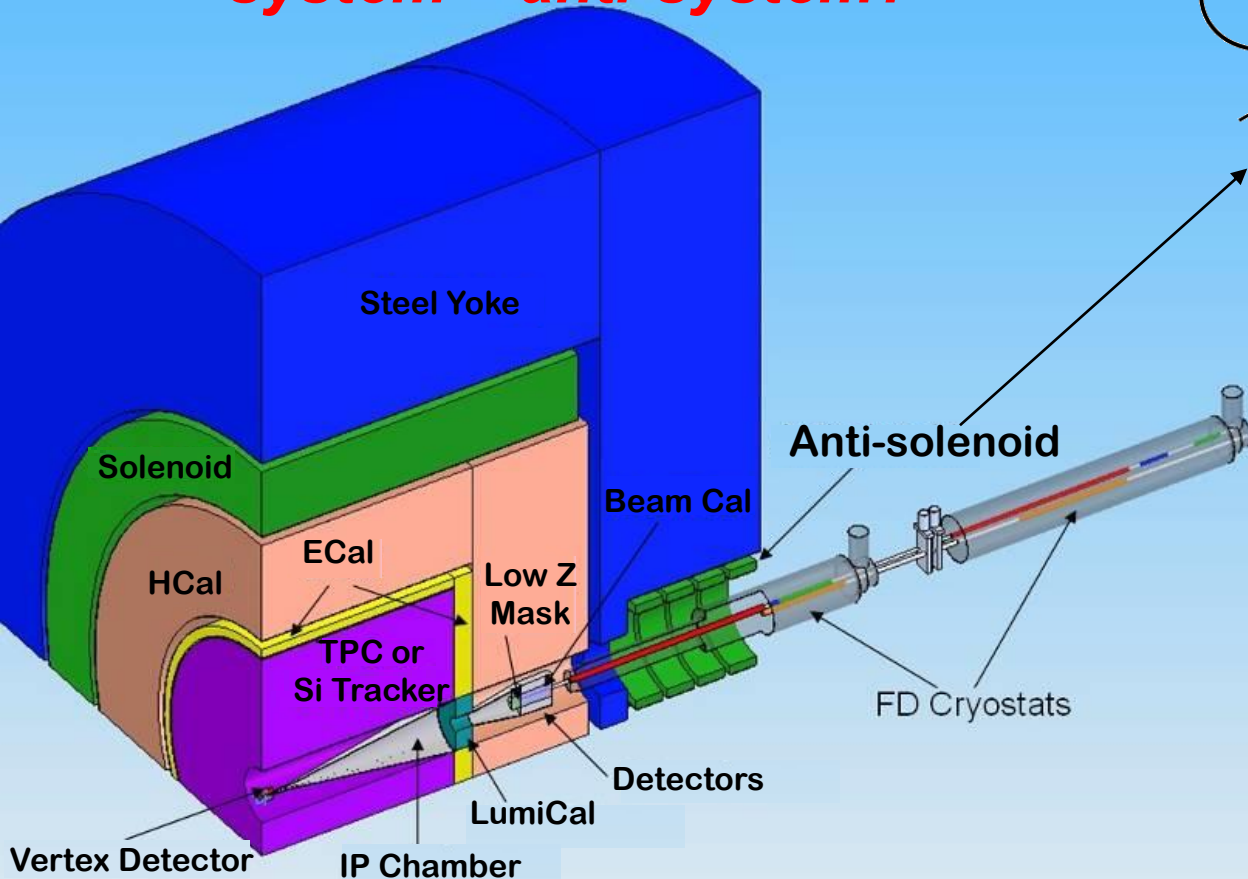
How to contain the magnetic flux?



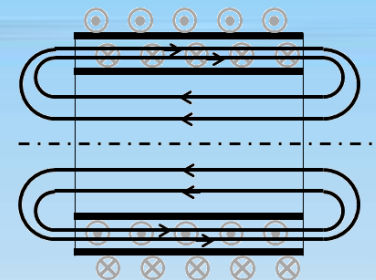
ILC Interaction Region...

Dual anti-solenoid is used, to cancel its external field – this makes it force-neutral

What if we try nested doll & system – anti-system?



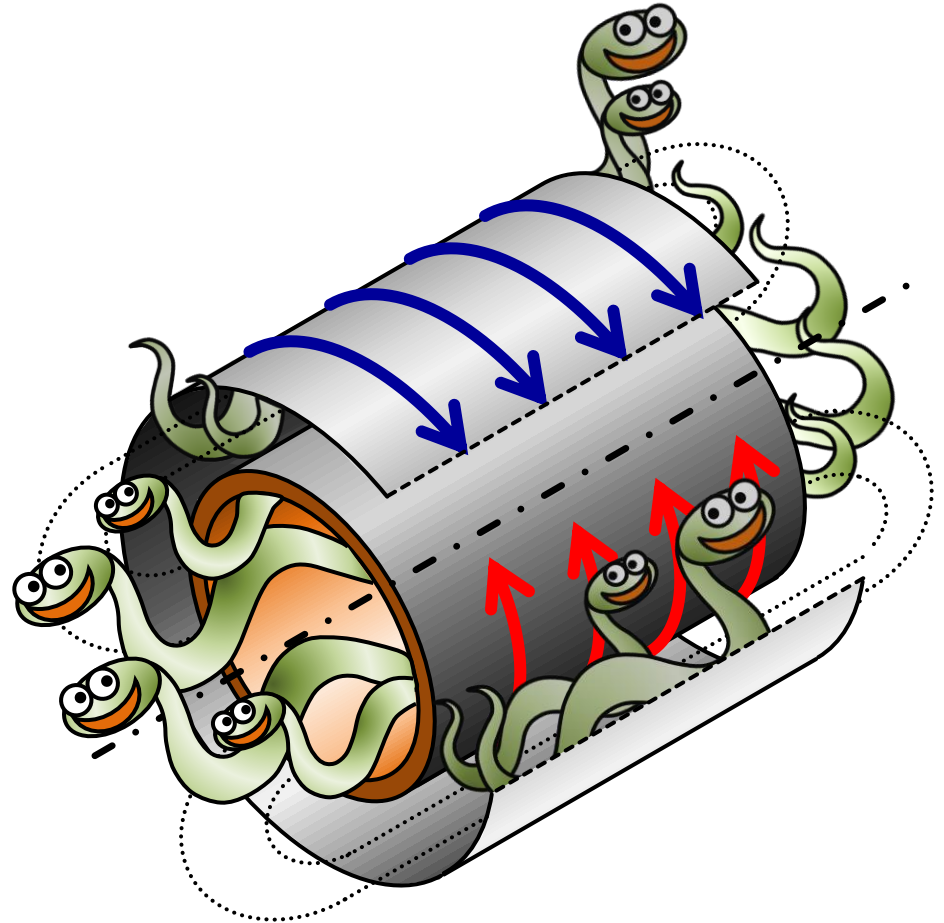
$$I_2 = -I_1 \cdot (R_1/R_2)^2$$



Synectics and use of analogies

- Use of analogies to generate fresh view

- ...
- empathic
(attempting to look at the problem identifying yourself with the object);
- ...
- **metaphorical**
(describing the problem in terms of **fairy-tales and legends**);



Synectics does not help

How to invent – evolution of the methods

- **Synectics is the limit of what can be achieved, maintaining the brute force method of exhaustive search**
 - **Indeed, why one would employ analogies and metaphors and irrational factors in order to come to a natural and universal formula “the action has to happen itself”**

How to invent – evolution of the methods

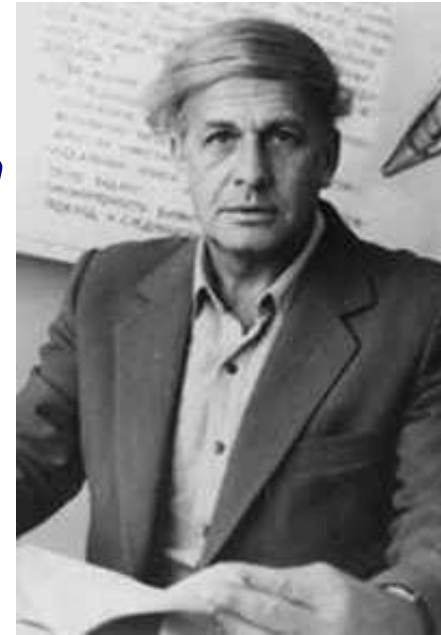
- Synectics is the limit of what can be achieved, maintaining the brute force method of exhaustive search
 - Indeed, why one would employ analogies and metaphors and irrational factors in order to come to a **natural and universal formula** “the action has to happen itself”
 - One should aim at such formula in any invention, armed with precise identification of physical contradiction – essence of TRIZ



Illustration by Sasha Seraia

How to invent – TRIZ

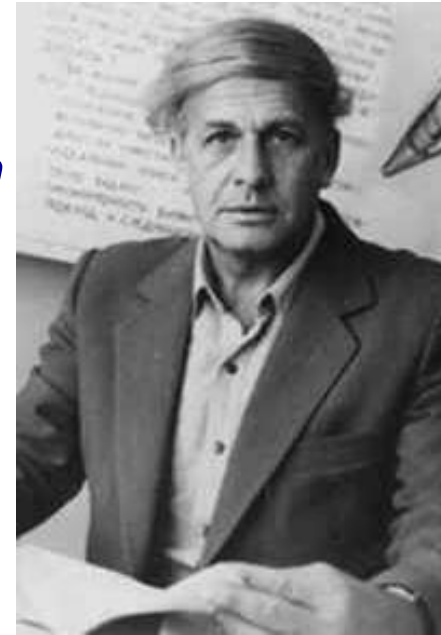
- **TRIZ – *Teoria Reshenia Izobretatelskikh Zadach***
= Theory of Inventive Problem Solving
- **Developed by Genrikh Altshuller in SU**
 - **Work in patent office in 1946**
 - **Analysed many patents, discovered patterns and identified what makes a patent successful**
 - **Formulated TRIZ in 1956-1985**



Genrikh Altshuller
(aka Altov) 1926-1998

How to invent – TRIZ

- **TRIZ – *Teoria Reshenia Izobretatelskikh Zadach***
= Theory of Inventive Problem Solving
- **Developed by Genrikh Altshuller in SU**
 - Work in patent office in 1946
 - Analysed many patents, discovered patterns and identified what makes a patent successful
 - Formulated TRIZ in 1956-1985
- **Four key discoveries of TRIZ:**
 - **The same Problems and Solutions appear again and again but in different industries**
 - **There is a recognisable Technological Evolution path for all industries**
 - **Innovative patents (23% of total) used science/engineering theories outside their own area/industry**
 - **An Innovative Patent uncovers and solves contradictions**



Genrikh Altshuller
(aka Altov) 1926-1998

How to invent more efficiently?

Forbes



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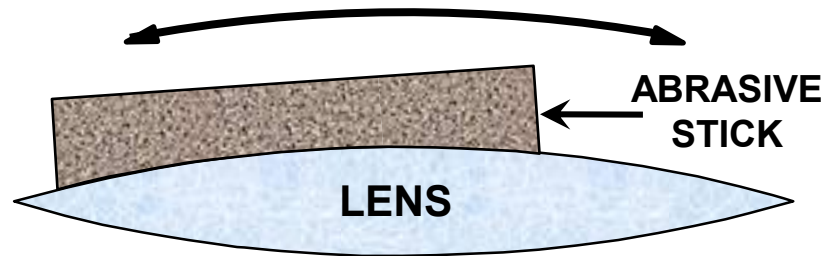
What Makes Samsung Such An Innovative Company?

*Why are we interested in this
in relation to science?
...wait a few more slides...*

But it was **TRIZ** that became the bedrock of innovation at Samsung. And it was introduced at Samsung by **Russian engineers** whom Samsung had hired into its Seoul Labs in the early 2000s.

In 2003 **TRIZ** led to 50 new patents for Samsung and in 2004 one project alone, a DVD pick-up innovation, saved Samsung over \$100 million. **TRIZ** is now an obligatory skill set if you want to advance within Samsung.

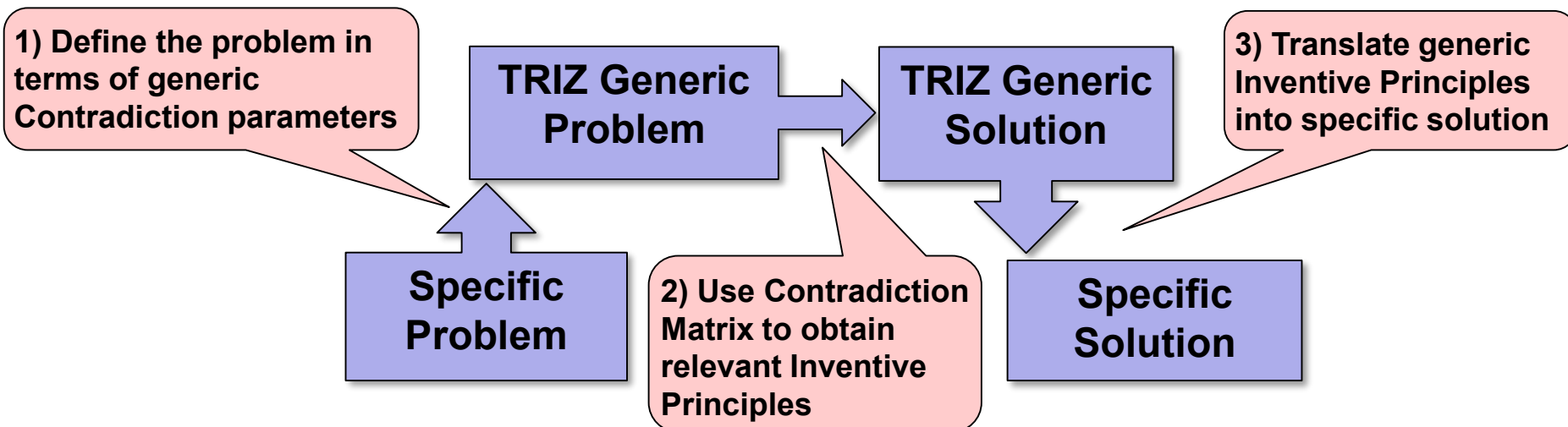
TRIZ in action - example



Problem: Lens polished – heat generated. Heat degrades optical properties. Existing cooling methods ineffective, as cannot achieve uniform cooling at each abrasive particle

To be improved: **SPEED**, What gets worse: **TEMPERATURE**

Has anyone else solved such contradiction?



Example: following J.Scanlan, School of Engineering Sciences, Univ. of Southampton

Elements of TRIZ contradiction matrix

1. Weight of moving object
2. Weight of stationary object
3. Length of moving object
4. Length of stationary object
5. Area of moving object
6. Area of stationary object
7. Volume of moving object
8. Volume of stationary object
9. Speed
10. Force (Intensity)
11. Stress or pressure
12. Shape
13. Stability of the object
14. Strength
15. Durability of moving object
16. Durability of non moving object
17. Temperature
18. Illumination intensity
19. Use of energy by moving object
20. Use of energy by stationary object

21. Power
22. Loss of Energy
23. Loss of substance
24. Loss of Information
25. Loss of Time
26. Quantity of substance/the
27. Reliability
28. Measurement accuracy
29. Manufacturing precision
30. Object-affected harmful
31. Object-generated harmful
32. Ease of manufacture
33. Ease of operation
34. Ease of repair
35. Adaptability or versatility
36. Device complexity
37. Difficulty of detecting
38. Extent of automation
39. Productivity

Only 39 Matrix parameters!!!

TRIZ Inventive Principles

1. Segmentation
2. Taking out
3. Local quality
4. Asymmetry
5. Merging
6. Universality
7. Russian dolls
8. Anti-weight
9. Preliminary anti-action
10. Preliminary action
11. Beforehand cushioning
12. Equipotentiality
13. "The other way round"
14. Spheroidality - Curvature
15. Dynamics
16. Partial or excessive actions
17. Another dimension
18. Mechanical vibration
19. Periodic action
20. Continuity of useful action

21. Skipping
22. Blessing in disguise
23. Feedback
24. Intermediary
25. Self-service
26. Copying
27. Cheap short-lived objects
28. Mechanics substitution
29. Pneumatics and hydraulics
30. Flexible shells and thin films
31. Porous materials
32. Colour changes
33. Homogeneity
34. Discarding and recovering
35. Parameter changes
36. Phase transitions
37. Thermal expansion
38. Strong oxidants
39. Inert atmosphere
40. Composite materials

Only 40 Principles !!!

TRIZ Principles and Contradiction matrix

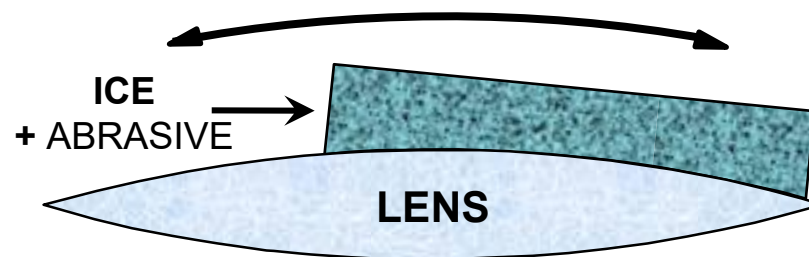
For our example with the lens:

Parameter that deteriorates						
Improving Parameter		...	9. Speed	...	17. Temperature	...
	...					
	9. Speed				2, 28, 30, 36	
	...					
	17. Temperature					
	...					

Suggested Principles that have solved similar Contradictions before

TRIZ in action - example

- Perform lookup* of TRIZ Matrix for this contradiction:
 - Improving 9: SPEED without damaging 17: TEMPERATURE
 - Find Principles to solve this contradiction:
 - 2. Taking out
 - 28. Mechanics substitution
 - 30. Flexible shells and thin films
 - 36. Phase transitions
- Use phenomena occurring during phase transitions (e.g. volume changes, loss or absorption of heat, etc.).

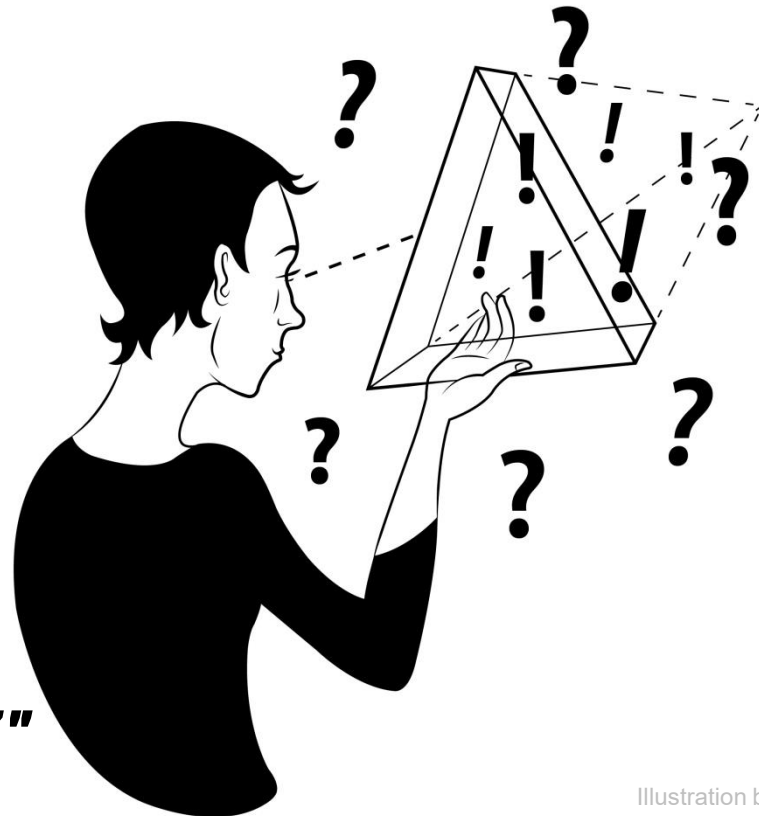


Abrasive + Ice - Inventive Principle 'Phase Transition'

*) E.g. at <http://www.triz40.com/>

TRIZ for Universities

Can TRIZ be useful in university education?



*Looking at the world
"through the prism of TRIZ"*

Illustration by Sasha Seraia

Can TRIZ be used in university

Yes, and very successfully **education?**

U.S. Particle Accelerator School
Education in Beam Physics and Accelerator Technology

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Current Program
USPAS sponsored by the University of California, Davis January 16-27, 2017 held in Rohnert Park, California
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Next Program
USPAS sponsored by Northern Illinois University June 12-23, 2017

USPAS class gets an oral talk at NAPAC 2016
Hearty congratulations to all!

A USPAS school project:
Compact ring-based X-ray source with on-orbit and on-energy laser-plasma injection
Marlene Turner, Aursalee Edelen, Andrei Seryi, Jeremy Cheatham, Osip Lishilin, Aakash Ajit Sahai, Brandon Zerbo, Andrew Lajoie, Chun Yan Jonathan Wong, Kai Shih, James Gerity, Gerard Lawler, Kookjin Moon.
Slides performed during the USPAS class "Unifying Physics of Accelerators, Lasers, and Plasmas" of Prof. Andrei Seryi. Teaching assistant: Andrew Sahai.

Class of graduate students, after one-week course on accelerators, lasers and plasma, and TRIZ, created a novel design and were invited to make a plenary invited presentation at the North American Particle Accelerator conference!

Can TRIZ be used in university education?

Yes. However, some critics:

It is not always possible to use prescriptive step-by-step methods with pre-defined tables of contradictions...

Expected critics: why only the first contradiction is addressed? Is it just a linear order correction?
How can TRIZ help to come to breakthrough ideas like theory of relativity? Etc...

Arguably, the way to teach TRIZ in universities should be different than in industrial companies...

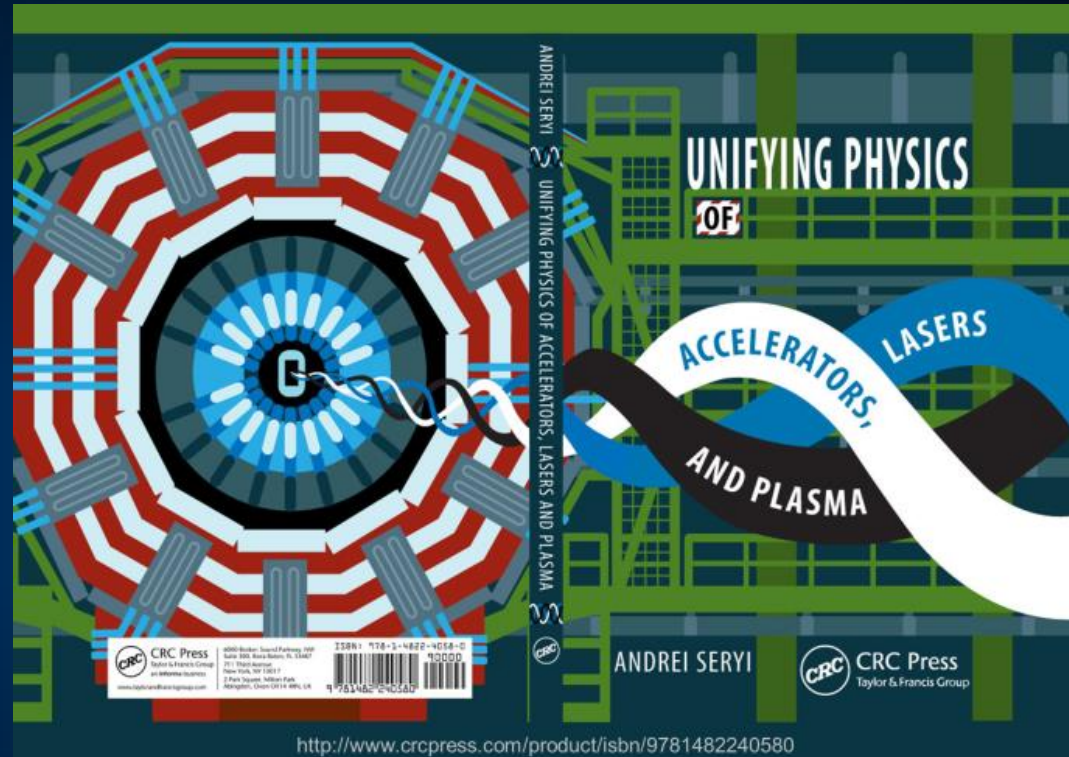
Maybe, the best way to introduce TRIZ to university is via the process of pro-active re-creation of TRIZ for science

TRIZ for universities

Can be very useful

Pro-active re-creation of TRIZ for university is attempted in this book:

- helps to connect different areas
- helps to learn inventiveness methods



However, this was just the first small step...

Creating TRIZ for science through the process of analysing and re-building TRIZ will also help us to study it proactively



Major components of TRIZ that should be kept for applications to university education (in extended & re-defined shape) are, to start with:

- inventive principles**
- laws of evolution of systems**

40 inventive principles in illustrations

- One can find many illustrations of inventive principles based on engineering examples
- On the next pages you will find illustrations based on **accelerator science and some other areas of science**
 - You will notice that some of the standard definitions of TRIZ principles are re-defined
 - Selected principles will be shown

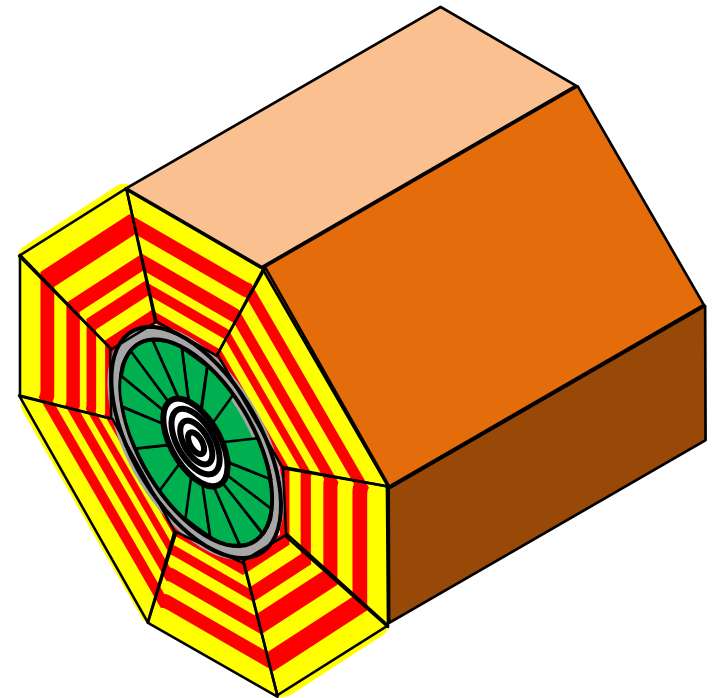
See more details in:

Accelerating Science TRIZ inventive methodology in illustrations
Elena Seraia, Andrei Seryi

arXiv:1608.00536 [physics.ed-ph]
<https://arxiv.org/abs/1608.00536>

7. Nested doll

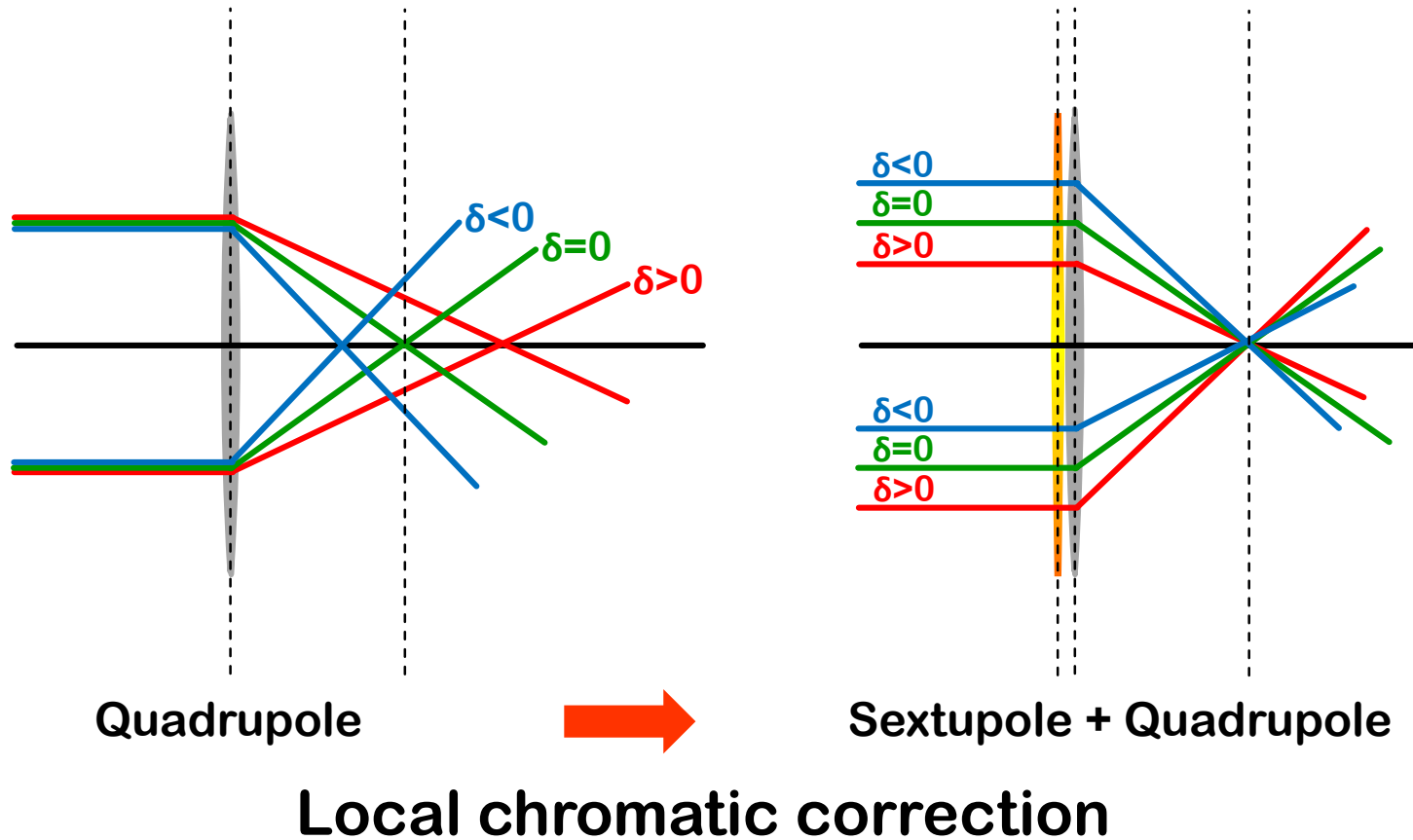
- Place one object inside another; place each object, in turn, inside the other.
 - Make one part pass through a cavity in the other.



High energy physics detectors

9. Preliminary anti-action

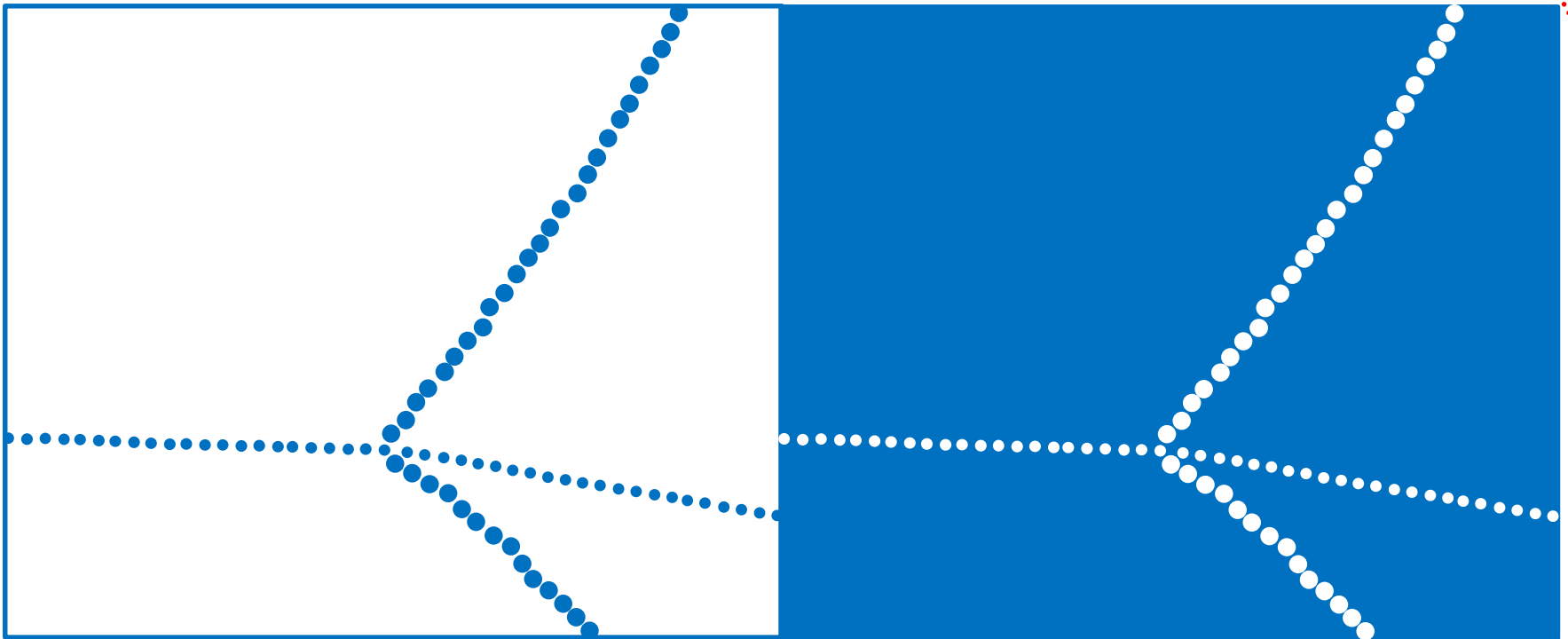
- If it will be necessary to do an action with both harmful and useful effects, this action should be replaced with anti-actions to control harmful effects.
- Create beforehand stresses in an object that will oppose known undesirable working stresses later on.



P. Raimondi, A. Seryi, PRL, 86, 3779 (2001)

13. The other way round

- Invert the action(s) used to solve the problem (e.g. instead of cooling an object, heat it).
- Make movable parts (or the external environment) fixed, and fixed parts movable.
 - Turn the object (or process) “upside down”.



Cloud and bubble chambers

35. Parameter changes

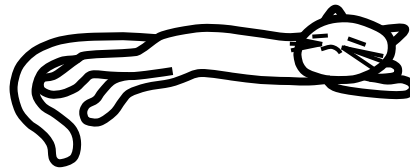
- Change an object's physical state (e.g. to a gas, liquid, or solid.)
 - Change the concentration or consistency.
- Change the degree of flexibility. Change the temperature.
 - Change volume to surface ratio, etc.



15° C



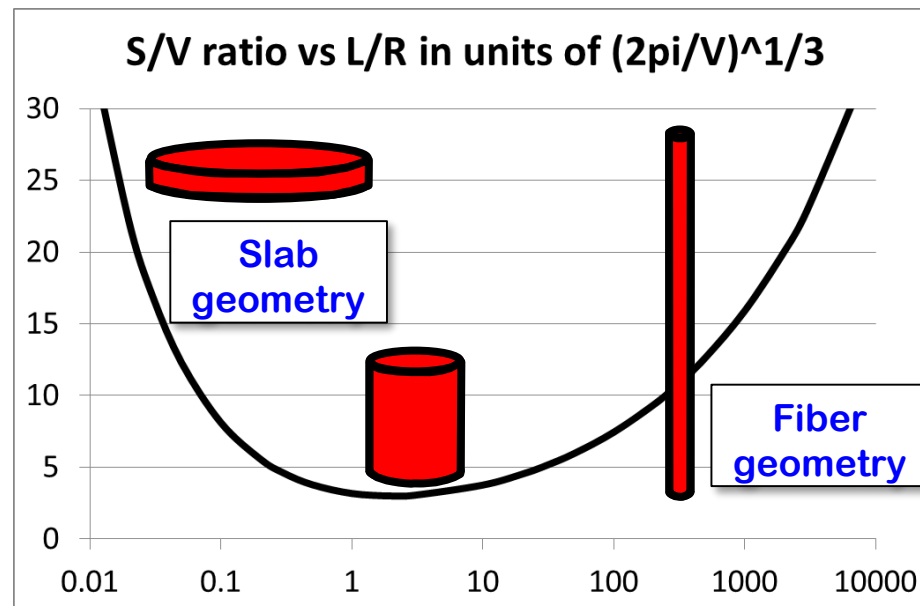
20° C



25° C



40° C



Fiber
lasers

...that was an example

... on how to make TRIZ more suitable for science

The principle “parameter change” can be interpreted very widely, and thus be very applicable to science

Define the “parameter” as the ratio of volume to surface area



The principle of changing the volume to surface ratio



The same volume,
but different surface area



The principle of changing the volume to surface ratio



The same volume,
but different surface area

The same principle is used in e^+e^- colliders, where "pancakes" are collided instead of "buns"



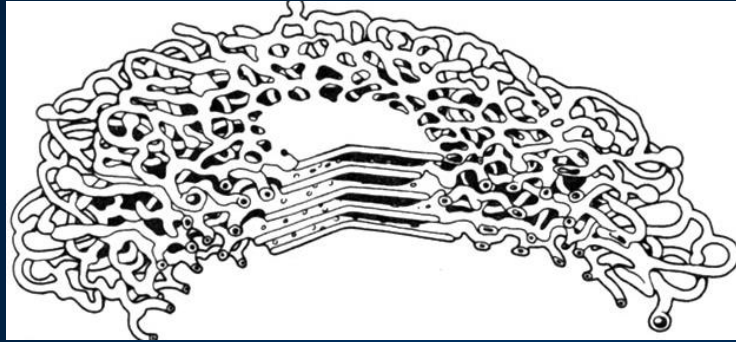
The principle of changing the volume to surface ratio – an example



The same volume, but different surface area and the different amount of information 😊

And could we suggest an example illustrating this principle, for instance, in biology?

The principle of changing the volume to surface ratio – examples



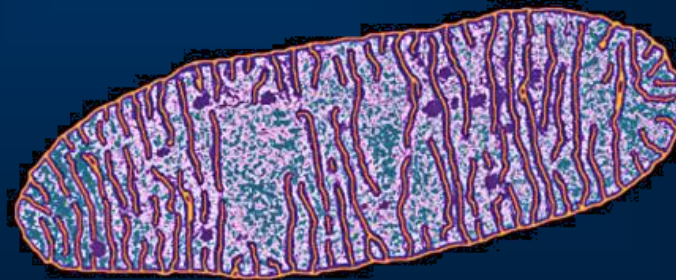
Golgi Apparatus



Brain



Algae



Mitochondria

Keeping the same volume but increasing the surface area to enhance the functionality

Inventive principles and fundamental symmetries

Including change of V/S into the principle “parameter change” connects it to fundamental symmetries, i.e. conservation laws of physics

$$\int_{\Delta V} d^3x \nabla \cdot \mathbf{A} = \oint_{\Delta S} \mathbf{A} \cdot d\mathbf{S}$$

Gauss theorem (divergence theorem): the total sources and sinks of a vectorial quantity, or the integral volume of its divergence, is equal to the net flux of this vectorial quantity across the volume boundary



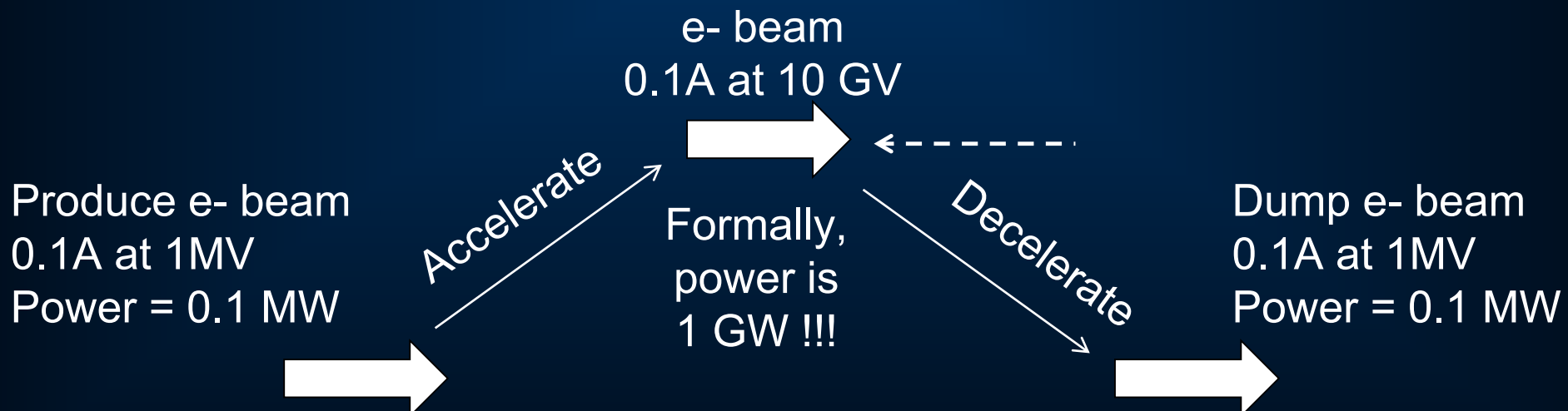
Further adjustments

What about quantum effects?

Can we (should we) include some inventive principles related to uncertainty principle, quantum entanglement, etc.?

Or what about energy recovery?

The method which enables many modern scientific instruments



Inventing sci. instruments

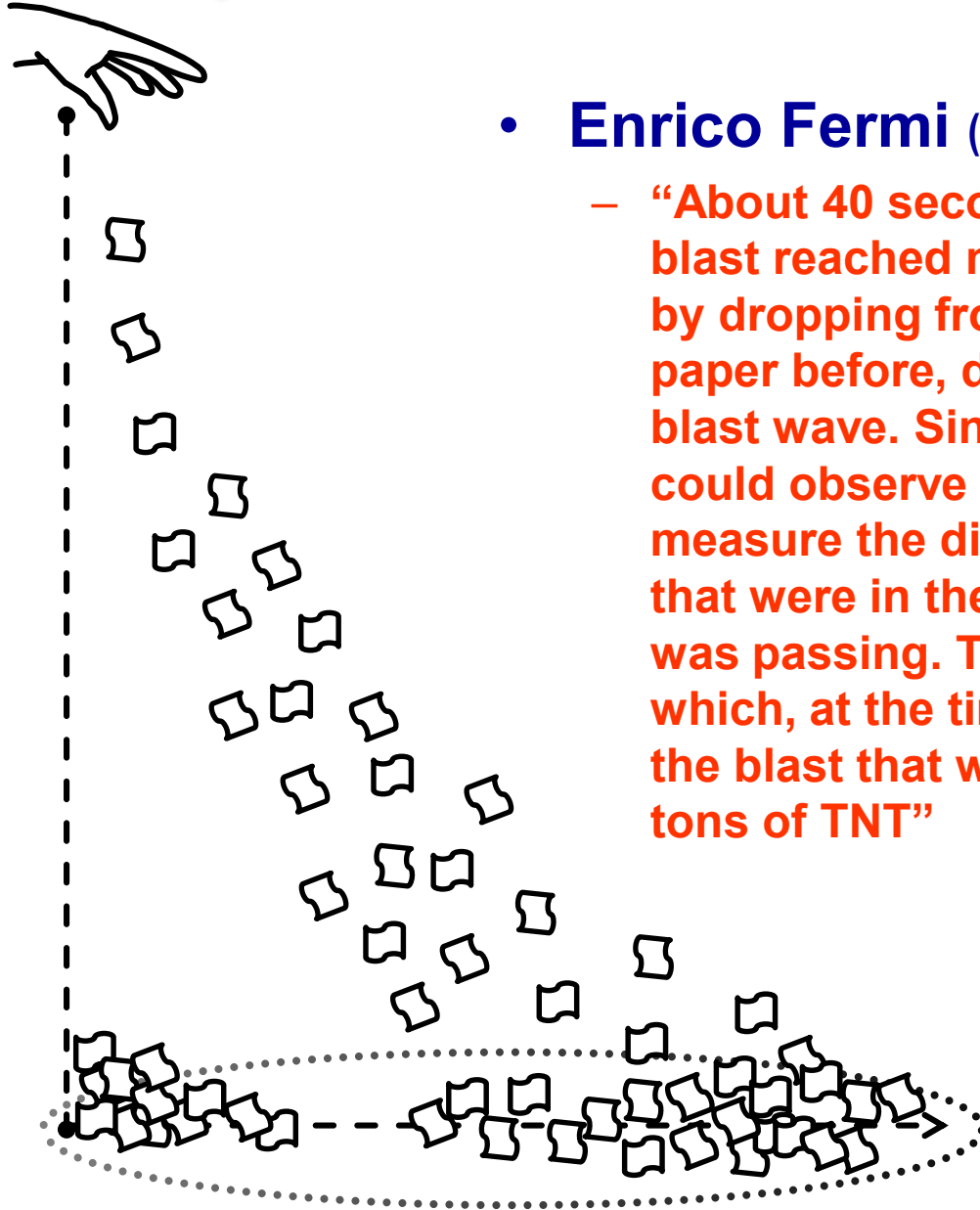
Even if one would define a set of TRIZ inventive principles that would include many approaches used in science, would it be sufficient?

No

What would be missing?

Most importantly – the art of estimations

Example of back-of-the-envelope estimations



- **Enrico Fermi** (who was ~10 miles from the Trinity test):
 - “About 40 seconds after the explosion the air blast reached me. I tried to estimate its strength by dropping from about six feet small pieces of paper before, during, and after the passage of the blast wave. Since, at the time, there was no wind I could observe very distinctly and actually measure the displacement of the pieces of paper that were in the process of falling while the blast was passing. The shift was about 2 1/2 meters, which, at the time, I estimated to correspond to the blast that would be produced by ten thousand tons of TNT”

Importance of back-of-the-envelope estimations

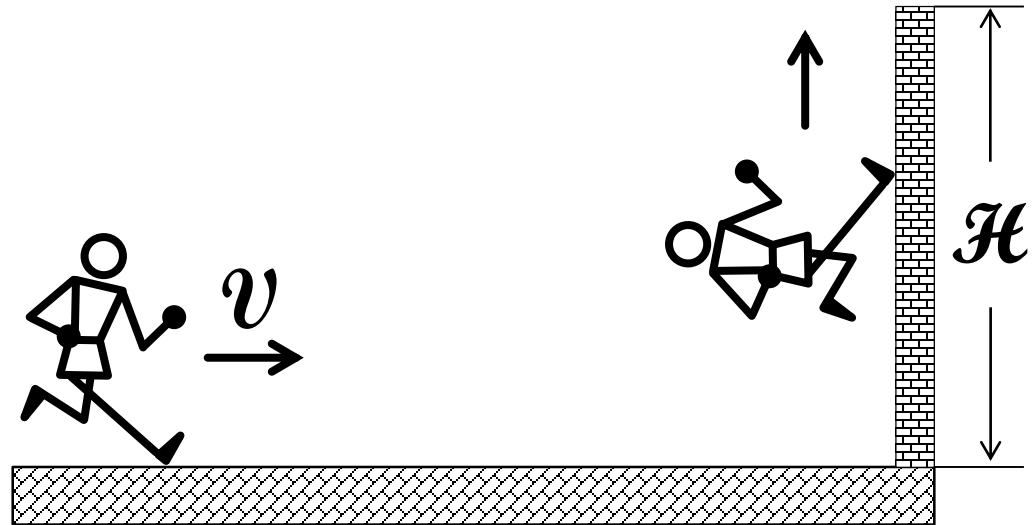
- They are important because
 - they help to quickly check if your idea is viable - obvious
 - but even more important: they allow to improve cross-disciplinary understanding of scientists from different fields, like biology and physics
- To train yourself on back-of-envelope estimations one can consider various questions
- They do not have to be necessarily serious ;-)
- But the estimates should be based on a physical effect that is considered most important for a given question
- Making an estimate would also allow us to make invention how to improve a system

Importance of back-of-the-envelope estimations

- They are important because
 - they help to understand things better - obvious
 - but even more important: they allow to improve cross-disciplinary understanding of scientists from different fields, like biology and physics

- What speed V is needed to reach height H and get to other side of the wall?

;-)



Importance of back-of-the-envelope estimations

- Estimate by requiring that during run along the wall the head would not fall to lower than half the height of the person...

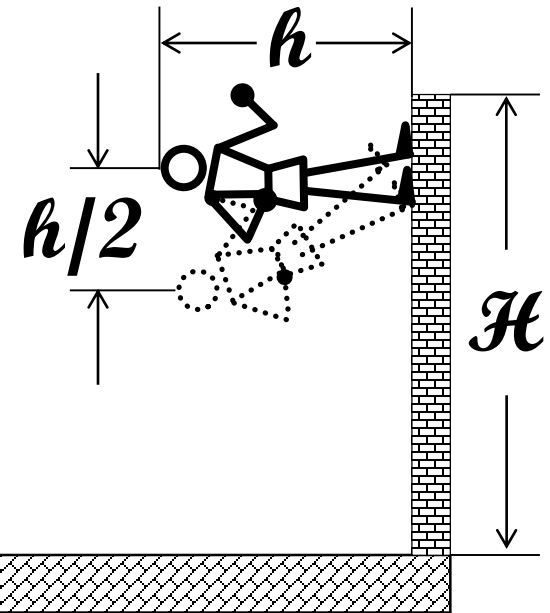
; -)

You will then find

$$V = H (g/h)^{1/2}$$

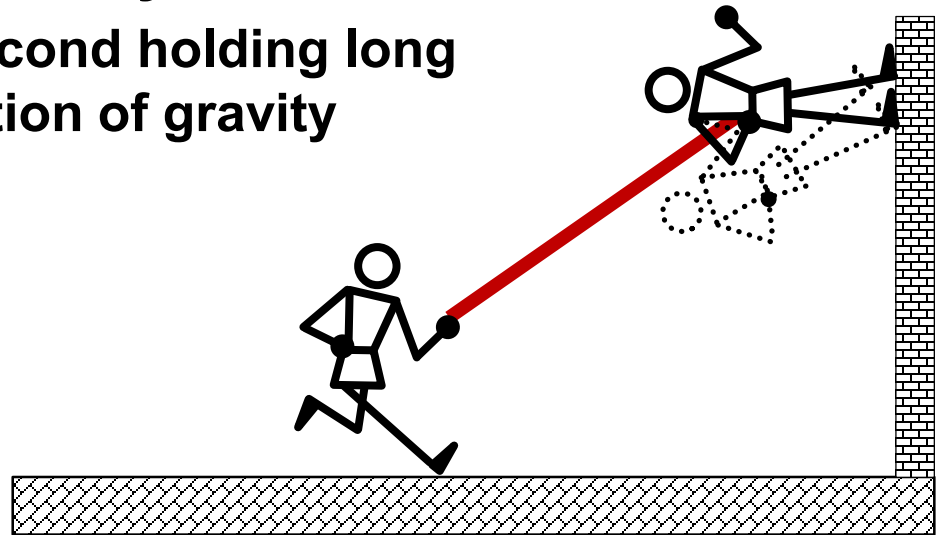
or, for $H=2\text{m}$

$$V \sim 4.7 \text{ m/s}$$



Applying an inventive principle

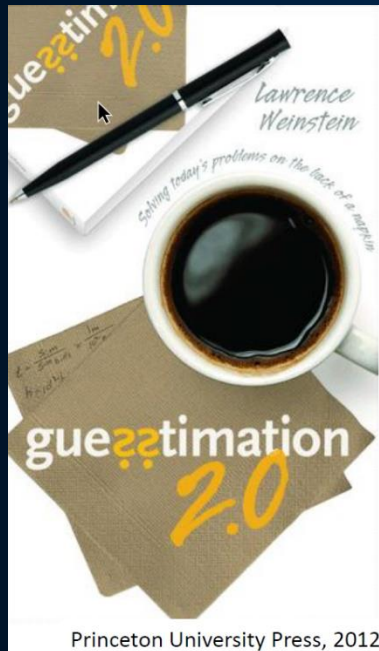
- Once we made the estimation and understood the main challenge, we can solve the problem
- Apply inventive principle
- In this case – #9 Preliminary anti-action
 - Two people running, second holding long stick to compensate action of gravity



The art of estimating

Enrico Fermi was known for his ability to back-of-envelope estimations

Many leading centers teach the art of estimating from school – e.g. the unique Phys-Math school in Novosibirsk

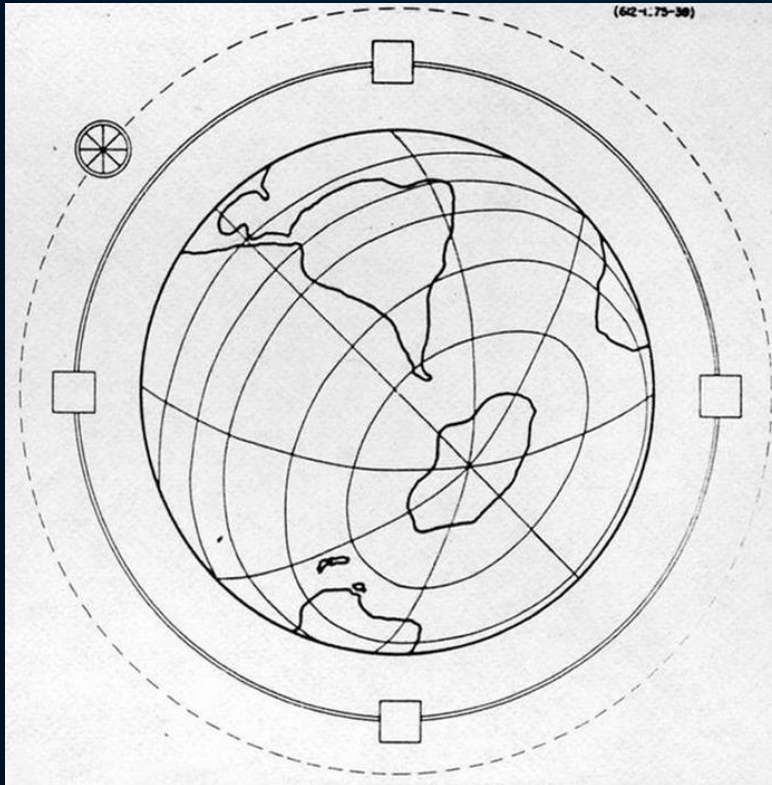


(Clockwise from top left) The Phys–Math School in Novosibirsk, Russia. Students deriving formulae during a class. Graduates of the 1963 class.

<http://cerncourier.com/cws/article/cern/69910>

There are books that can help to master the art of estimations, e.g. “Guesstimation 2.0” by Lawrence Weinstein (Old Dominion University)

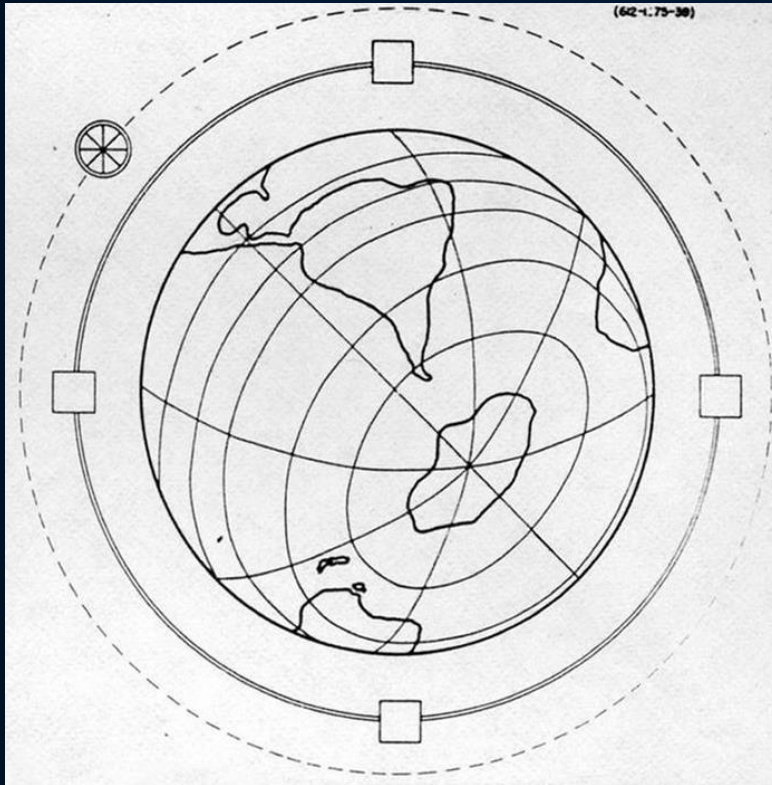
Evolution of accelerators



In 1954 Enrico Fermi presented, in his lecture, a vision of an accelerator that would encircle the Earth, and would attain highest possible energies

Would this be indeed a natural evolution of accelerators?

Evolution of accelerators



Enrico Fermi Earth accelerator, 1954

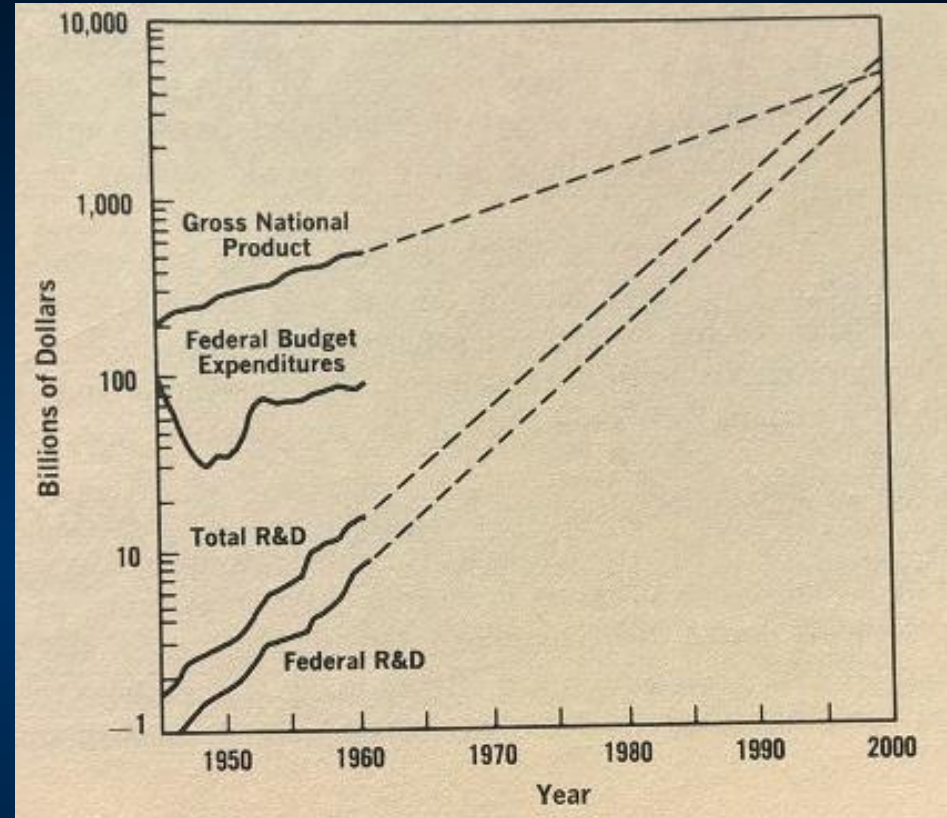
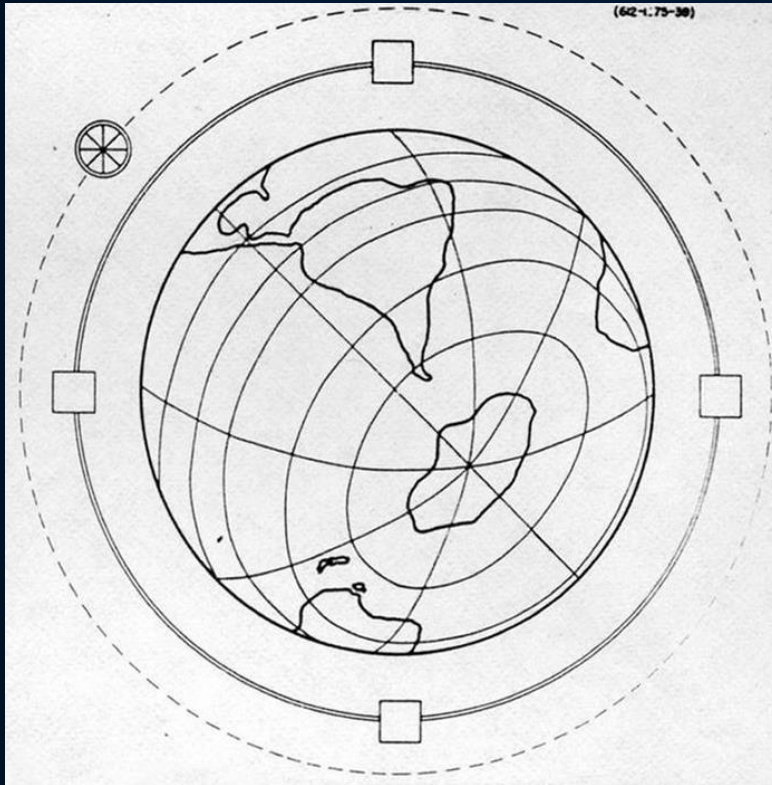


Fig 6, GNP and R&D: Failure of naïve extrapolation. "The Year 2000", 1968, K. Herman, A. Wiener

Would this be indeed a natural evolution of accelerators?
No. And not only because R&D budget is now not growing faster than GDP

Evolution of accelerators



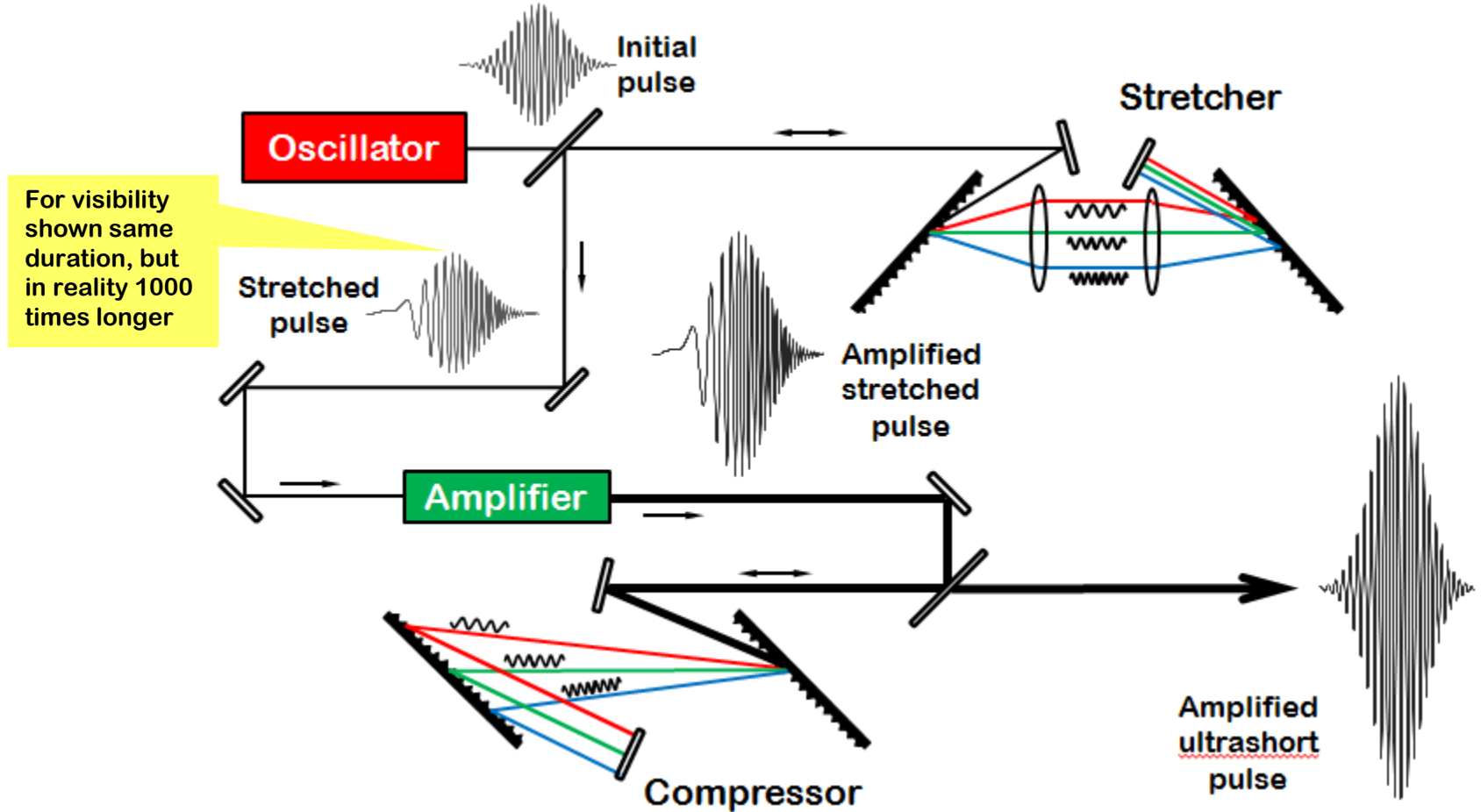
Enrico Fermi Earth accelerator, 1954

Would this be indeed a natural evolution of accelerators?

No.

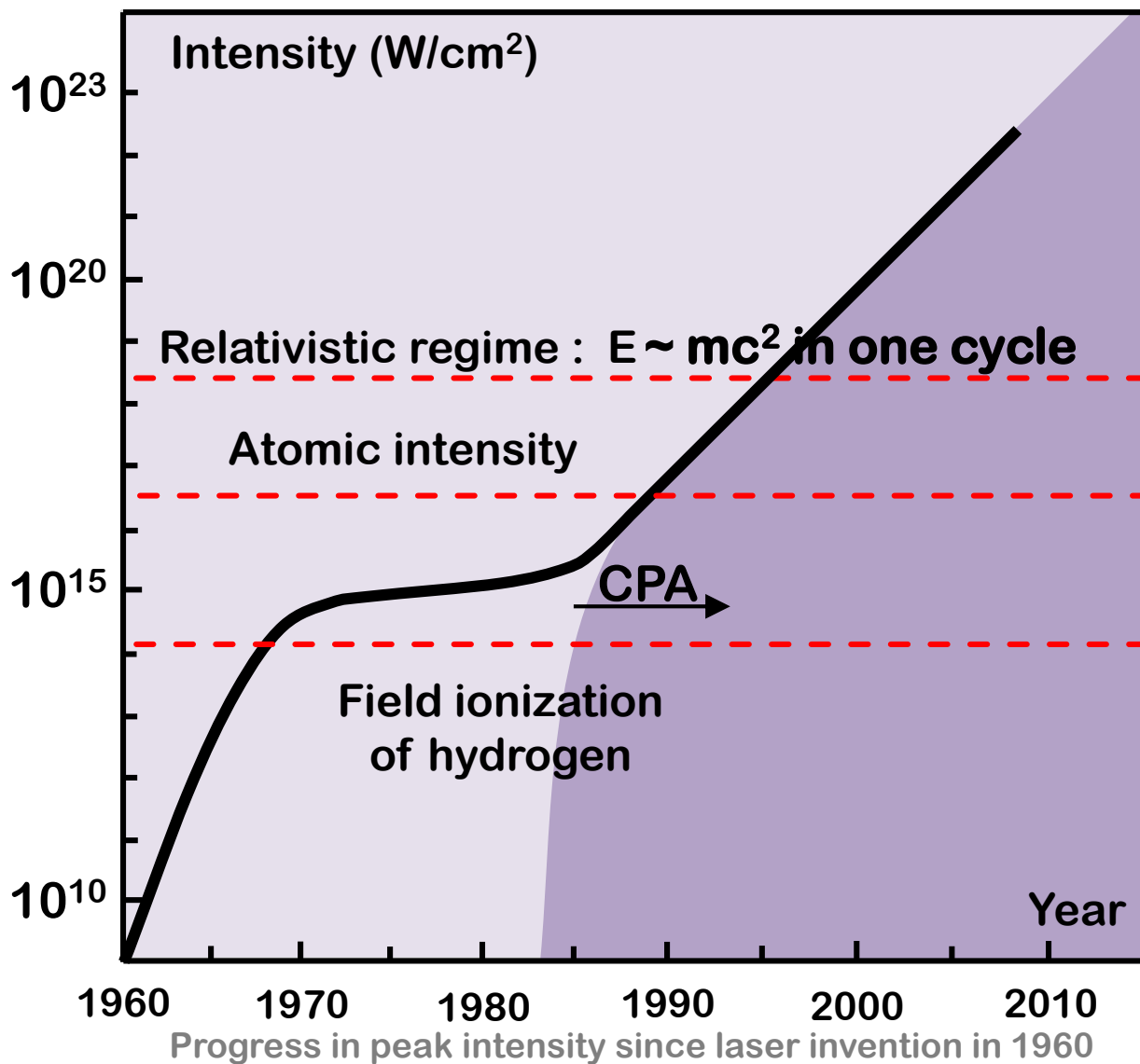
Increasing the size or base of the experiment, to increase precision, with proportional or even faster increase of the cost, would unlikely be accepted by governments and society

CPA – Chirped Pulse Amplification



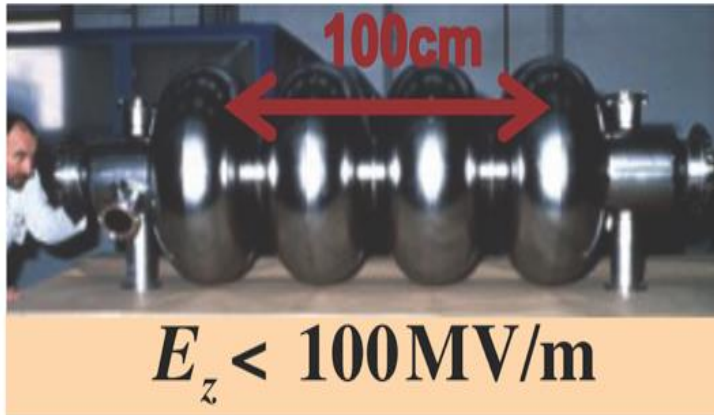
- CPA: pulse stretching and compressing using time-energy correlation

CPA invention: exponential growth of laser power

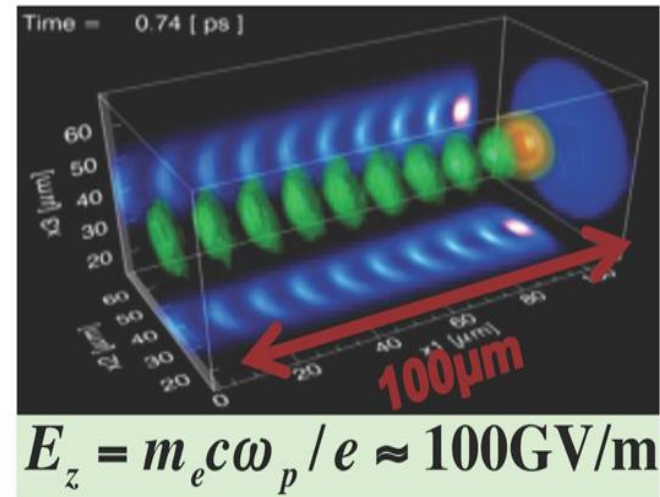
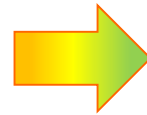


27. Cheap short-living objects

- Replace an expensive object with a multiple of inexpensive objects, comprising certain qualities (such as service life, for instance).



Accelerating structure,
metal (normal conductive
or super-conductive)



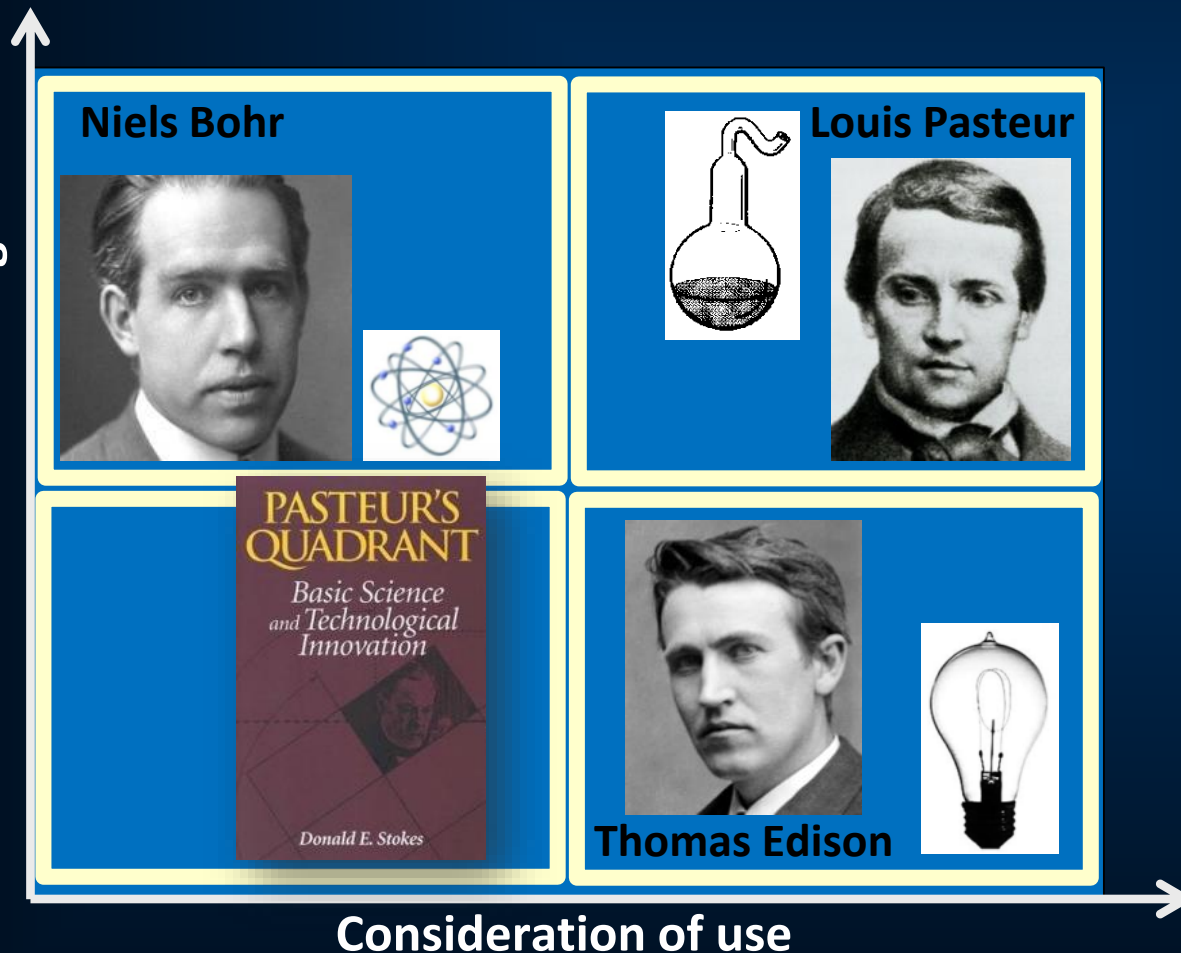
“Accelerating structure”
produced on-the-fly in
plasma by laser pulse

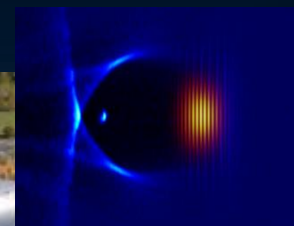
Plasma acceleration

**Let's now talk about evolution of
synchrotron light sources and FELs**

including those based on plasma acceleration

**But first, let's define some metric which allow us
to evaluate and compare the importance of
different directions of research**

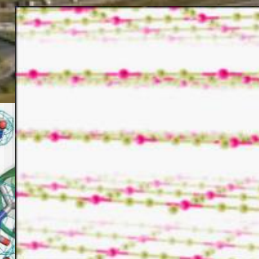




Niels Bohr



Louis Pasteur



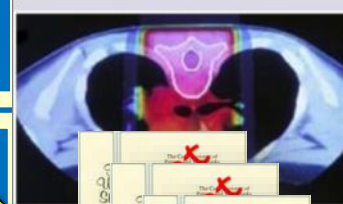
Accelerator Science & Technology

PASTEUR'S QUADRANT

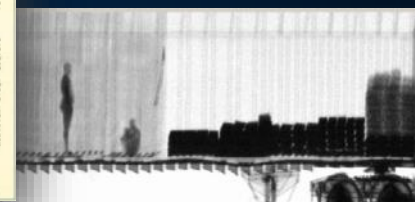
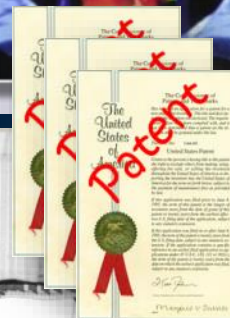
Basic Science
and Technological
Innovation

Donald E. Stokes

Thomas Edison

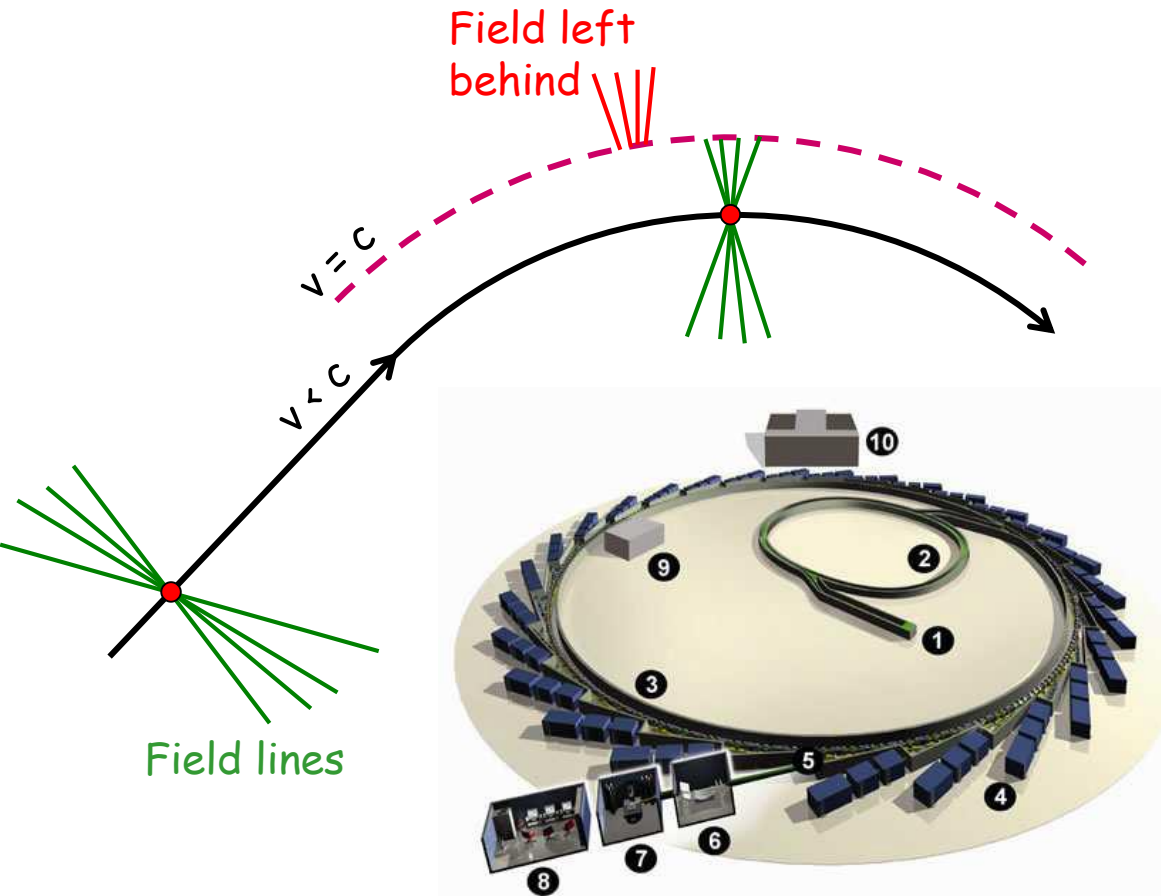


Protons/Ions

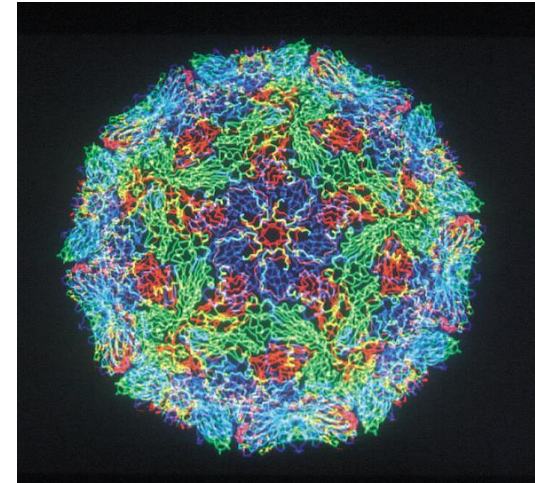


Synchrotron radiation light sources

Synchrotron Radiation (SR) caused by leaving part of fields behind when the beam moves along the curve



Synchrotron radiation can be useful



Example of a structure of a virus decoded using SR



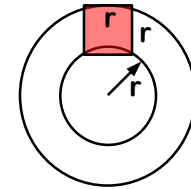
Diamond SR source at Harwell, UK

Synchrotron radiation *on-the-back-of-the-envelope* – power loss

Energy in the field left behind (radiated !):

$$W \approx \int E^2 dV$$

The field $E \approx \frac{e}{r^2}$ the volume $V \approx r^2 dS$



Energy loss per unit length:

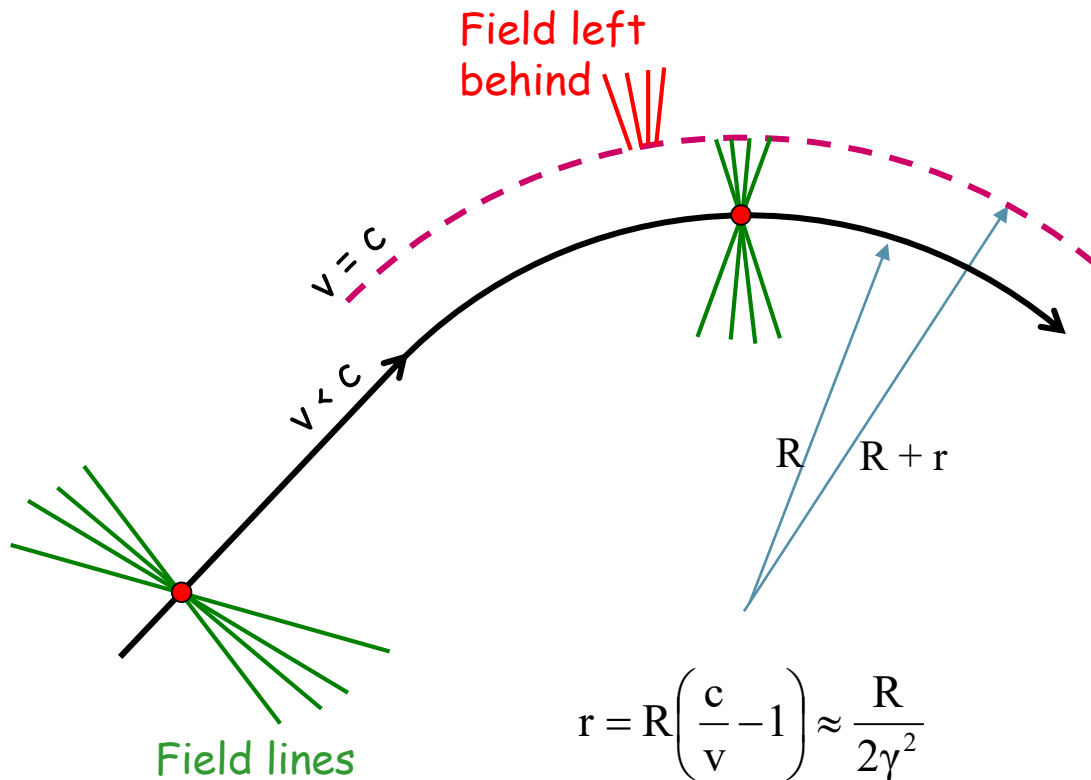
$$\frac{dW}{dS} \approx E^2 r^2 \approx \left(\frac{e}{r^2} \right)^2 r^2$$

Substitute $r \approx \frac{R}{2\gamma^2}$ and get an estimate:

$$\boxed{\frac{dW}{dS} \approx \frac{e^2 \gamma^4}{R^2}}$$

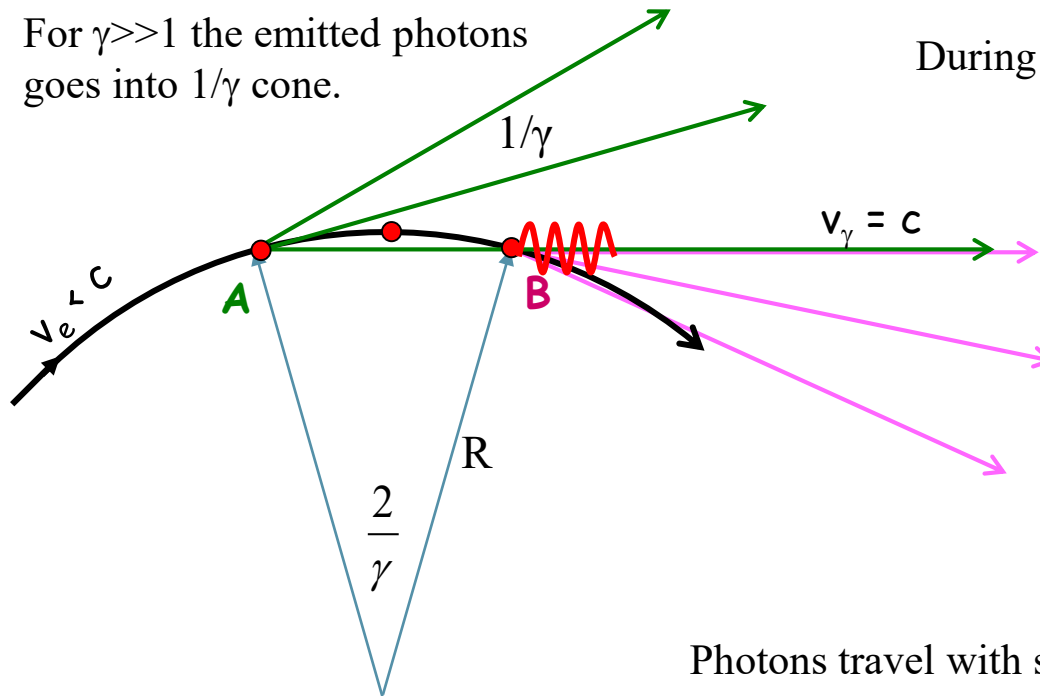
Gaussian units on this page!

Compare with exact formula: $\frac{dW}{dS} = \frac{2}{3} \frac{e^2 \gamma^4}{R^2}$



Synchrotron radiation *on-the-back-of-the-envelope* – photon energy

For $\gamma \gg 1$ the emitted photons go into $1/\gamma$ cone.



During what time Δt the observer will see the photons?



Photons emitted during travel along the $2R/\gamma$ arc will be observed.

Photons travel with speed c , while particles with v .

At point B, separation between photons and particles is

$$dS \approx \frac{2R}{\gamma} \left(1 - \frac{v}{c} \right)$$

Therefore, observer will see photons during $\Delta t \approx \frac{dS}{c} \approx \frac{2R}{c\gamma} (1 - \beta) \approx \frac{R}{c\gamma^3}$

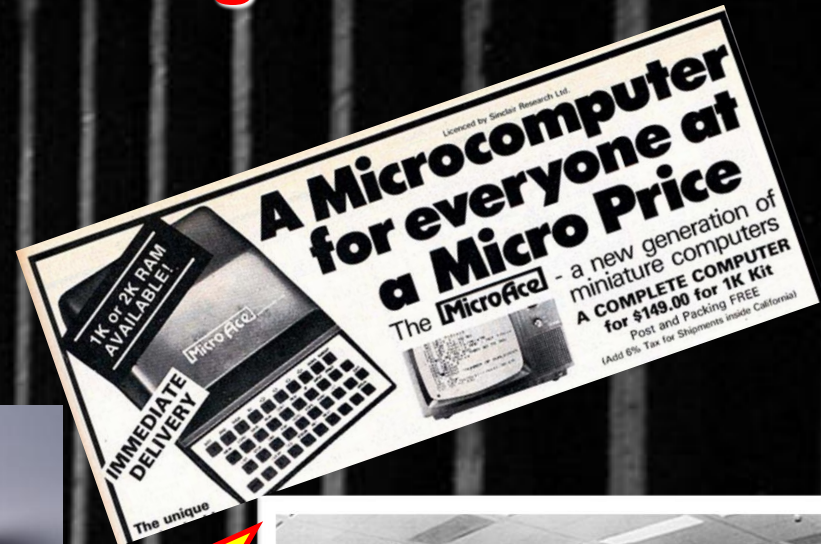
Estimation of characteristic frequency $\omega_c \approx \frac{1}{\Delta t} \approx \frac{c\gamma^3}{R}$

Compare with exact formula: $\omega_c = \frac{3}{2} \frac{c\gamma^3}{R}$

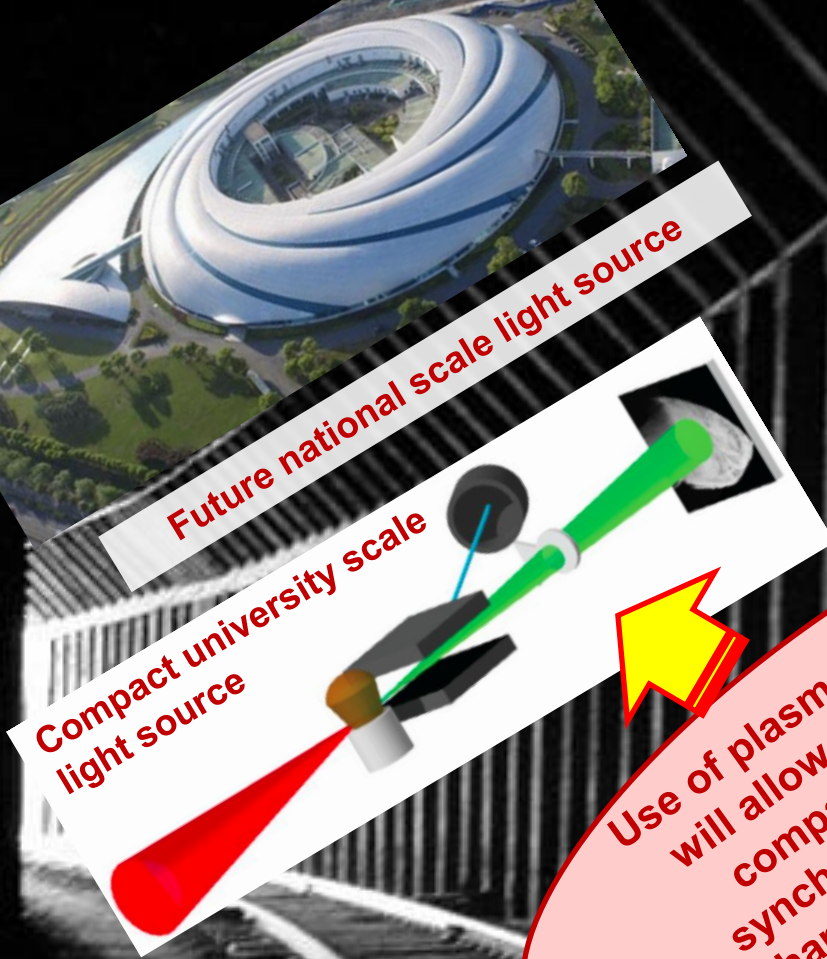
Evolution of computers and light sources



"IBM bringing out a personal computer would be like teaching an elephant to tap dance" cca. 1981



Evolution of computers and light sources

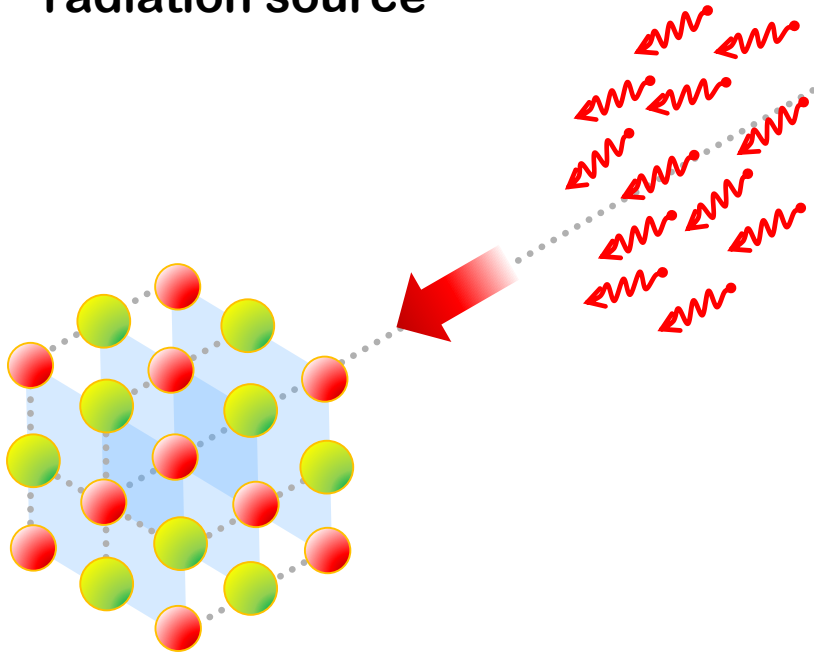


Use of plasma acceleration will allow to create very compact sources of synchrotron radiation – change of the paradigm

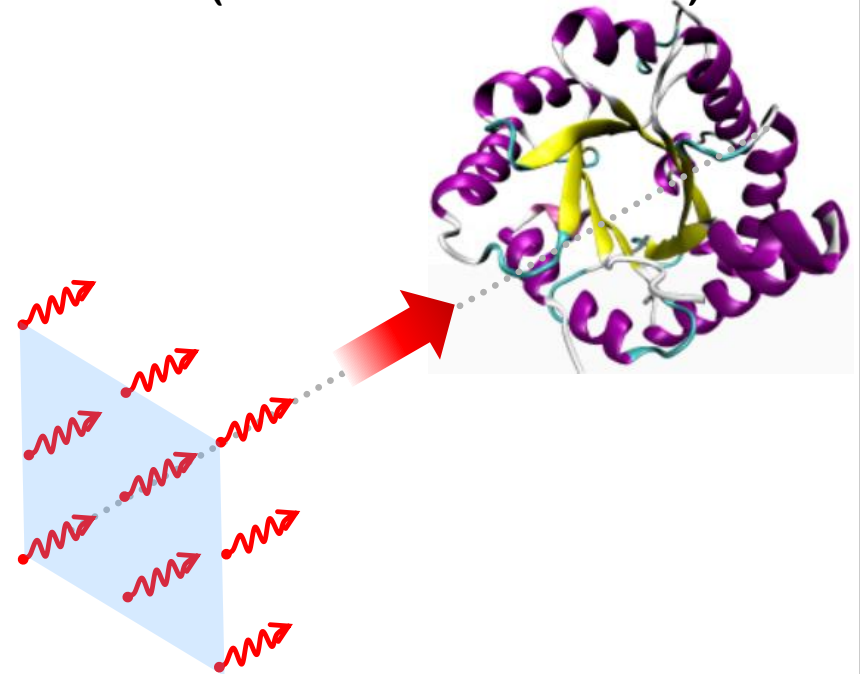


Incoherent SR => coherent

Incoherent synchrotron radiation source



Coherent synchrotron radiation source (Free Electron Laser)



Era of studies of crystal structures by incoherent sources of X-rays

Era of studies of non-crystalline structures by coherent sources of X-rays

...and also this is an inventive principle “the other way around”

Further evolution of light sources

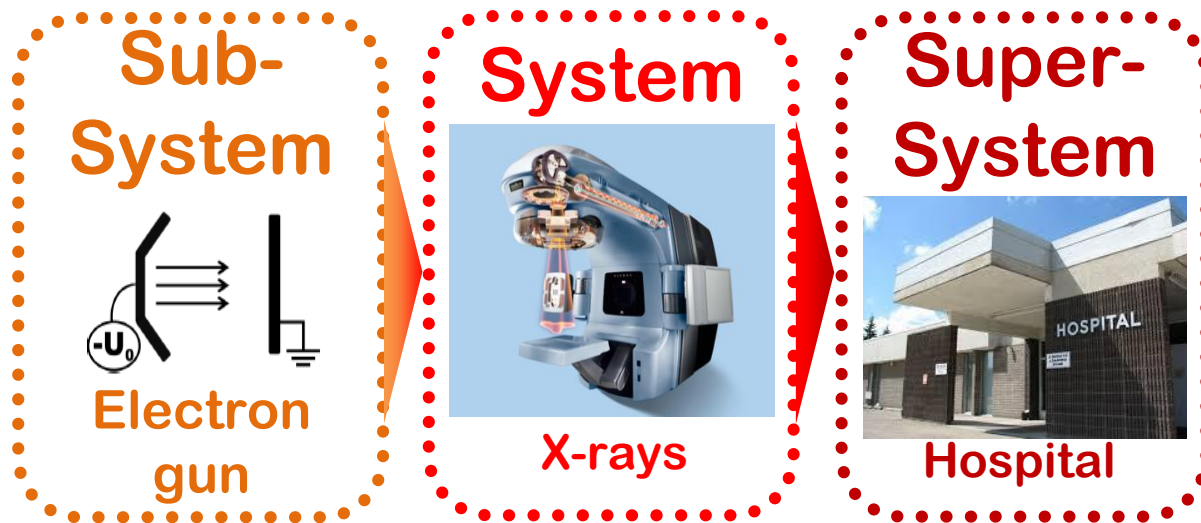
Let's assume that laser-plasma FEL is working

What are long-terms perspectives and evolution of light sources then?

Let's apply the TRIZ general laws of evolution

Transition to a super-system

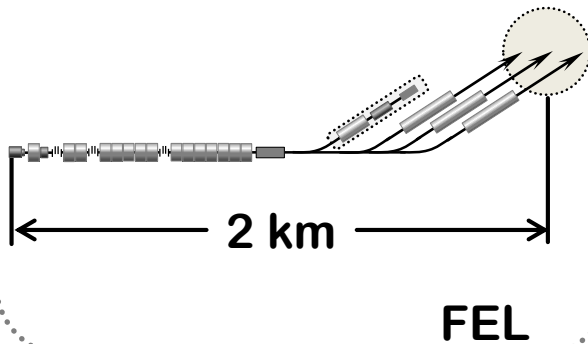
- **Kinematic laws (standard TRIZ)**
 - The law of transition to a super-system



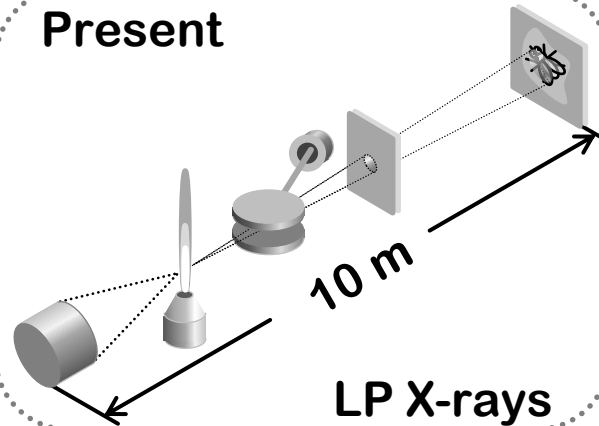
“a system exhausting possibilities of further significant improvement is included in a super-system as one of its parts”

FEL evolution forecast

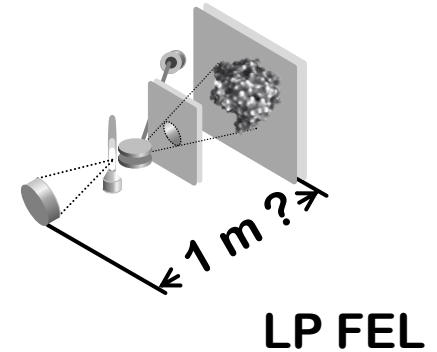
Past



Present

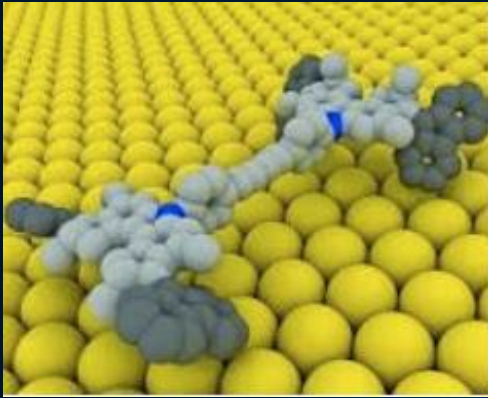


Future



FEL will be so compact and developed that it can become part of another system, and that system in turn part of super-system

Nobel prize 2016 – molecular machines



Pierre Sauvage, J. Fraser Stoddart, and Ben L. Feringa, Chemistry Nobel Prize 2016

TRIZ evolution laws allow to predict what parts of molecular machine would be invented next:

■ Static Laws

- The law of the completeness of the parts of the system
 - 4 parts: engine, transmission, working unit, control element
- The law of energy conductivity of the system
 - every technical system is a transformer of energy and it should circulate freely and efficiently through its 4 main parts

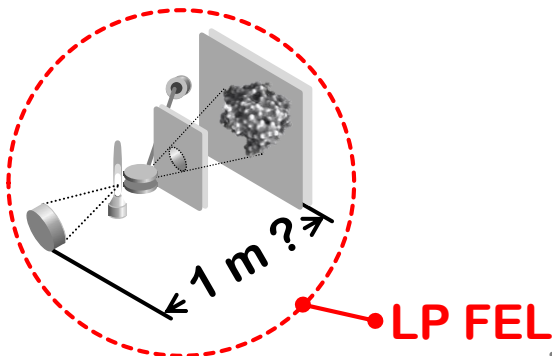
**Most importantly –
these machines can become part of another super-system**

FEL and molecular machine becomes part of another system

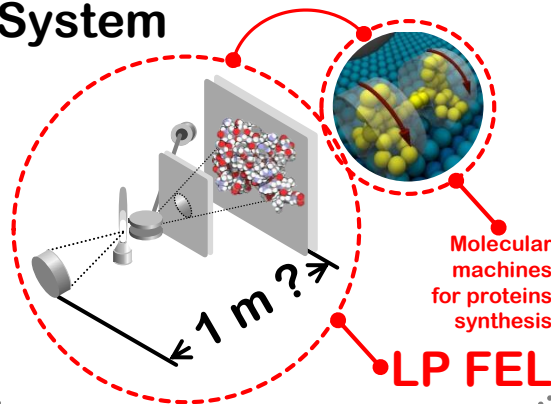
FEL is part of system where it analyses proteins synthesized by molecular machine

FEL is part of super-system where it analyses proteins synthesized by molecular machine, while the entire super-system produces patient-tailored molecular machines for DNA repair

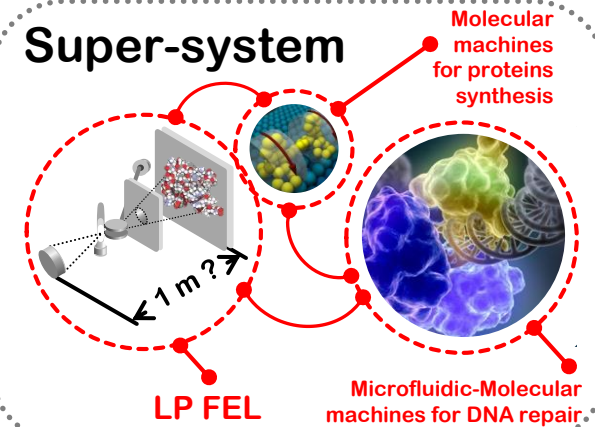
Sub-system



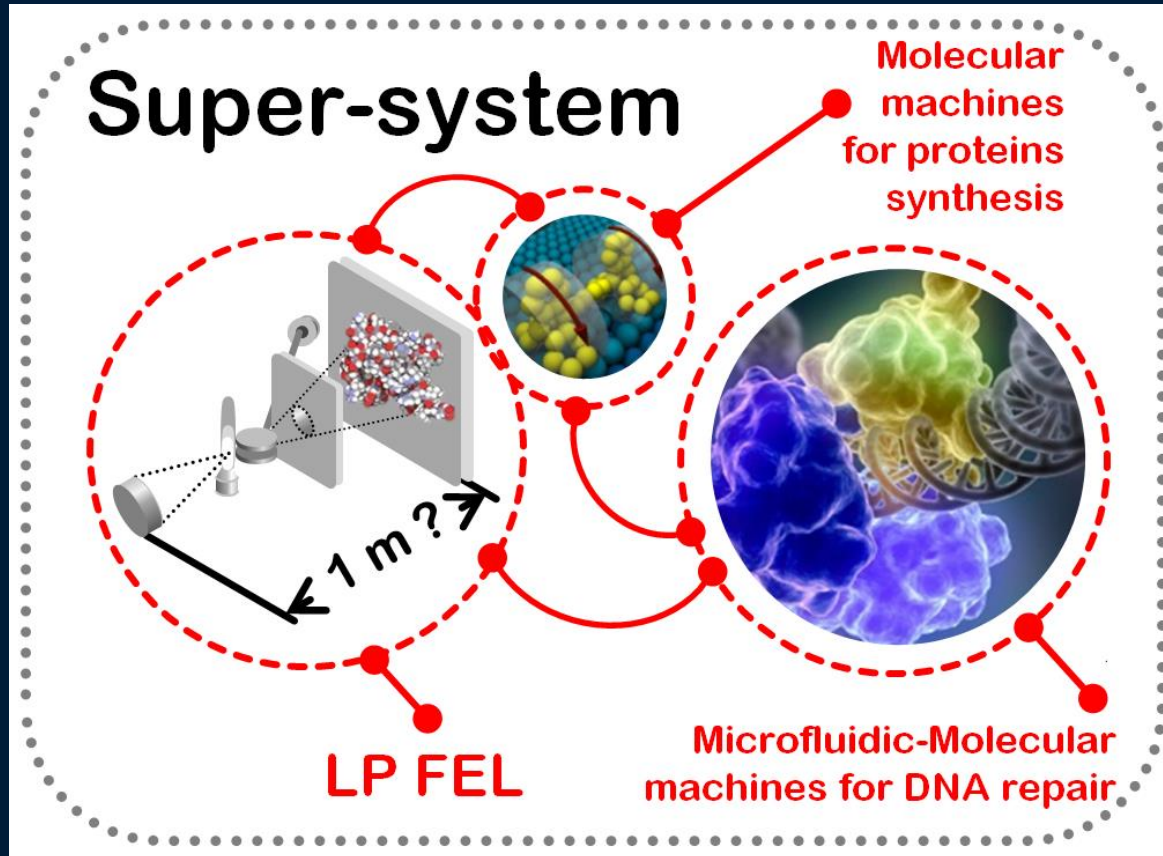
System



Super-system



Make this dream – with help of Breakthrough By Design approach – a reality!



Laser plasma FEL is part of super-system where it analyses proteins synthesized by molecular machine, while the entire super-system produces patient-tailored molecular machines for DNA repair

Make this dream – with help of Breakthrough By Design approach – a reality!

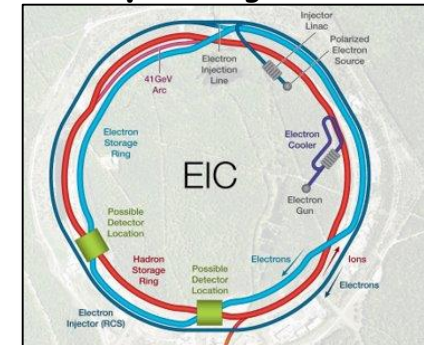
Sub-system



System



Super-system



There are similar challenging and forward looking ideas based on technologies developed for the Electron-Ion Collider

Reiterate the main message

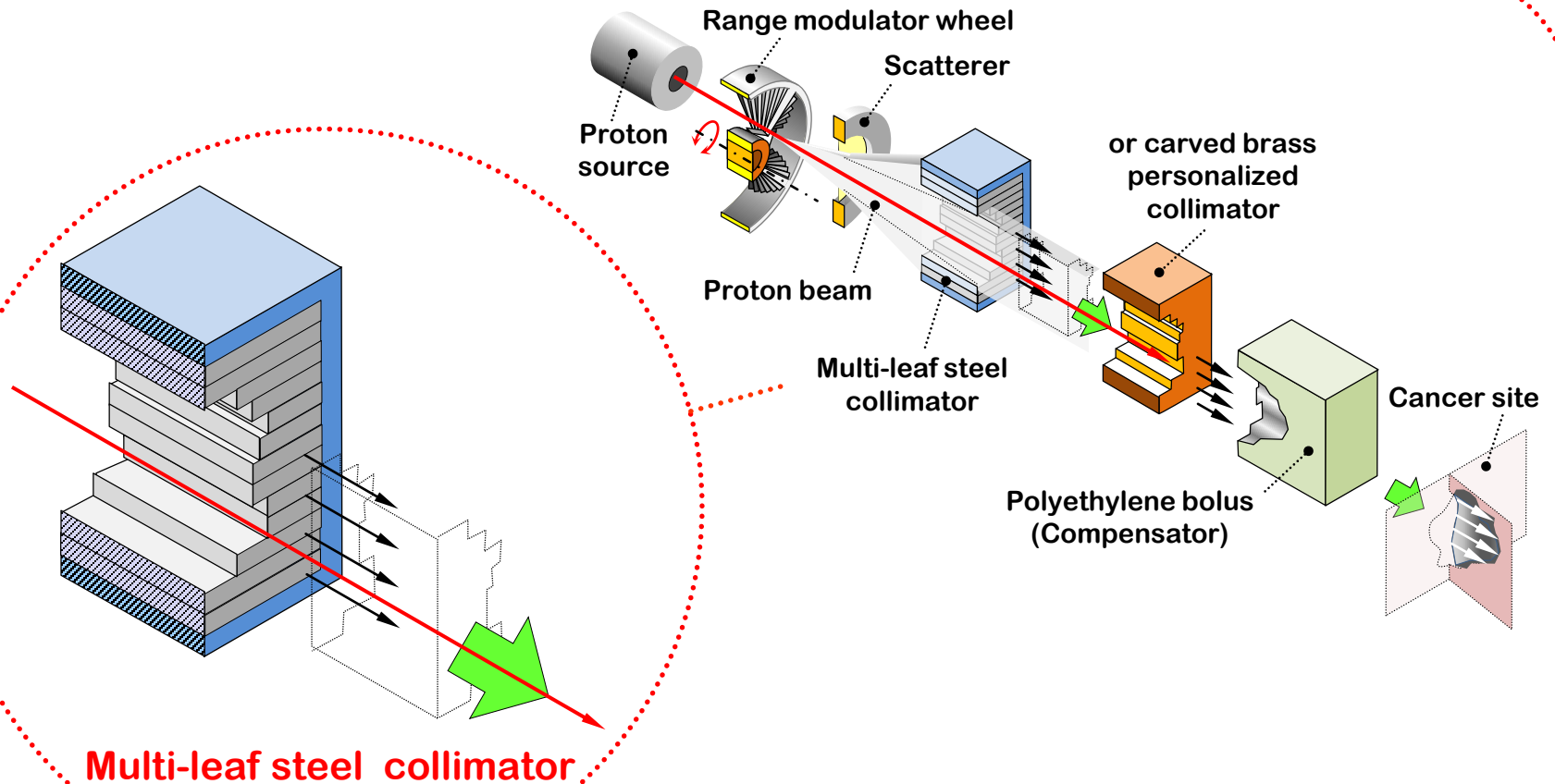
- **Breakthrough By Design approach – re-formulated TRIZ combined with the art of estimations – for connecting different areas of science & engineering**
- **Step by step development and implementation at various levels**
 - Undergraduate, master, PhD courses, technical/industrial courses
 - With Natural Sciences, Engineering, Humanities
 - With nearby industry
 - cross silos, stronger connection between science and the needs of society
- **Target audience**
 - Undergraduates
 - Graduate student
 - Technical experts in the area
 - Veterans who are re-training for new career
- **Possible implementation – via traineeships or institutes**
 - aim: connect different areas of science & industries with enhanced impact on all
 - composition: researchers, postdocs, industrial affiliates, innovation facilitators
 - connections: industry, science, business and investor angels
 - impact on science and industry, impact on society

Let's look at some areas of accelerator S&T via inventive principles

**We will see how much we can fit into the remaining time
of this 2-hrs tutorial**

1. Segmentation

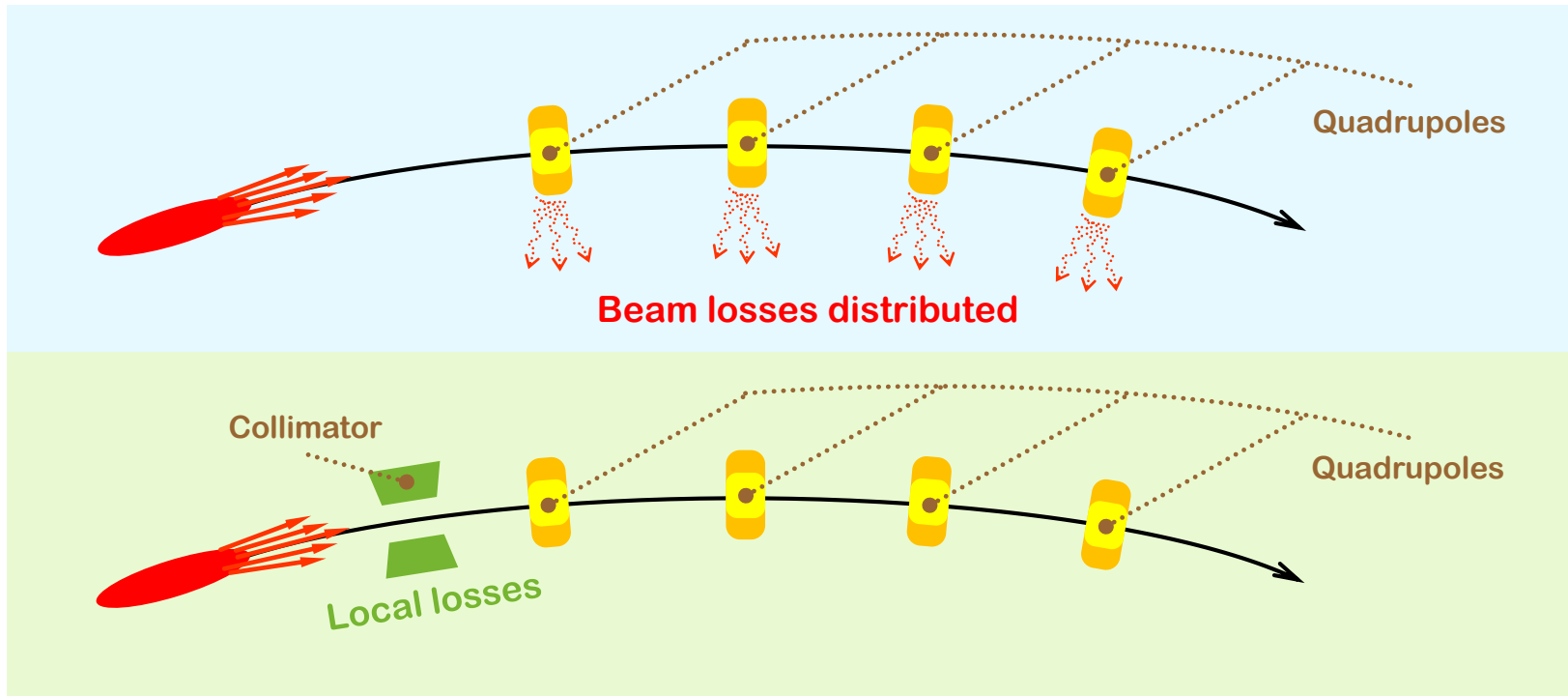
- Divide an object into independent parts.
 - Make an object easy to disassemble.
- Increase the degree of fragmentation or segmentation.



Proton therapy

2. Taking out

- Separate an interfering part or property from an object;
- Single out the only necessary part (or property) of an object.



Collimation of the beam to localize beam losses

3. Local quality

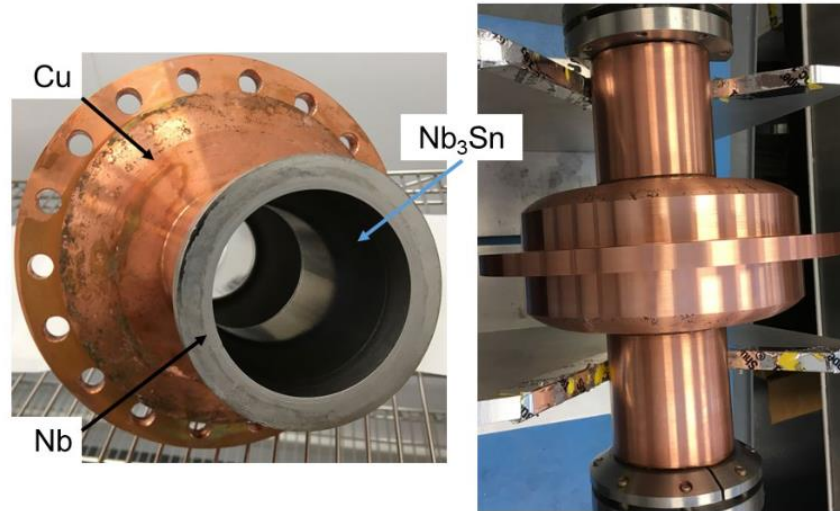
- Change an object's structure from uniform to non-uniform, change an external environment (or external influence) from uniform to non-uniform.
- Make each part of an object function in conditions most suitable for its operation.
 - Make each part of an object fulfill a different and useful function.



Nb coated copper cavity

Enzo Palmieri, A.A.Rossi, R. Vaglio, “Experimental Results on Thermal Boundary Resistance for Nb and Nb/Cu”, Science, Oct 2014

Cu-Nb-Nb₃Sn cavities & TRIZ #3 local quality



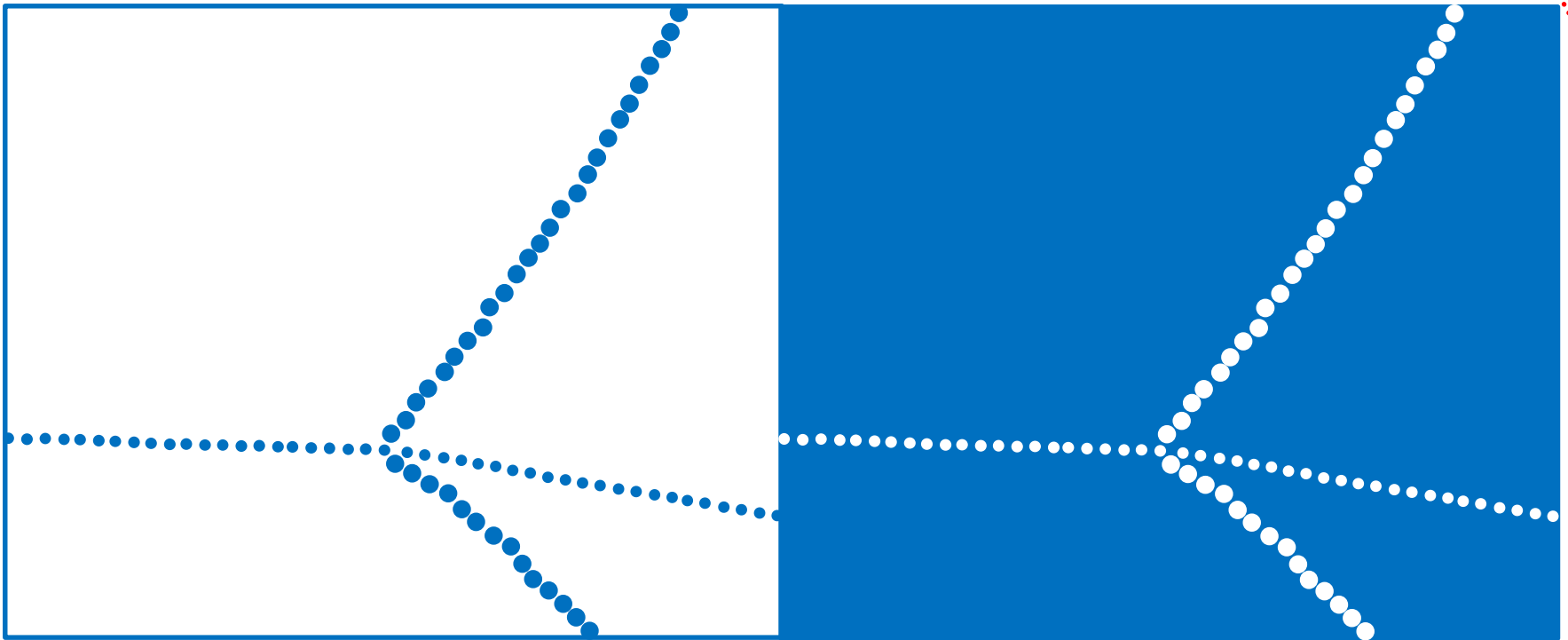
- **Efforts toward conduction-cooled cavities**
 - Nb cavities covered outside with Cu, and inside with Nb₃Sn
 - Can work at 4K
 - Cu provide thermal sink
 - Suitable for cryo-coolers, stand-alone compact linac for applications
 - Efforts at FNAL, Cornell, Jefferson Lab, Euclid

G. Ciovati, G. Cheng, U. Pudasaini, R. Rimmer, Supercond. Sci. Technol. 33, 07LT01 (2020)

R. Dhuley et al., IOP Conf. Ser.: Mater. Sci. Eng. 755, 012136 (2020)

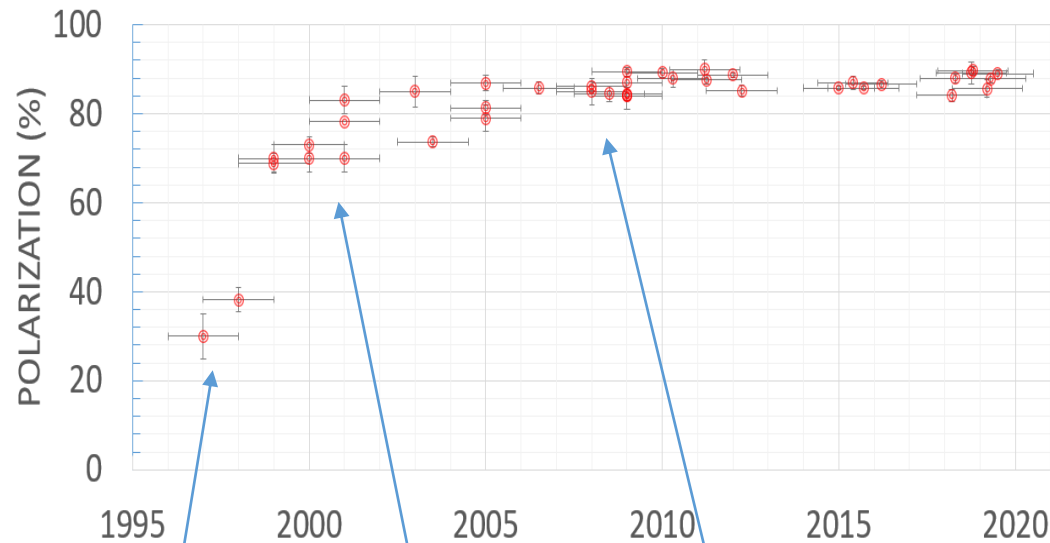
13. The other way round

- Invert the action(s) used to solve the problem (e.g. instead of cooling an object, heat it).
- Make movable parts (or the external environment) fixed, and fixed parts movable.
 - Turn the object (or process) “upside down”.



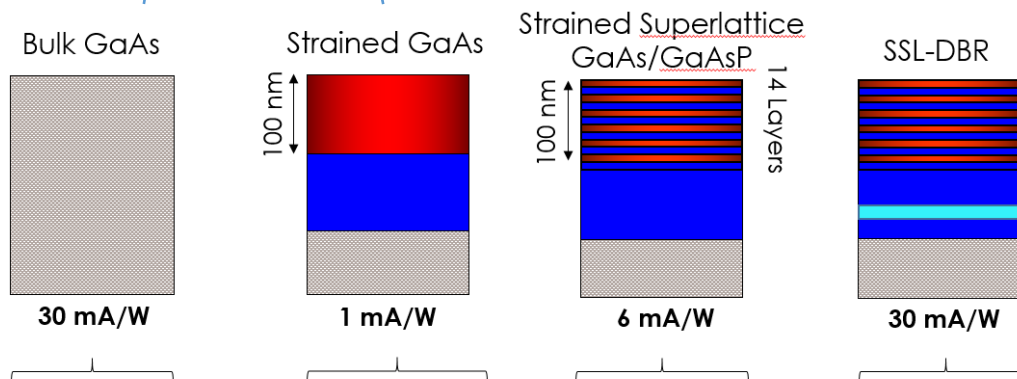
Cloud and bubble chambers

#3 Local Quality & Improving Polarized Source Performance



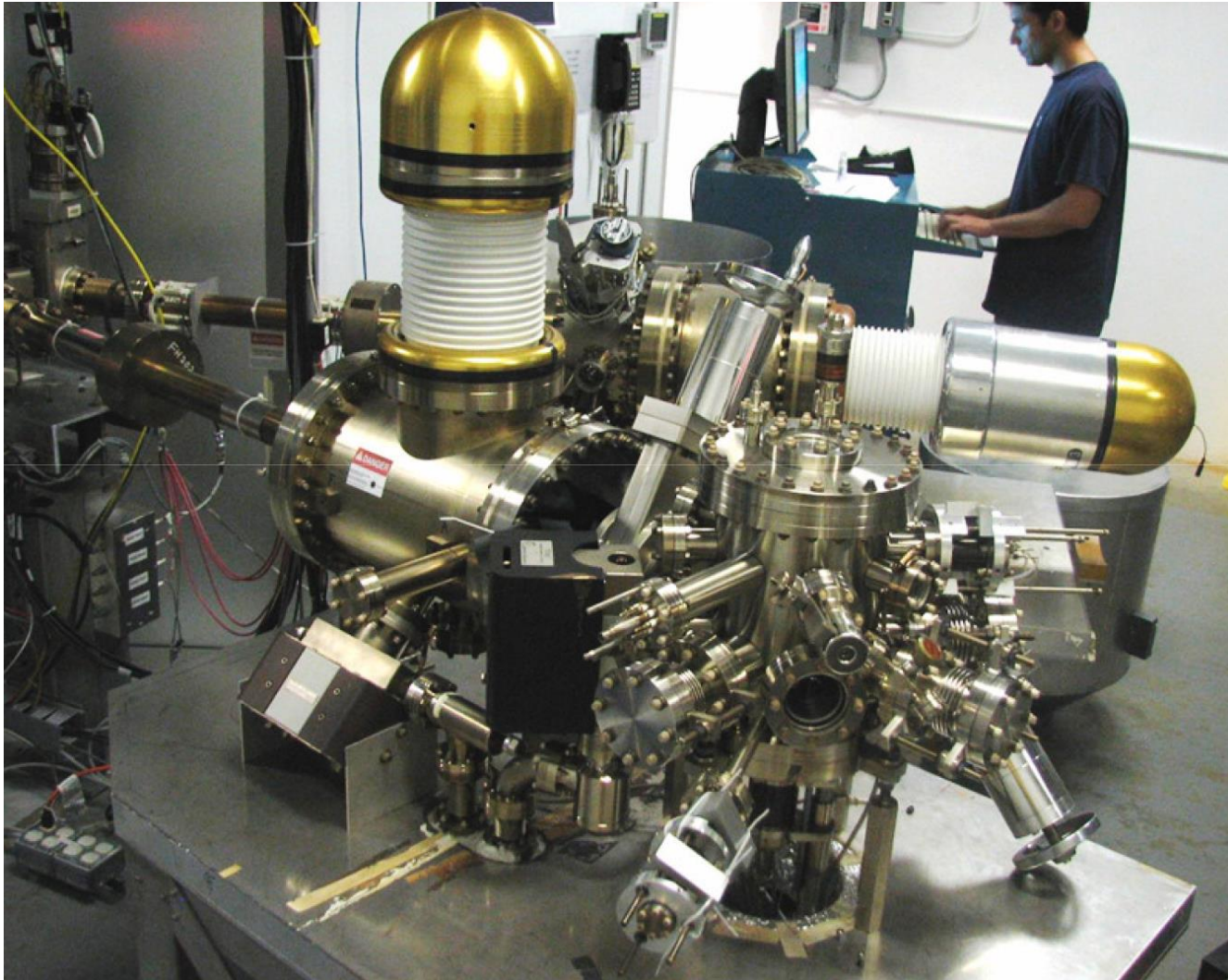
- Technology advances for e-beam polarization improvements
- New experiments demanded longer lifetime of cathodes and new design of guns
- ILC played a stimulating role for development of new ILC/CEBAF “Inverted gun”

Inventive principle #3
“Local quality”



P_{e-}	35%	35%	75%	75%	85%	89%	?
I_{e-}	30μA	100μA	50μA	100μA	150μA	200μA	
	1995	1998	1999	2000	2004	2017	

CEBAF load-lock polarized gun – installed in 2007



100kV gun. Lifetime limited to 30C. Path to higher lifetime and beam quality – higher voltage

#13 – New CEBAF/ILC “Inverted” gun ~2007-2009

“Inverted” Gun



Present Ceramic

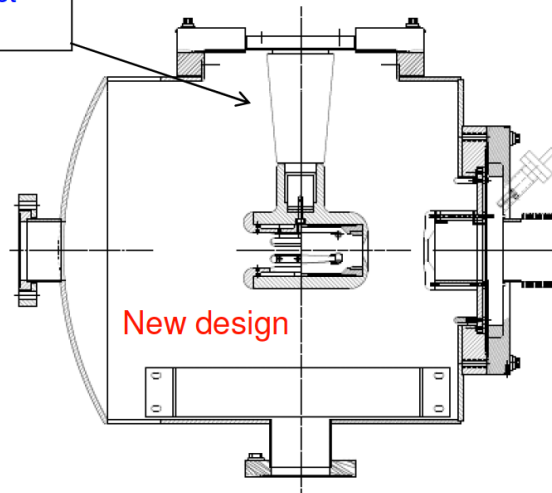
- Exposed to field emission
- Large area
- Expensive (~\$50k)

Medical x-ray technology



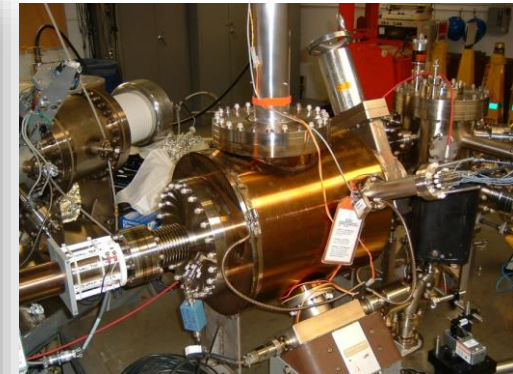
New Ceramic

- Compact
- ~\$5k



neg modules

Move away from “conventional” insulator used on most GaAs photoguns today – expensive, months to build, prone to damage from field emission.



Inverted
CEBAF/ILC Gun#1
installed at CEBAF,
July 2009

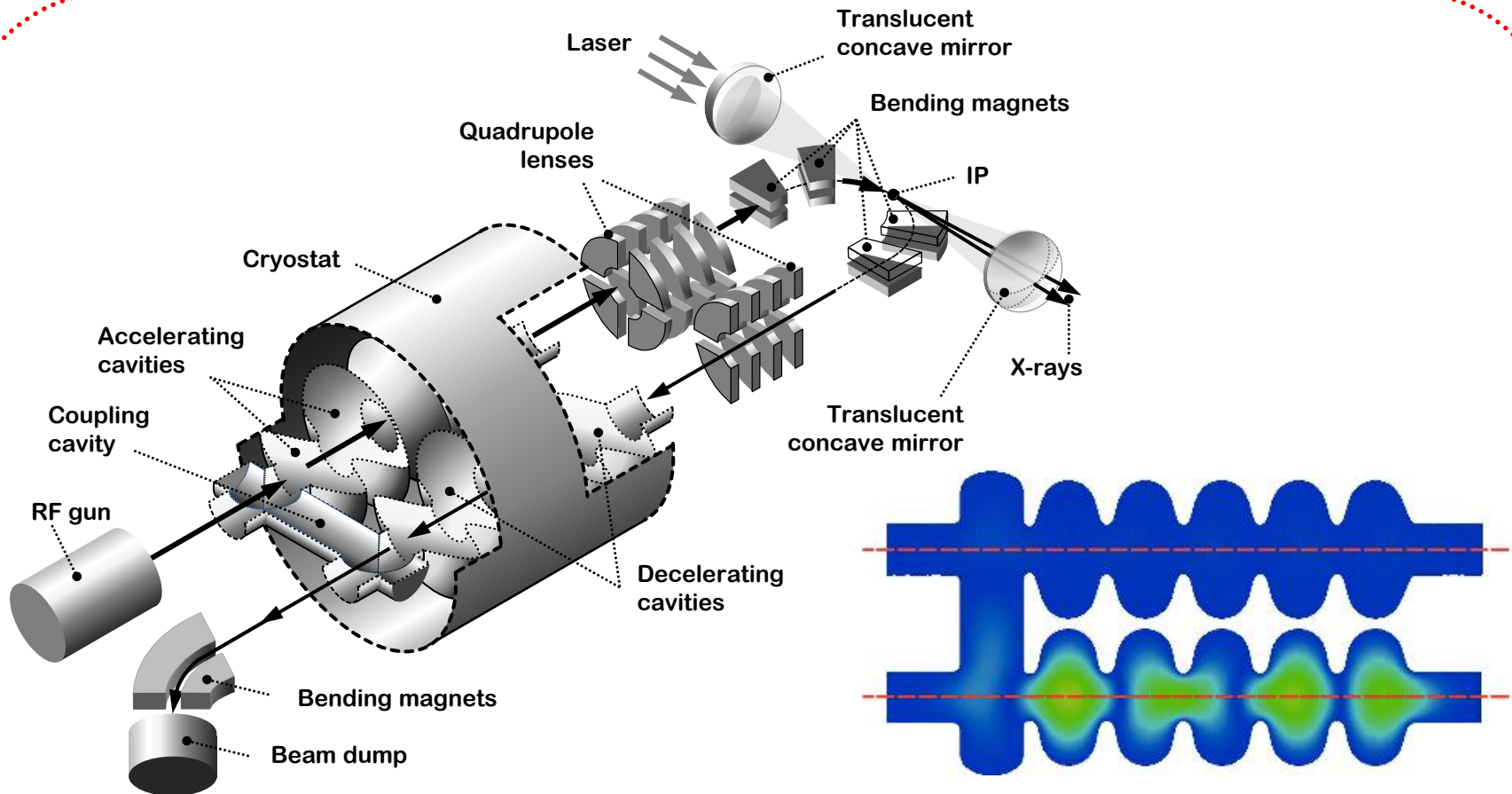
Higher voltage,
higher lifetime

200 kV for CEBAF,
350 kV for ILC

Inventive principle #13
“The other way around”

4. Asymmetry

- Change the shape of an object from symmetrical to asymmetrical.
 - If an object is asymmetrical, increase its degree of asymmetry.



**Cavities are slightly different to resonate on the main mode
but be decoupled for all higher order modes**

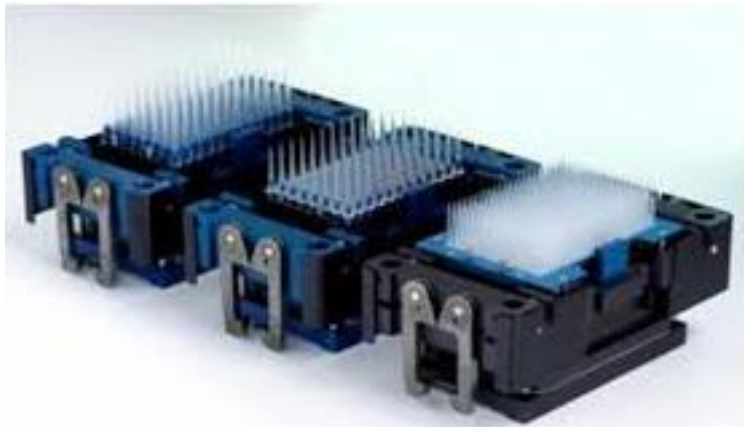
R. Ainsworth, G. Burt, I. V. Konoplev, A. Seryi, arXiv:1509.03675, 2015

5. Merging

- Bring closer together (or merge) identical or similar objects, assemble identical or similar parts to perform parallel operations.
- Make operations contiguous or parallel; bring them together in time.



Single-channel and
Multi-channel (8- and 12-) pipettes



96- or 384-channel Modular Dispense
Technology™ (MDT) dispense heads.
PerkinElmer Janus.

Illustration: PerkinElmer

#5 Merging – and fiber lasers combination

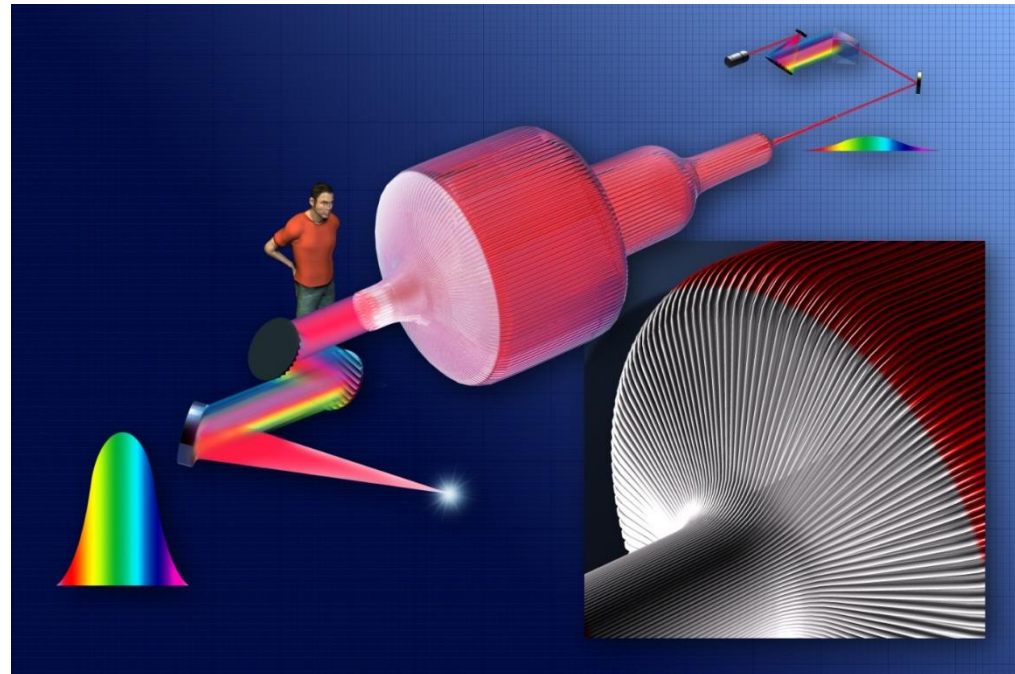
- Commercial fiber lasers reach 100 kW in CW
- Wall plug efficiency > 40%
- But low energy pulsed or per fiber



Research on combining many fibre lasers (short pulses!) together for high rep rate, high energy laser systems.



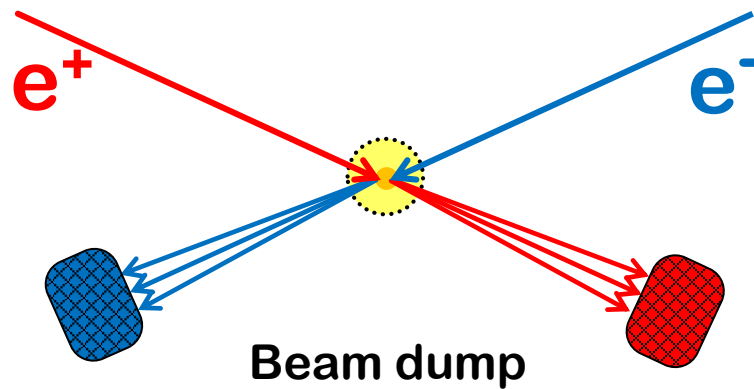
Phase control and combine
100s – 1000s fibres



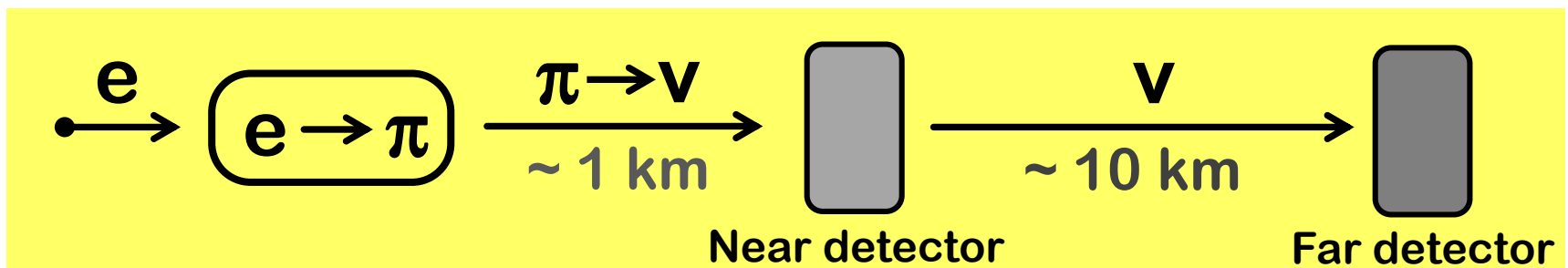
“The future is fibre accelerators”, Gerard Mourou, Bill Brocklesby, Toshiki Tajima & Jens Limpert, Nature Photonics 7, 258–261 (2013)

6. Universality

- Make a part or object perform multiple functions; eliminate the need for other parts.



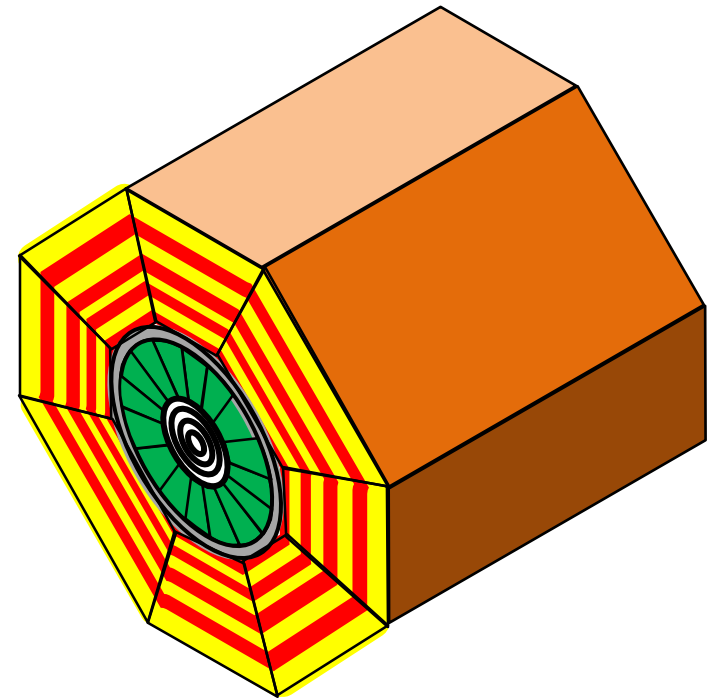
Make beam dump of linear collider to be sub-critical reactor to generate power or make neutrino factory out of it



I.F. Ginzburg, arXiv:1411.3295, 2014

7. Nested doll

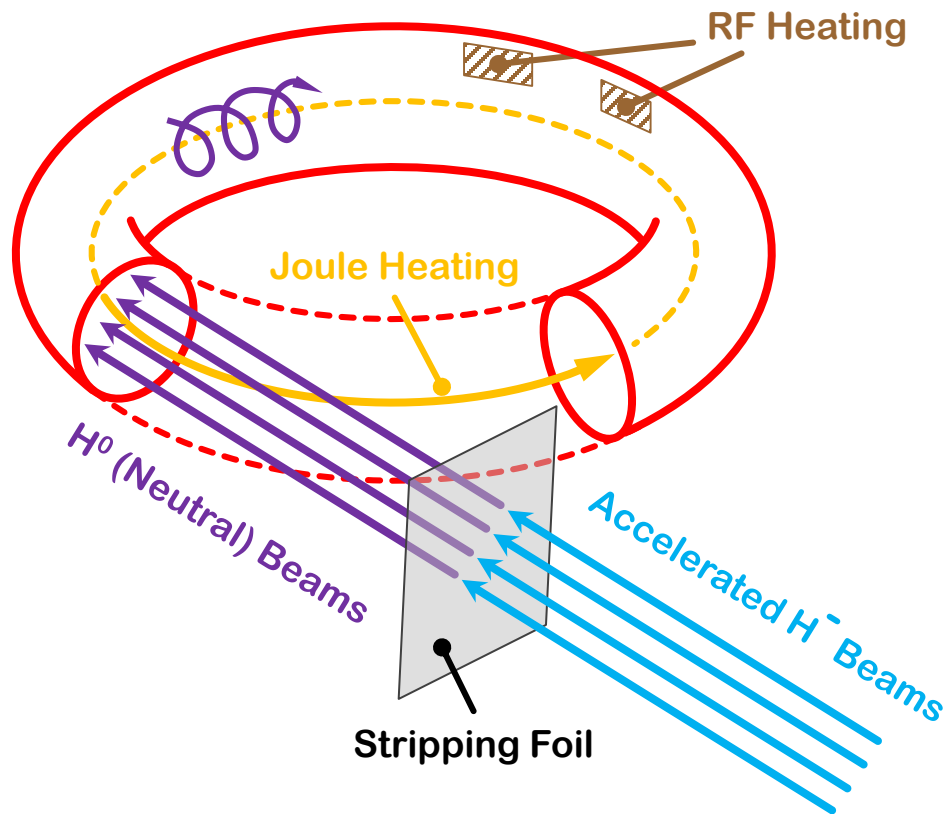
- Place one object inside another; place each object, in turn, inside the other.
 - Make one part pass through a cavity in the other.



High energy physics detectors

8. Anti-weight force

- To compensate for the weight of force on an object, merge it with other objects that provide compensating force.
- To compensate for the weight of force on an object, make it interact with the environment (e.g. use aerodynamic, hydrodynamic, buoyancy and other forces).



Heating of plasma with neutral beams

Wien filter and TRIZ #8 Anti-force

Wien filter

From Wikipedia, the free encyclopedia

Not to be confused with [Wiener filter](#) or [Wien bridge](#).

A **Wien filter** also known as **velocity selector** is a device consisting of perpendicular **electric** and **magnetic** fields that can be used as a **velocity filter** for charged particles, for example in **electron microscopes** and spectrometers.

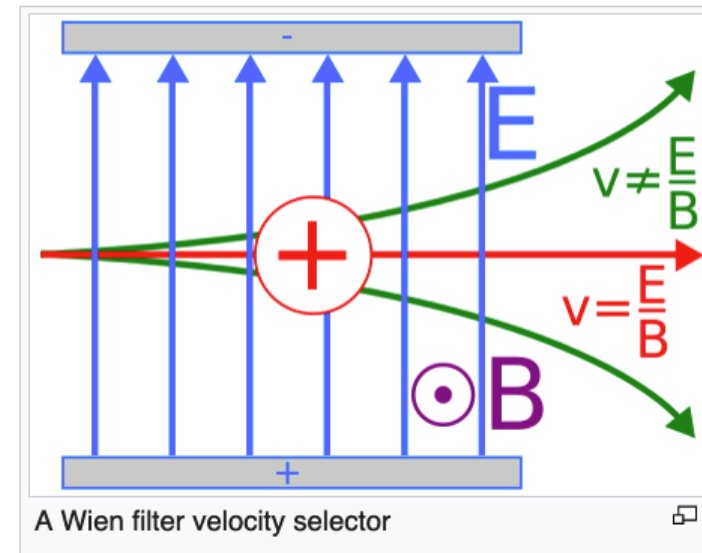
^{[1][2]} It is used in **accelerator mass spectrometry** to select particles based on their speed. The device is composed of **orthogonal** electric and magnetic fields, such that particles with the correct speed will be unaffected while other particles will be deflected. It is named for **Wilhelm Wien** who developed it in 1898 for the study of **anode rays**.^{[3][4]} It can be configured as a charged particle energy analyzer, **monochromator**, or **mass spectrometer**.^{[1][2]}

Theory ^{[[edit](#)]}

Any charged particle in an electric field will feel a force proportional to the charge and field strength such that $\vec{F} = q\vec{E}$, where F is force, q is charge, and E is electric field strength. Similarly, any particle moving in a magnetic field will feel a force proportional to the velocity and charge of the particle. The force felt by any particle is then

https://en.wikipedia.org/wiki/Wien_filter

Since only B acts on spin, can use it for spin manipulation

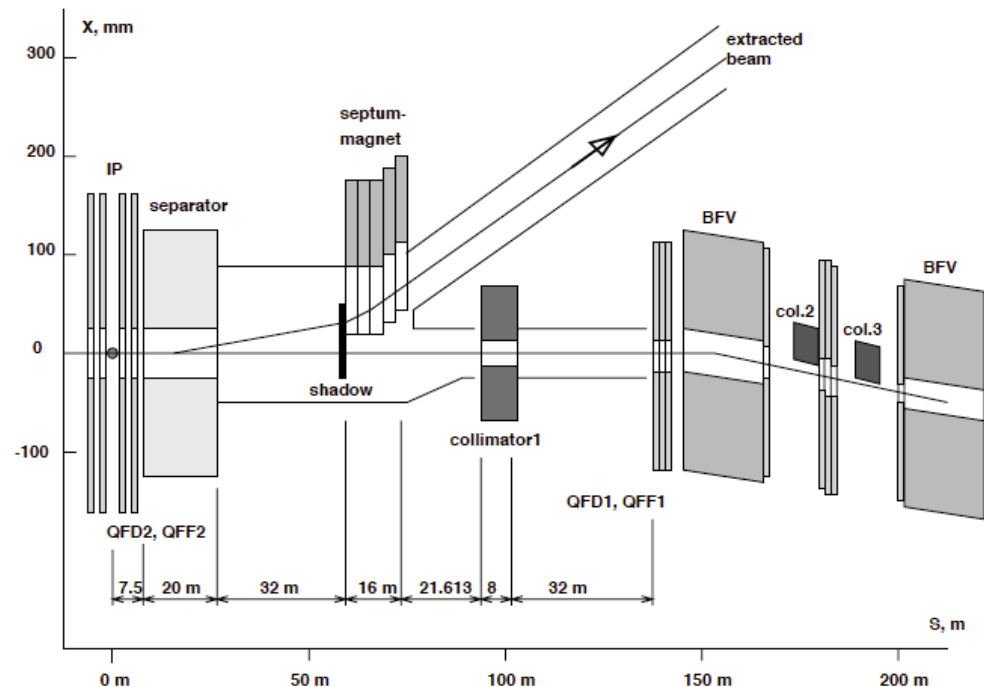


TESLA collider design and TRIZ #8 Anti-force

- TESLA used head on collisions of e^+ and e^- beams
- How to separate beams after collisions?
 - Beams are very disrupted after e^+e^- collision, and cannot propagate down through the incoming apertures

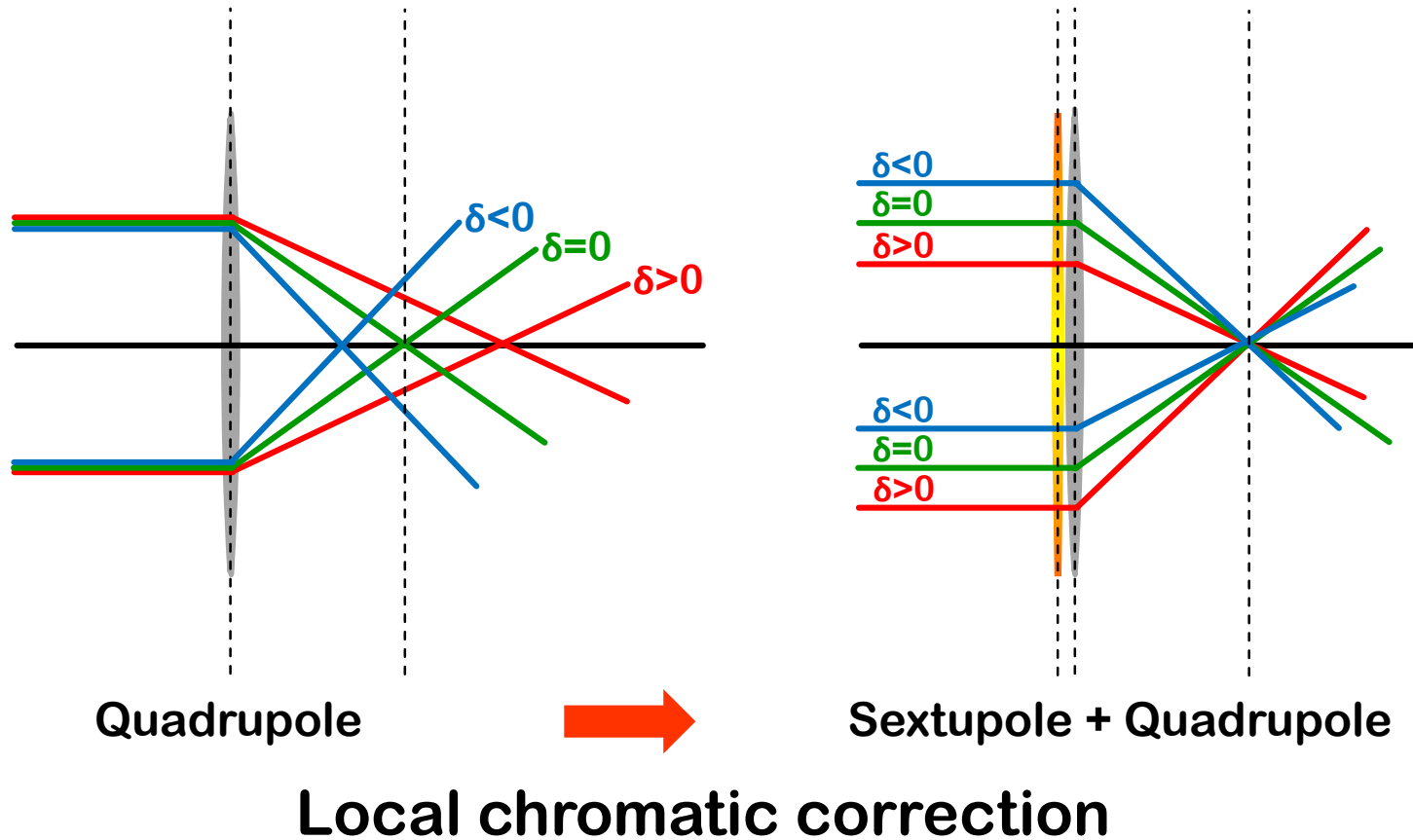


The answer was to use separator with combined E and B field. It does not act on incoming e^+ or e^- beams, but bends the outgoing beams



9. Preliminary anti-action

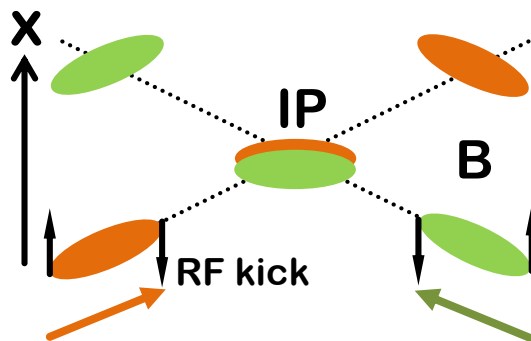
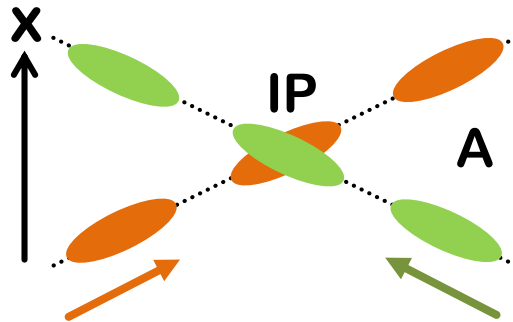
- If it will be necessary to do an action with both harmful and useful effects, this action should be replaced with anti-actions to control harmful effects.
- Create beforehand stresses in an object that will oppose known undesirable working stresses later on.



P. Raimondi, A. Seryi, PRL, 86, 3779 (2001)

10. Preliminary action

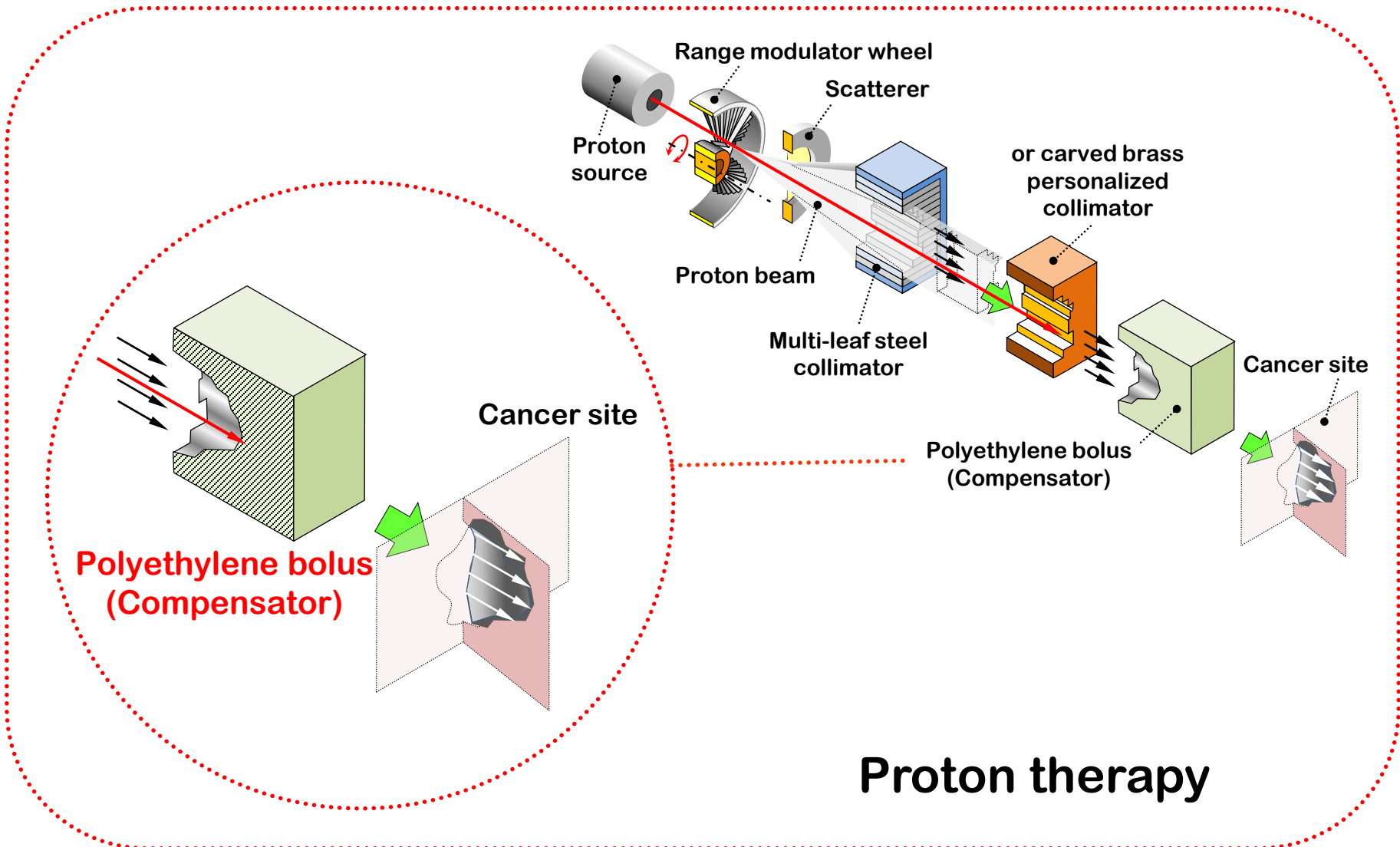
- Perform, before it is needed, the required change of an object (either fully or partially).
- Pre-arrange objects such that they can come into action from the most convenient place and without losing time for their delivery.



Crabbed collisions

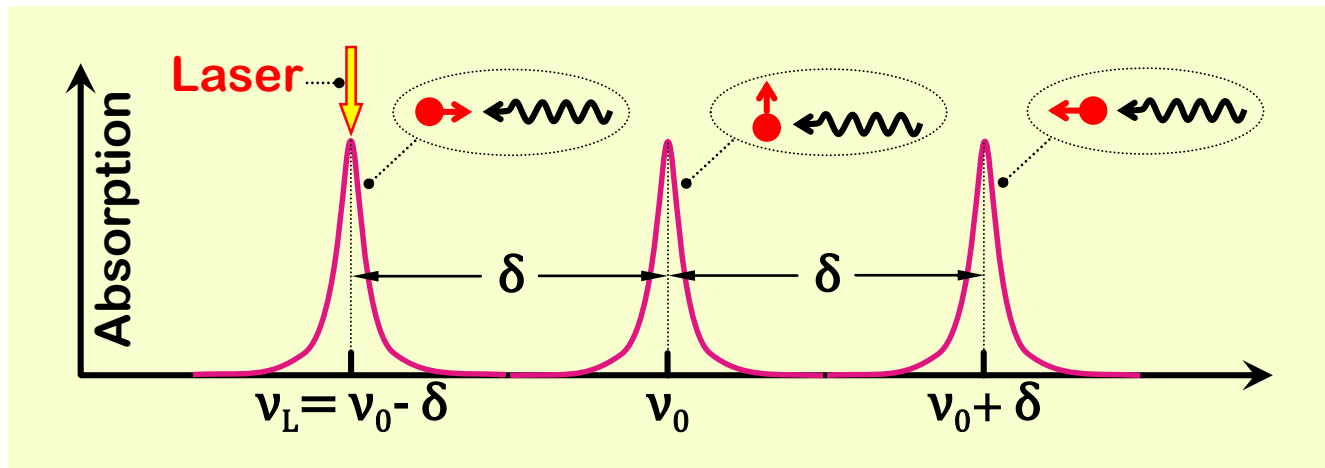
11. Beforehand cushioning

- Prepare emergency means beforehand to compensate for the relatively low reliability of an object.



12. Equipotentiality

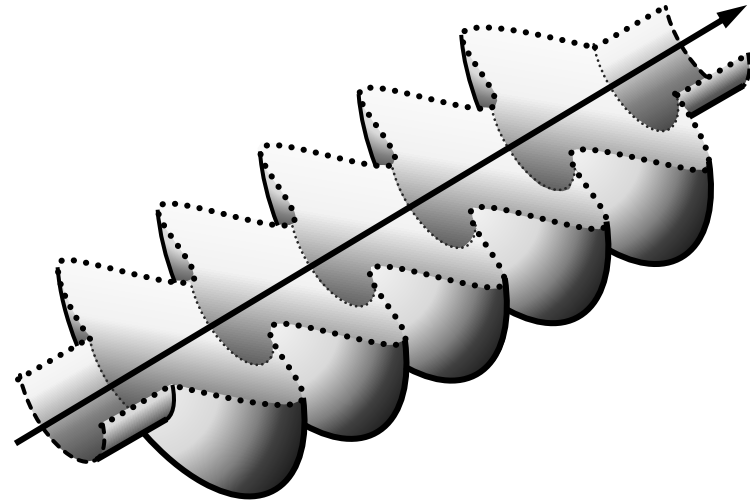
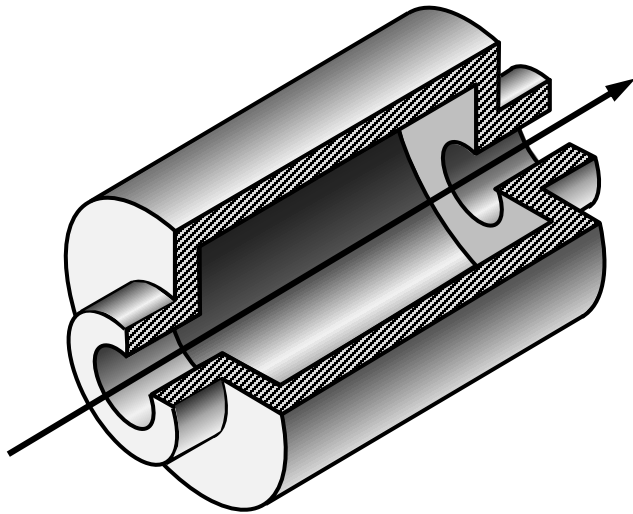
- In a potential field, limit position changes (e.g. change operating conditions to eliminate the need to raise or lower objects in a gravity field).



Laser cooling

14. Spheroidality – Curvature

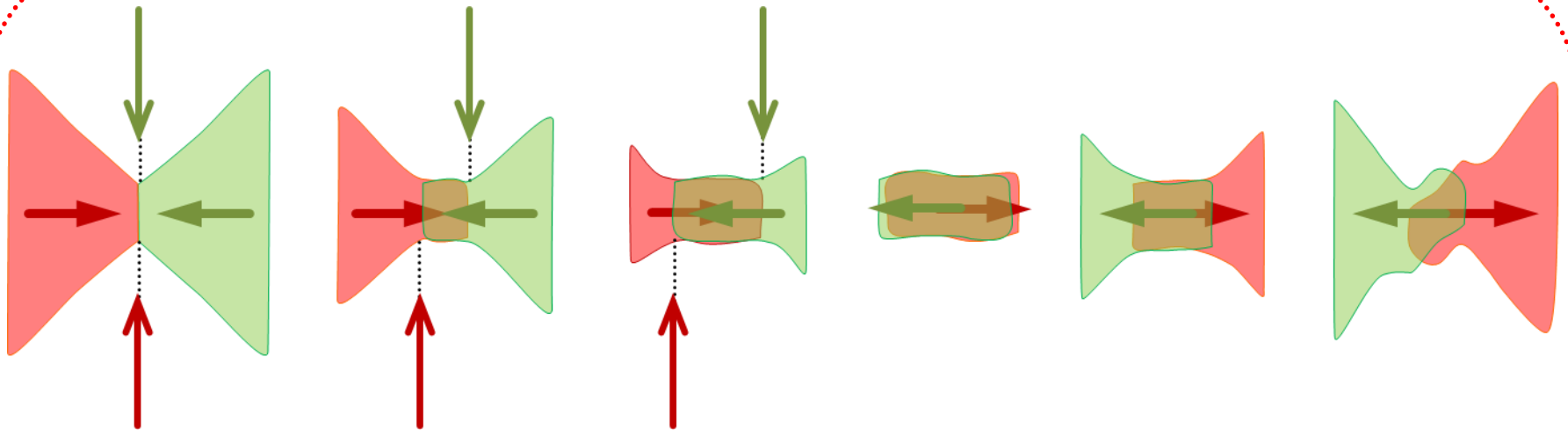
- Instead of using rectilinear parts, surfaces, or forms, use curvilinear ones; move from flat surfaces to spherical ones; from parts shaped as a cube (parallelepiped) to ball-shaped structures.
 - Use rollers, balls, spirals, domes.
- Go from linear to rotary motion, use centrifugal forces.



Pill-box and crab-cavity

15. Dynamics

- Allow (or design) the characteristics of an object, external environment, or process to change to be optimal or to find an optimal operating condition.
 - Divide an object into parts capable of movement relative to each other.
 - If an object (or process) is rigid or inflexible, make it movable or adaptive.

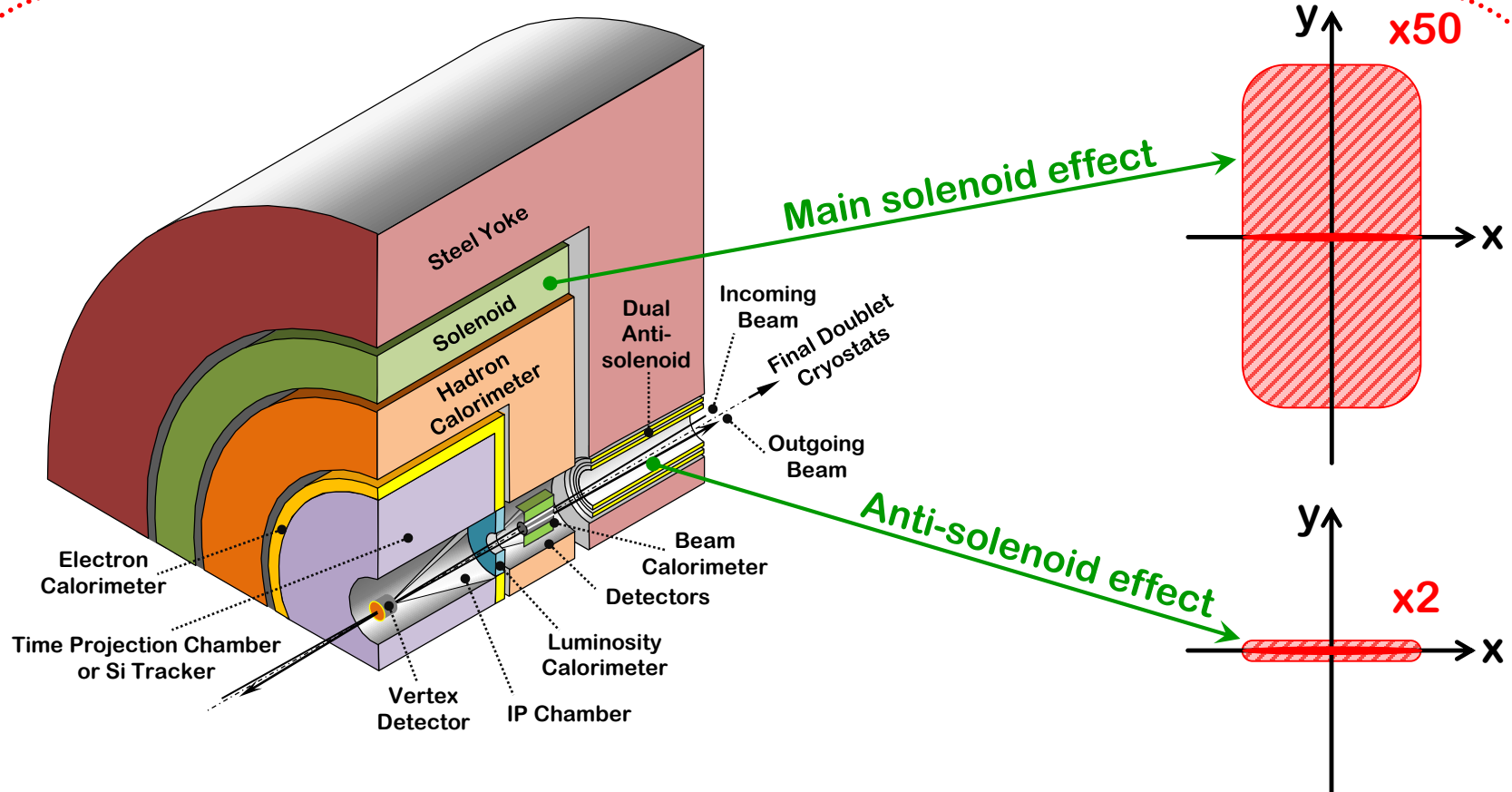


Travelling focus

V. Balakin, 1991

16. Partial or excessive actions

- If 100 percent of an object is hard to achieve using a given solution method then, by using “slightly less” or “slightly more” of the same method, the problem may be considerably easier to solve.

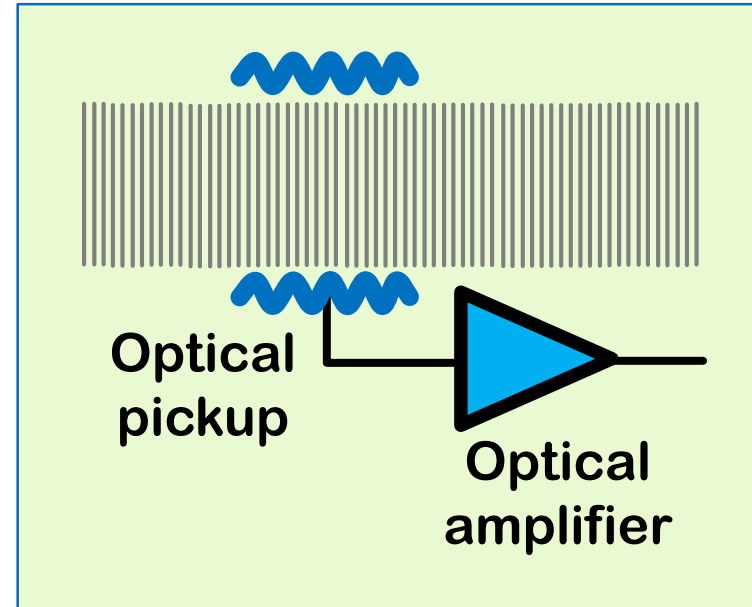
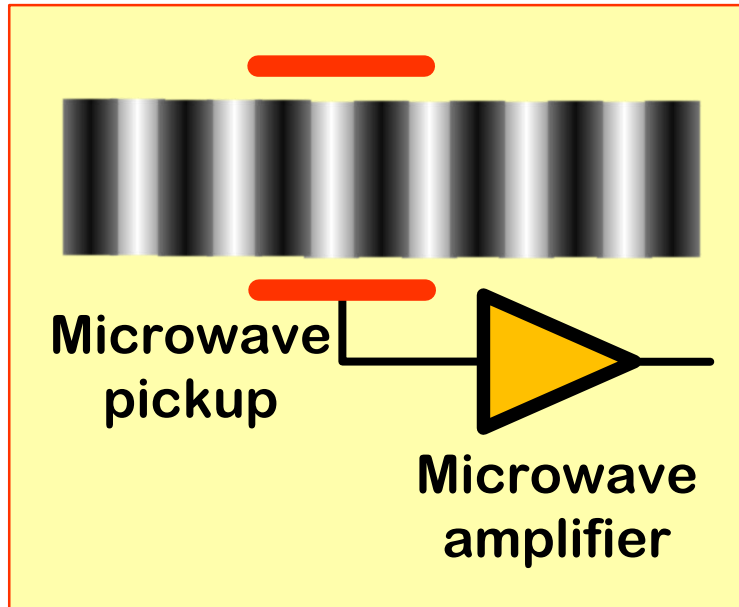


Huge coupling due to overlap of solenoid with Final Doublet quads
=> partial compensation by weak anti-solenoid

Y. Nosochkov, A. Seryi, PRSTAB, 8, 021001 (2005)

18. ~~Mechanical vibration~~ Oscillations and resonances

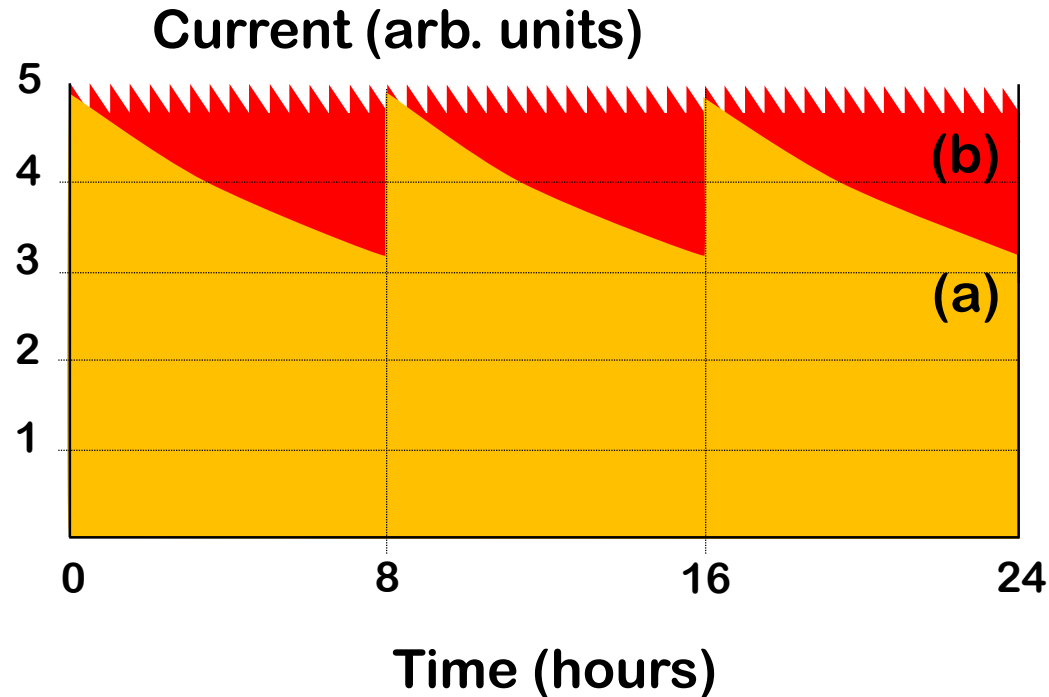
- Cause an object to oscillate or vibrate.
- Increase its frequency (even up to the ultrasonic from microwave to optical).
 - Use an object's resonant frequency.
 - Use piezoelectric vibrators instead of mechanical ones.
- Use combined ultrasonic and electromagnetic field oscillations.



Stochastic cooling => optical stochastic cooling

20. Continuity of useful action

- Carry on work continuously; make all parts of an object work at full load, all the time.
- Eliminate all idle or intermittent actions or work.



Top off injection

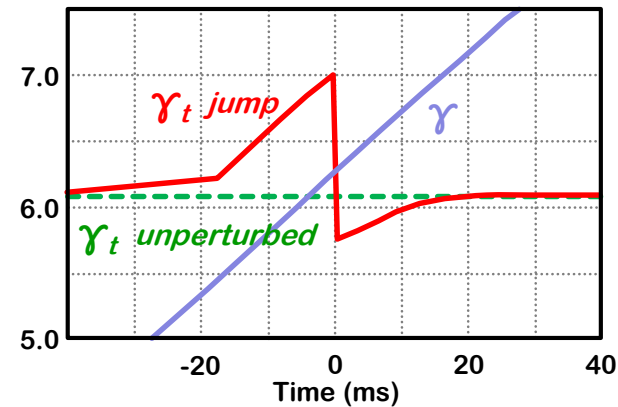
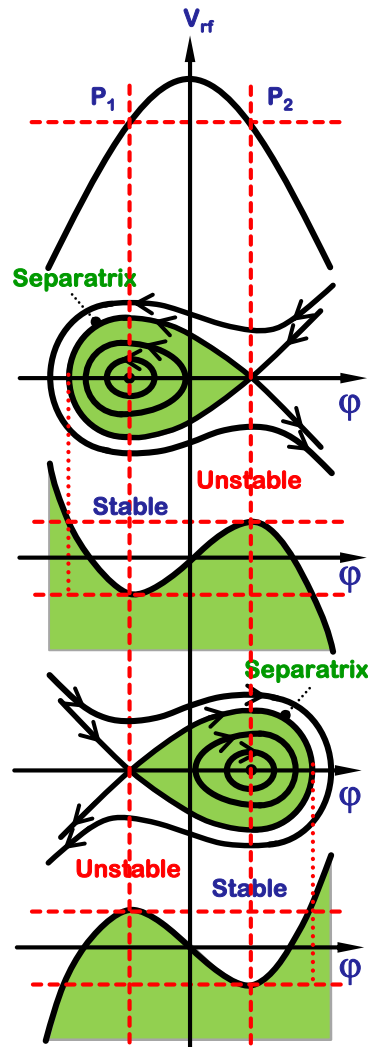
21. Skipping

- Conduct a process , or certain stages (e.g. destructible, harmful or hazardous operations) at high speed.



21. Skipping

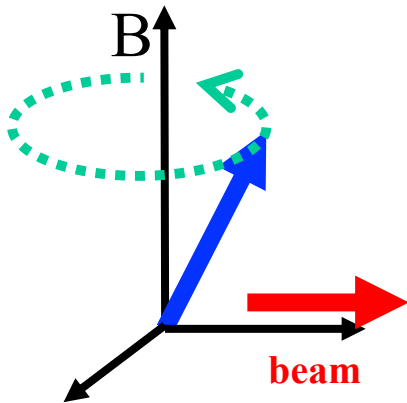
- Conduct a process, or certain stages (e.g. destructible, harmful or hazardous operations) at high speed.



Crossing
transition energy
with γ_t jump
technique

Polarization preservation

- **Spin motion in accelerator: spin vector precesses around its guiding field along the vertical direction**



- Spin tune Q_s : number of precessions in one orbital revolution: $Q_s = \gamma G$
 - Anomalous g- factor for proton $G = 1.793$

Depolarization due to resonances:

Imperfection resonances:

$$Q_s = n$$

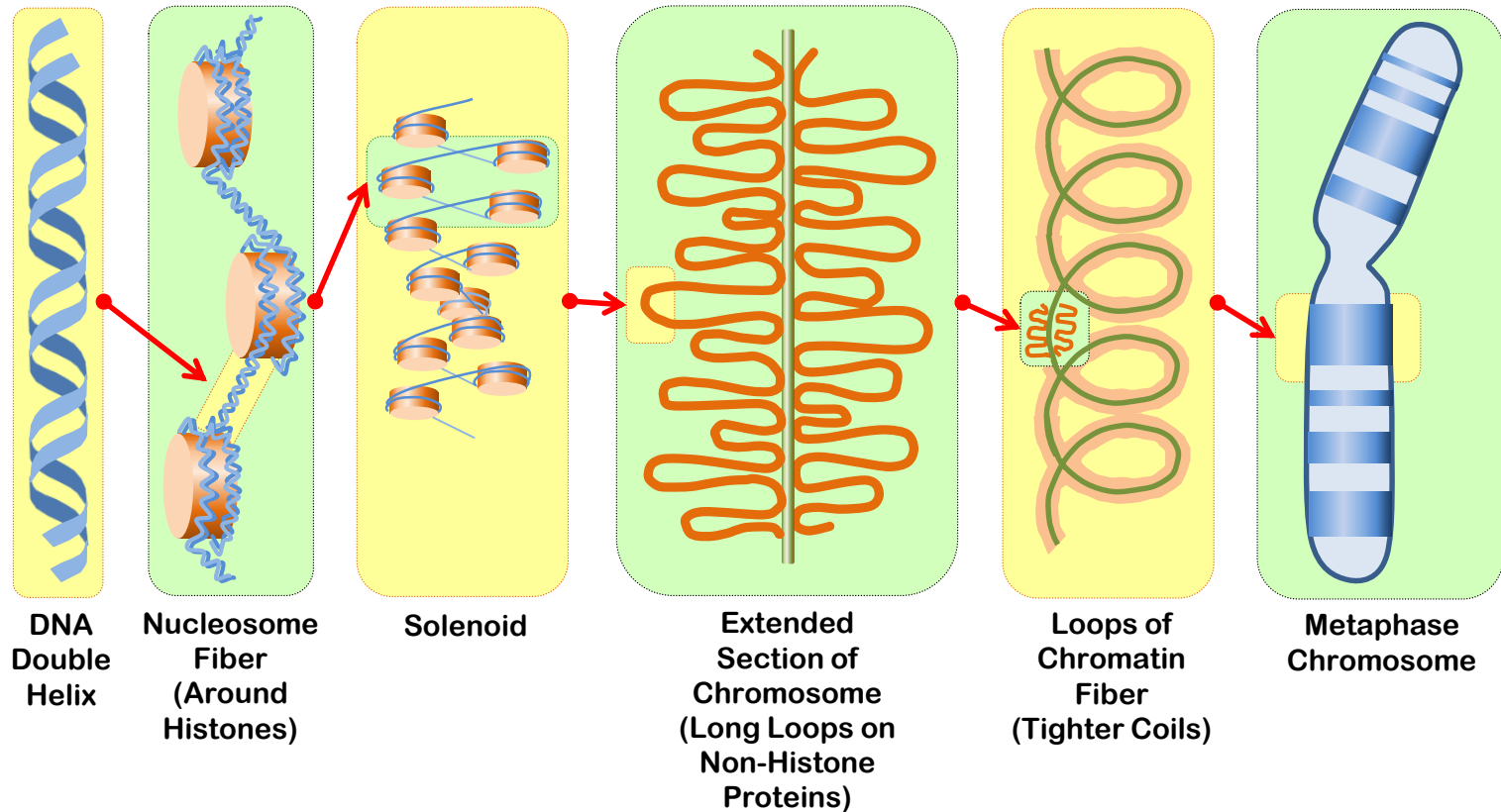
Intrinsic resonances:

$$Q_s = nP \pm Q_y$$

Here n – integer, P – number of superperiods

17. Another dimension

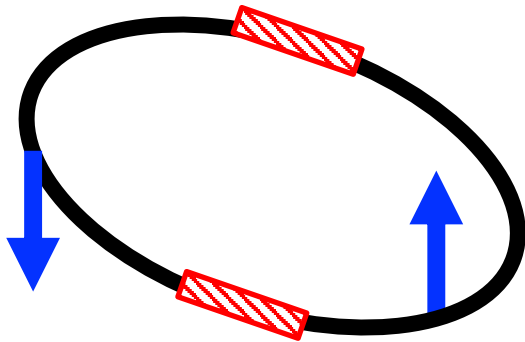
- To move into an additional dimension.
- Use a multi-story arrangement of objects instead of a single-story arrangement.
 - Tilt or re-orient the object, lay it on its side.
 - Use “another side” of a given area.



DNA packaging levels

Polarization preservation – Siberian snakes

- **Siberian snakes – special (e.g. helical) magnets that rotate spin (preserving orbit outside)**



Polarization kinematics of particles in storage rings.

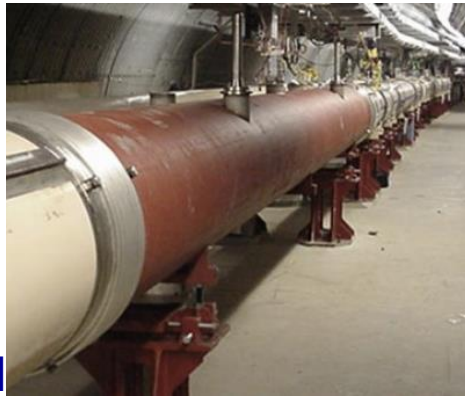
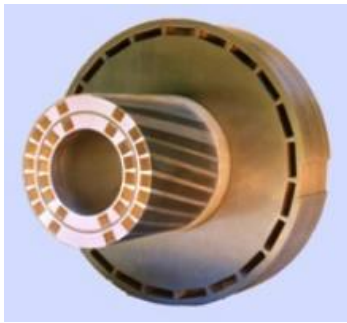
Ya.S. Derbenev, A.M. Kondratenko (Novosibirsk, INP) Jun 1973.

Zh.Eksp.Teor.Fiz.64:1918-1929,1973

- **Full Siberian snakes flip spin 180 degrees. Two full snakes make $Q_s = 1/2$**
 - **Two full snakes control:**
 - Intrinsic resonances
 - Imperfection resonances
- **Partial Siberian snake**
 - **Break coherent build up of perturbation of spin**
 - Some control of imperfection resonances

Polarization preservation – Siberian snakes

- Siberian snakes in RHIC – two full snake than make $Q_s = \frac{1}{2}$

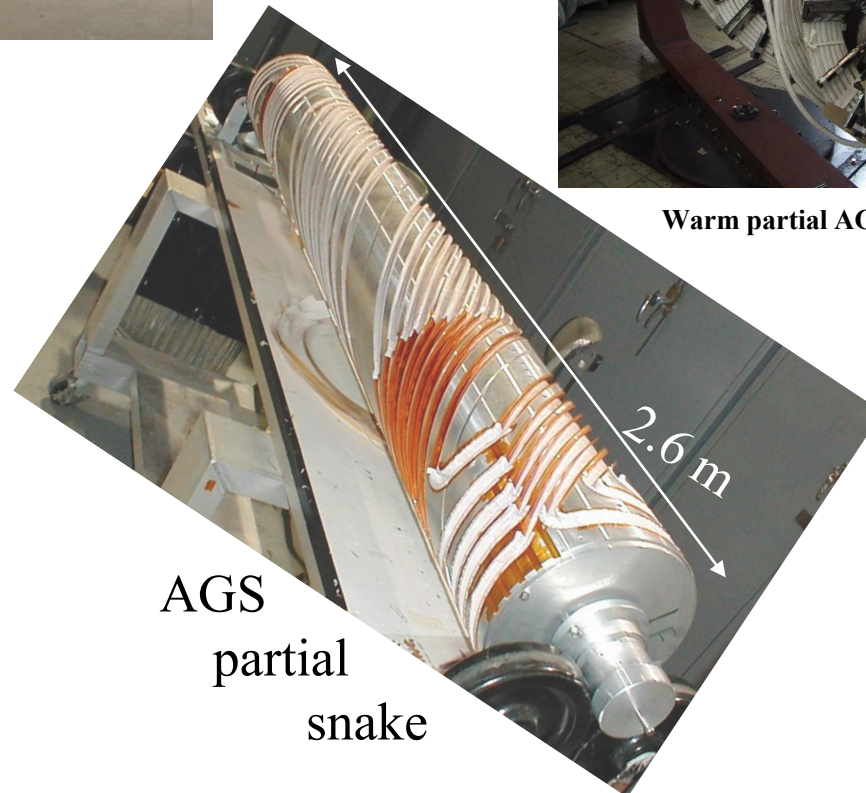


RHIC snake: 4T, 2.4m/snake, 360° twist, 100mm aperture

First Polarized Proton Collisions at RHIC. T, Roser, et al, AIP Conference Proceedings 667, 1 (2003)

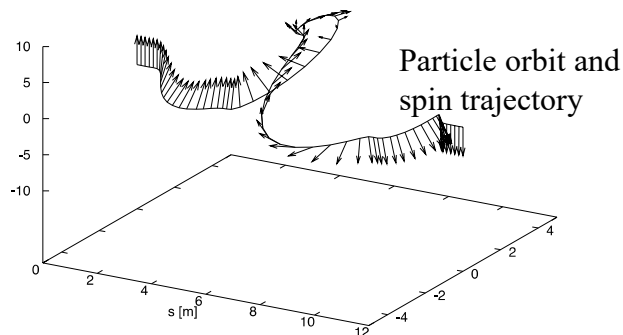
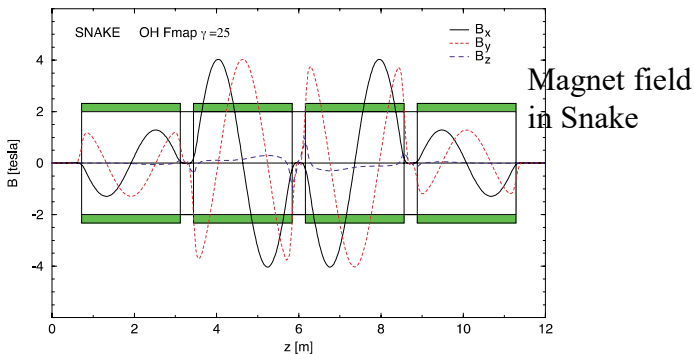


Warm partial AGS Snake



AGS
partial
snake

- AGS – partial



EIC Hadron Polarization

- Existing p Polarization in RHIC achieved with “Siberian snakes”
- Near term improvements will increase proton polarization in RHIC from 60% to 80%
- ^3He polarization of >80% measured in source
- 80% polarized ^3He in EIC will be achieved with six “snakes”,
- Acceleration of polarized Deuterons in EIC 100% spin transparent
- Need **tune jumps** in the hadron booster synchrotron



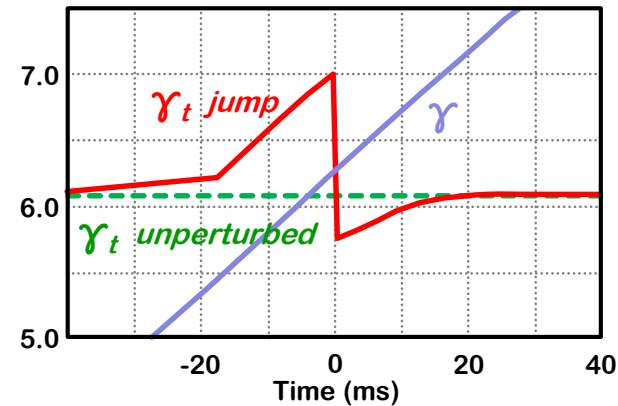
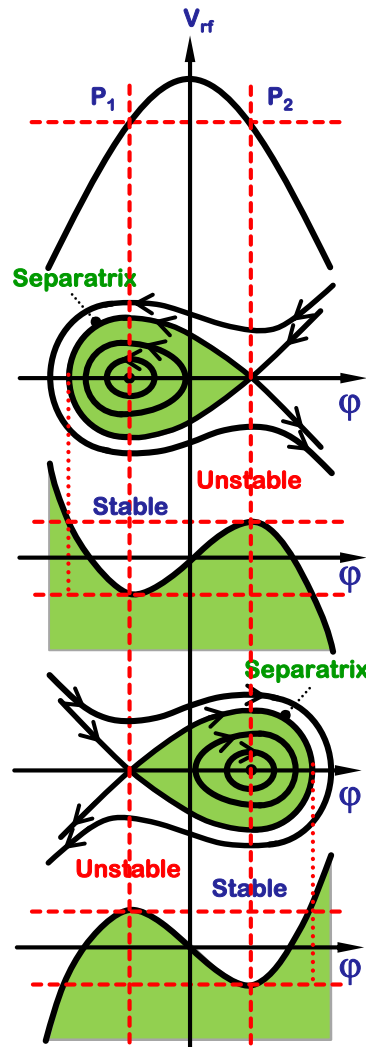
Electron beam ion
source EBIS with
polarized ^3He
extension

TRIZ inventive principle #21

- Conduct a process , or certain stages (e.g. destructible, harmful or hazardous operations) at high speed.

21. Skipping

Crossing transition energy with γ_t jump technique



Tune jump for polarization preservation conceptually similar

See more examples in “Accelerating Science TRIZ inventive methodology in illustrations”

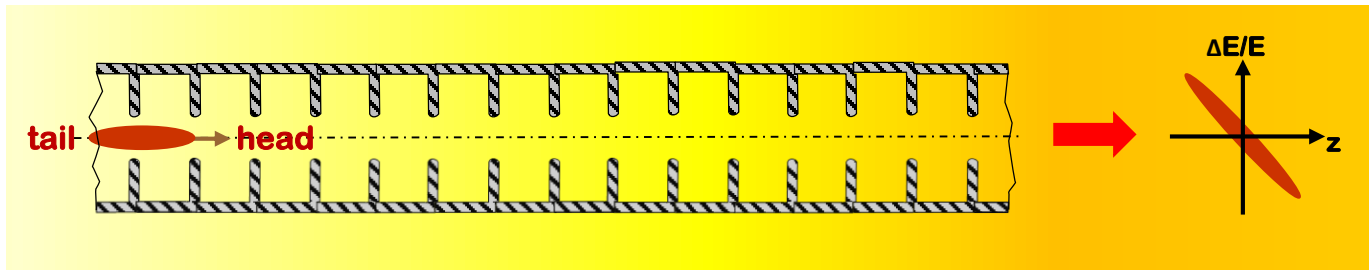
[arXiv:1608.00536](https://arxiv.org/abs/1608.00536)

“Unifying Physics of Accelerators, Lasers and Plasma” (CRC Press 2015) – in Open Access:

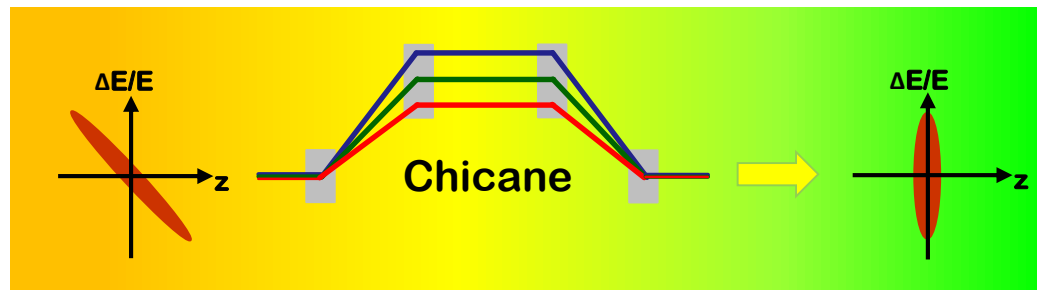
<https://doi.org/10.1201/b18696>

22. "Blessing in disguise" or "Turn Lemons into Lemonade"

- Use harmful factors (particularly, harmful effects of the environment or surroundings) to achieve a positive effect.
- Eliminate the primary harmful action by adding it to another harmful action to resolve the problem.
- Amplify a harmful factor to such a degree that it is no longer harmful.



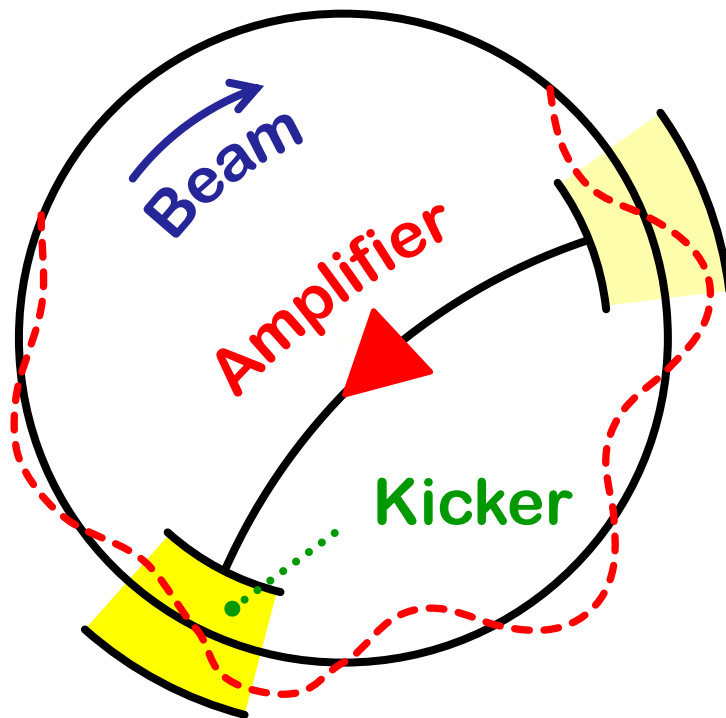
Wakefields in long Linac are harmful.
They can be made useful by compressing bunch for use in FEL.



Linac wake for FEL use

23. Feedback

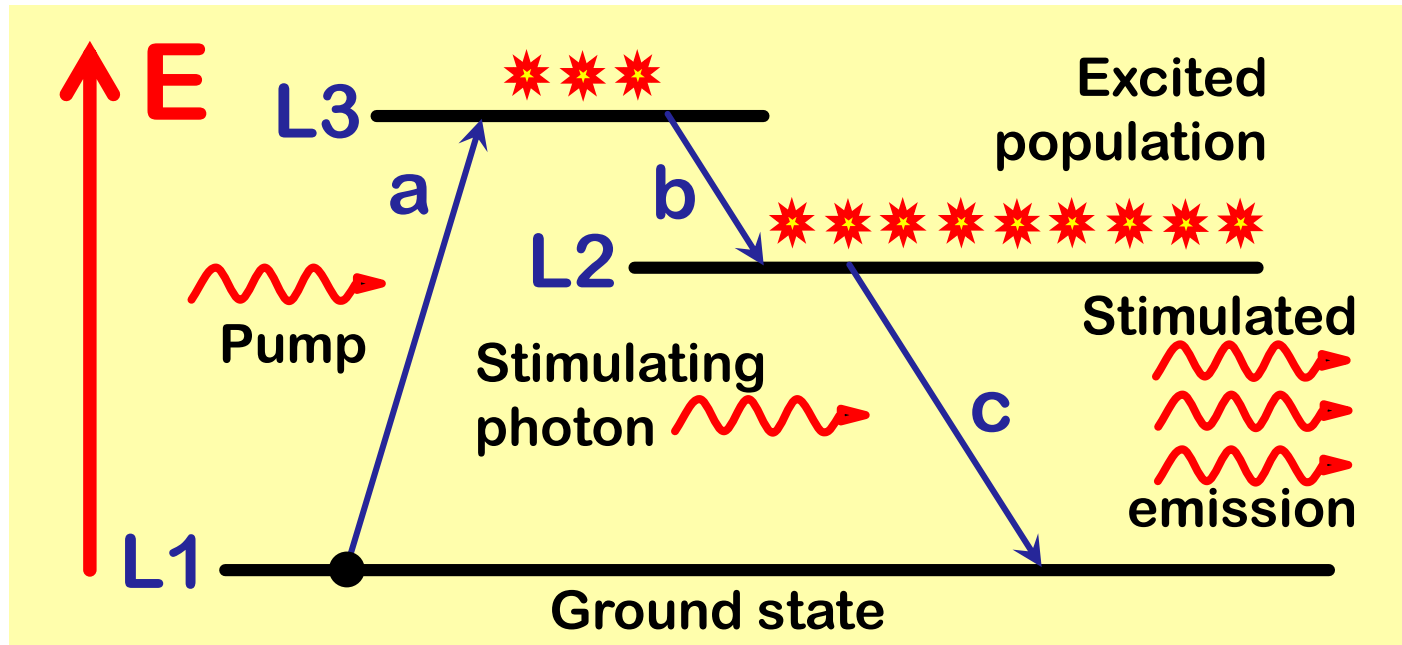
- Introduce feedback (referring back, cross-checking) to improve a process or action.
 - If feedback is already used, change its magnitude or influence.



Stochastic cooling

24. Intermediary

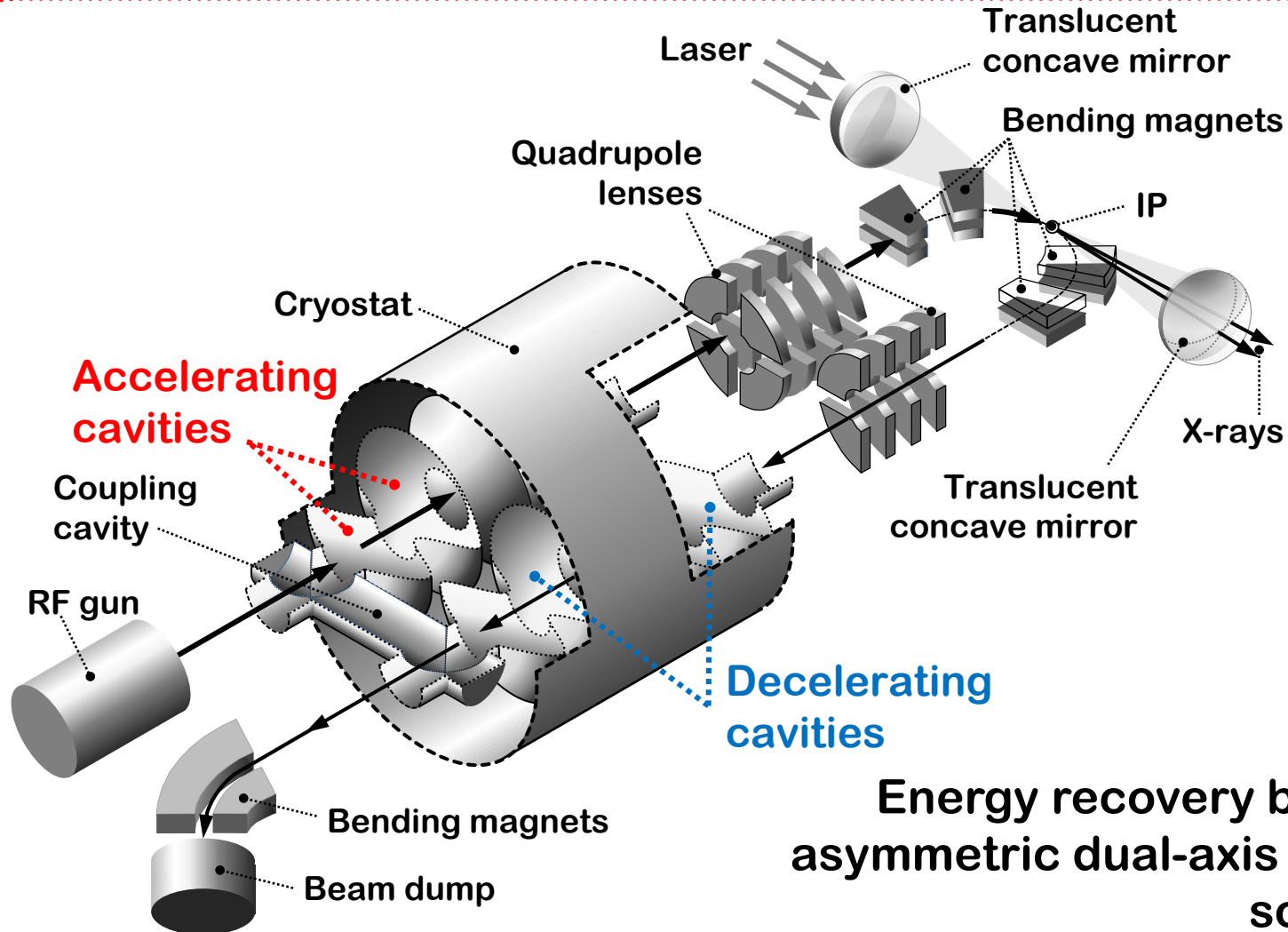
- Use an intermediary carrier object or intermediary process.
- Merge one object temporarily with another (which can be easily removed).



Three-level laser

25. Self service

- Make an object serve itself by performing auxiliary helpful functions
 - Use waste resources, energy, or substances.

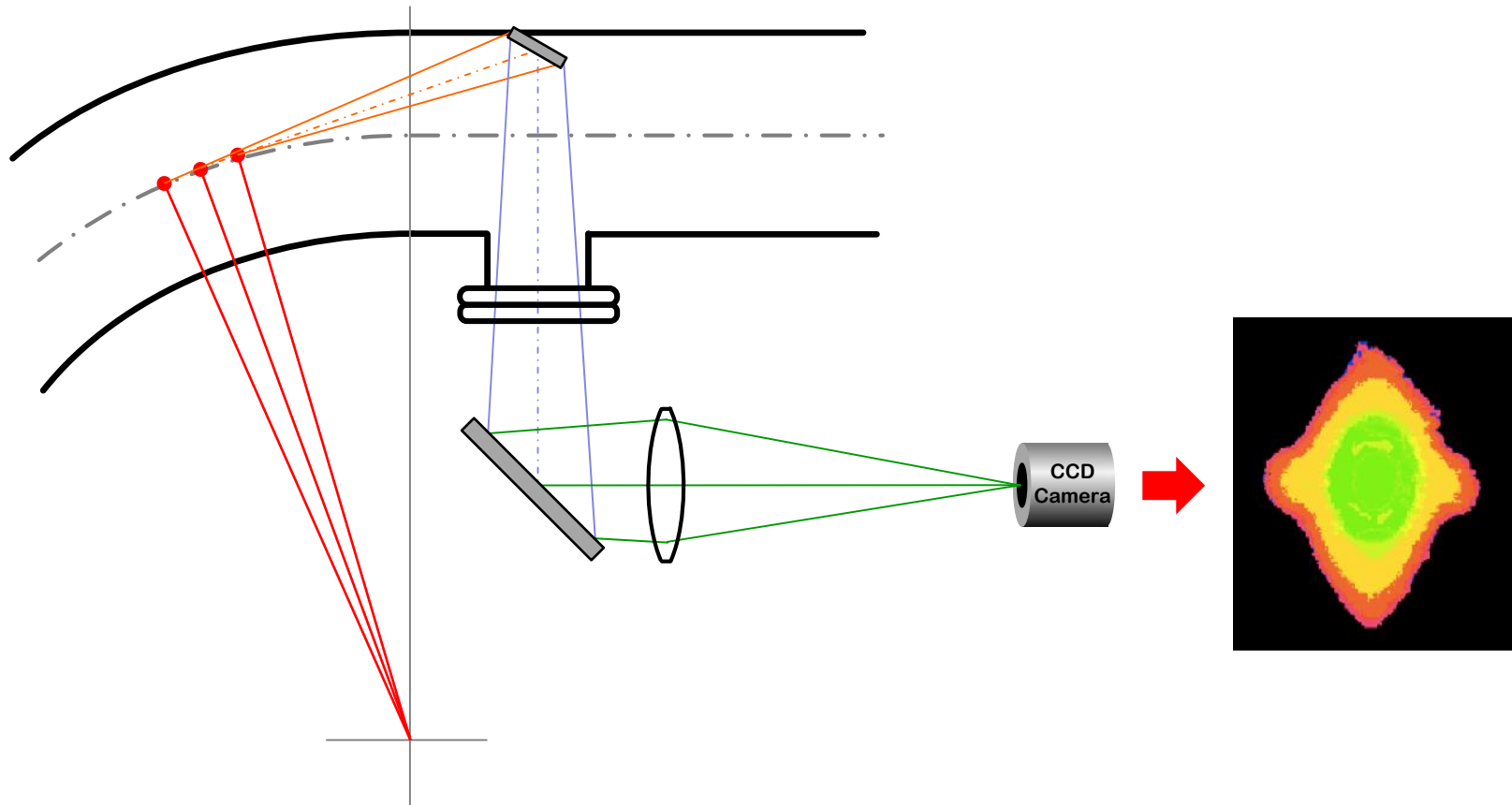


Energy recovery based
asymmetric dual-axis X-ray
source

R. Ainsworth, G. Burt, I. V. Konoplev, A. Seryi, arXiv:1509.03675, 2015

26. Copying

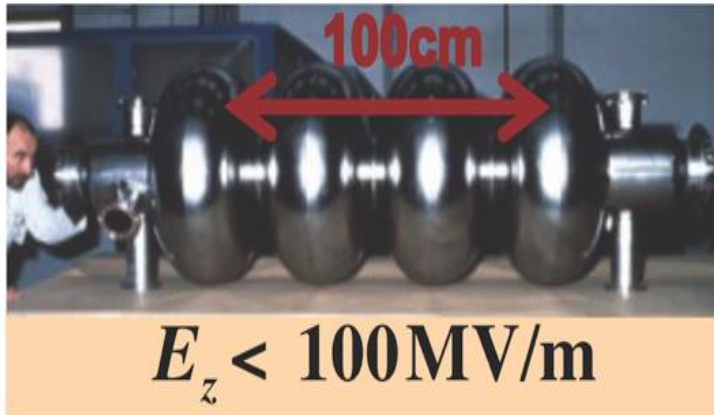
- Instead of an unavailable, expensive, fragile object, use simpler and inexpensive copies.
- Replace an object, or process with optical copies.
- If visible optical copies are already used, move to infrared or ultraviolet copies.



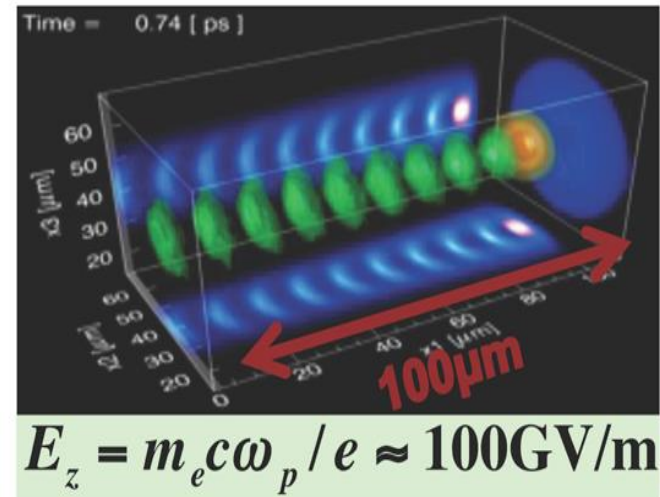
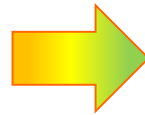
Synchrotron radiation profile monitor

27. Cheap short-living objects

- Replace an expensive object with a multiple of inexpensive objects, comprising certain qualities (such as service life, for instance).



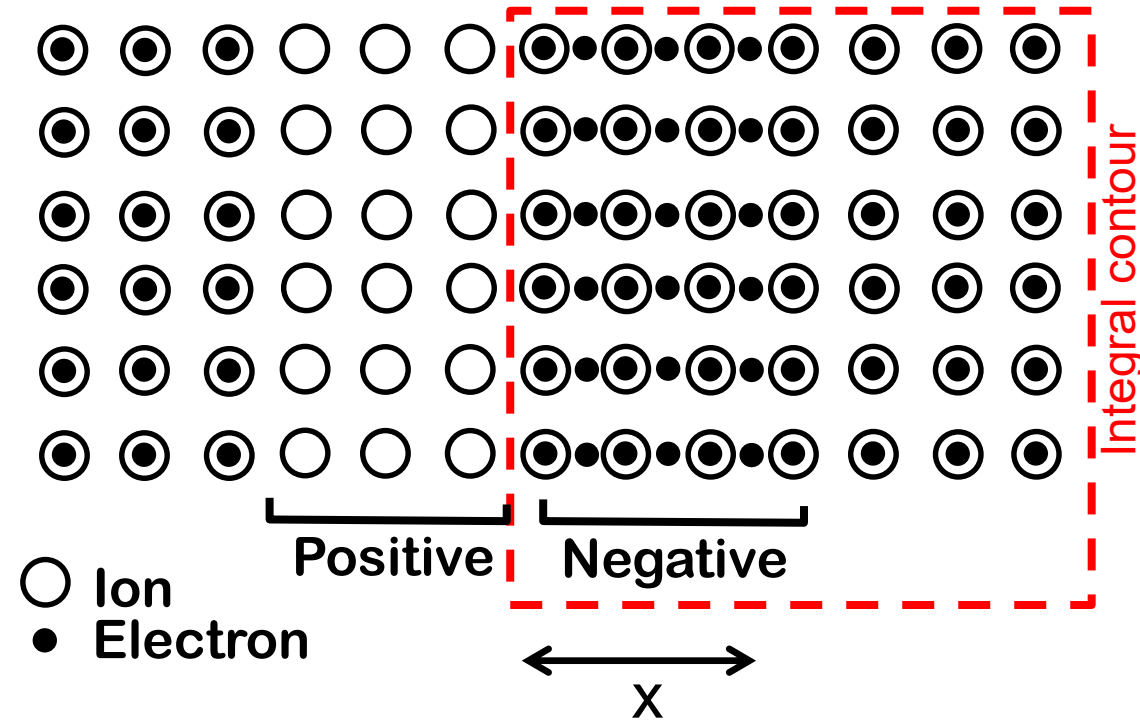
Accelerating structure,
metal (normal conductive
or super-conductive)



“Accelerating structure”
produced on-the-fly in
plasma by laser pulse

Plasma acceleration

Plasma oscillations



Derive ω_p

$$\oint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV$$

$$E = \frac{nex}{\epsilon_0}$$

$$F = m \frac{d^2 x}{dt^2} = -eE = -\frac{ne^2 x}{\epsilon_0}$$

Oscillation frequency: $\omega_p^2 = \frac{ne^2}{\epsilon_0 m}$

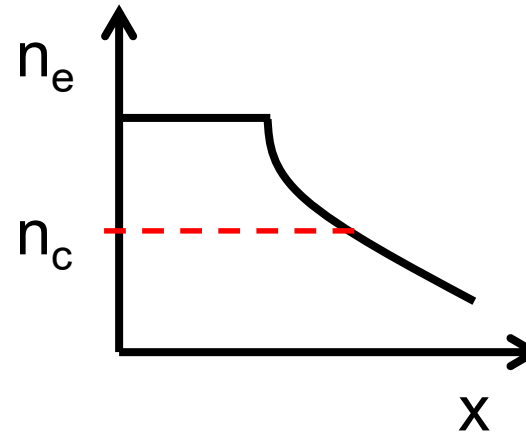
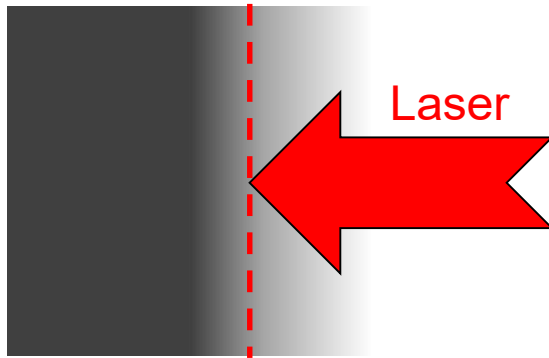
use: $r_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{m_e c^2}$

to rewrite as: $\omega_p^2 = 4\pi n c^2 r_e$

Useful formula $f_p \sim 9000 n^{1/2}$ (n in cm^{-3})

Critical density and surface

When laser hits a target its surface is heated, and plasma is formed
Plasma expands into the vacuum, and its density drops



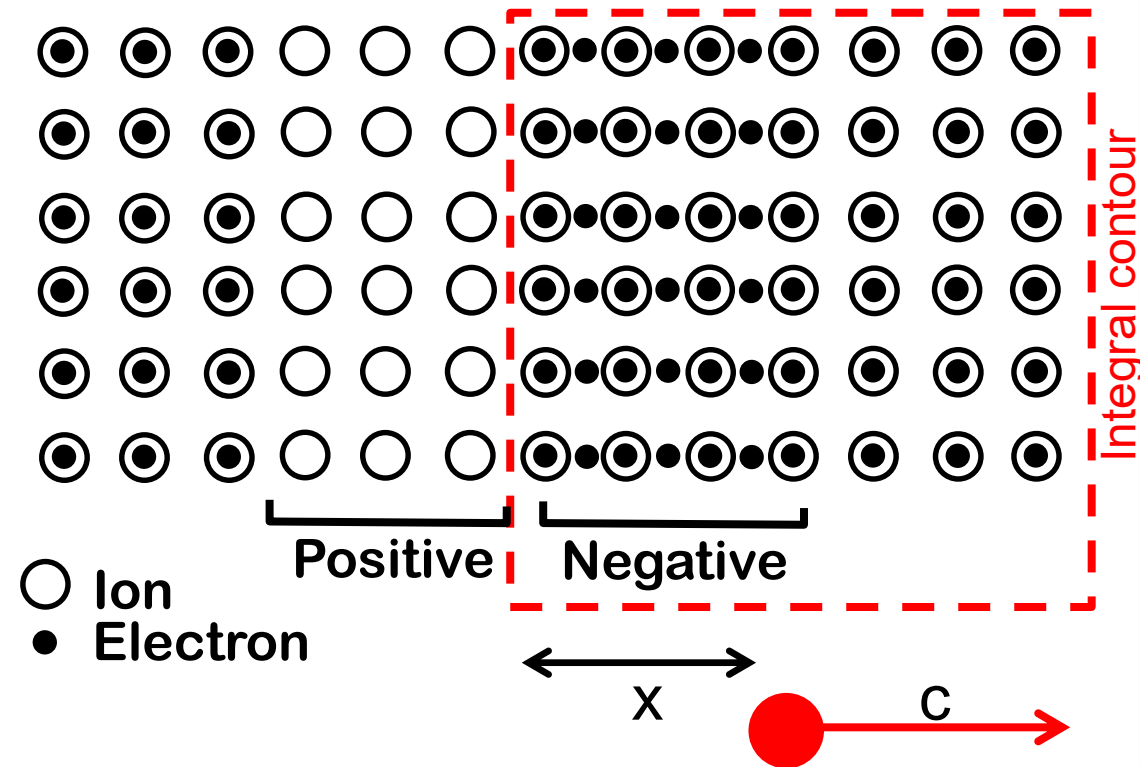
If ω_p is larger than laser frequency ω , then the plasma electrons can move fast enough to screen the laser

Therefore, laser penetrates only to the point where $\omega_p < \omega$

$$n_c = \omega^2 / (4\pi c^2 r_e)$$

The critical density is thus

Maximum field in plasma



Recall how we derived ω_p

$$\oint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV \quad E = \frac{nex}{\epsilon_0}$$

$$F = m \frac{d^2x}{dt^2} = -eE = -\frac{ne^2x}{\epsilon_0}$$

Oscillation frequency: $\omega_p^2 = \frac{ne^2}{\epsilon_0 m}$

use: $r_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{m_e c^2}$

to rewrite as: $\omega_p^2 = 4\pi n c^2 r_e$

Assume wave is excited by object moving with c

If total charge separation achieved in plasma, max field estimated taking $x \sim \lambda_p \sim \frac{c}{\omega_p}$

Thus $E_{\max} \sim \frac{nec}{\epsilon_0 \omega_p} = \frac{mc\omega_p}{e}$ or $eE_{\max} \cong mc^2 \frac{\omega_p}{c}$

Use $f_p \sim 9000 n^{1/2}$ (n in cm^{-3}) $\rightarrow eE_{\max} \approx 1 \frac{\text{eV}}{\text{cm}} \cdot n^{1/2} (\text{cm}^{-3})$

1 GeV/cm for
plasma 10^{18} cm^{-3}

How to excite plasma

- We see that GeV/cm require plasma with $n=10^{18} \text{ cm}^{-3}$

$$\lambda_p = \frac{c}{f_p} \rightarrow \lambda_p \approx 0.1 \text{ mm} \sqrt{\frac{10^{17} \text{ cm}^{-3}}{n}}$$

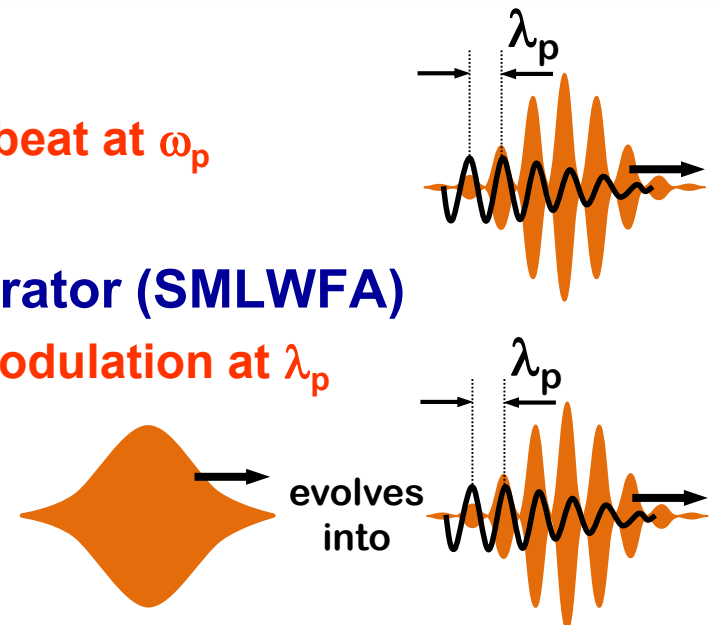
- Thus, short sub-ps pulses needed for plasma excitation
- In absence of short laser pulses other methods suggested:

- **Plasma Beat Wave Accelerator (PBWA)**

- Two laser pulses of closer frequencies beat at ω_p

- **Self-Modulated Laser Wakefield Accelerator (SMLWFA)**

- Instability in a long laser pulse cause modulation at λ_p



J.M. Dawson, Phys. Rev. Lett. 43, 267 (1979)

How to excite plasma

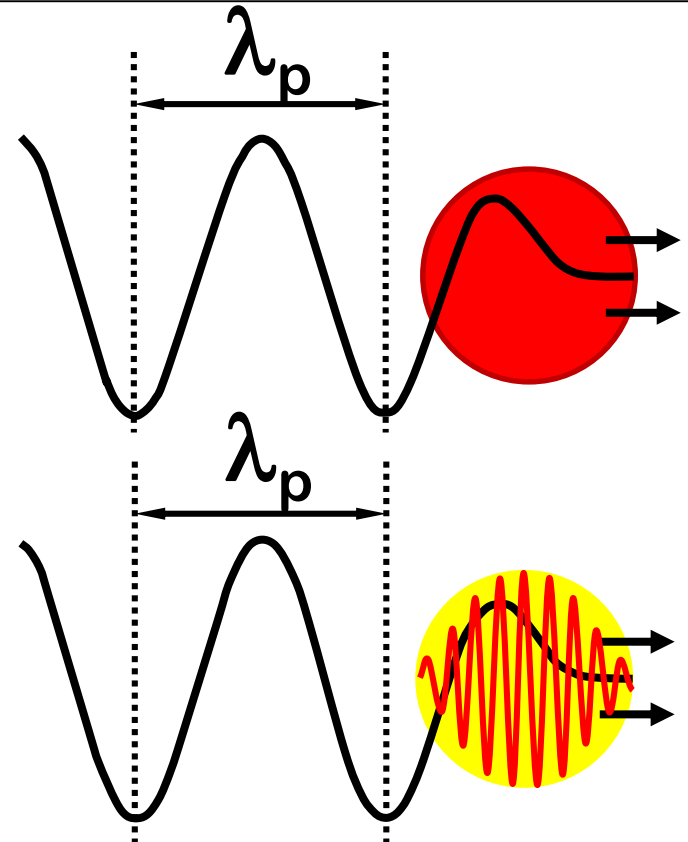
- Availability of short sub-ps pulses of laser or beams stimulated rapid progress of plasma acceleration

- **Plasma Wakefield Accelerator (PWFA)**

- A short high energy particle bunch

- **Laser Wakefield Accelerator (LWFA)**

- A short laser pulse of high intensity



Laser pulse of high intensity

Laser intensity (in vacuum)

$$I = \frac{1}{2} \epsilon_0 E_{\max}^2 c \quad (\text{SI})$$

$$I = \frac{1}{8\pi} E_{\max}^2 c \quad (\text{Gaussian})$$

Fields in practical units:

$$E_{\max} \left[\left(\frac{\text{V}}{\text{cm}} \right) \right] \cong 2.75 \times 10^9 \left(\frac{I}{10^{16} \text{ W/cm}^2} \right)^{1/2}$$

$$B_{\max} [\text{Gauss}] \cong 9.2 \times 10^6 \left(\frac{I}{10^{16} \text{ W/cm}^2} \right)^{1/2}$$

(useful to remember that 300 V/cm is ~same as 1 Gauss)

Compare with field in a hydrogen atom. Bohr radius and field:

$$a_B = \frac{\hbar^2}{me^2} = 5.3 \times 10^{-9} \text{ cm}$$

$$E_a = \frac{e}{a_B^2} = \frac{e}{4\pi\epsilon_0 a_B^2} \approx 5.1 \times 10^{11} \frac{\text{V}}{\text{m}}$$

(Gaussian)

(SI)

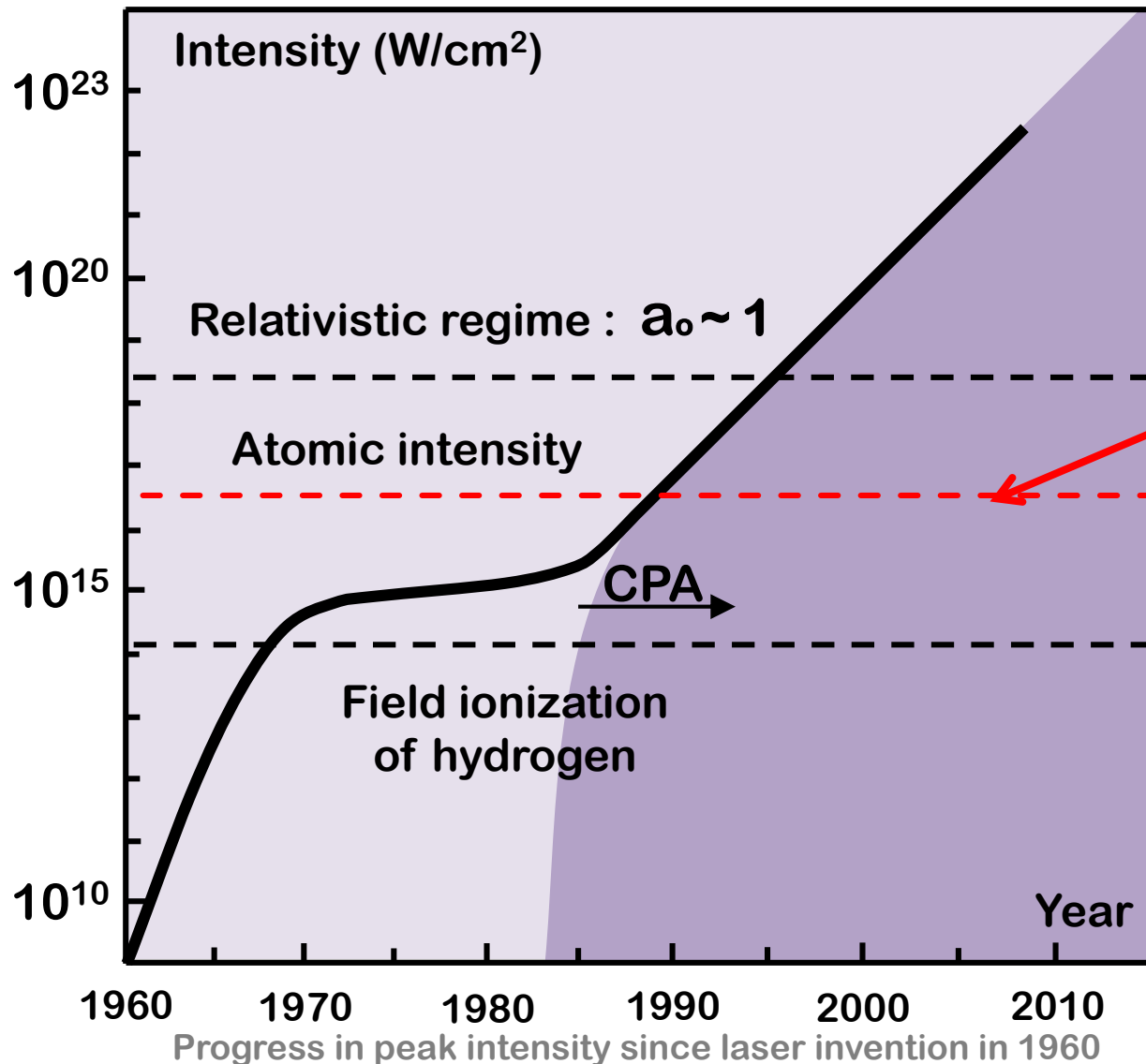
(Recall $\epsilon_0 \approx 8.8 \cdot 10^{-12} \frac{\text{A}^2 \cdot \text{s}^4}{\text{kg} \cdot \text{m}^3}$)

Atomic intensity

$$I_a = \frac{\epsilon_0 c E_a^2}{2} \cong 3.51 \times 10^{16} \frac{\text{W}}{\text{cm}^2}$$

A laser with intensity higher than that will ionize gas immediately

Laser intensity

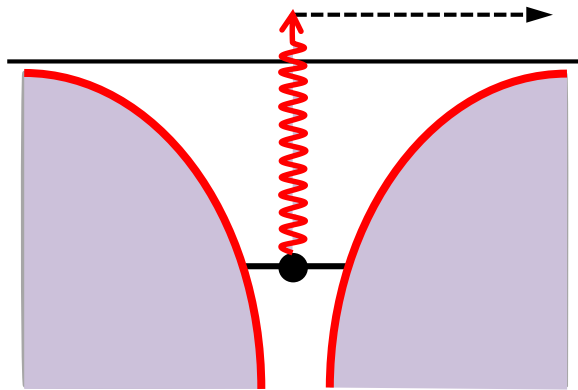


$$I_a = \frac{\epsilon_0 c E_a^2}{2} \cong 3.51 \times 10^{16} \frac{\text{W}}{\text{cm}^2}$$

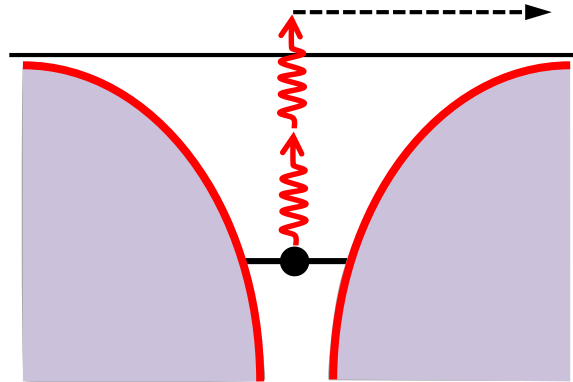
- In fact, ionization can occur well below this threshold due to:

- multi-photon effects;
- tunneling ionization

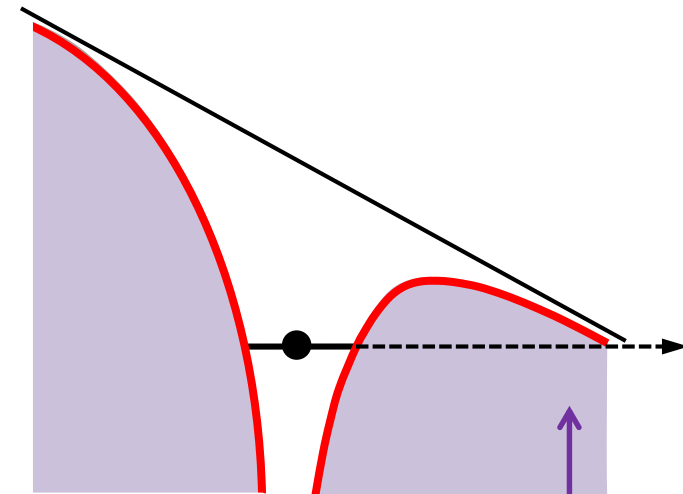
Types of ionization



Direct ionization



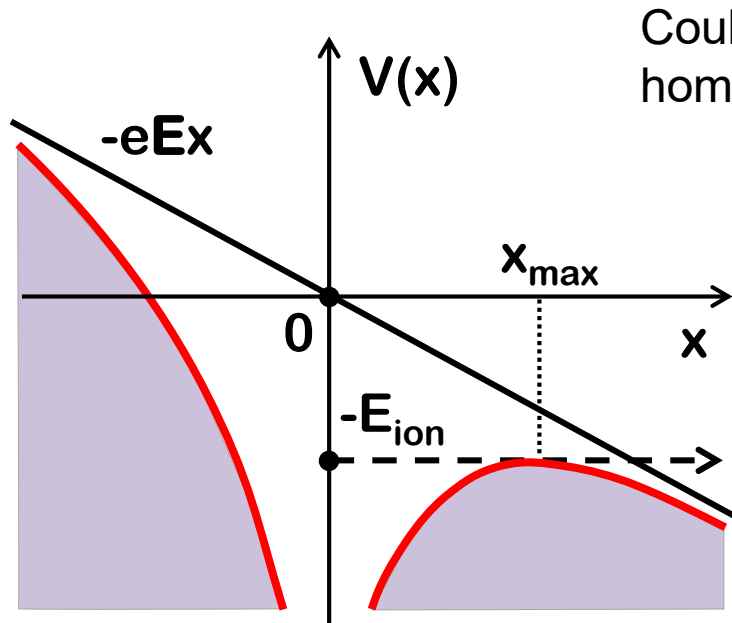
Multi-photon ionization



Tunneling ionization

With even more laser intensity – barrier suppression ionization (BSI)

Laser intensity for barrier suppression ionization



Coulomb potential of hydrogen atom distorted by homogeneous field ε :

$$V(x) = -\frac{e^2}{x} - e\varepsilon x$$

Find position of the maximum:

$$x_{\text{max}} = (e / \varepsilon)^{1/2}$$

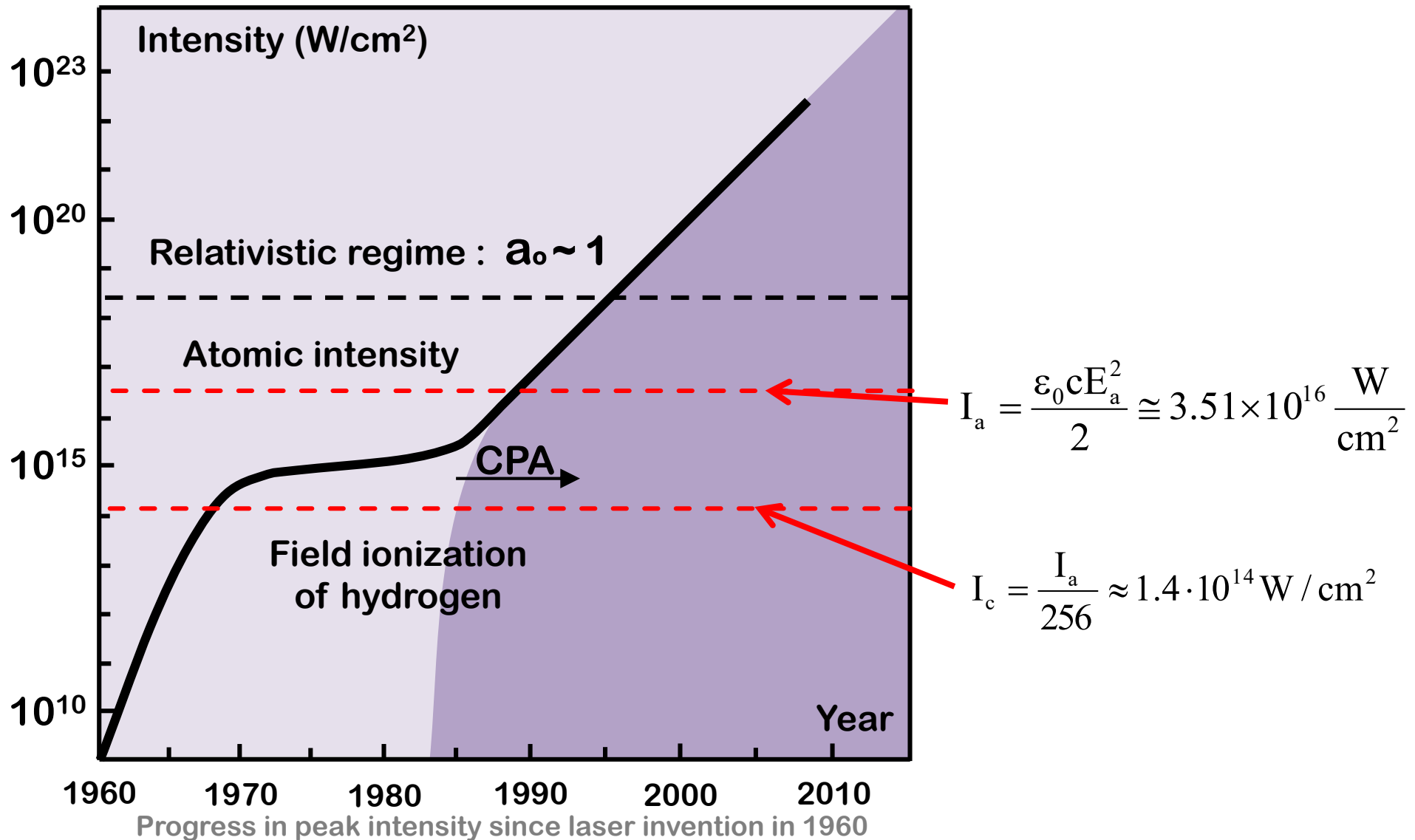
Equate potential value at max to hydrogen ionization potential

$$V(x_{\text{max}}) = 2(e^3 \varepsilon)^{1/2} = E_{\text{ion}} = \frac{e^2}{2a_B} \approx 13.6 \text{ eV}$$

The critical field for hydrogen is therefore $\varepsilon_c = \frac{e}{16a_B^2} = \frac{E_a}{16}$

Which corresponds to intensity $I_c = \frac{I_a}{256} \approx 1.4 \cdot 10^{14} \text{ W / cm}^2$

Laser intensity



Normalized vector potential

The laser field can be written in terms of the vector potential of the laser field $\mathbf{A}$ as

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{c \partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}$$

For linearly polarized field

$$\mathbf{A} = A_0 \cos(kz - \omega t) \mathbf{e}_\perp$$

We see that $E_0 = \frac{A_0 \omega}{c}$

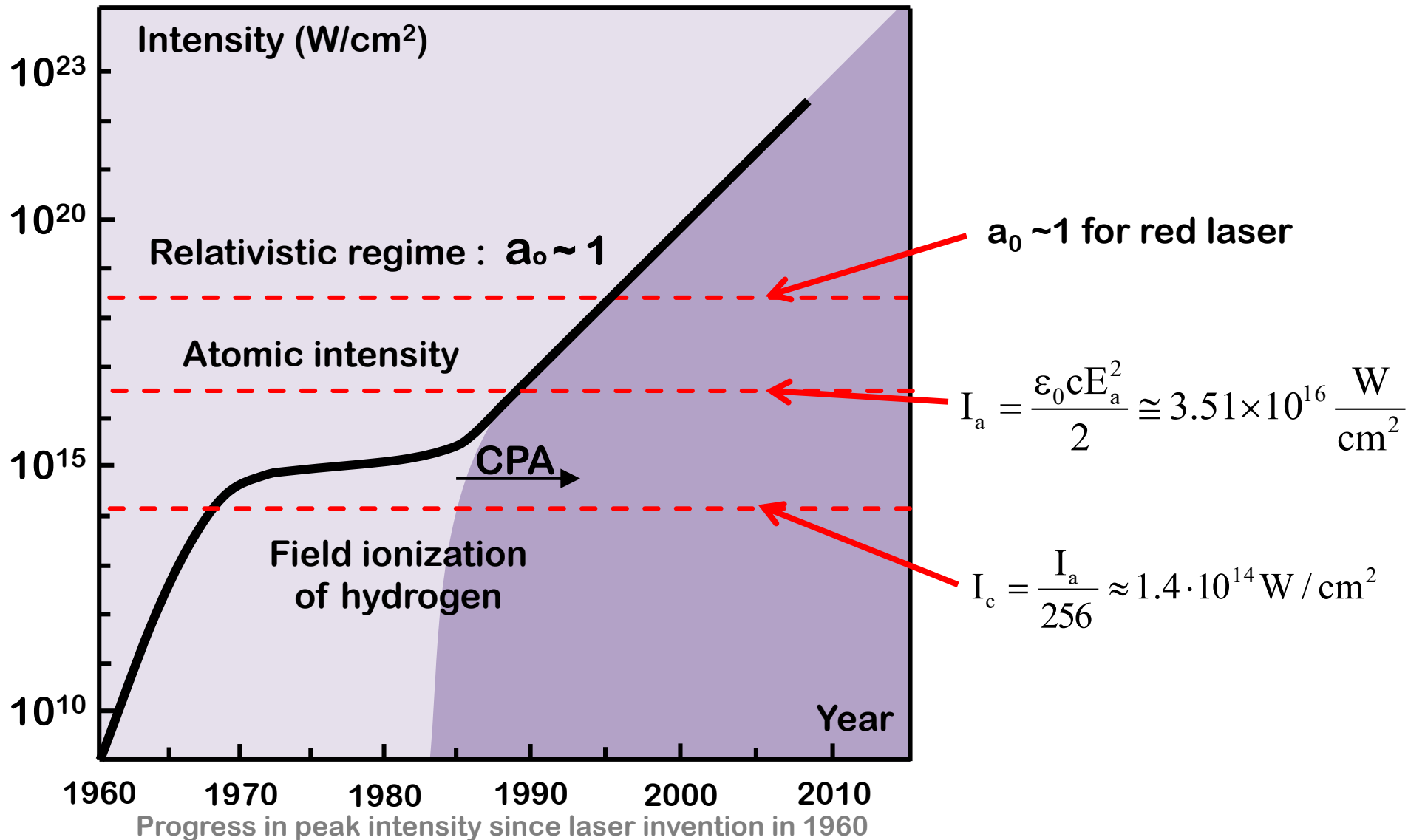
Compare momentum gained by e- in one cycle of laser field $e E \Delta t \cong \frac{e E}{\omega}$ with $m_e c$

We see that it is useful to define the normalized vector potential as $\mathbf{a} = \frac{e \mathbf{A}}{m_e c^2}$ with amplitude $a_0 = \frac{e E_0}{m_e \omega c}$

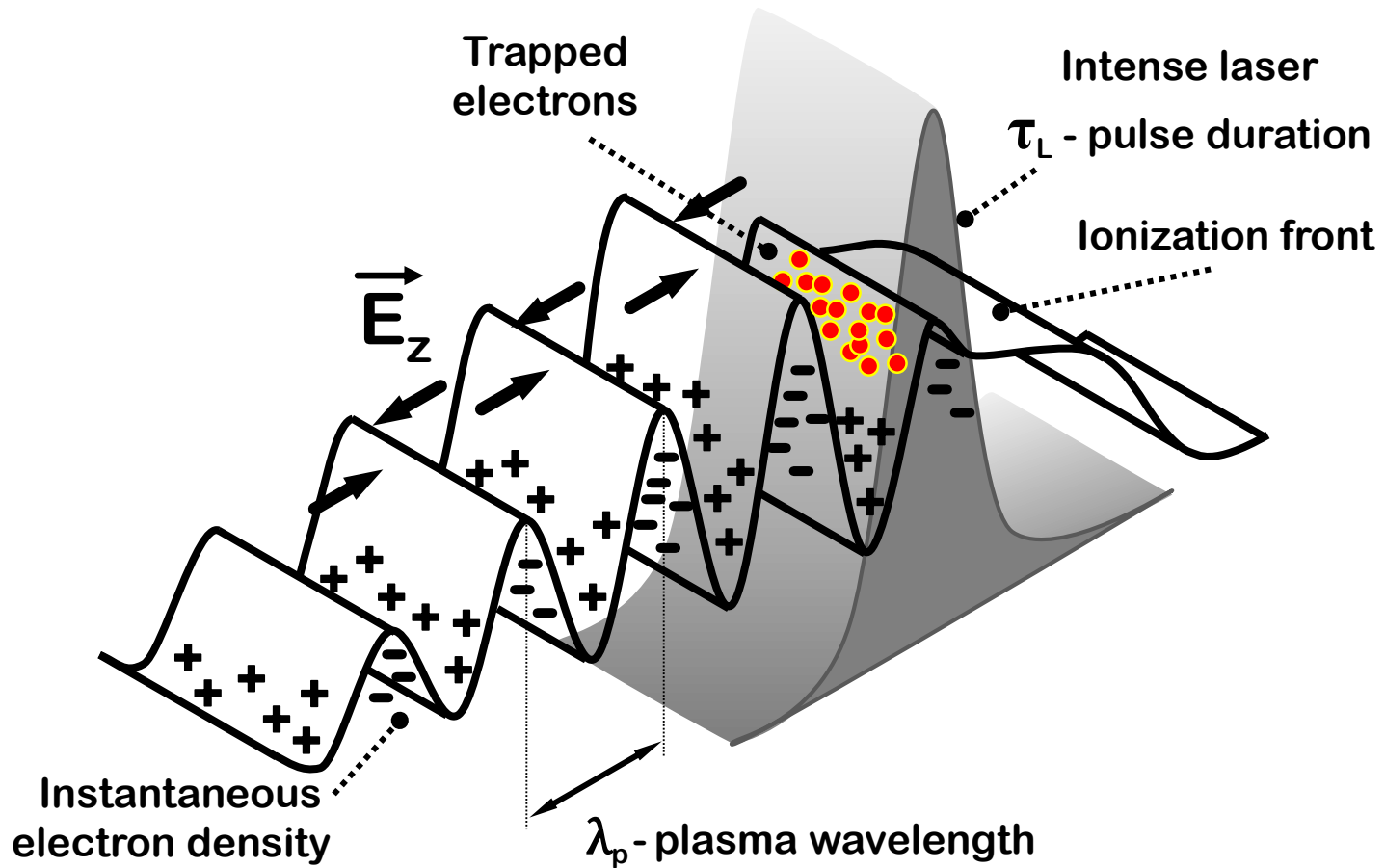
The amplitude a_0 will indicate if the electron motion in laser field relativistic
 $a_0 \gg 1$ – relativistic, $a_0 \ll 1$ – non relativistic

In practical units $a_0 \approx \left(\frac{I [\text{W} / \text{cm}^2]}{1.37 \cdot 10^{18}} \right)^{\frac{1}{2}} \cdot \lambda [\mu\text{m}]$ where $\lambda = \frac{2\pi c}{\omega}$

Laser intensity



Laser acceleration - conceptually



- **Note in particular**

- **Ionization front starting at the front tail of laser**
- **Laser pulse length similar of shorter than plasma wavelength**
- **Electrons trapped in the first bubble**

Formation of bubble – ponderomotive force

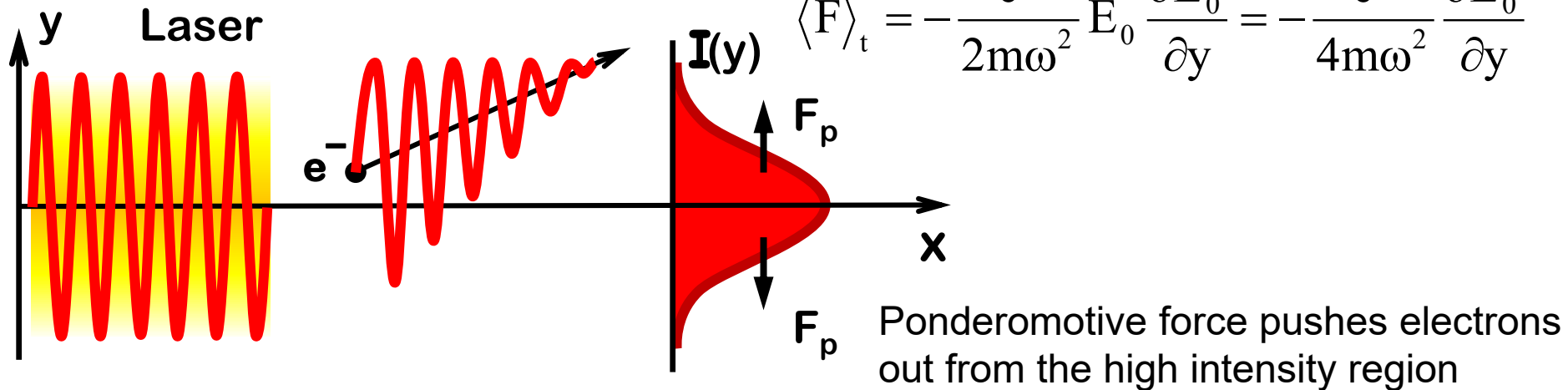
First, assume laser field homogeneous: $E = E_0 \cos(\omega t)$

Motion of electron: $\ddot{y} = \frac{F}{m} = \frac{eE}{m} \Rightarrow y = -\frac{eE_0}{m\omega^2} \cos(\omega t)$

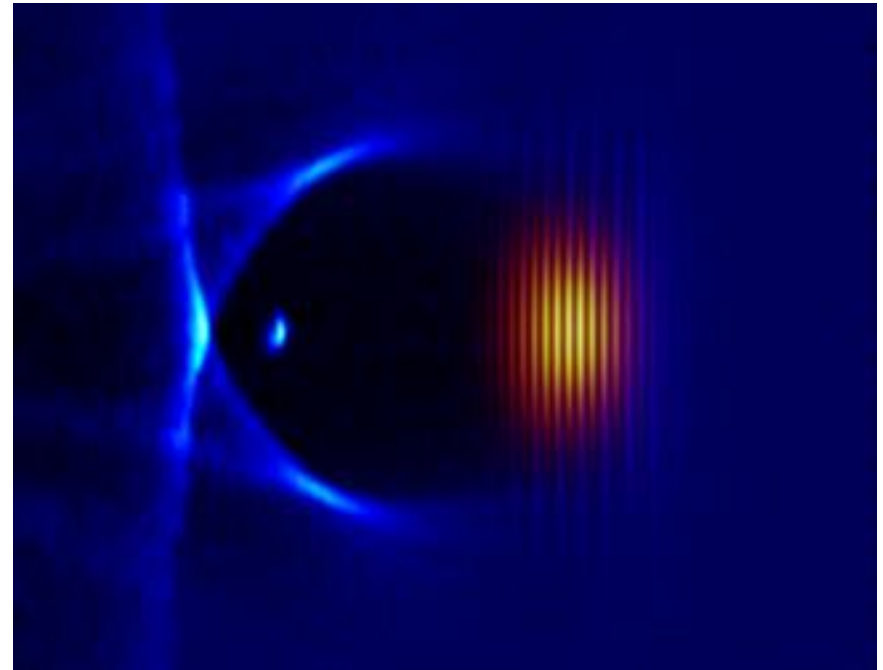
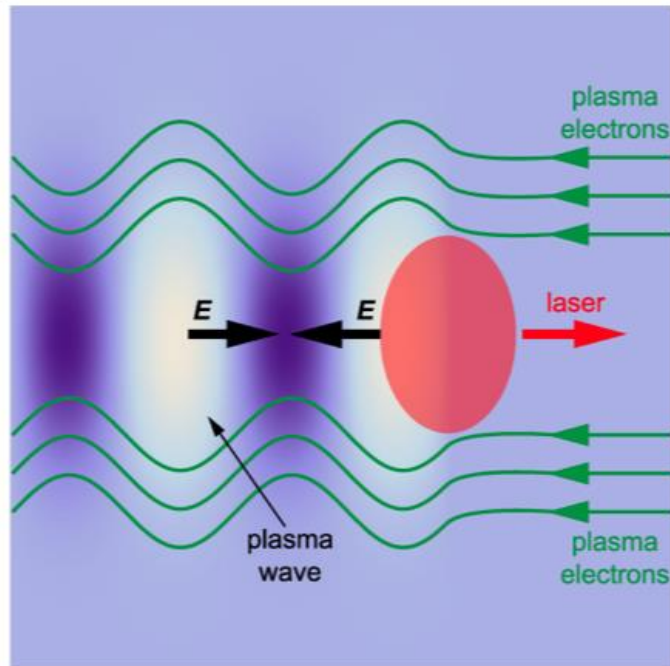
Now, assume E has gradient in y: $E = E_0(y) \cos(\omega t) \approx E_0 \cos(\omega t) + y \frac{\partial E_0}{\partial y} \cos(\omega t)$

Find time average of force acting on e-: $\langle F \rangle_t = \left\langle -\frac{eE_0}{m\omega^2} \cos(\omega t) \cdot \frac{\partial E_0}{\partial y} \cos(\omega t) \right\rangle_t$

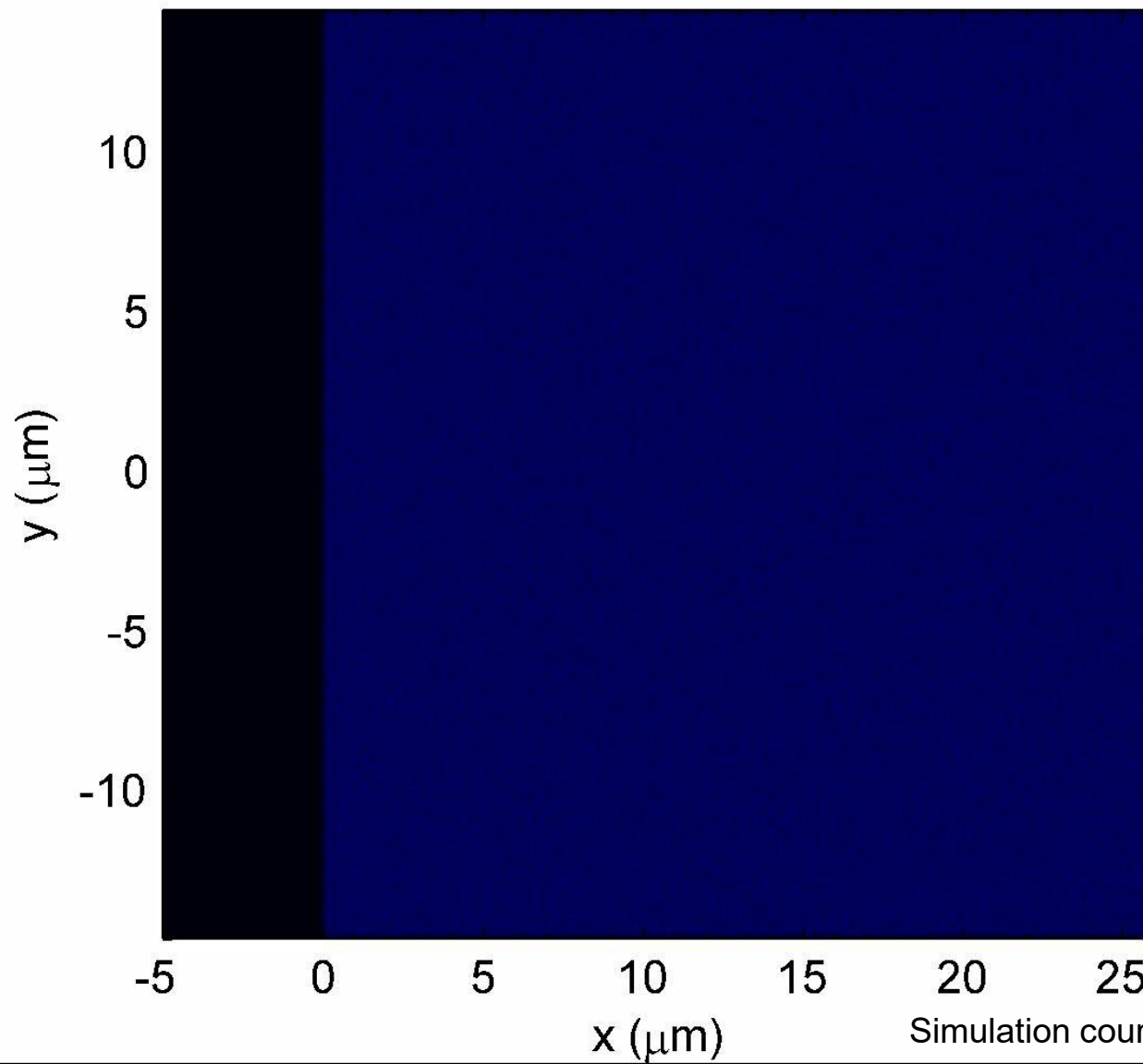
$$\langle F \rangle_t = -\frac{e^2}{2m\omega^2} E_0 \frac{\partial E_0}{\partial y} = -\frac{e^2}{4m\omega^2} \frac{\partial E_0^2}{\partial y}$$



Laser-Driven Plasma Acceleration



- **Ponderomotive force of short (50fs), intense ($10^{18} \text{ W cm}^{-2}$) laser pulse expels plasma electrons while heavier ions stay at rest**
- **Electrons attracted back to ions, forming a bubble (blow-out regime) and setting up plasma wave which trails laser pulse**
- **Electric fields within plasma wave of order 100 GV/m formed**

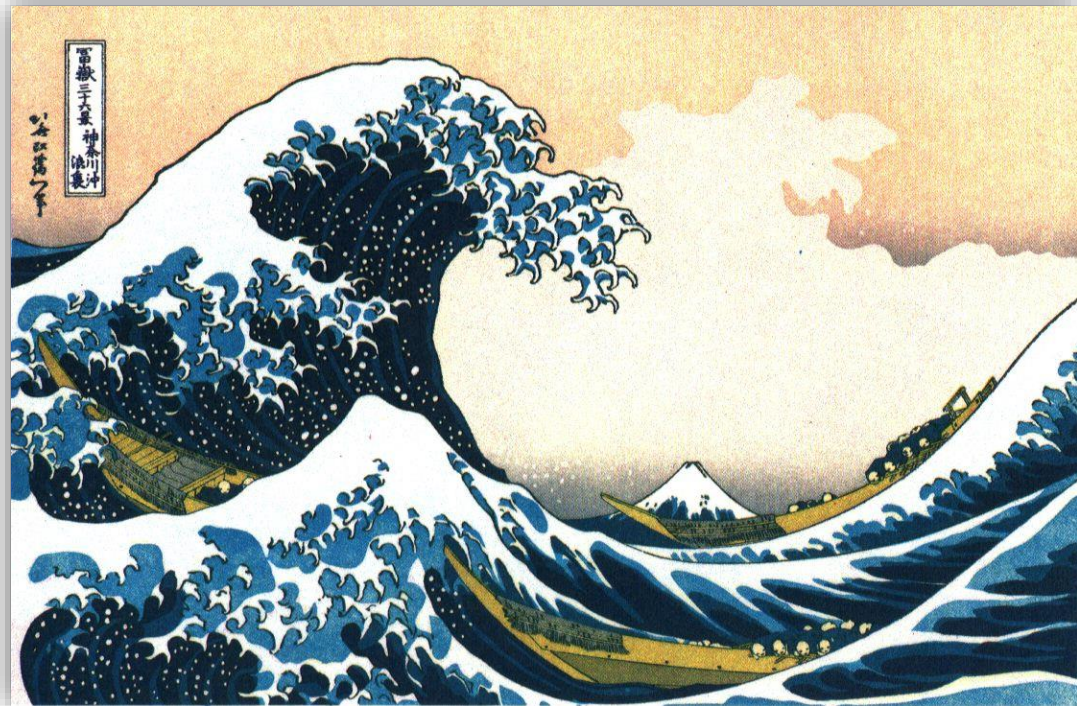


Simulation courtesy Prof Simon Hooker

How e- gets into the bubble – wave breaking

- **Wave breaking**

- Self-injection of background plasma electrons to the wake when some particles outrun the wake



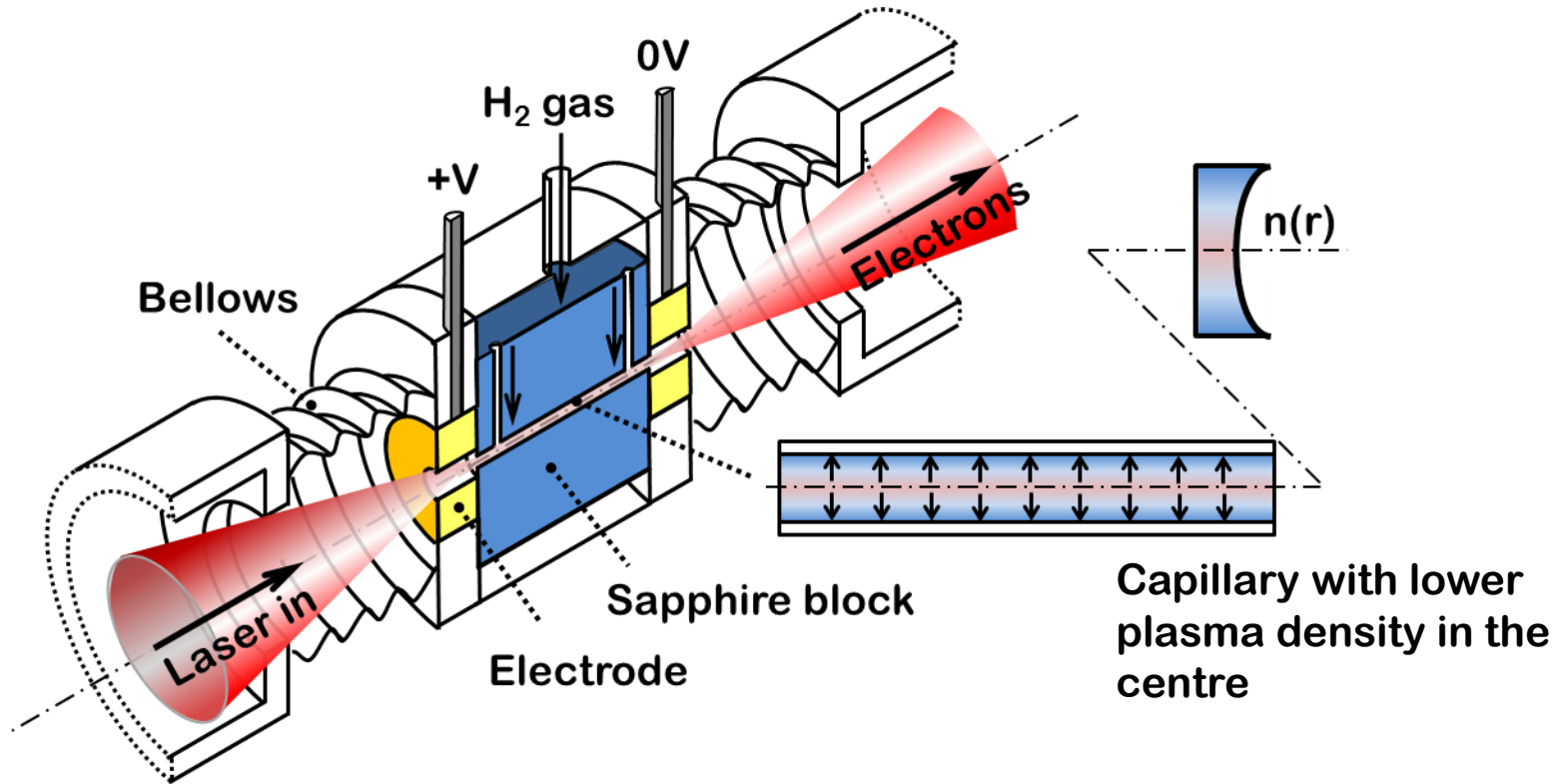
- **Other methods**

- External injection (difficult for so short bunches)
- Methods which involve two laser pulses and mix of two gases with different ionization potential

Importance of laser guidance

- As laser propagates through the gas/plasma, several competing effects are important
 - **Dephasing**
 - **Depletion**
 - **Longitudinal compression by plasma waves**
 - **Self focusing**
 - Including relativistic effect – electrons of plasma at centre become relativistic and have higher mass
 - **Diffraction**
 - Small laser beam ($\sim 30\mu\text{m}$) will diffract very fast
 - Includes ionization caused diffraction (centre where intensity is higher ionized first)
- A possible solution – create a channel with plasma density profile $n(r)$ to guide laser
 - **A particular solution – capillary discharge channel developed in Oxford**

Importance of laser guidance

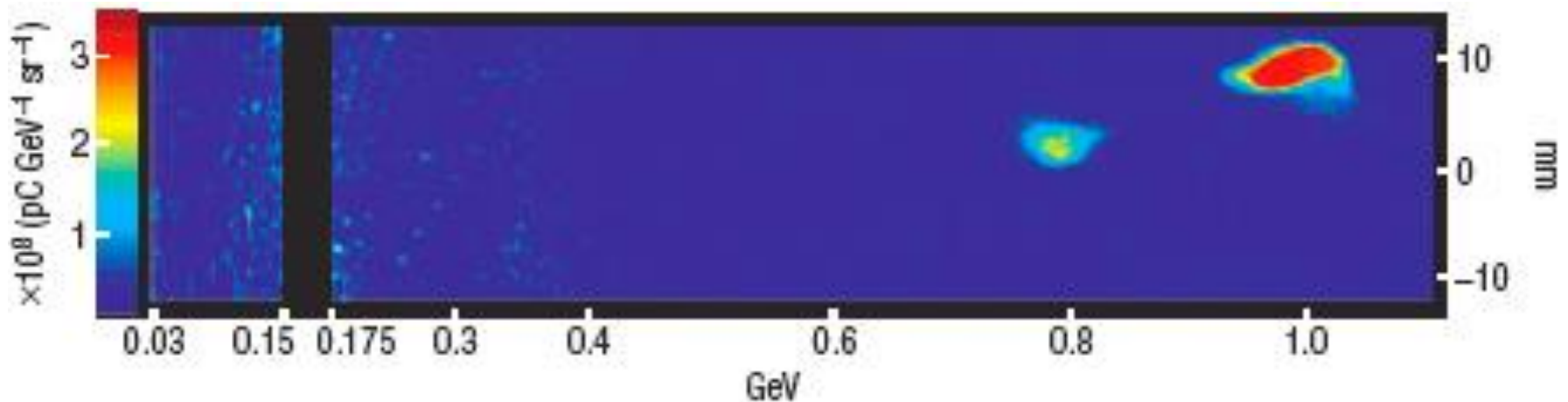


Capillary channel designed by Prof Simon Hooker

- **Capillary channel allowed exceeding 1GeV laser plasma acceleration for the first time**

First ever 1 GeV from laser plasma accelerator

- 1 GeV acceleration & monoenergetic beam
 - Use of guiding capillary was essential



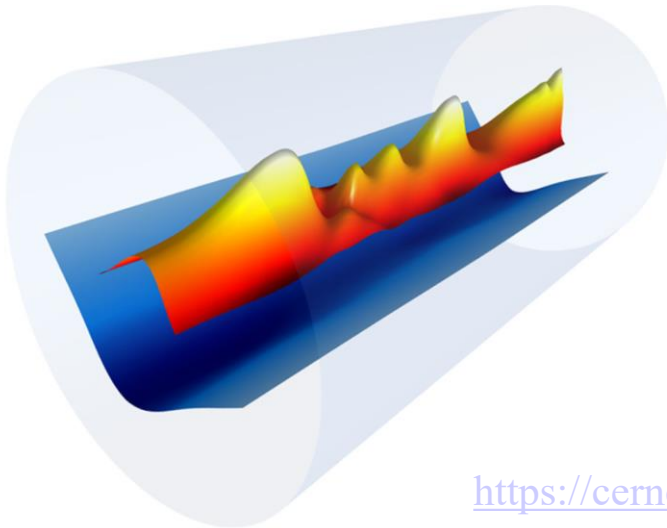
1GeV acceleration in just 3cm of plasma

W. Leemans, B. Nagler, A. Gonsalves, C. Toth, K. Nakamura, C. Geddes, E. Esarey, C. B. Schroeder, & S. Hooker, *Nature Physics* 2006

Plasma density $2.7 \times 10^{18} \text{ cm}^{-3}$, 40 TW laser with 10^{18} W/cm^2

Multi-GeV from laser plasma accelerator

- **7.8 GeV acceleration & monoenergetic beam**
 - Use of guiding capillary was essential, and increasing the focusing strength of a capillary discharge waveguide using laser inverse bremsstrahlung heating

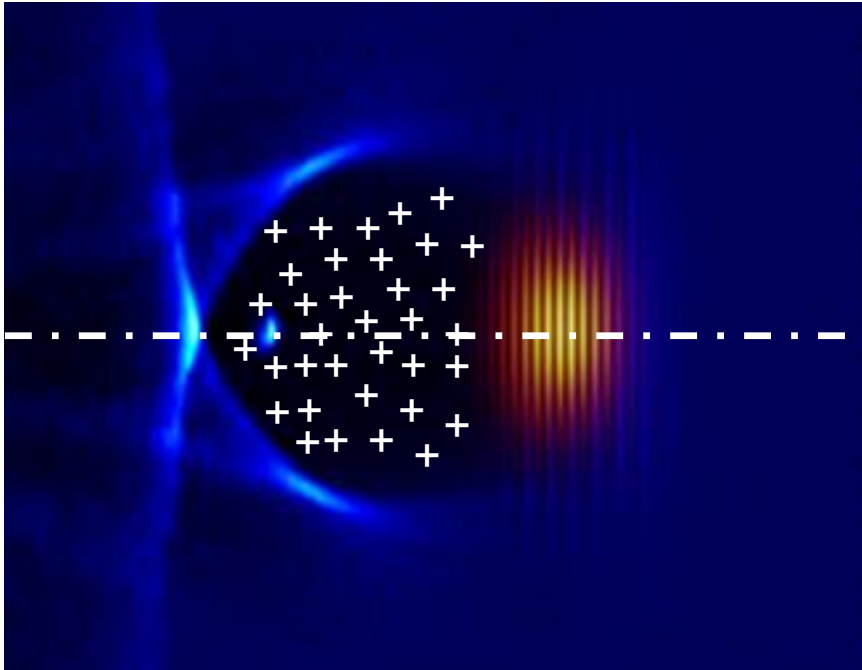


The plasma channel's electron density profile (blue) formed inside a sapphire tube (grey) with the combination of an electrical discharge and ultrashort laser pulse (red, orange, and yellow). Credit: G Bagdasarov/A Gonsalves/J-L Vay and CERN Courier

<https://cerncourier.com/a/bella-sets-new-record-for-plasma-acceleration/>

A Gonsalves *et al.* 2019 *Phys. Rev. Lett.* **122** 084801.

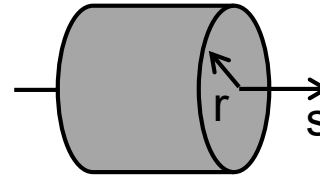
Transverse fields in the bubble



The ions are heavy and are inside of the bubble. They produce focusing force.

$$\oint \mathbf{E} \cdot d\mathbf{S} = 4\pi \int \rho dV \quad (\text{Gaussian})$$

Assume cylindrical symmetry



Focusing force $eE = 2\pi n e^2 r$

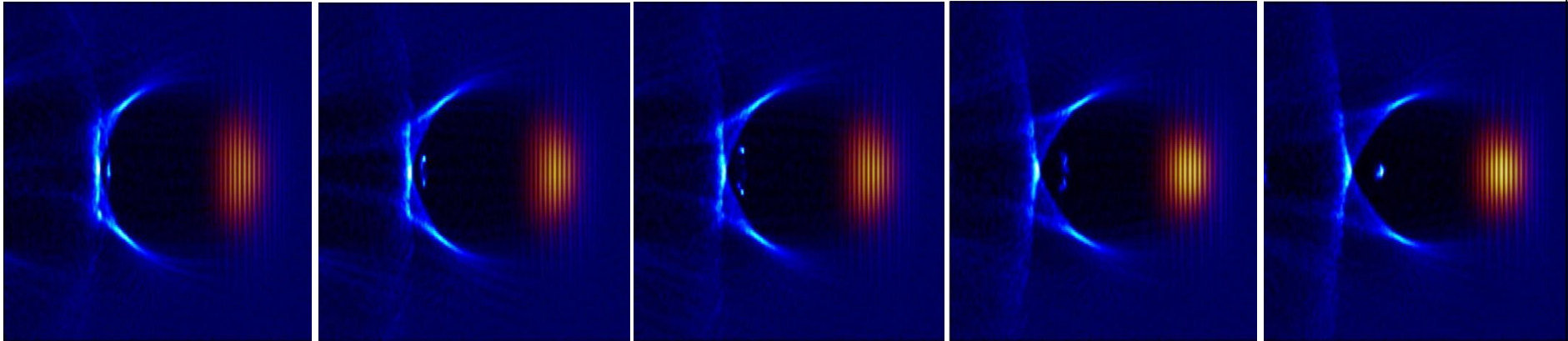
Assume electron is relativistic with γ

It will oscillate in this field as

$$\frac{d^2 r}{ds^2} = \frac{2\pi n e^2 r}{\gamma m c^2} = \frac{\omega_p^2}{2\gamma c^2} r$$

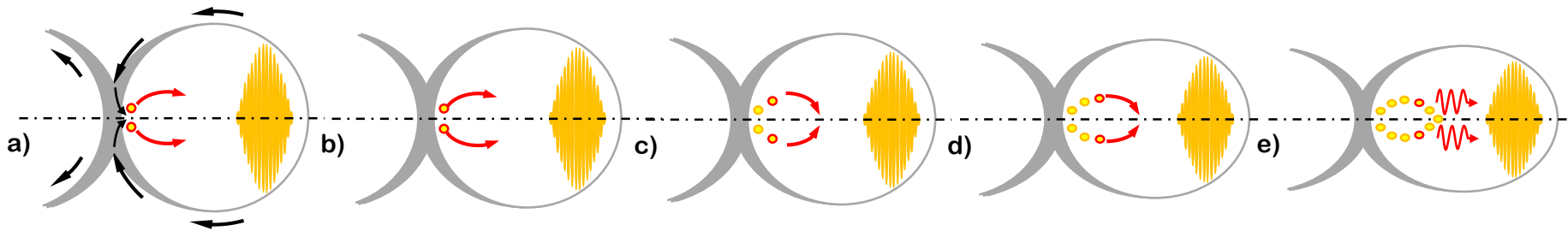
The period of oscillation is therefore $\lambda = \sqrt{2\gamma} \lambda_p$

Betatron radiation



- **Strong radial electric field within plasma wave cause transverse oscillation of electron bunch**
- **Generates bright betatron radiation in 1- 100 keV range**
- **Let's estimate parameters of this radiation**

Betatron radiation



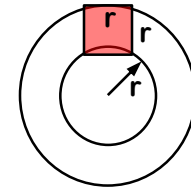
- **Strong radial electric field within plasma wave cause transverse oscillation of electron bunch**
- **Generates bright betatron radiation in 1- 100 keV range**
- **Let's estimate parameters of this radiation**

Synchrotron radiation *on-the-back-of-the-envelope* – power loss

Energy in the field left behind (radiated !):

$$W \approx \int E^2 dV$$

The field $E \approx \frac{e}{r^2}$ the volume $V \approx r^2 dS$



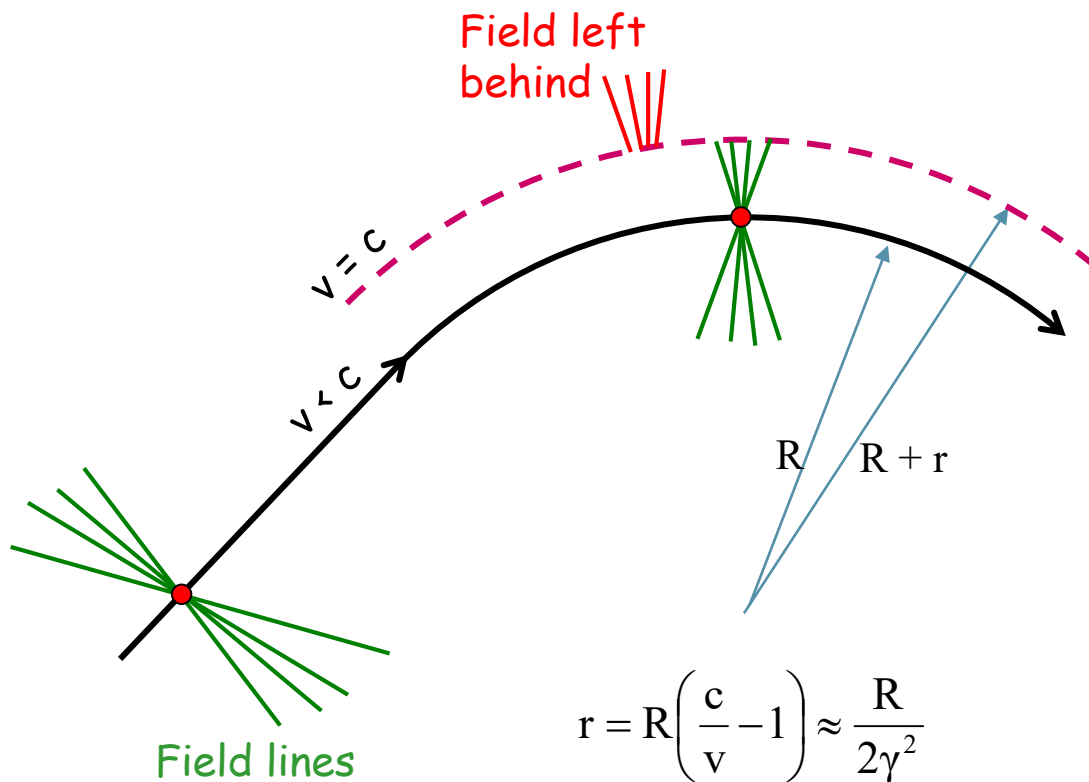
Energy loss per unit length:

$$\frac{dW}{dS} \approx E^2 r^2 \approx \left(\frac{e}{r^2} \right)^2 r^2$$

Substitute $r \approx \frac{R}{2\gamma^2}$ and get an estimate:

$$\boxed{\frac{dW}{dS} \approx \frac{e^2 \gamma^4}{R^2}}$$

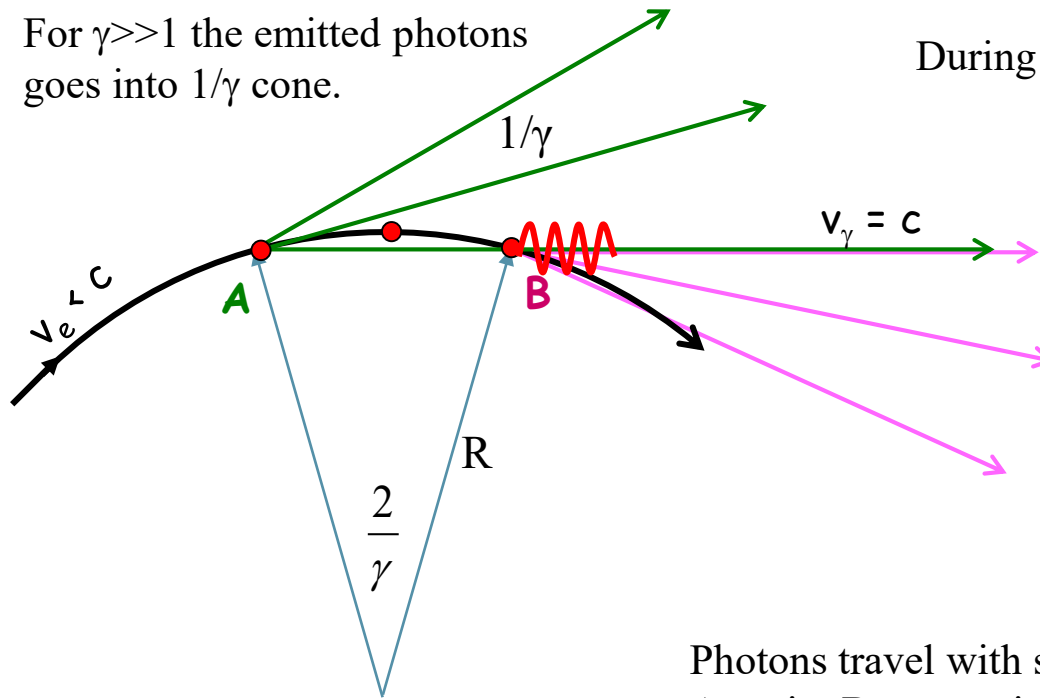
Compare with exact formula: $\frac{dW}{dS} = \frac{2}{3} \frac{e^2 \gamma^4}{R^2}$



Gaussian units on this page!

Synchrotron radiation *on-the-back-of-the-envelope* – photon energy

For $\gamma \gg 1$ the emitted photons go into $1/\gamma$ cone.



During what time Δt the observer will see the photons?



Photons emitted during travel along the $2R/\gamma$ arc will be observed.

Photons travel with speed c , while particles with v .

At point B, separation between photons and particles is

$$dS \approx \frac{2R}{\gamma} \left(1 - \frac{v}{c} \right)$$

Therefore, observer will see photons during $\Delta t \approx \frac{dS}{c} \approx \frac{2R}{c\gamma} (1 - \beta) \approx \frac{R}{c\gamma^3}$

Estimation of characteristic frequency $\omega_c \approx \frac{1}{\Delta t} \approx \frac{c\gamma^3}{R}$

Compare with exact formula: $\omega_c = \frac{3}{2} \frac{c\gamma^3}{R}$

Synchrotron radiation *on-the-back-of-the-envelope* – number of photons

We estimated the rate of energy loss : $\frac{dW}{dS} \approx \frac{e^2 \gamma^4}{R^2}$ And the characteristic frequency $\omega_c \approx \frac{c \gamma^3}{R}$

&

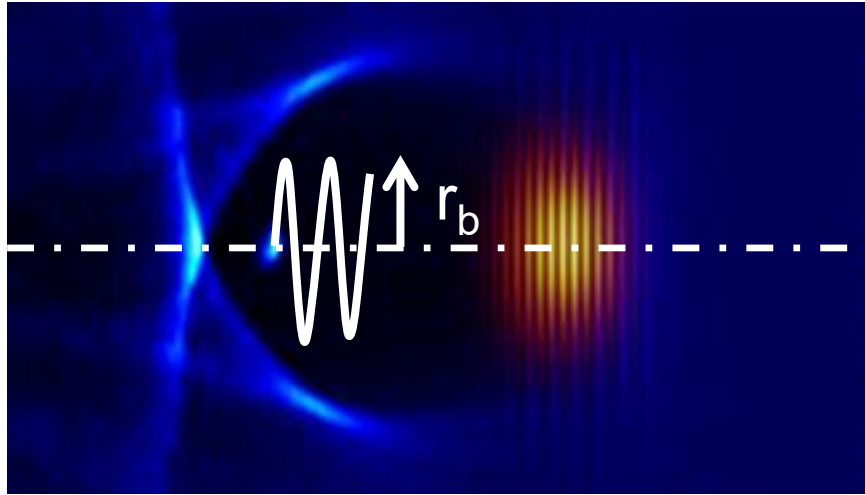
The photon energy $\varepsilon_c = \hbar \omega_c \approx \frac{\gamma^3 \hbar c}{R} = \frac{\gamma^3}{R} \lambda_e mc^2$ where $r_e = \frac{e^2}{mc^2}$ $\alpha = \frac{e^2}{\hbar c}$ $\lambda_e = \frac{r_e}{\alpha}$

=>

Number of photons emitted per unit length $\frac{dN}{dS} \approx \frac{1}{\varepsilon_c} \frac{dW}{dS} \approx \frac{\alpha \gamma}{R}$ (per angle θ : $N \approx \alpha \gamma \theta$)

Gaussian units on this page!

Estimations of betatron radiation



We found that relativistic electron with γ will oscillate in the field of ions as

$$\frac{d^2 r}{ds^2} = \frac{2\pi n e^2 r}{\gamma m c^2} = \frac{\omega_p^2}{2\gamma c^2} r$$

period of oscillation is $\lambda = \sqrt{2\gamma} \lambda_p$

If amplitude of oscillation is r_b
then the radius of curvature
of the trajectory is

$$R = \frac{\lambda^2}{4\pi^2 r_b}$$

Substitute and get the radius of curvature as $R = \frac{\gamma \lambda_p^2}{2\pi^2 r_b}$

Substitute into $\omega_c = \frac{3}{2} \frac{c \gamma^3}{R}$ and get estimation of radiation wavelength

$$\lambda_c = \frac{\lambda_p^2}{3\pi \gamma^2 r_b}$$

Use $\frac{dN}{dS} \approx \frac{\alpha \gamma}{R}$ to estimate N_γ photons emitted per λ

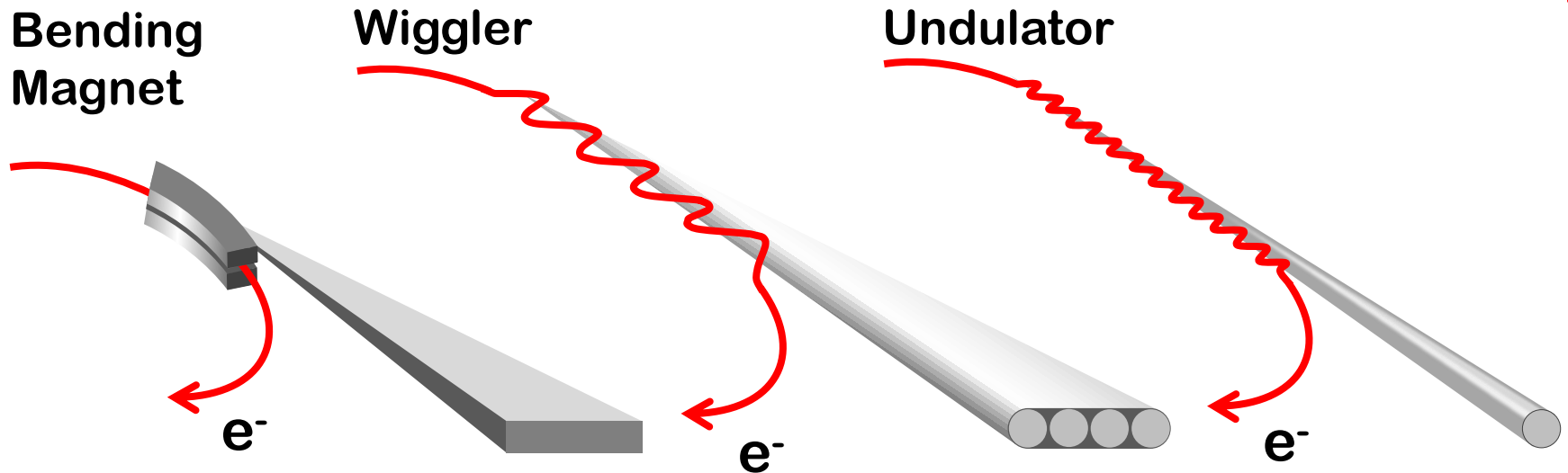
$$N_\gamma \approx \sqrt{2\gamma} \cdot 2\pi^2 \alpha \frac{r_b}{\lambda_p}$$

Assume 1GeV ($\gamma=2E3$), $\lambda_p=0.03\text{mm}$, $r_b=0.001\text{mm} \Rightarrow \lambda_c=0.25 \text{ A}$ or $\sim 50 \text{ keV}$
and N_γ per λ is ~ 0.3

Many hard photons!

19. Periodic action

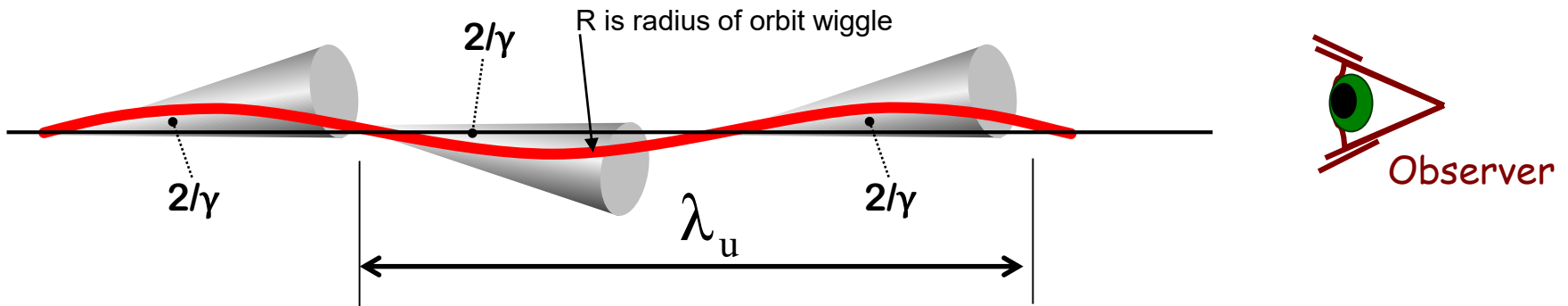
- Instead of continuous action, use periodic or pulsating actions.
- If an action is already periodic, change the periodic magnitude or frequency.
 - Use pauses between impulses to perform a different action.



Devices for generation of synchrotron radiation

FEL – coherent radiation conditions

1)



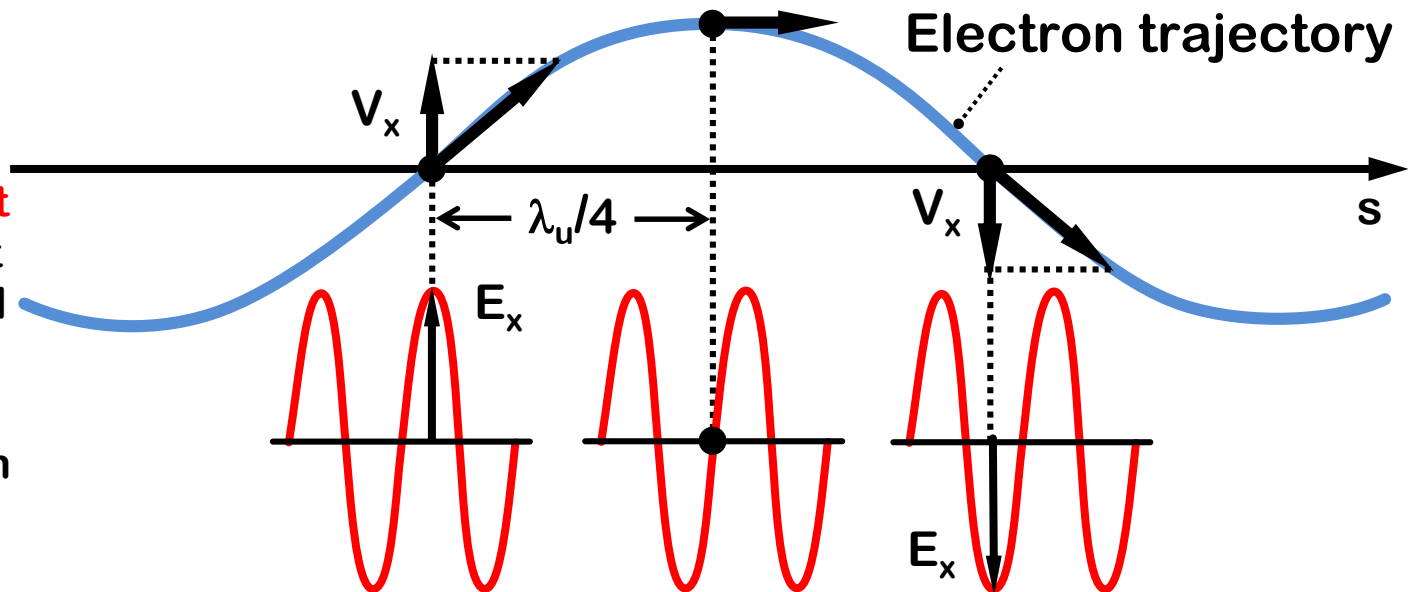
If $2R/\gamma \gg \lambda_u/2 \Rightarrow$ regime where entire wiggling trajectory contribute to radiation

Define $K \sim \gamma \lambda_u/R$

$K \ll 1$ – undulator regime

2)

Condition for **resonant energy transfer** is that EM wave slips forward with respect to electron by $\lambda/2$ per half period of electron trajectory, i.e.



Plasma acceleration FELs – first lasing

“experimental demonstration of undulator radiation amplification in the exponential-gain regime by using electron beams based on a laser wakefield accelerator. The amplified undulator radiation, which is typically centred at 27 nanometres and has a maximum photon number of around $1\text{E}10$ per shot, yields a maximum radiation energy of about 150 nanojoules. In the third of three undulators in the device, the maximum gain of the radiation power is approximately 100-fold, confirming a successful operation in the exponential-gain regime”

Free-electron lasing at 27 nanometres based on a laser wakefield accelerator, Wentao Wang, et al., 516 | Nature | Vol 595 | 22 July 2021

<https://www.nature.com/articles/s41586-021-03678-x>



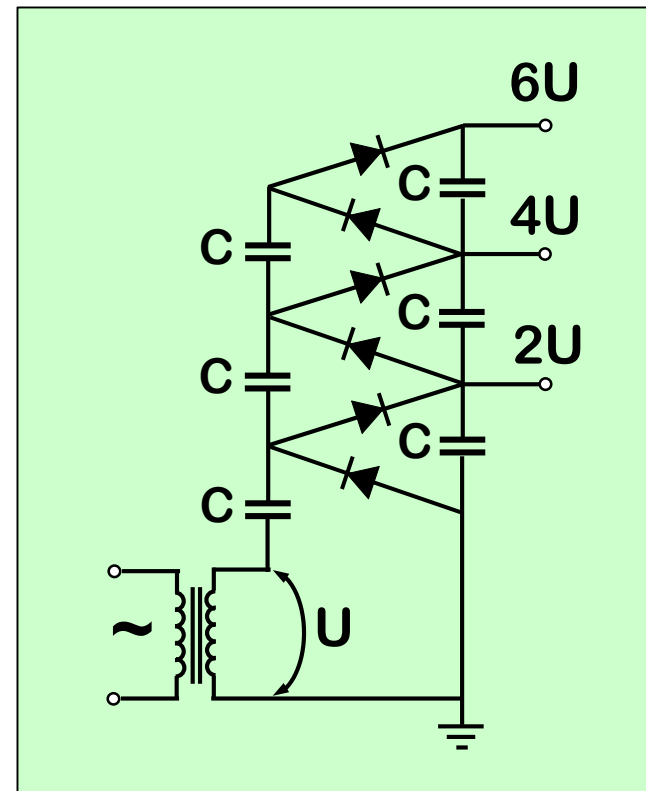
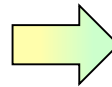
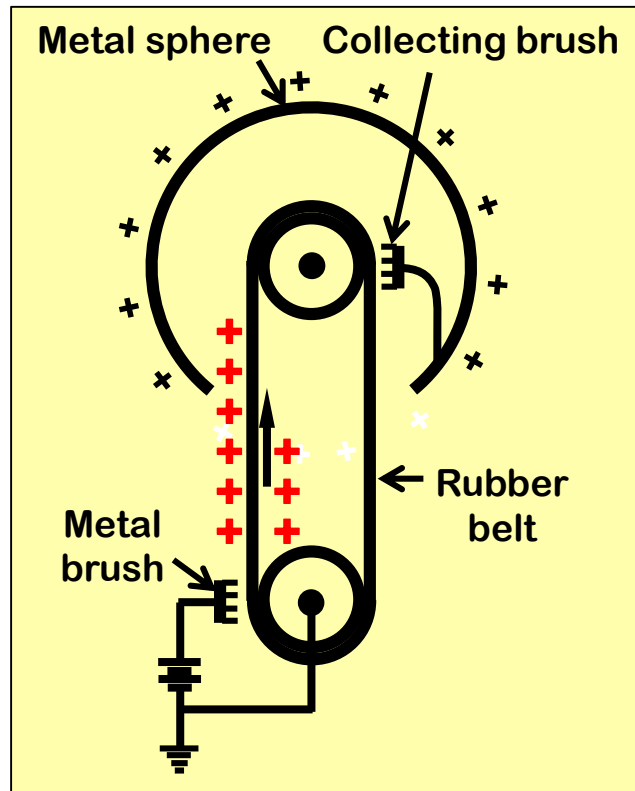
“experimental evidence of FEL lasing by a compact (3-cm) particle-beam-driven plasma accelerator. ... observation of narrow-band amplified radiation in the infrared range with typical exponential growth of its intensity over six consecutive undulators”

Free-electron lasing with compact beam-driven plasma wakefield accelerator, R. Pompili, D. Alesini, ...M. Ferrario, Nature volume 605, pages 659–662 (2022)

<https://www.nature.com/articles/s41586-022-04589-1>

28. Mechanics substitution

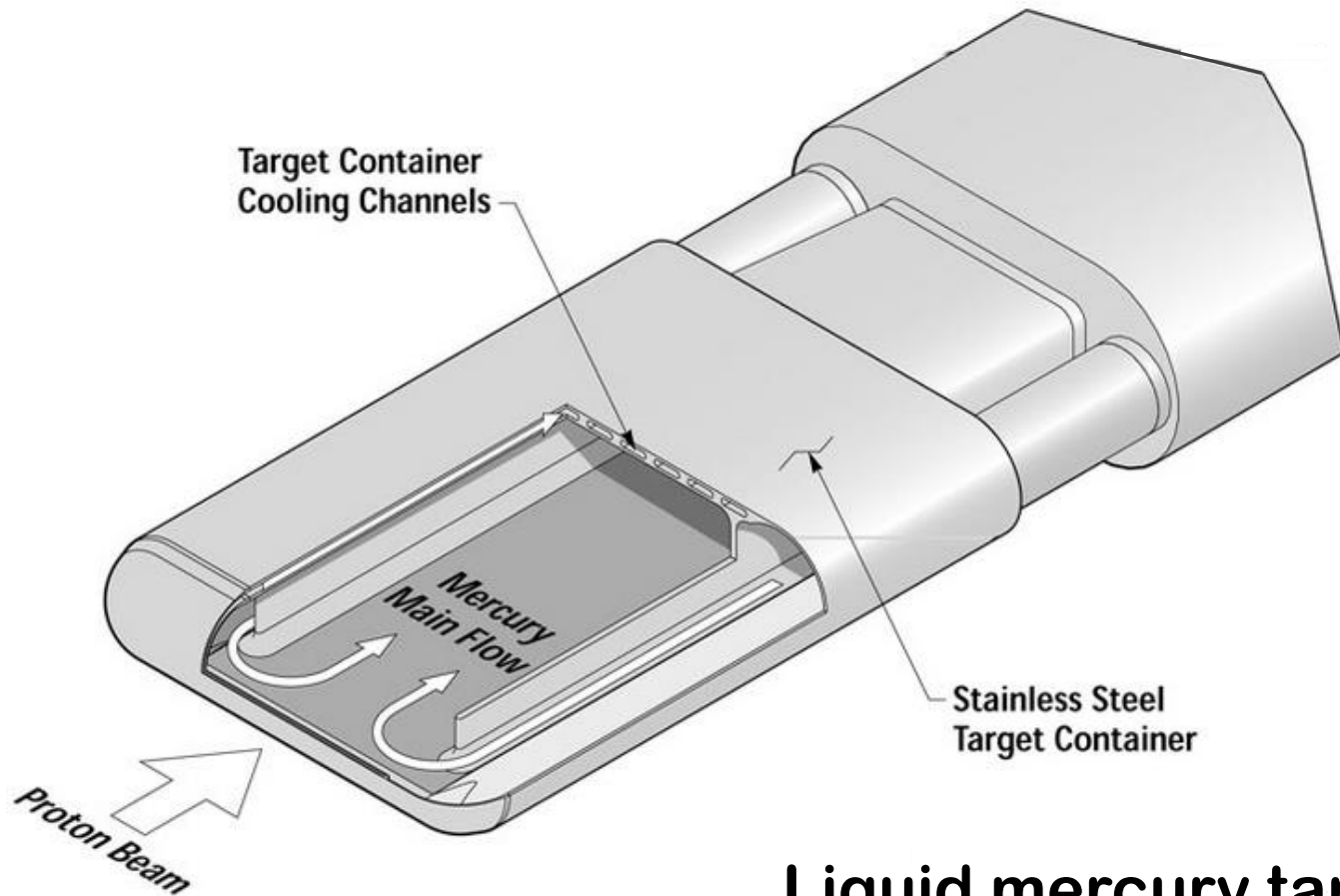
- Replace a mechanical means with a sensory (optical, acoustic, taste or smell) means.
- Use electric, magnetic and electromagnetic fields to interact with the object.
- Change from static to movable fields, from unstructured fields to those having structure.
- Use fields in conjunction with field-activated (e.g. ferromagnetic) particles.



Van der Graaf to Cockroft-Walton generator

29. Pneumatics and hydraulics

- Use gas and liquid parts of an object instead of solid parts (e.g. inflatable, filled with liquids, air cushion, hydrostatic, hydro-reactive).

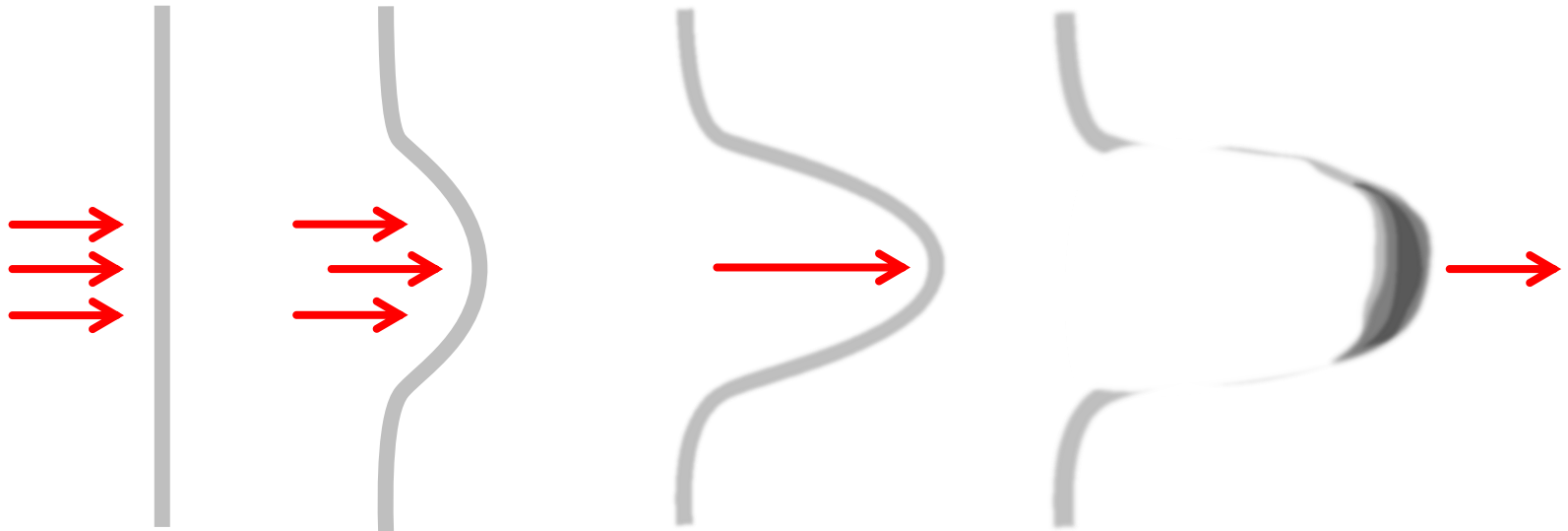


Liquid mercury target

Illustration: ORNL

30. Flexible shells and thin films

- Use flexible shells and thin films instead of three dimensional structures
- Isolate the object from the external environment using flexible shells and thin films.



Light sail laser-plasma ion acceleration

31. Porous materials

Make an object porous or add porous elements (inserts, coatings, etc.).
If an object is already porous, use the pores to introduce a useful substance or function.

Membranes made with ion beams

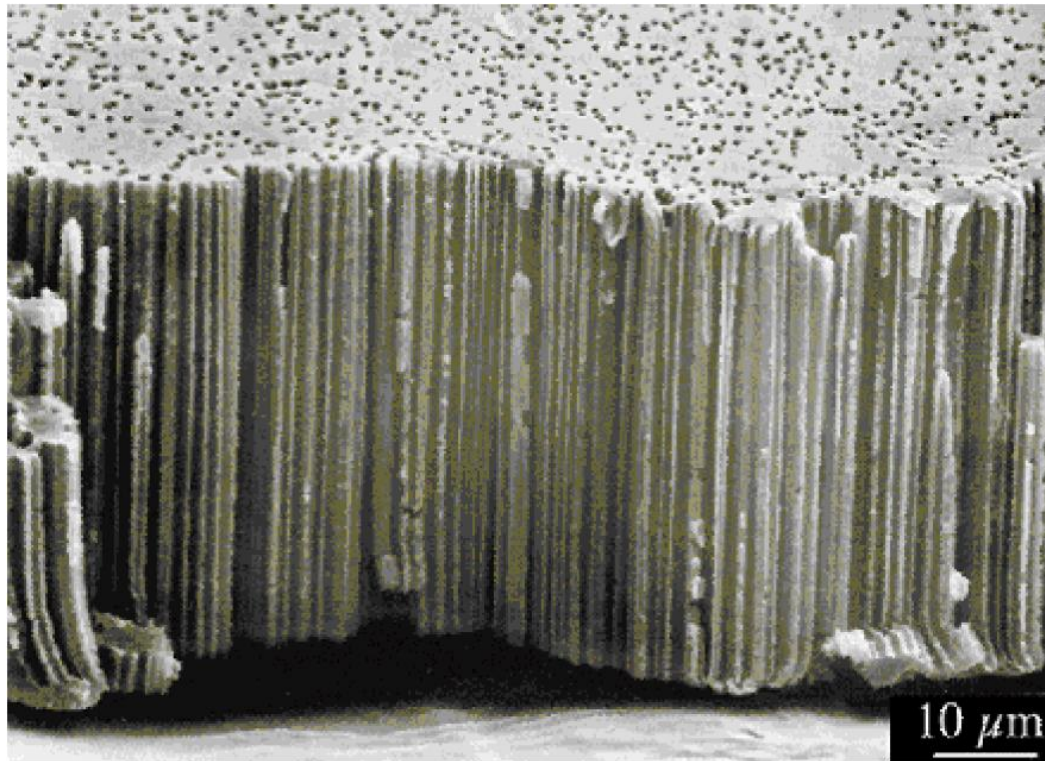
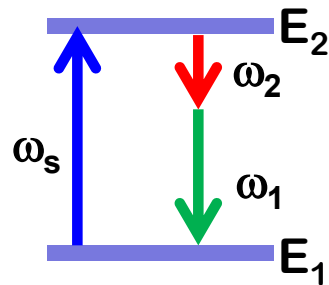


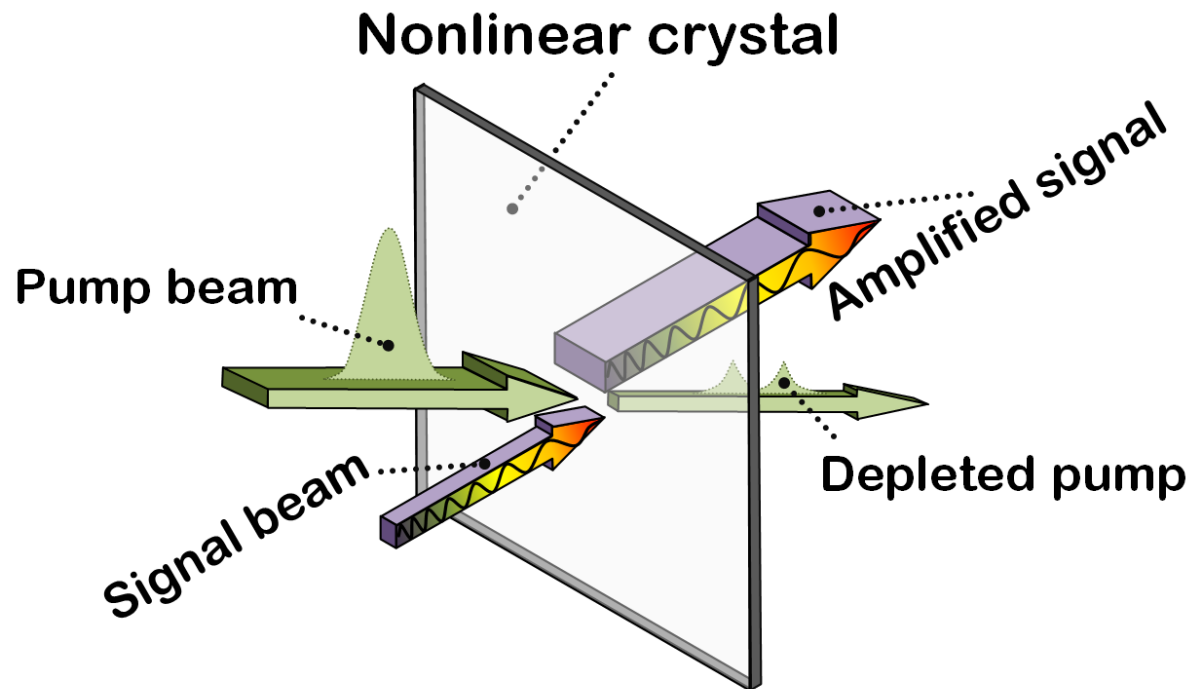
Illustration from “Engines of Discovery: A Century of Particle Accelerators”, A. Sessler and T. Wilson, 2007

32. Color changes

- Change the color of an object or its external environment.
- Change the transparency of an object or its external environment.
- To improve observability of things that are difficult to see, use colored additives or luminescent elements.
- Change the emissivity properties of an object subject to radiant heating.



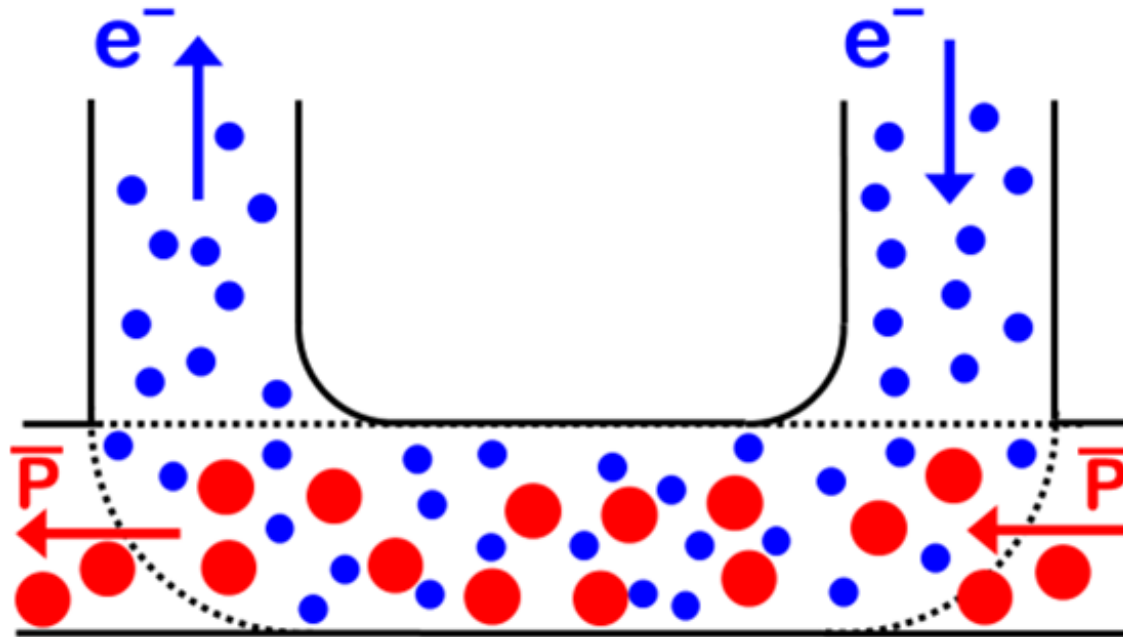
Optical
Parametric
process



Optical Parametric Chirped Pulse Amplification - OPCPA

33. Homogeneity (*Similia similibus curantur*)

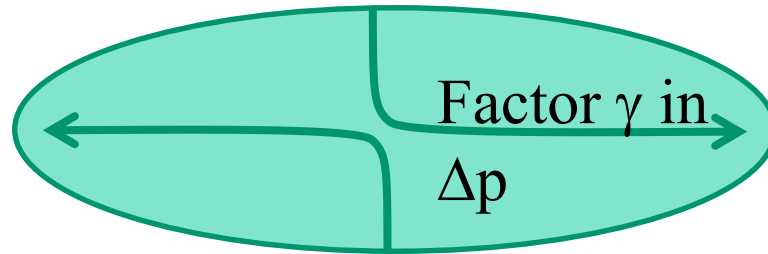
- Make objects interacting with a given object of the same material (or material with identical properties).



Electron cooling

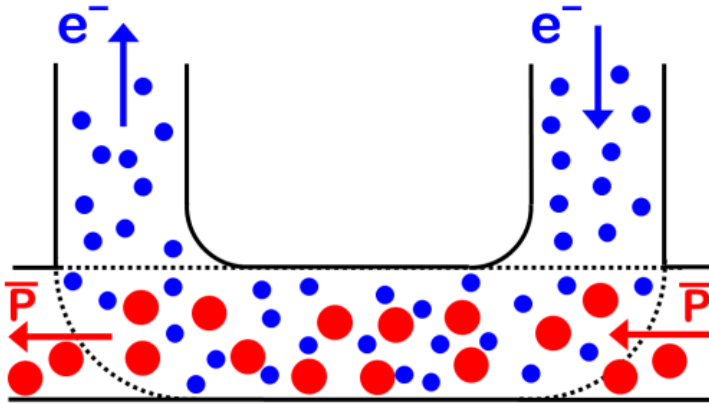
The need for beam cooling – IBS

- Intrabeam Scattering (IBS): Lorentz boosted Coulomb scattering inside bunches

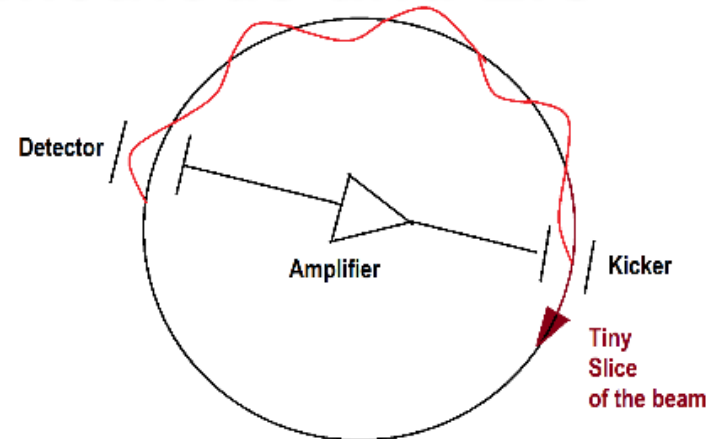


- Higher charge and smaller emittances increase IBS growth rate
 - IBS can be partially mitigated by reducing dispersion and increasing energy spread
- IBS rates for EIC parameters ~2 hour
- Beam cooling methods needed to counteract IBS

Conventional cooling methods and EIC



Electron cooling



Stochastic cooling

Cooling time:

$$\tau \propto \frac{A}{Z^2} \frac{\gamma^2}{4\pi r_p r_e n_e c \eta \Lambda_c} \left(\frac{\gamma \mathcal{E}_n}{\beta_c} + \sigma_p^2 \right)^{3/2}$$

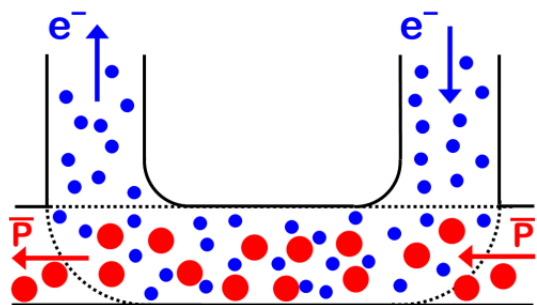
$$\tau \propto C/\sigma_z * N/\Delta F$$

Cooling gets much weaker at higher energy

Cooling gets much weaker for denser bunch

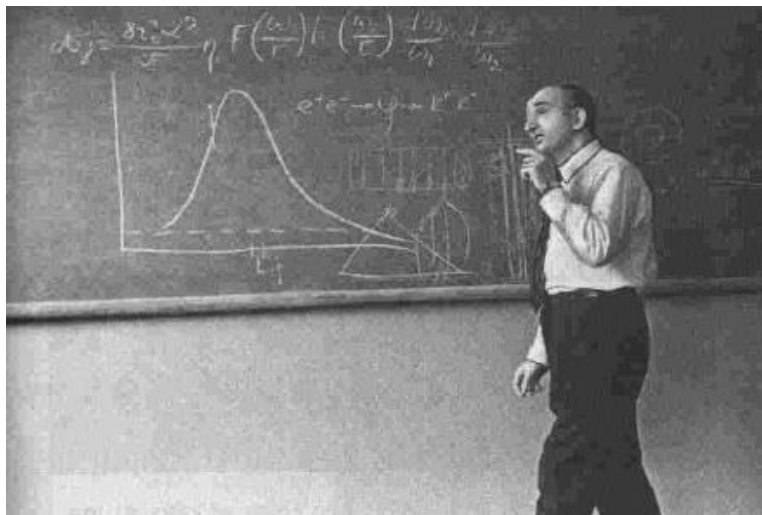
- Either of these methods, if scaled to EIC parameters and stay within technically feasible range, will not provide sufficient cooling – they would be too weak
- For EIC “Strong” hadron cooling is needed – cooling that will provide sufficiently high cooling rate for proton bunches at EIC parameters

Electron Cooling



Electron cooling concept

When electron cooling idea was first presented (1966), the common opinion of the community was – “brilliant idea, but unfortunately non-realistic”



Budker G.I., Effective method of damping particle oscillations at proton antiproton storage rings, Atomic Energy 1967, v.22, №5, p.346



First e-cooler at INP, Novosibirsk, ~1974

Magnetized Electron Cooling

Initial measurements at INP show cooling time of 17 s (protons, 65 MeV)

There was expectation that protons will cool to equilibrium temperature of 1000K (cathode temperature)

$$\frac{MV_i^2}{2} = \frac{mV_e^2}{2} = T_{equilibrium}$$

However, after alignment improvement of the magnetic system, the cooling time became 0.05 s, consistent with electron beam temperature 1K

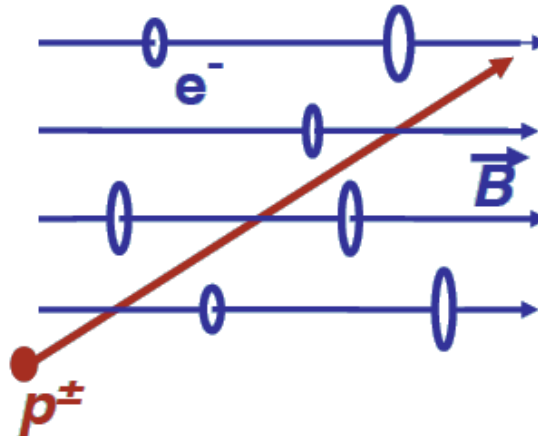
*They called it
“fast electron
cooling”*

Reasons:

1) Longitudinal T of electrons flattens due to acceleration:

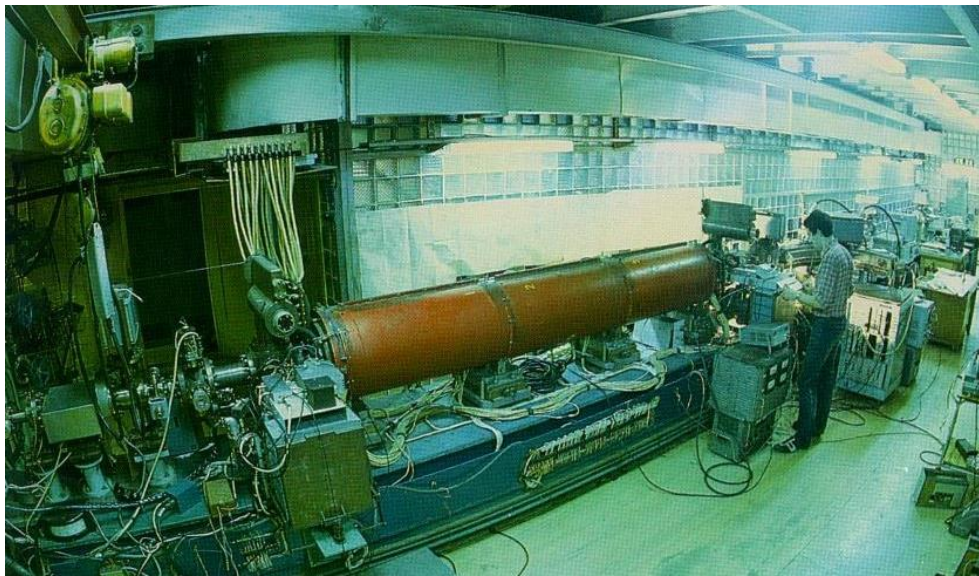
$$T_{||} = \frac{T_{Cathode}^2}{\beta^2 \gamma^2 mc^2}$$

2) Transverse T of electrons does not play any role if Larmor radius $\ll n^{-1/3}$



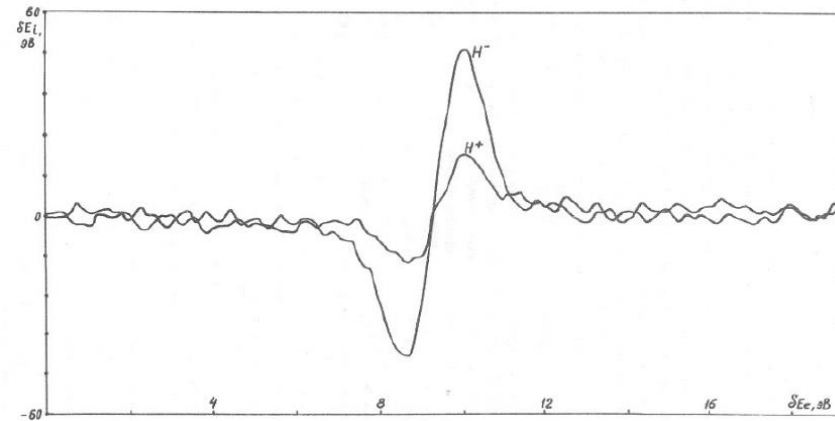
The magnetization effects in electron cooling,
Ya. Derbenev, A. Skrinsky, Rus. Plasma
Physics, v.4 (1978) 492

Single Pass Electron Cooling Experiment



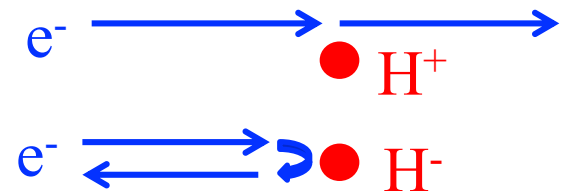
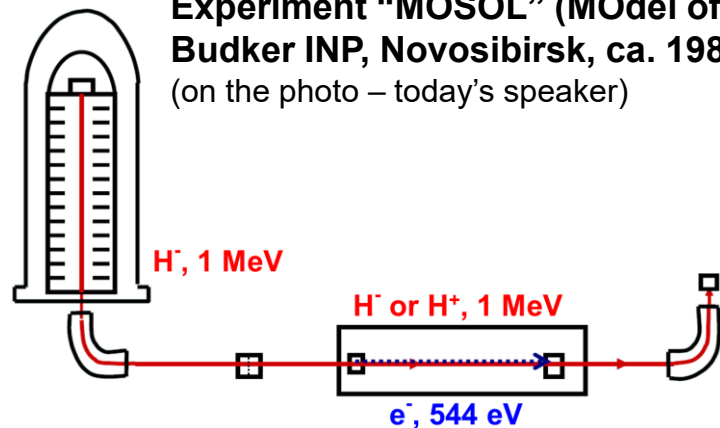
Experiment “MOSOL” (MModel of SOLenoid) – Budker INP, Novosibirsk, ca. 1986
(on the photo – today’s speaker)

Magnetization effects allowed to observe the difference of the e-cooling friction force (which is normally $\sim e^2 Z^2$) on the charge of the particle



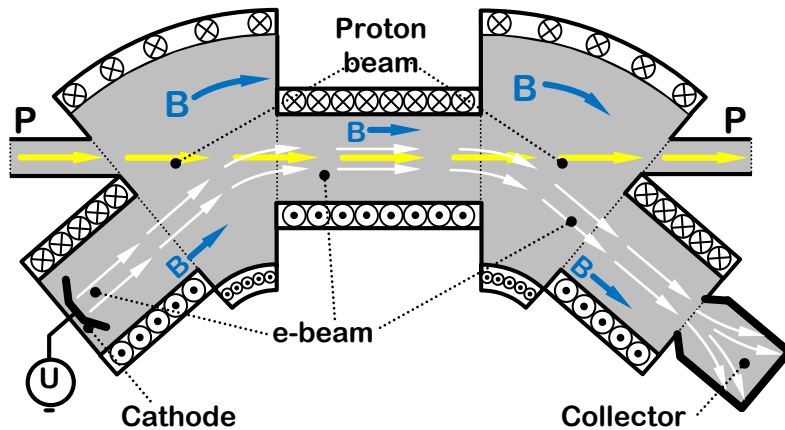
Experiment at MOSOL revealed large difference in cooling force for positive and negative particles

Reason: in magnetized case and low relative velocity, the negative ion reflects the electron, making a large momentum transfer, while for positive ion the electron is first attracted and then pulled back, minimizing momentum transfer



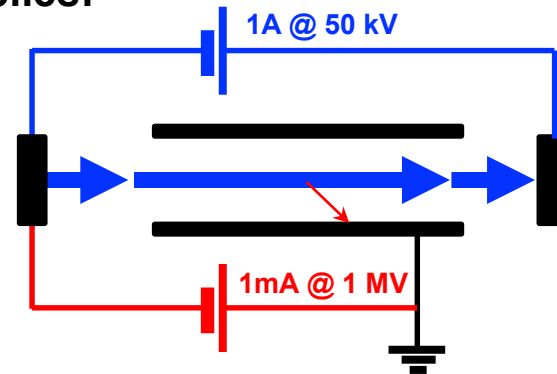
Electron Cooling & Energy Recovery

- Standard electron cooling use energy recovery
 - For example, if we need 1A @ 1MeV electron beam*, it does not mean we need 1 MW power supply



Typical scheme of a standard electron cooler for low energy range (~several tens MeV of p energy)

- Typical arrangements of e-cooler power supplies:

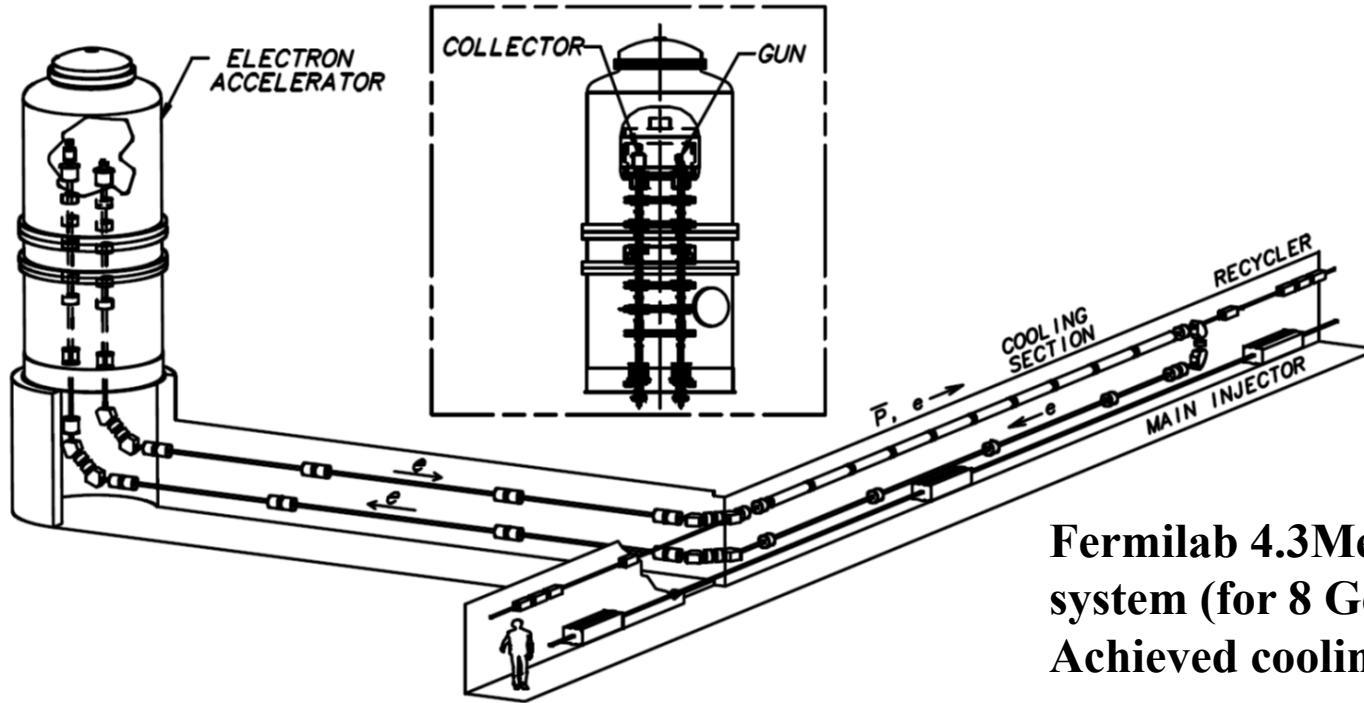


- Losses of 1A e-beam due to interaction with p-beam or scattering are low
- Thus, power of 1MV power supply is defined by e-beam losses and can be much lower than 1MW, just 1kW in example above

* Numbers are for illustration only

Taking Electron Colling to higher energy

- Energy recovery is even more important for high energy electron cooling



Fermilab 4.3 MeV electron cooling system (for 8 GeV antiprotons)
Achieved cooling times ~ 0.5 hours

- The electron cooling time has a very unfavourable beam energy scaling $\sim \gamma^{2.5}$
- Mitigating scaling dependence by a) increasing cooling section length; b) higher electron current – has practical limits
- For 41-257 GeV energy of EIC proton beam – standard electron cooling would be extremely challenging

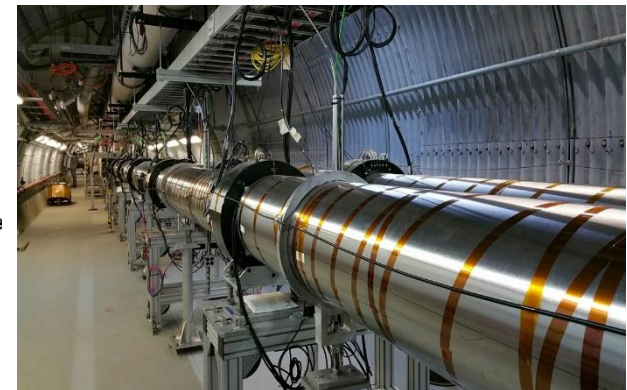
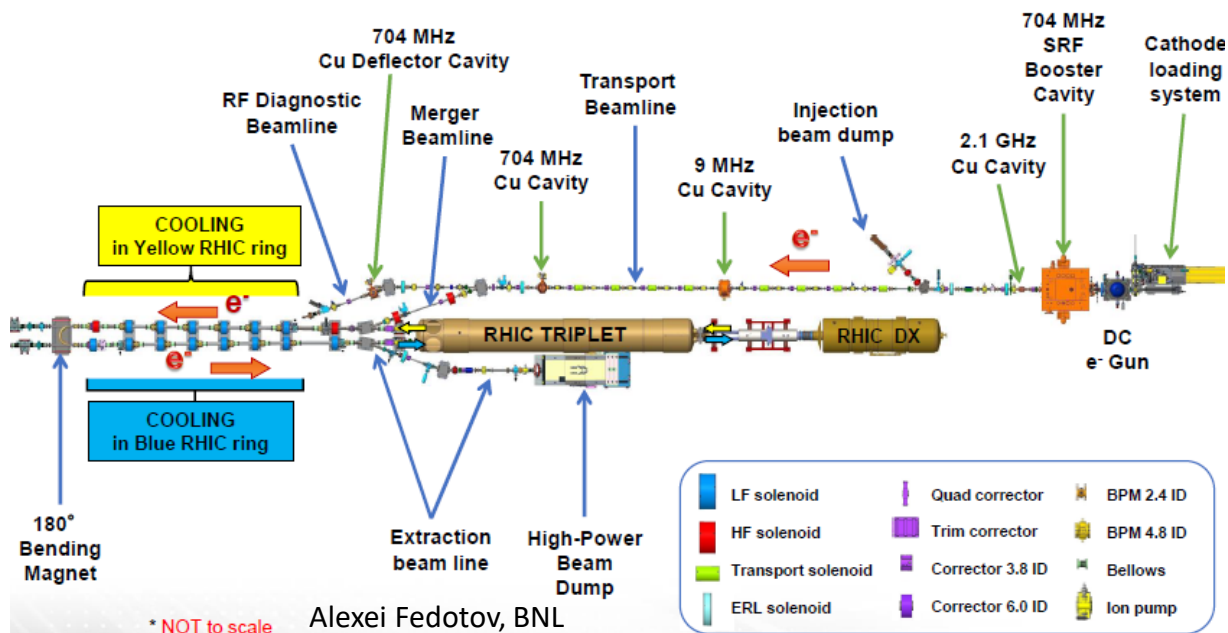
Getting electron cooling to higher energy

Low-Energy RHIC electron Cooler (LEReC) at BNL:

- First e-cooler based on the RF acceleration of e-beam (of up to 2.6 MeV energy)
- **Observation of first cooling using bunched electron beam on April 5, 2019**
- LEReC will be used in RHIC Beam Energy Scan II for Low energy ($\sqrt{s_{NN}} = 7.7, 9.1, 11.5, 14.5, 19.6$ GeV) Au+Au runs using electron cooling to increase luminosity
- Cooling using bunched electron beam produced with RF acceleration is new, and opens the possibility of electron cooling at high beam energies

LEReC Accelerator

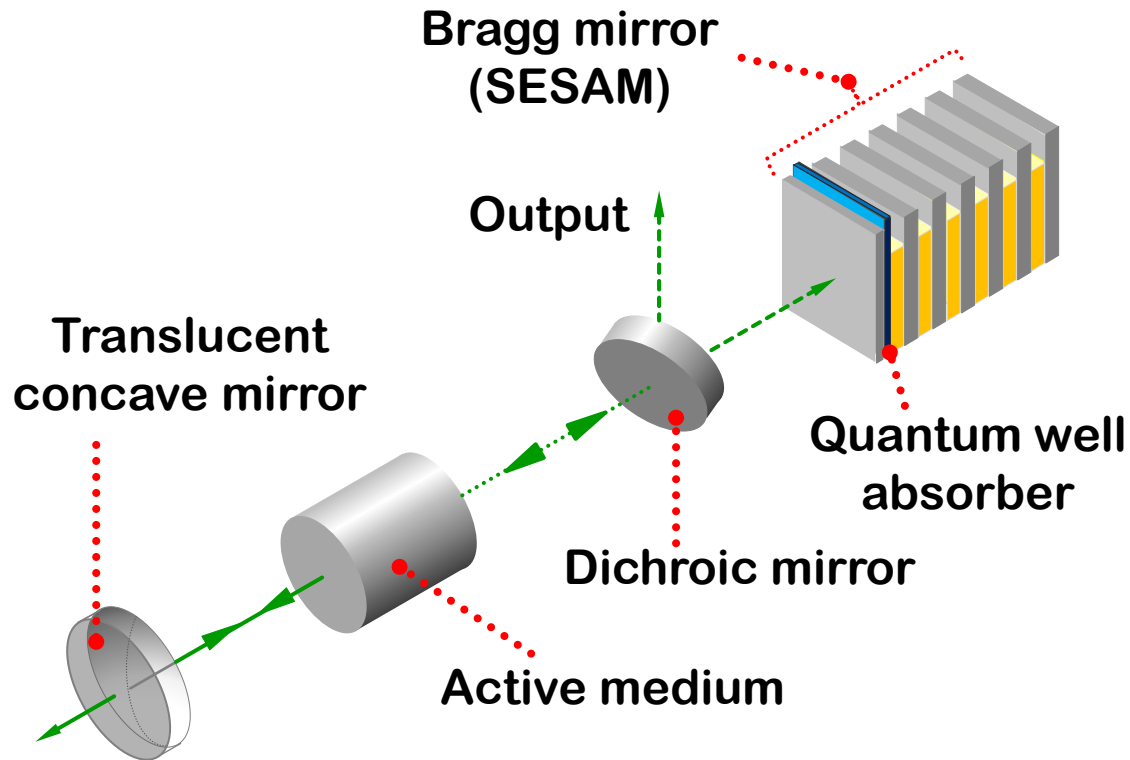
(100 meters of beamlines with the DC Gun, high-power fiber laser, 5 RF systems, including one SRF, many magnets and instrumentation)



LEReC approach can be used for EIC as injection energy pre-cooler. However, at collision energy enhanced/strong cooling mechanism is needed.

34. Discarding and recovering

- Make portions of an object that have fulfilled their functions go away (discard by dissolving, evaporating, etc.) or modify these directly during operation.
- Conversely, restore consumable parts of an object directly in operation.



Semiconductor Saturable Absorber Mirror - SESAM

35. Parameter changes

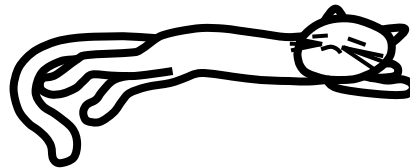
- Change an object's physical state (e.g. to a gas, liquid, or solid.)
 - Change the concentration or consistency.
 - Change the degree of flexibility.
 - Change the temperature.



15° C



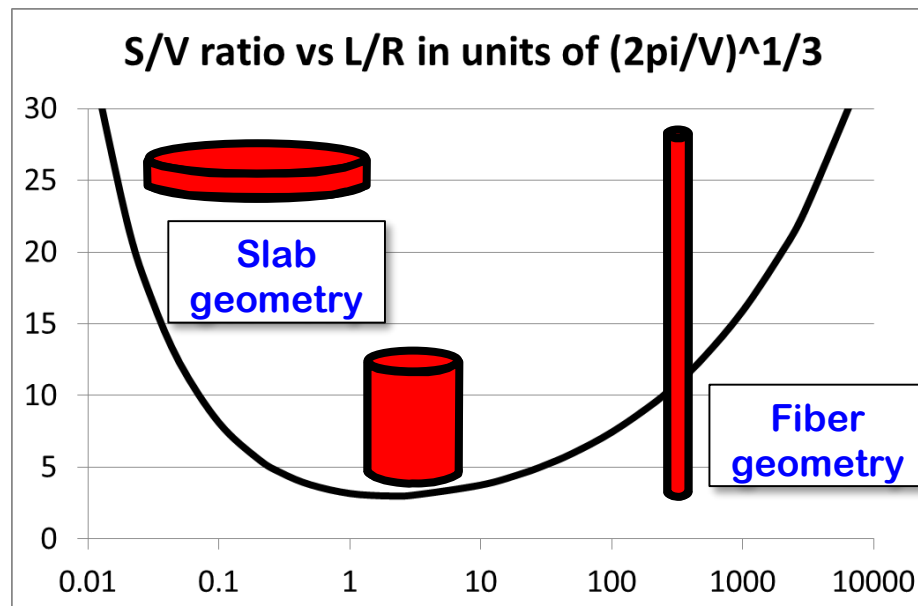
20° C



25° C



40° C

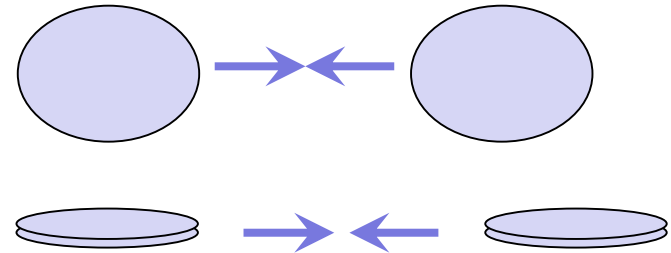


Fiber
lasers

So, we saw that

colliding beams...

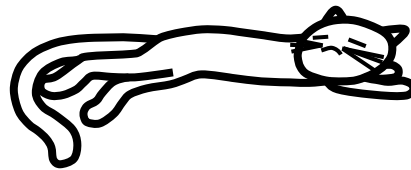
cats...



15° C



22° C



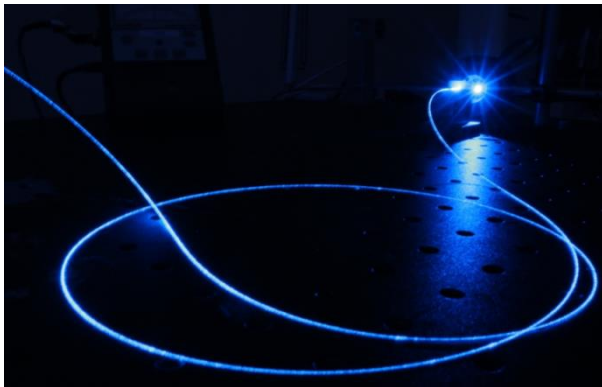
25° C



40° C

The cat intuitively knows the inventive principle of surface to volume ratio

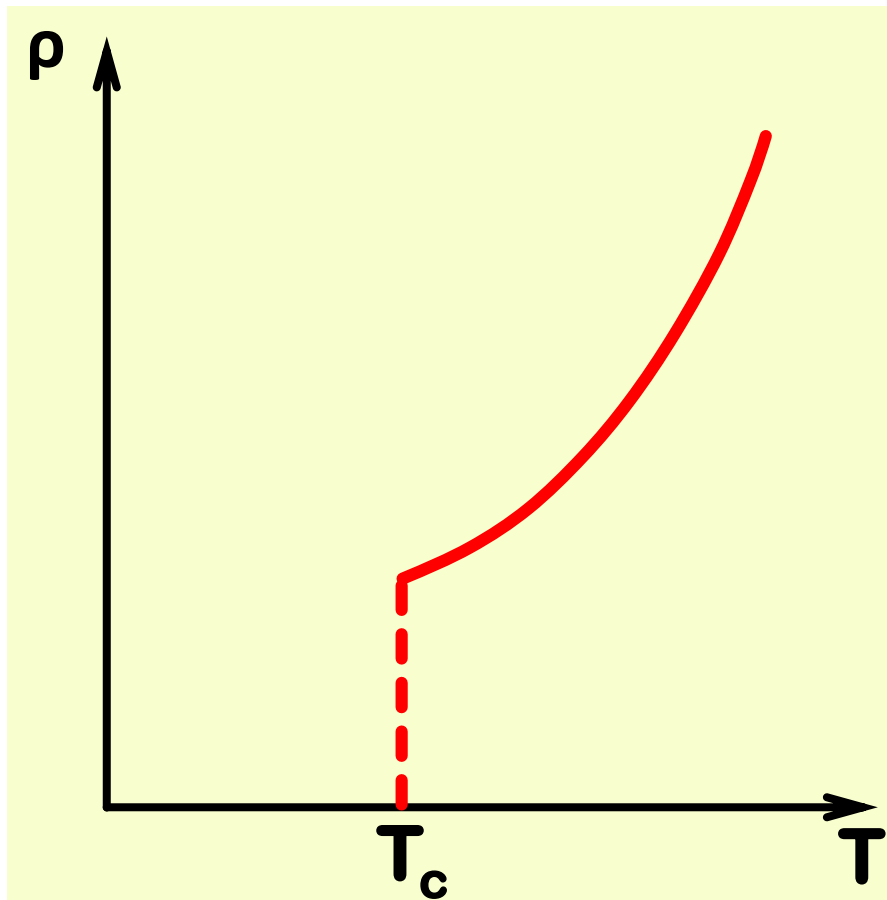
and fibre lasers...



***...are connected via
TRIZ inventive
principles!***

36. Phase transitions

- Use phenomena occurring during phase transitions (e.g. volume changes, loss or absorption of heat, etc.).

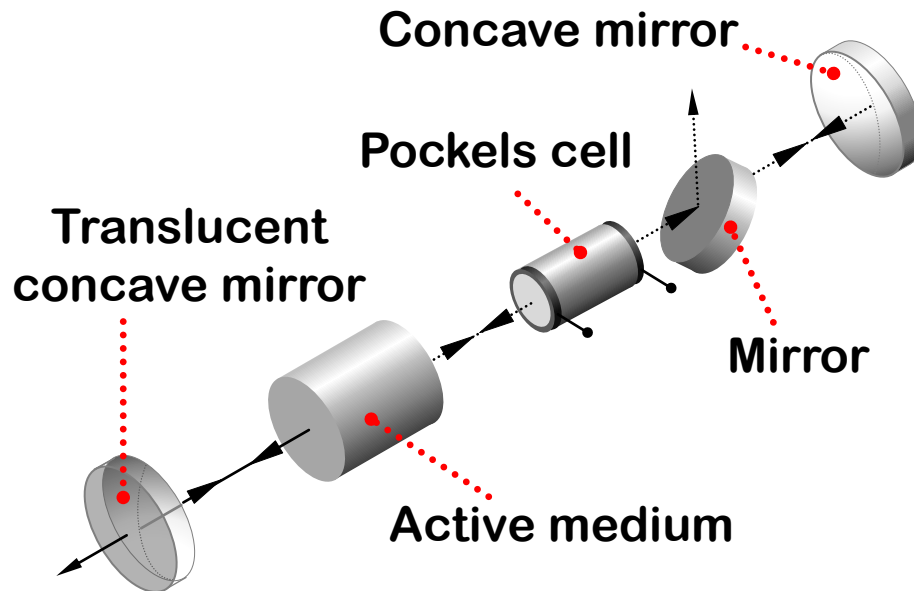


Phase transition

Superconductivity

37. Thermal or electrical expansion or property change

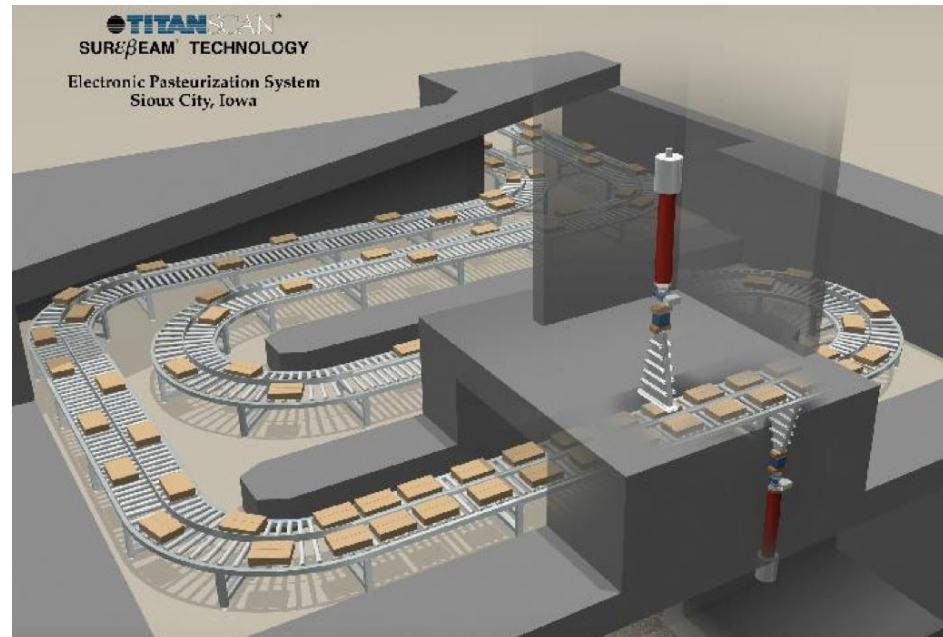
- Use thermal or electrical expansion (or contraction) or other property change of materials.
- If thermal or electrical expansion (property change) is being used, use multiple materials with different coefficients of thermal expansion (property change).



**Electro-optic effect —
dependence of optical
properties
of objects such as
absorption or refraction
(Pockels effect)
on the applied electric
field**

38. Strong oxidants

- Replace common air with oxygen-enriched air.
 - Replace enriched air with pure oxygen.
 - Expose air or oxygen to ionizing radiation.
 - Use ionized oxygen.
- Replace ozonized (or ionized) oxygen with ozone.



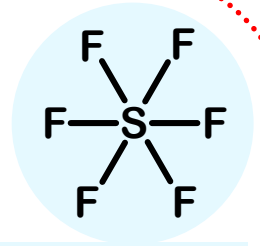
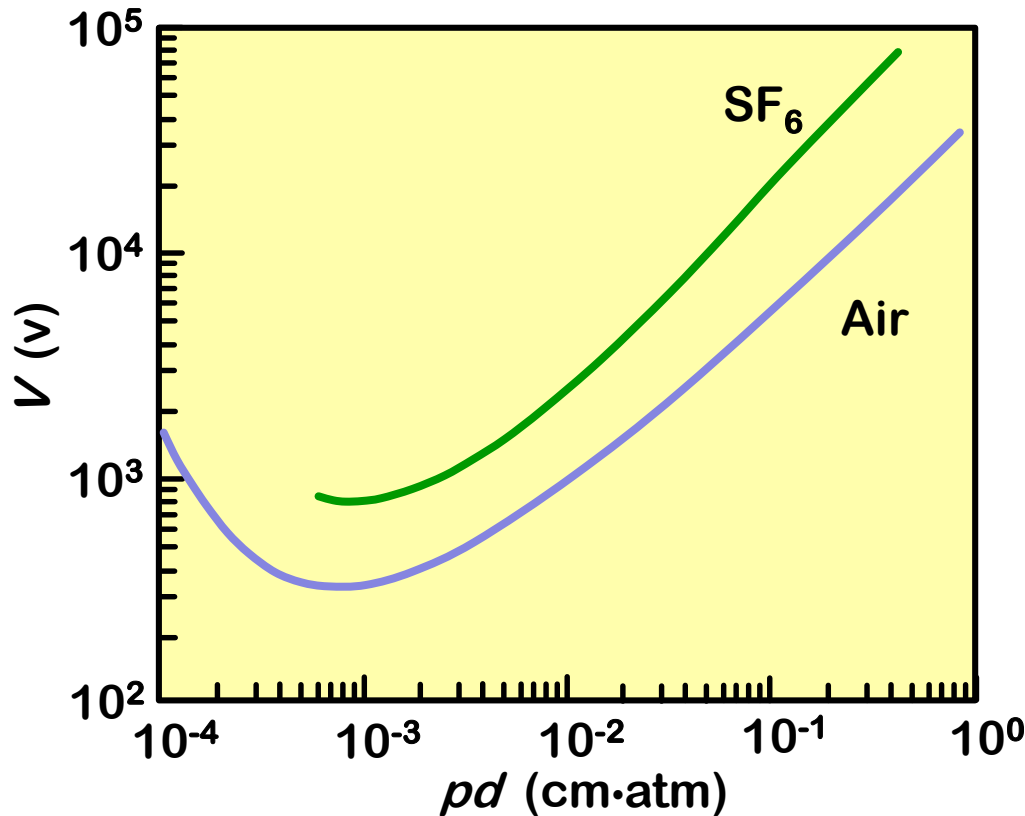
TITANSCAN*
SUR&BETAM TECHNOLOGY
Electronic Pasteurization System
Sioux City, Iowa

**Irradiation of food
for sterilisation**

Illustration: TITANSCAN

39. Inert atmosphere

- Replace a normal environment with an inert one.
- Add neutral parts, or inert additives to an object.



Sulfur hexafluoride (SF_6 or Elegas) is a colorless non-flammable gas with excellent electric insulating and arc-quenching capacity. It is widely used in the fields of electric, laser, medical, meteorological, freezing, fire-fighting, chemical, military, space aviation, nonferrous metallurgy and physical research areas.

40. Composite materials

- Change from uniform to composite (multiple) materials.

Ion beam surface treatment

Hardening an artificial knee joint using ion implantation

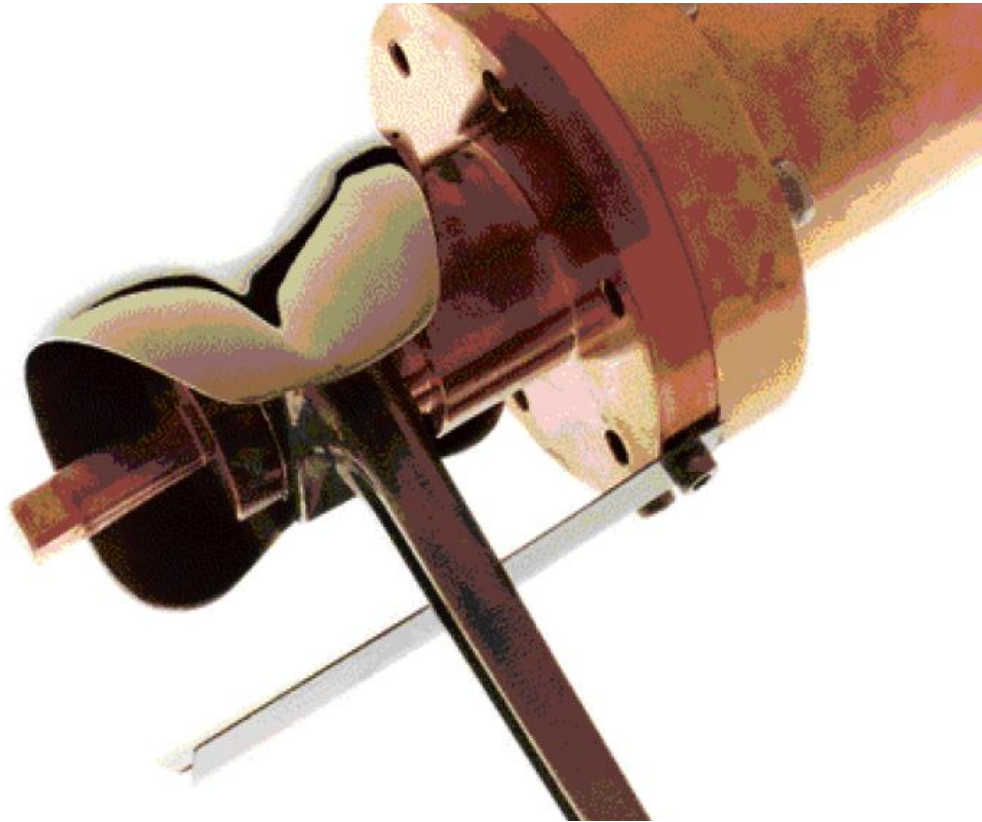


Illustration from “Engines of Discovery: A Century of Particle Accelerators”
A. Sessler and T. Wilson, 2007



“The greater danger for most of us lies not in setting our aim too high and falling short; but in setting our aim too low, and achieving our mark”

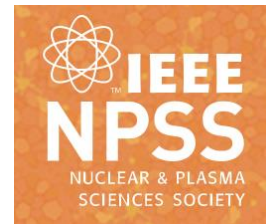
Michelangelo

In combination with the art of estimations, TRIZ can be very useful for university education and research

**As an inspiration, as a very efficient toolbox, as a way to connect different disciplines, as a new way to see the world –
Breakthrough By Design approach**

Acknowledgements

- **Jefferson Lab**
 - Continuing support of graduate educational program
- **Old Dominion University**
 - Materials presented here will be used as part of VITA graduate and undergraduate courses
 - VITA – Virginia Innovative Traineeship in Accelerators
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- **NA-PAC 2022**
 - For the opportunity to present this tutorial
- **IEEE NPSS Distinguished Lecture Program**
 - <https://ieee-npss.org/distinguished-lecturers/>
- **My co-author for the 1st and oncoming 2nd edition of the book “Unifying Physics of Accelerators, Lasers and Plasma” Elena Seraia**
 - <https://www.unifyingphysics.com/>
- **CERN for “eBook for all!” program that enabled conversion of the 1st edition of “Unifying Physics...” to open access**



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- Particle Accelerator Science & Technology (*Particle Accelerator Conference*)
- Plasma Science & Applications (*Intl. Conference on Plasma Science*)
- Pulsed Power Science & Technology (*Pulsed Power Conference*)
- Radiation Effects (*Nuclear and Space Radiation Effects Conference*)

Look for the IEEE Membership booth!

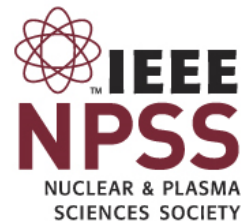
NPSS Journals

IEEE Transactions on Nuclear Science

IEEE Transactions on Plasma Science

IEEE Transactions on Medical Imaging

IEEE Transactions on Radiation and Plasma Medical Sciences



Tutorial materials

- Slides are available at
 - <https://www.unifyingphysics.com/>



- See section Resources
- You can also access the 1st edition of the book which is now Open Access

Thank you for your attention!

*And thanks to all colleagues for materials
used in these slides*